MN150012EN

PXM 3000 User Manual









Powering Business Worldwide

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Please read this manual carefully before installation, operation, and maintenance of PXM 3000 meter.

The following symbols in this manual and on PXM 3000 meters are used to provide warning of danger or risk during the installation and operation of the meters.

Safety Precautions

All safety codes, safety standards, and/or regulations must be strictly observed in the installation, operation, and maintenance of this device.

⚠ WARNING

The warnings and cautions included as part of the procedural steps in this document are for personnel safety and protection of equipment from damage. An example of a typical warning call-out is shown above. This will help to ensure that personnel are alert to warnings that may appear throughout the document. In addition, cautions are all upper case and boldfaced as shown below.

⚠ WARNING

Completely read and understand the material presented in this document before attempting installation, operation, or application of the equipment. Only qualified persons should be permitted to perform any work associated with the equipment. The wiring, installation and application use instructions presented in this document must be followed precisely. Failure to do so could cause permanent equipment damage, bodily injury, or death.

⚠ WARNING

Do not attempt to install or perform maintenance on equipment while it is energized. Death, severe personal injury, or substantial property damage can result from contact with energized equipment. Always verify that no voltage is present before proceeding with the task, and always follow generally accepted safety procedures.

Eaton is not liable for the misapplication or misinstallation of its products.

Congratulations!

You have purchased an advanced, versatile, multifunction power meter. This meter can work as a remote terminal unit (RTU) that contributes to your system's stability and reliability by providing real-time power quality monitoring and analysis.

When you open the package, you will find the following items.

	QTY
1. PXM 3000 Meter	1
2. Terminal Blocks	3
3. Installation clips	4
4. Rubber Gasket	1
5. Product Disk (Manual, Warranty, Software)	1
6. Additional documentation (Quick Setup Guide, Calibration Certificate)	2

To avoid complications, please read this manual carefully before installation and operation of the PXM 3000 meter.

1. Introduction

1.1 Meter Overview

Powerful Multifunction Power Meter

The PXM 3000 multifunction digital power meter is designed using modern MCU and DSP technology. It integrates three-phase energy measurement and display, energy accumulation, power quality analysis, malfunction alarms, data logging, and network communication. A vivid LCD display with large characters and time-of-use programmable backlight provides a clear real-time data readout.

Ideal for Electric Automation SCADA Systems

The PXM 3000 meter is the ideal choice for replacing traditional, analog electric meters. In addition to providing clear real-time readings on the front of the meter, it can also be used as a remote terminal unit (RTU) for monitoring and controlling for a SCADA system. Users can access all measurement parameters via the standard RS-485 communication port.

Energy Management

The PXM 3000 meter is able to measure bidirectional, four quadrants kWh and kvarh. It provides maximum/minimum records for power usage and power demand parameters. All power and energy parameters can be viewed remotely via software in order to easily monitor various parameters. In addition, measurement tables can be viewed from the free PXM 3000 software.

Remote Power Control

The PXM 3000 is designed for measuring and monitoring power quality parameters. Since different I/O modules can be added to the meter, this expands the capabilities and provides a very flexible platform for using the meter as a distributed RTU, for metering, monitoring, and remote controlling, all in one unit.

Power Quality Analysis

Utilizing digital signal processing (DSP) technology, the PXM 3000 meter provides high accuracy power quality analysis. The meter continuously updates metering results and allows the users to access the meter online to monitor parameters such as voltage and current THD, harmonics up to the 31st, voltage crest factor, current K factor, and voltage and current unbalance factor, etc.

Data Logging

The PXM 3000 meters contain eight megabytes of onboard memory for data logging and historical trending. Since each meter contains a real-time clock, all events and logged data will be time stamped.

Time of Use (TOU - PXM 3000)

The user can assign up to four different tariffs (sharp, peak, valley, and normal) to different time periods within a day according to the billing requirements. The meter will calculate and accumulate energy to different tariffs, according to the meter's internal clock timing and TOU settings.

Power Quality Event Logging

When a power quality event happens, such as voltage sag and swell, etc., the PXM 3000 will record the timestamp and the triggering condition of the event. It can save up to 50,000 power quality events.

Waveform Capture

PXM 3000 contains another 8 megabytes of on board memory for power quality event logging and waveform capture. PXM 3000 can record 100 groups of voltage and current waveforms. It logs at 64 points per cycle. (200 groups of voltage and current waveforms in case of 64 points per cycle and 100 groups of voltage and current waveform in case of 128 points per cycle) It provides the waveform record of 10 cycles before and after the triggering point. It also supports a settable triggering condition.

1.2 Areas of Application

Power Distribution Automation ..... Electric Switch Gear and Control Panels

Industry Automation ..... Building Automation

Energy Management Systems ..... Marine Applications

Renewable Energy

1.3 Functionality

Multifunction

PXM 3000 meters provide powerful data collecting and processing functions. In addition to measuring various parameters, the meter is able to perform demand metering, harmonic analysis, max/min statistic recording, out-of-limit alarms, energy accumulation, and data logging.

High Accuracy

Accuracy of Voltage and Current is 0.1%, True-RMS.

Accuracy of Power and Energy is 0.2%, while monitoring all four quadrants.

Compact and Easy to Install

This meter can be installed into a standard ANSI C39.1
(4" Round) or an IEC 92 mm DIN (Square) cut out. With the 51 mm (2") depth after mounting, the PXM 3000 meter can be installed in a small cabinet. Note that I/O modules add 19.5 mm (0.77in) depth (see Figure 55). Maximum of three modules can be installed. Mounting clips are used for easy installation and removal.

Easy to Use

All metering data and setting parameters can be accessed using the front panel keys or via the communication port. Settings parameters are stored in the EEPROM so that content will be preserved when the meter is powered off.

Multiple Wiring Modes

The PXM 3000 series meter can be used in medium voltage (with PTs), low voltage, three-phase three-wire, three-phase four-wire, and single-phase systems using different wiring mode settings.

High Safety, High Reliability

PXM 3000 series meter was designed according to industrial standards. It can run reliably under high power disturbance conditions. This meter has been fully tested for EMC and safety compliance in accordance with UL and IEC standards.

Table 1. Function Comparison of PXM 3000 Series Meters. • **Function;** ⊙ **Option;** **Blank NA**

Category		Item	Parameters	PXM 3000
Metering	Real-time Metering	Phase Voltage	V1, V2, V3, VInavg	•
		Line Voltage	V12, V23, V31, VIlavg	•
		Current	I1, I2, I3, In, Iavg	•
		Power	P1, P2, P3, Psum	•
		Reactive Power	Q1, Q2, Q3, Qsum	•
		Apparent Power	S1, S2, S3, Ssum	•
		Power Factor	PF1, PF2, PF3, PF	•
		Frequency	V1	•
		Load Features	Load Features	•
		Four Quadrant Powers	Four Quadrant Powers	•
	Energy & Demand	Energy	Ep_imp, Ep_exp, Ep_total, Ep_net	•
		Reactive Energy	Eq_imp, Eq_exp, Eq_total, Eq_net	•
		Apparent Energy	Es	•
		Demand	Dmd_P, Dmd_Q, Dmd_S, Dmd_I1, Dmd_I2, Dmd_I3	•
TOU	Time of Use	Energy/Max Demand	TOU, 4 Tariffs, 12 Seasons, 14 Schedules	•
	Daylight Saving Time	Two Formats Adjust	Month/Day/Hour/Minute; Month/Week/First few weeks/ Hour/Minute	•
Monitoring	Waveform Capture (Function not available for 400 Hz)	Voltage and Current Waveform	Trigger, Manual, DI change, Sag/Dips, Swell, Over Current	128 samples per cycle
	Power Quality Function not available for 400 Hz	Voltage Unbalance Factor	U_unbl	•
		Current Unbalance Factor	I_unbl	•
		Voltage THD	THD_V1,THD_V2,THD_V3, THD_Vavg	•
		Current THD	THD_I1, THD_I2, THD_I, THD_Iavg	•
		Individual Harmonics	Harmonics 2nd to 31st	•
		Voltage Crest Factor	Crest Factor	•
		TIF	THFF	•
		Current K factor	K Factor	•
	Statistics	MAX with Time Stamp MIN with Time Stamp	Each Phase of V & I; Total of P, Q, S,PF & F; Demand of P,Q & S; Each Phase THD of V & I; Unbalance factor of V & I	•

Note: This manual uses U and V interchangeably for voltage.

(continued)

Table 1. Function Comparison of PXM 3000 Series Meters. • **Function;** ⊙ **Option;** **Blank NA**

Category		Item	Parameters	PXM 3000
Others	Alarm	Over/Under Limit Alarm	V,I,P,Q,S,PF,V_THD & I_THD Each Phase and Total or Average; Unbalance Factor of V & I; Load Type; Analog Input of Each Channel	•
	Power Quality Event Logging (Function not available for 400 Hz)	SAG/DIPS,SWELL	Voltage	•
	Data Logging	Data Logging 1 Data Logging 2 Data Logging 3	F, V1/2/3/lnavg, V12/23/13/lavg, I1/2/3/n/avg, P1/2/3/sum, Q1/2/3/sum, S1/2/3/sum, PF1/2/3, PF, U_unbl, I_unbl, Load Type, Ep_imp, Ep_exp, Ep_total, Ep_net, Eq_imp, Eq_exp, Eq_total, Eq_net, Es, THD_V1/2/3/avg, THD_I1/2/3/avg, Harmonics 2nd to 63rd, Crest Factor, THFF, K Factor, Sequence and Phase Angles, DI Counter, AI, AO, Dmd P/Q/S, Dmd I1/2/3	•
	Onboard Memory Size	Memory	Bytes	1.5GB
	Communication	RS485 Port,Half Duplex, Optical Isolated	Modbus®-RTU Protocol	•
	Time	Real Time Clock	Year, Month, Date, Hour, Minute, Second	•
	400Hz Type:	Only Supports Full-wave (True RMS) Energy, Supports 2nd ~ 15th Individual Harmonics.		•
Option Module	I/O Option	Switch Status (DI)	Digital Input (Wet)	•
		Power Supply for DI	24 Vdc	•
		Relay Output (RO)	NO, Form A	•
		Digital Output (DO)	Photo-MOS	•
		Pulse Output (PO)	By using DO	•
		Analog Input (AI)	0(4)~20mA, 0(1)~5V	•
		Analog Output (AO)	0(4)~20mA, 0(1)~5V	•
	Communication	RS485 Module	Modbus-RTU Protocol	•
		RS485 Module	Additional Modbus RTU	•

Note: This manual uses U and V interchangeably for voltage.

2. Installation

⚠ CAUTION

Installation of the meter must only be performed by qualified personnel, who follow standard safety precautions through the installation procedures. The personnel should have appropriate training and experience with high voltage devices. Appropriate safety gloves, safety glasses, and protective clothing are recommended. Follow NFPA 70E guidelines for protective equipment.

The meter and its I/O output channels are NOT designed as primary protection devices and shall NOT be used as primary circuit protection or in an energy-limiting capacity. The meter and its I/O output channels can only be used as secondary protection. AVOID using the meter under situations where failure of the meter cause injury or death. AVOID using the meter for any application where risk of fire may occur.

⚠ WARNING

During normal operation, hazardous voltages will be present. This includes terminals and any connected current transformers (CTs) and potential transformers (PTs), input and output (I/O) modules, and their circuits. All primary and secondary circuits can, at times, produce lethal voltages and currents.

⚠ WARNING

Do not attempt to install or perform maintenance on equipment while it is energized. Death, severe personal injury, or substantial property damage can result from contact with energized equipment. Always verify that no voltage is present before proceeding with the task, and always follow generally accepted safety procedures.

Eaton is not liable for the misapplication or misinstallation of its products.

⚠ WARNING

All meter terminals should be inaccessible after installation.

DO NOT perform Dielectric (HIPOT) test to any inputs, outputs, or communication terminals. High voltage testing may damage electronic components of the meter.

Applying more than the maximum voltage to the meter and/or its modules will permanently damage the product. Please refer to the specifications for all devices before applying voltages.

When removing the meter for service, use shorting blocks and fuses for voltage leads and power supply to prevent hazardous voltage conditions or damage to the CTs. CT grounding is optional.

NOTE

The PXM3000 components are maintenance free once properly commissioned. There are no user serviceable components or features.

Cleaning of the various PXM3000 device housings should only be done with power and mains disconnected. A clean dry rag can be used to remove dust. No liquids should be used.

⚠ CAUTION

If the equipment is used in a manner not specified by the manufacturer, the protection provided by the equipment may be impaired.

⚠ CAUTION

No preventive maintenance or inspection is required for safety. Any maintenance or repair should be performed by the factory.

⚠ WARNING

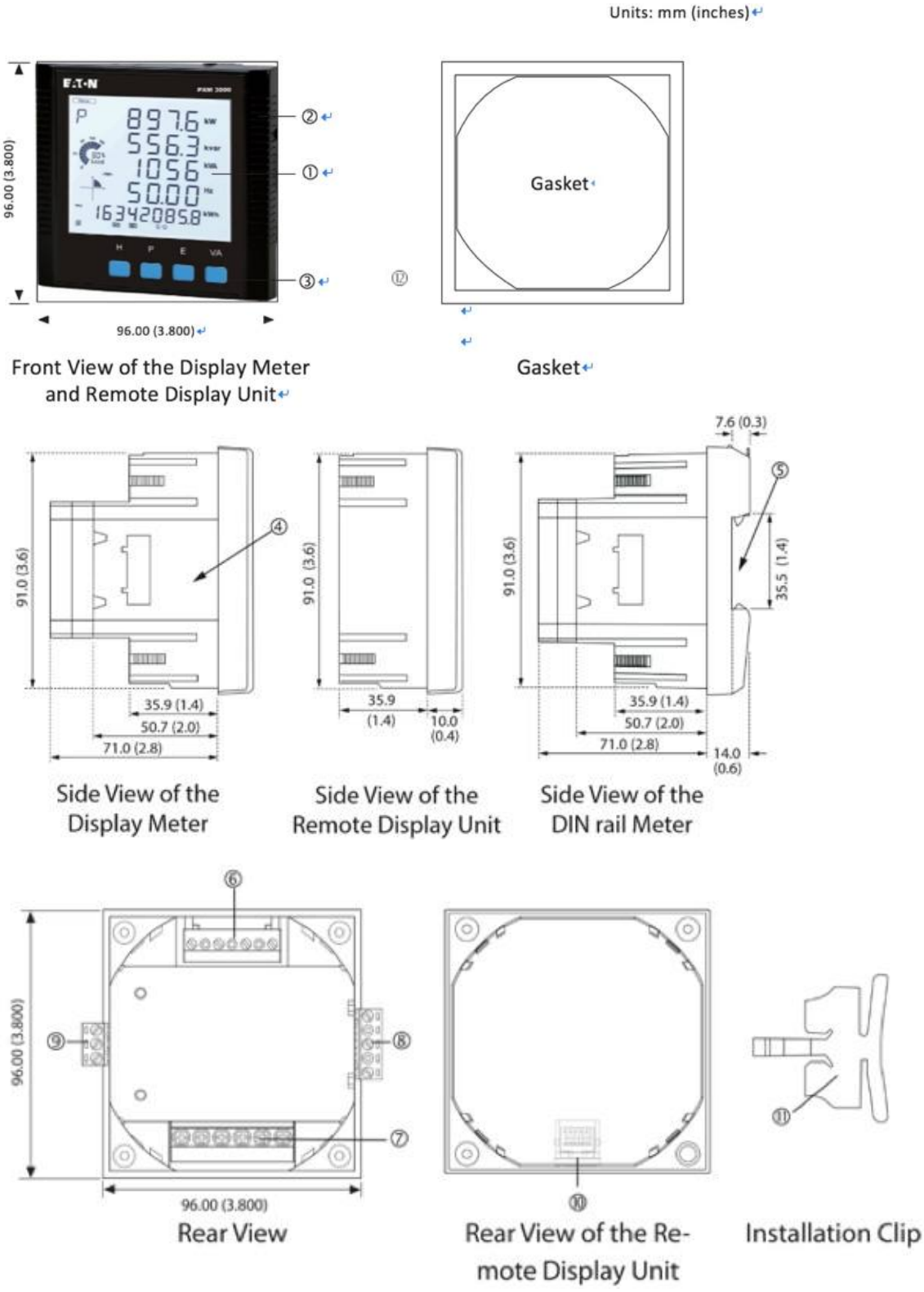
DISCONNECT DEVICE: The following part is considered the equipment disconnect device.

A switch or circuit-breaker shall be included in the installation. The switch shall be in close proximity to the equipment and within easy reach of the operator. The switch shall be marked as the disconnecting device for the equipment.

The installation method is introduced in this chapter. Please read this chapter carefully before beginning installation.

2.1 Appearance and Dimensions

Figure 1. Appearance and Dimensions of the PXM 3000 Meter.



Note: Unit: mm (inches).

Note:

Table 2. Part Name of the PXM 3000 Meter.

Part Name	Description
① LCD Display	Large, bright white backlight LCD display.
② Front Casing	Visible portion (for display and control) after mounting onto a panel.
③ Key	Four keys are used to interact with the meter.
④ Enclosure	The PXM 3000 meter enclosure is made of high strength, non-combustible plastic.
⑤ DIN rail	Used for Installation - 35 mm rail of the DIN rail meter.
⑥ Voltage Input Terminals	Used for voltage input.
⑦ Current Input Terminals	Used for current input.
⑧ Power Supply Terminals	Used for control power input.
⑨ Communication Terminals	Communication output.
⑩ Interface	Used to link the remote display unit and the DIN rail meter.
⑪ Installation Clip	Used for mounting the meter to the panel.
⑫ Gasket	Insert the gasket in between the meter and the cutout to cover up gaps from the round hole.

2.2 Installation Methods

Environmental

Before installation, please check the environment, temperature, and humidity to ensure optimum placement of the PXM 3000 for performance.

Temperature

Operation: -25˚C to 70˚C (-13˚F to 158˚F).

Storage: -40˚C to 85˚C (-40˚F to 185˚F).

Humidity

5% to 95% non-condensing.

The PXM 3000 meter should be installed in a dry and dust free environment. Avoid exposing the meter to excessive heat, radiation, and high electrical noise sources.

Installation Steps

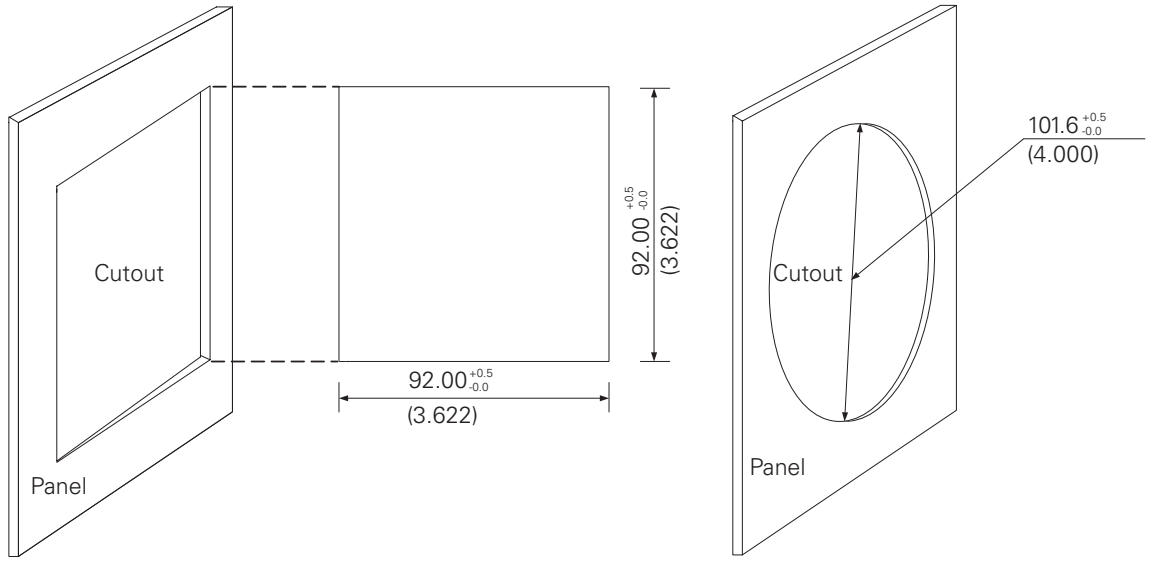
⚠ CAUTION

Before drilling or cutting, take the appropriate steps to ensure that drill shavings or metal chips from drilling and cutting do not fall onto or into any components mounted to the enclosure door or within the enclosure. Drill shavings and metal chips can cause short circuits and damage to the components within the enclosure.

The PXM 3000 meter can be installed into a standard ANSI C39.1 (4" round) or an IEC 92 mm DIN (square) form.

Step 1. Cut a square or round hole on the panel of the switch gear. The cutout size is shown in Figure 2.

Figure 2. Panel Cutout.

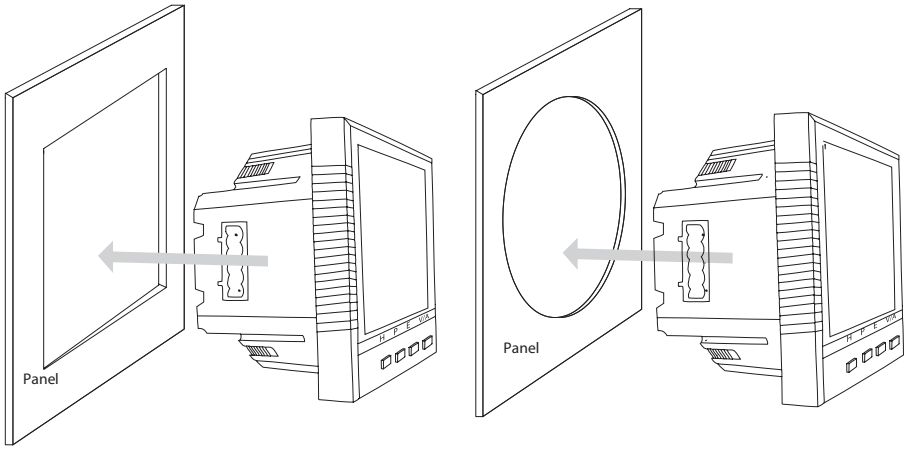


Units: mm (inches).

Step 2. Remove the clips from the meter and insert the meter into the square hole from the front side.

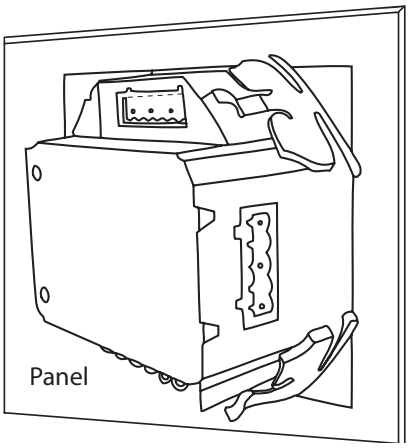
Note: Optional rubber gasket should be installed on the meter before inserting the meter into the cut out to cover up gaps from the round hole and to ensure IP ratings.

Figure 3. Installing Meter Into Cutout.



Step 3. Install clips on the back side of the meter and secure tightly to ensure the meter is tightly clamped to the panel.

Figure 4. Use the Clips to Mount the Meter on the Panel.



Note: The display meter and the remote display unit have the same installation method. The DIN rail meter is simply installed on a 35 mm DIN rail.

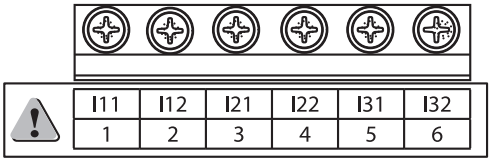
2.3 Wiring

2.3.1 Terminal Strips

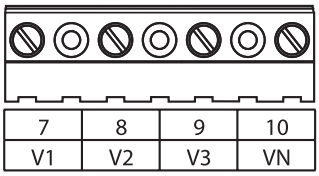
There are four terminal strips at the back of the PXM 3000 meter. The three- phase voltage and current are represented by 1, 2, and 3 respectively. These numbers have the same meaning as A, B, and C, or R, S, and T used in other literature.

Figure 5. Terminal Strips of PXM 3000 Meter.

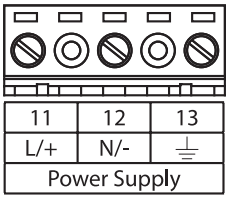
Current Input Terminal Strip

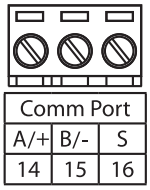


Voltage Input Terminal Strip



Power Supply Terminal Strip - Communication Terminal Strip





⚠ WARNING

Do not attempt to install or perform maintenance on equipment while it is energized. Death, severe personal injury, or substantial property damage can result from contact with energized equipment. Always verify that no voltage is present before proceeding with the task, and always follow generally accepted safety procedures.

⚠ WARNING

Switch main power off and wait five (5) minutes before making any connection or disconnection on the device. Danger of explosion!

Eaton is not liable for the misapplication or misinstallation of its products.

Safety Earth Connection

Before setting up the meter's wiring, please make sure that the switchgear has an earth ground terminal. Connect the meter's earth ground terminal to that of the switchgear.. The following ground terminal symbol is used in this user manual.

Figure 6. Safety Earth Symbol.



2.3.2 Power Requirement

Control Power

Standard: 100~300 Vac (50/60 Hz) or 100-300 Vdc.

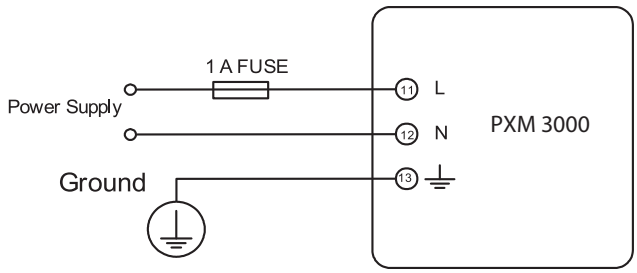
The correct option must be chosen for the application. Please see the ordering information appendix for further details.

The meter's typical power consumption is very low and can be supplied by an independent source or by the measured load line. If the control power source is not reliable, an uninterruptible power supply (UPS) should be considered. Terminals for the control power supply are 11, 12, and 13 (L, N, and Ground). A switch or circuit-breaker shall be included in a building installation. It shall be in close proximity to the equipment, within easy reach of the operator, and shall be marked as the disconnecting device for the equipment.

Note: Make sure the control power ground terminal 13, is connected to earth ground.

Note: Make sure the control power supplied to the meter matches the meter's power supply ratings.

Figure 7. Power Supply.



A fuse (typical 1 A/250 Vac) should be used in the auxiliary power supply loop. Terminal 13 must be connected to the ground terminal of the switchgear. In very rare applications where a power quality problem exists in the power supply, an isolated transformer or EMC filter should be used in the control power supply loop.

Meter power supply input accepts AWG 14 maximum wire size.

Voltage Input

Maximum input voltage for the PXM 3000 meter shall not exceed 400LN/690LL VAC rms for three-phase or 400LN VAC rms for single-phase.

A potential transformer (PT) must be used for medium voltage systems. A typical secondary output for PTs is 100 or 120 V. Note that the PT secondary voltage setting defaults to 220. Make sure to change this to the appropriate PT setting when setting up the meter. Please make sure to select an appropriate PT to maintain the measurement accuracy of the meter. When connecting using the star configuration wiring method, the PT's primary side rated voltage should be approximately equal to the phase to neutral voltage of the system to utilize the full range of the PT. When connecting using the delta configuration wiring method, the PT's primary side rated voltage should be approximately equal to the line to line voltage of the system. A fuse (typical 1 A/250 Vac) should be used in the voltage input loop. The wire for voltage input is AWG 16-12 or
1.3-2.0 mm^2.

Note: In no circumstance should the secondary of the PT be shorted. The secondary of the PT should be grounded. Please refer to the wiring diagram section for further details.

Current Input

Current transformers (CTs) are required in most engineering applications. A typical current rating for the secondary side of the CT is 5 A (standard) or 1 A (optional). Please refer to the ordering information appendix for further details. CTs must be used if the system rated current is over 5 A. A CT with 0.5% accuracy or better, with a rating over 2.5 VA is recommended to ensure the meter's accuracy. The wire length between the CTs and meter shall not exceed the burden rating of the CT to insure accuracy.

Meter Current inputs accept AWG 15-10 or 1.5-2.5 mm^2.

⚠ WARNING

When power is on, the secondary side of the CT should not be open circuit in any circumstance. There should not be any fuse or switch in the CT loop. One end of the CT loop should be connected to Ground.

Vn Connection

Vn is the reference point of the PXM 3000 meter voltage input. Low wire resistance helps to improve the measurement accuracy. Different system wiring modes require different Vn connection methods. Please refer to the wiring diagram section for more details.

3-Phase Wiring Diagram

This meter can satisfy almost any kind of three-phase wiring diagrams. Please read this section carefully before choosing the optimum wiring configuration for your power system.

The voltage and current input wiring mode can be set separately in the meter parameter setting process. The voltage wiring mode can be set as 3-phase 4-wire Wye (3 LN), 3-phase 3-wire direct connection (3 LL), 3-phase 3-wire open delta (2 LL), single-phase 2-wire (1 LN), and single-phase 3-wire (1 LL). The current input wiring mode can be set as 3 CT, 2 CT, and 1 CT. The voltage mode can be grouped with the current mode as 3 LN-3 CT (3 CT or 2 CT), 3 LL-3 CT, 2 LL-3 CT, 2 LL- 2 CT, 1 LL-2 CT, 1 LN-1 CT.

2.3.3 Voltage Input Wiring

3-Phase 4-Wire Wye Mode (3 LN)

The 3-phase 4-wire wye mode is commonly used in low voltage electric distribution power systems. For voltage lower than 400 LN/690 LL Vac, a power line can be connected directly to the meter's voltage input terminal as shown in Figure 8.

For high voltage systems (over 400 LN/690 LL Vac), PTs are required as shown in Figure 9. The meter should be set to 3 LN for both voltage levels.

Figure 8. 3 LN Direct Connection.

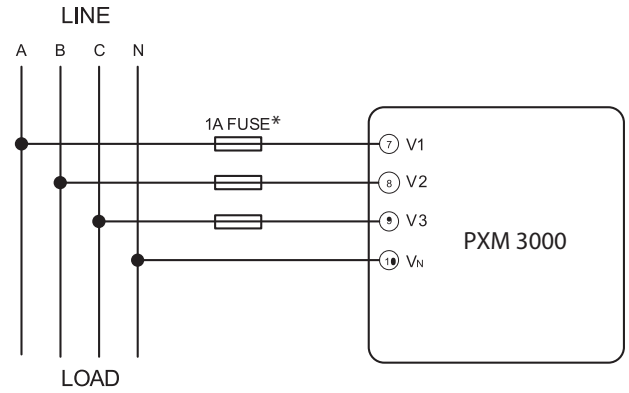
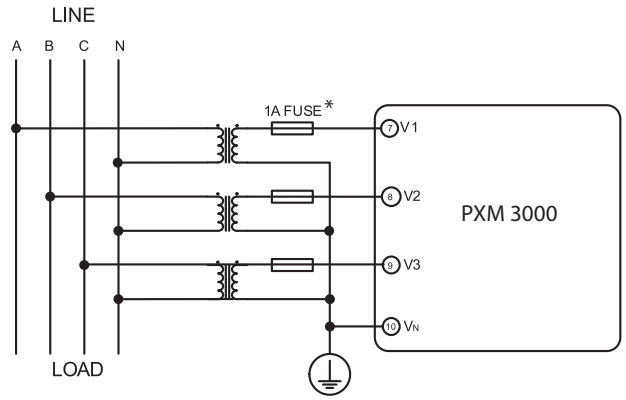


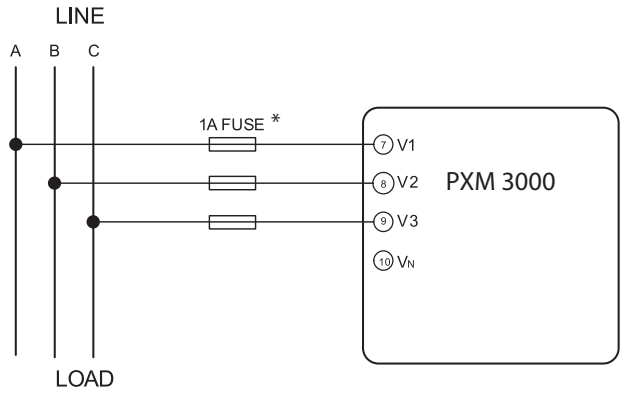
Figure 9. 3 LN with 3 PT.



3-Phase 3-Wire Direct Connection Mode (3 LL)

In a 3-phase 3-wire system, power line A, B, and C are connected to V1, V2, and V3 directly. Leave Vn disconnected. The voltage input mode of the meter should be set to 3 LL.

Figure 10. 3 LL 3-Phase 3-Line Direct Connection.

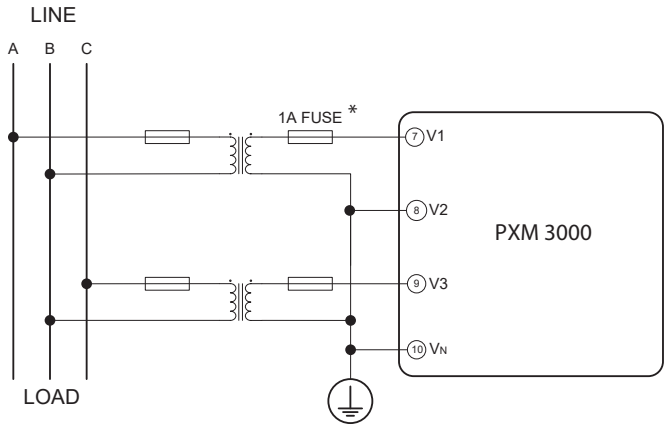


* 1A fuse typical

3-Phase 3-Line open Delta Mode (2 LL)

The open delta wiring mode is often used in high voltage systems. V2 and Vn are connected together in this mode. The voltage input mode of the meter should be set to 2 LL for this voltage input wiring mode.

Figure 11. 2 LL with 2PTs.



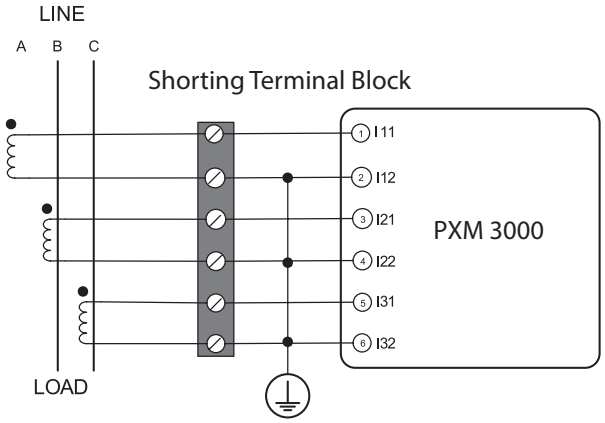
* 1A fuse typical

2.3.4 Current Input Wiring

3 CT

The 3CT current wiring configuration can be used when either 3 CTs are connected (as shown in Figure 12) or 2 CTs are connected (as shown in Figure 13) to the system. In either case, there is current flowing through all three current terminals.

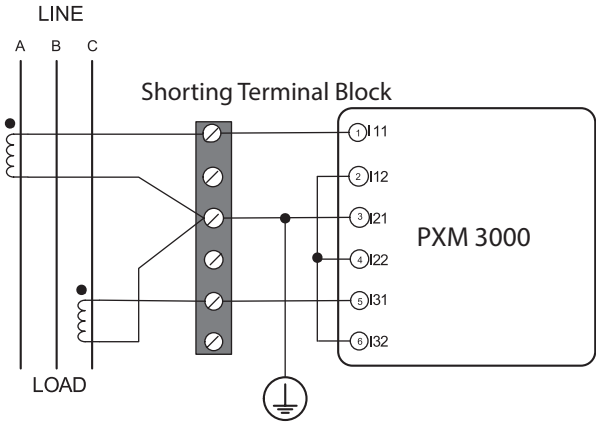
Figure 12. 3 CTs A.



Note: For meters used with voltage input current sensors, unused channels need to be tied to ground as shown in the figures. If meters are used with amperage input current sensors, then the unused channels do not need to be tied to ground. This note applies to Figures 12, 14, and 15.

Shorting terminal block not required when used with voltage input current sensors.

Figure 13. 3 CTs B.



Note: Shorting terminal block not required when used with voltage input current sensors.

This applies to all figures on this page

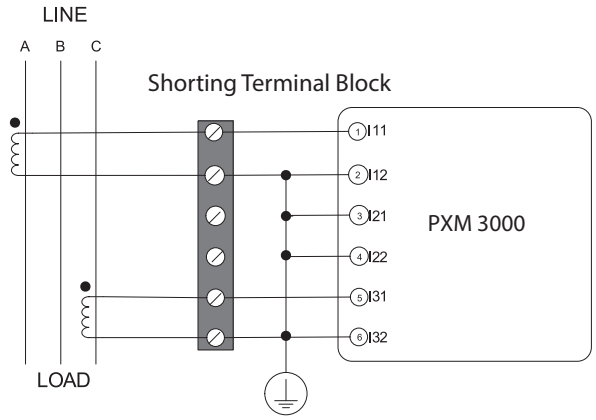
2 CT

For Figures 14 and 15, the meter should be set to the I2 value which is calculated from formula:

$$i1 + i2 + i3 = 0.$$

The current input mode of the meter should be set to 2CT .

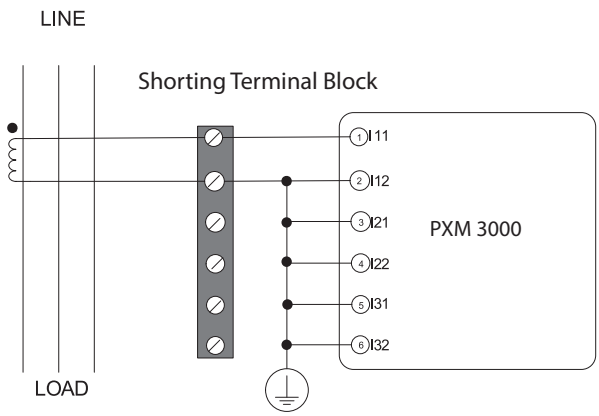
Figure 14. 2 CT.



See Note in Figure 12.

1CT

Figure 15. 1 CT.



See Note in Figure 12.

2.3.5 Frequently Used Wiring Configurations

In this section, the most common voltage and current wiring combinations are shown in different diagrams. In order to display measurement readings correctly, please select the appropriate wiring diagram for your setup and application.

1. 3 LN, 3 CT with 3 CTs.

Figure 16. Three-phase, Three-wire (3LL, 3CT)

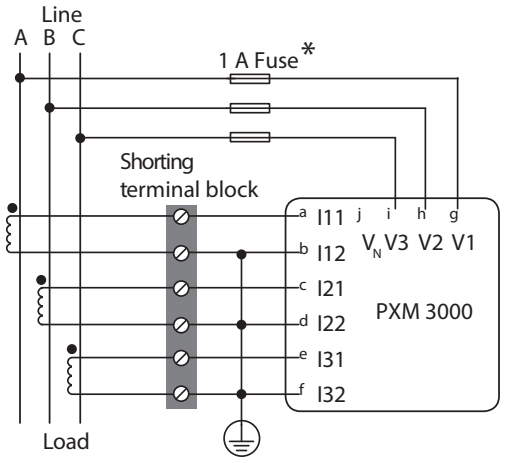
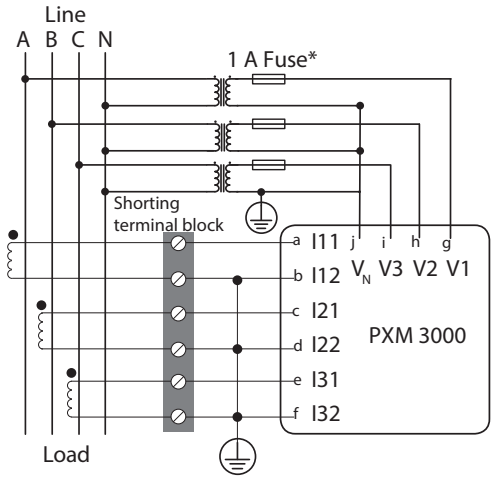
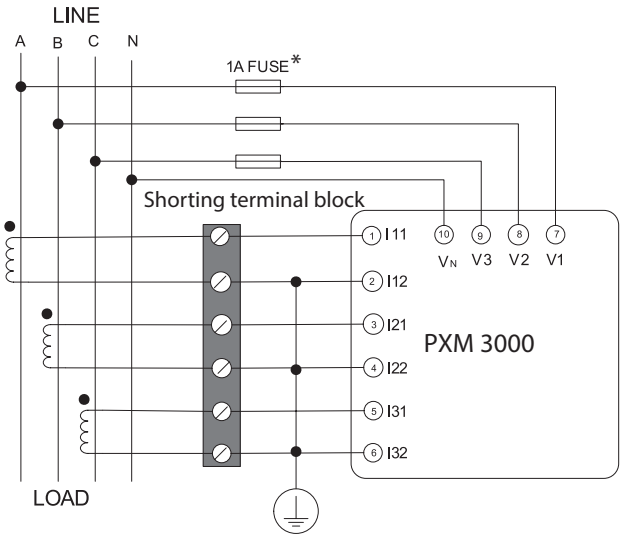


Figure 17. Three-phase, Four-wire with PT (3LN, 3CT)



* 1A fuse typical.

Figure 18. 3 LN, 3 CT.

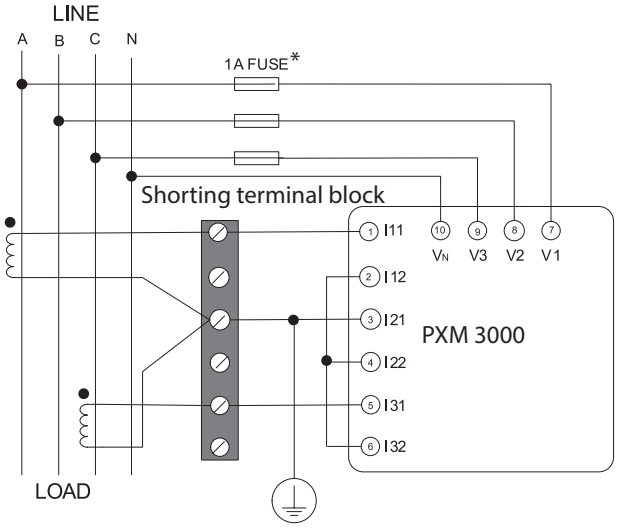


Note: Shorting terminal block not required when used with voltage input current sensors.

This applies to all figures on this page

2. 3 LN, 3 CT with 2 CTs.

Figure 19. 3 LN, 3 CT with 2 CTs.
Note: The sum of I1 and I3 is used as I2 input.

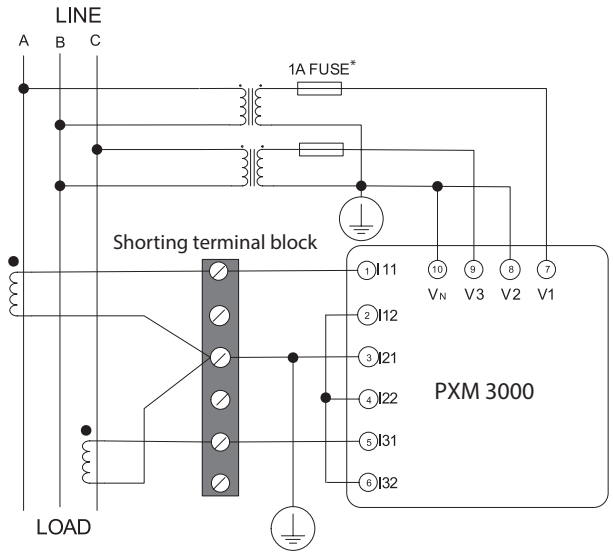


Note: For meters used with voltage input current sensors, unused channels need to be tied to ground as shown in the figures. If meters are used with amperage input current sensors, then the unused channels do not need to be tied to ground.

* 1A fuse typical.

3. 2 LL, 3 CT

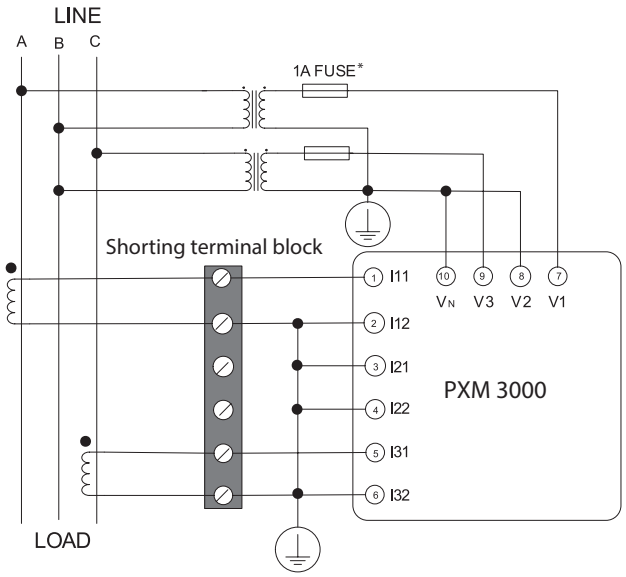
Figure 20. 2 LL, 3 CT.



Note: For meters used with voltage input current sensors, unused channels need to be tied to ground as shown in the figures. If meters are used with amperage input current sensors, then the unused channels do not need to be tied to ground.

4. 2 LL, 2 CT

Figure 21. 2 LL, 2 CT.



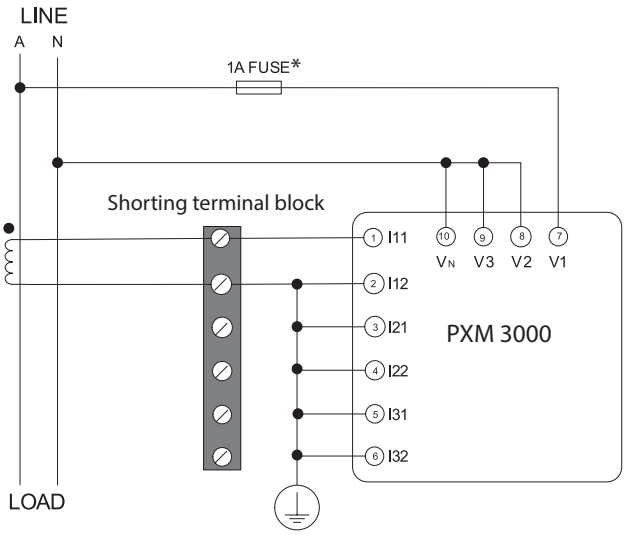
Note: Shorting terminal block not required when used with voltage input current sensors.

This applies to all figures on this page

* 1A fuse typical.

5. 1 LN, 1 CT (Wiring Mode Setting 1 LN, 1 CT).

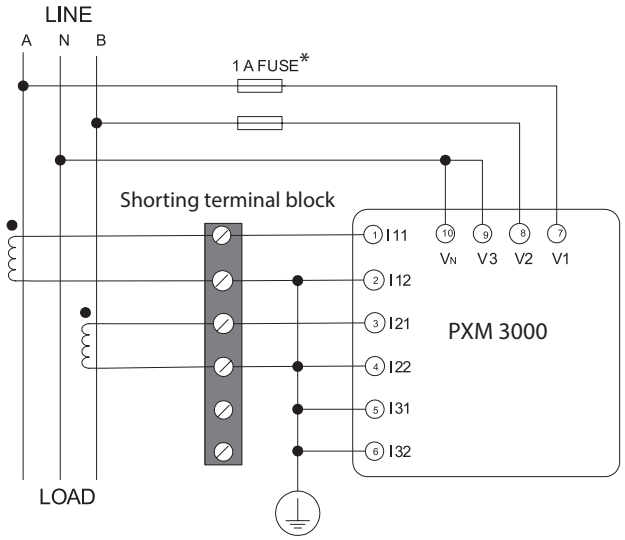
Figure 22. Single Phase 2 Lines.



Note: For meters used with voltage input current sensors, unused channels need to be tied to ground as shown in the figures. If meters are used with amperage input current sensors, then the unused channels do not need to be tied to ground.

6. 1 LL, 2 CT (Wiring Mode Setting 1 LL, 2 CT)

Figure 23. Single Phase 3 Lines.



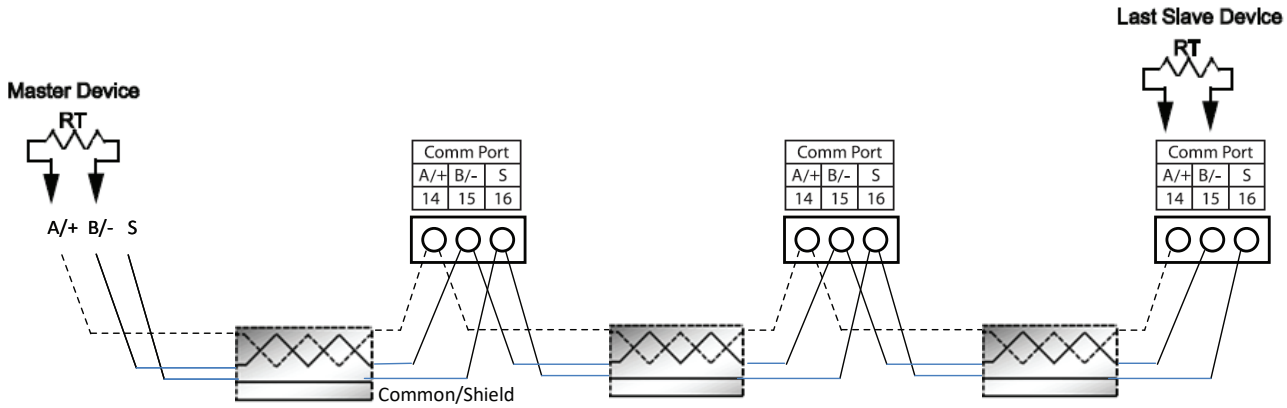
* 1A fuse typical.

2.3.6 Communication

PXM 3000 meter uses RS-485 serial communication and the Modbus-RTU protocol. The terminals for communication are A, B, and S (14, 15, and 16). A is differential signal +, B is differential signal -, and S is connected to the shield of the twisted pair cables. Up to 32 devices can be connected on a RS-485 bus. Use good quality shielded twisted pair cable, AWG 22 (0.5 mm2) or higher. The overall length of the RS-485 cable connecting all devices should not exceed 1200 m (4000 ft). The PXM 3000 meter is used as a slave device for masters such as a PC, PLC, Data Collector or RTU.

If the master does not have a RS-485 communication port, a converter (such as a RS-232/ RS-485 or a USB/RS-485 converter) will be required. Do not use "Star" and "Tee" connections, daisy chain only. The shield of each segment of the RS-485 cable must be connected to the ground at one end only.

Figure 24. RS485 Daisy Chain Connection.



Avoid branching in a "T" configuration. Branches should begin at the source where possible.

Keep communication cables away from sources of electrical noise whenever possible.

When using a long communication cable to connect several devices, an anti signal reflecting resistor (typical value 120 Ω-300 Ω/0.25 W) is normally added to the end of the cable next to the last meter if the communication quality is distorted.

2.4 Quick Start Guide

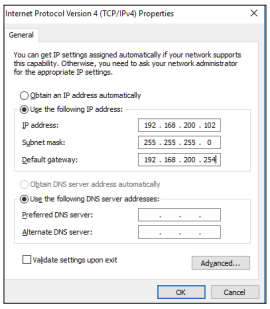
Configuration with USB

Factory installed ethernet module



Connect the PXM3000 to laptop using the USB (micro USB to USB) cable. The RNDIS driver needs to be installed to use the micro USB port on the PXM3000 meter as ethernet port. The default IP for the PXM3000 meter is set to 192.168.200.101. To connect to the meter, the IPv4 settings for the laptop need to be changed as follows.

1. Open the control panel.
2. Click on the Network and Sharing Center icon.
3. In the view your active networks section, click the local area connection or ethernet link, or click the internet connection name next to connections.
4. On the local area connection or ethernet status window, click the properties button.
5. In the properties window, select the internet protocol version 4 (TCP/IPv4) connection and click the properties button. The connection you select depends on the type of internet protocol in use for your internet or network connection.
6. In the Internet Protocol Properties window, select the Use the following IP address option if not already selected. Specify the new IP address as 192.168.200.XXX. You also need to specify the subnet mask as 255.255.255.0 and default gateway as 192.168.200.XXX.



The meter can now be configured using the Chrome browser with a default URL https://192.168.200.101.

The default login credentials for the URL are as follows

Username: admin

Password: admin

Settings mode

Meter settings

Without communications, press V/A and H together to enter the mode selection screen. Press P or E to move the cursor left or right respectively to the "Setting" tab then press V/A. Before accessing the parameter setting mode, a four digit password is required every time. The default password is 0000.



Four options are available in the parameter selection page. "SYS" for system parameter, "I/O" for I/O module parameter, "NET" for Ethernet module parameter, and "ALM" for alarm parameter. Press P or E to move the cursor down or up respectively. Press V/A to enter the selected parameter settings page.

Wiring diagram, PT, and CT settings

Enter "SYS" page, press P or E to scroll through setting pages, press V/A to change a parameter. Scroll to page S04 for voltage wiring, S05 for current wiring, S06 for primary side PT ratio, S07 for secondary side PT ratio, S08 for primary side CT ratio, S09 for secondary side CT ratio. For PT and CT ratio settings, press P or E to increase or decrease digit by 1 respectively, press H to move the cursor to the next digit, press V/A to accept change.

Communication

The meter has a standard RS485 communication port. It supports Modbus and DNP3.0 protocol. You can modify the protocol in "SYS". Enter "SYS" page and scroll to page S35 to set the protocol. Press V/A to accept the change. The meter supports dual communication through both the RS485 serial port and ethernet port.

Serial communication settings

Serial communication terminals are A, B, S (14, 15, 16). A is the differential signal +, B is the differential signal –, and S is shield. Up to 32 devices can be connected on a RS485 bus. The overall length of the bus cannot exceed 1200 m (4000 ft). When multiple meters are connected serially on the same RS485 bus, each meter shall have a different device address. Enter "SYS" page and scroll to page S01 to set the device address. This address can be any integer between 1 and 247.

Baud rate can be selected from one of the seven values: 1200, 2400, 4800, 9600, 19200, 38400 and 115200 bps. PXM3000 needs to set S31 Parity to one of the followings: Odd, Even, None1 (stop bit 1), None2 (stop bit 2). By default, it is set as None1.

2.4 Quick Start Guide (continued)

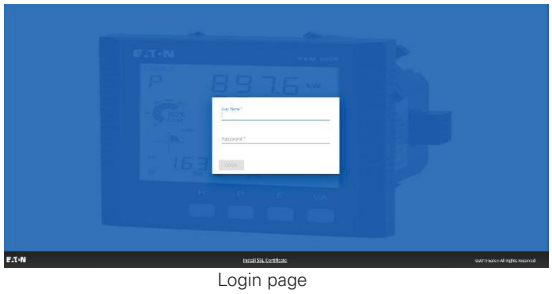
Ethernet



RJ45 ethernet

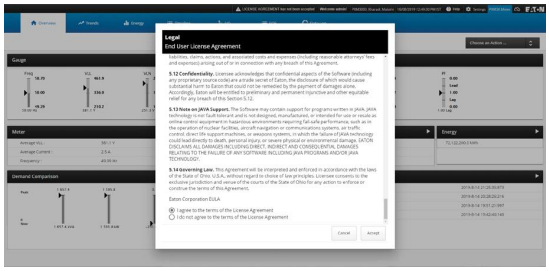
The built-in website for the PXM3000 can be accessed by entering the meter's IP address as the URL in the Chrome web browser (https://<IP Address>). To know the IP address of the meter, connect PXM3000 over the ethernet using the RJ45 ethernet port. Enter "NET" page on the meter display and scroll to page N01 to set the DHCP Setting to "AUTO". Press V/A to accept the change. Scroll to page N02 to check the IP Address of the PXM3000 meter. The default login credentials for the URL are as follows:

Username: admin
Password: admin



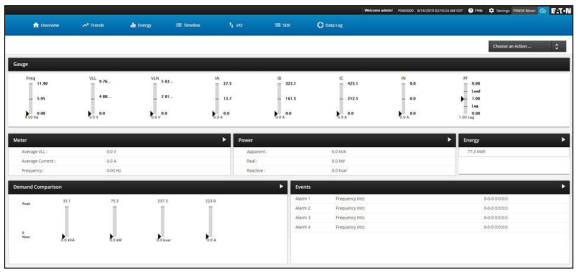
Login page

The first-time login would require installing the security certificate and acceptance of the license agreement.



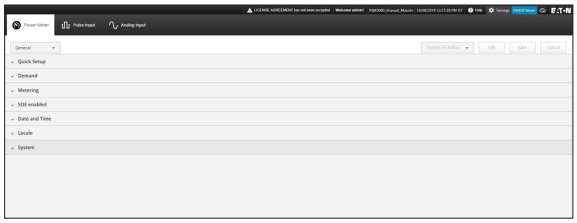
End user license agreement

Once logged in, the meter overview page opens. From this page, you can navigate to the trends, energy, timeline, I/O, SOE, data-log pages to access basic metering information as well as wave - forms, trending graphs, harmonic tables, phasors, alarms, data logs and other functions. The navigational buttons are in a row at the top of the screen.



Overview page

PXM3000 can also be configured using the settings page of the web UI. The settings page is the administrative interface for the meter. It can be used to initially configure the meter for operation or to fine-tune the configurations set before.



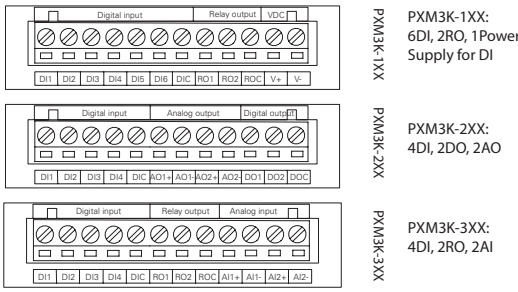
Settings page

In the settings page, parameters such as system parameters, expanded I/O module parameters, alarm parameters, data-log parameters and TOU parameters, can be read and modified. Select the type of settings from the dropdown and save the changes to configure the meter.

I/O options (if equipped)

Three types of IO modules with different IO combinations are available:

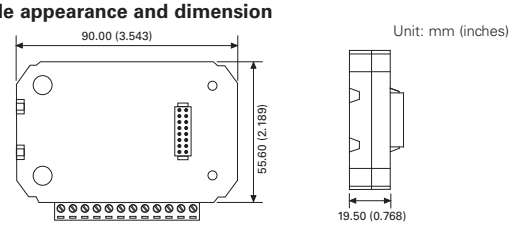
2.4 Quick Start Guide (continued)



Note: Maximum three modules for one meter, only one communication module for one meter, and maximum of two same type IO module for one meter.

IO parameters can be set or viewed either from the meter front or through communication using utility software. Please refer to IO modules user's manual chapter 3 for operation details.

I/O module appearance and dimension



Digital input (DI)

External power supply (20~160 Vac/dc) is required for the digital inputs. DI can be used as pulse counter, to monitor switch status, or to monitor sequence of events.

Digital output (DO)

Two modes available: alarm output and energy pulse output. One module only supports one mode at a time. In alarming mode, conditions can be set using the utility software. In energy pulse mode, output parameters can be set from the meter front or from the utility software.

Relay output (RO)

Two modes available: control (latch or pulse) and alarm output (latch only). One module only supports one mode at a time. In control mode, relays can be switched on and off from the utility software. In alarming mode, conditions can be set using the utility software.

Analog output (AI)

Each module only supports one type of input (either current option or voltage option). Tracking object and upper/lower limit can be set from the front panel control keys or from the utility software.

Analog output (AO)

Each module only supports one type of output (either current option or voltage option). Tracking object and upper/lower limit can be set from the front panel control keys or from the utility software.

DIN rail installation

Except the LCD display and the front panel control keys, the meter with DIN rail mount option includes the same functions as the panel mount version. The default device address and the default baud rate of the DIN rail meter are 1 and 9600 respectively. Those two default values will always be used for the -first minute after the meter is powered on. The device address and the baud rate of the meter will change to the user defined values after the first minute.

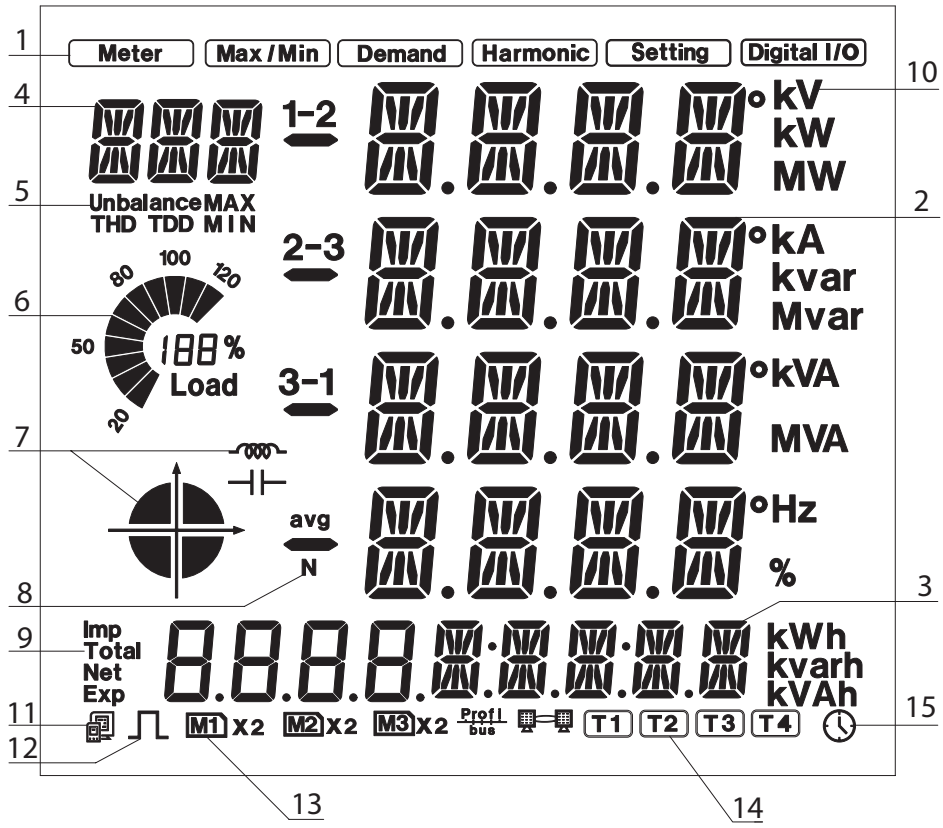
3. Meter Display and Parameter Settings

Details of the human-machine interface (HMI) of the meter will be described in this chapter. This includes viewing real-time metering data and setting parameters using different key combination.

3.1 Display Panel and Keys

The front of the PXM 3000 meter consists of an LCD screen and four control keys. All the display segments are illustrated in Figure 25. Users should note that under normal conditions, all segments will never be displayed simultaneously.

Figure 25. All Display Segments.



SN	Display	Description
1	Display mode indication	Shows different modes on the display area. "Meter" for real-time measurement; "Max/Min" to indicate maximum and minimum readings; "Demand" for power demand data; "Harmonic" for harmonic data; "Setting" for parameters setting; and "Digital I/O" for expanded IO module data.
2	Four lines of "88888" digits in the metering area	Main display area: displays metering data such as voltage, current, power, power factor, frequency, unbalance, phase angle, etc. Displays statistics such as maximum and minimum, demand data, display settings, and expanded I/O data.
3	Four "8" and five "88888" digits	Displays energy data and real-time clock. Also used for the setting mode and digital I/O mode display.
4	Three "888" digits	Item Icons: "U" for voltage; "I" for current; "P" for active power; "Q" for reactive power; "S" for apparent power; "PF" for power factor; "F" for frequency; "∠" for phase angles; "DMD" for demand; "Mxx" for expanded IO module type; and display setting page number.
5	Unbalance, THD, TDD, MAX, MIN	Item Icons: "Unbalance" for unbalance of the voltage and current; "THD" for total harmonics distortion; "TDD" for total demand distortion; "MAX" for maximum, and "MIN" for minimum.
6	Load rate	Displays the percentage of load current to the nominal current.
7	Four quadrant icon Load type icon	: quadrant of the system power. : inductive load; ⊣⊢ : capacitive load.
8	1-2, 2-3, 3-1, avg, N	1, 2, 3 for 3-phase A, B, C; 1-2, 2-3, 3-1 for 3-phase line-to-line AB, BC, CA; avg for average, and N for neutral.
9	Energy icon: Imp, Total, Net, Exp	Imp: import energy. Exp: export energy. Total: absolute sum of Imp and Exp energy. Net: algebraic sum of Imp and Exp energy.
10	Units measured	Voltage: V, kV; Current: A, kA; Active power: kW, MW; Reactive power: kvar, Mvar; Apparent power: kVA, MVA; Frequency: Hz; Active energy: kWh; Reactive energy: kvarh; Apparent energy: kVAh; Percentage: %; Phase angle: °.
11	Communication icon	No icon: no communication. One icon: query sent. Two icons: query sent and response received.
12	Energy pulse output indicator	No icon: no pulse output. With icon: icon blinks when sending pulse output.
13	Expanded I/O module indicator M1 x2 M2 x2 M3 x2	M1: one PXM1K-1 connected. M1x2: two PXM1K-1 connected. None: no PXM1K-1 connected. M2: one PXM1K-2 connected. M2x2: two PXM1K-2 connected. None: no PXM1K-2 connected. M3: one PXM1K-3 connected. M3x2: two PXM1K-3 connected. None: no PXM1K-3 connected.
14	T1 T2 T3 T4	Current tariff.
15	Time icon	Time display .

There are four keys on the front panel, labeled H, P, E, and V/A from left to right. Use these four keys to read real-time metering data, set parameters, and navigate the meter.

Note: If the LCD backlight is off, pressing any key will turn the backlight on.

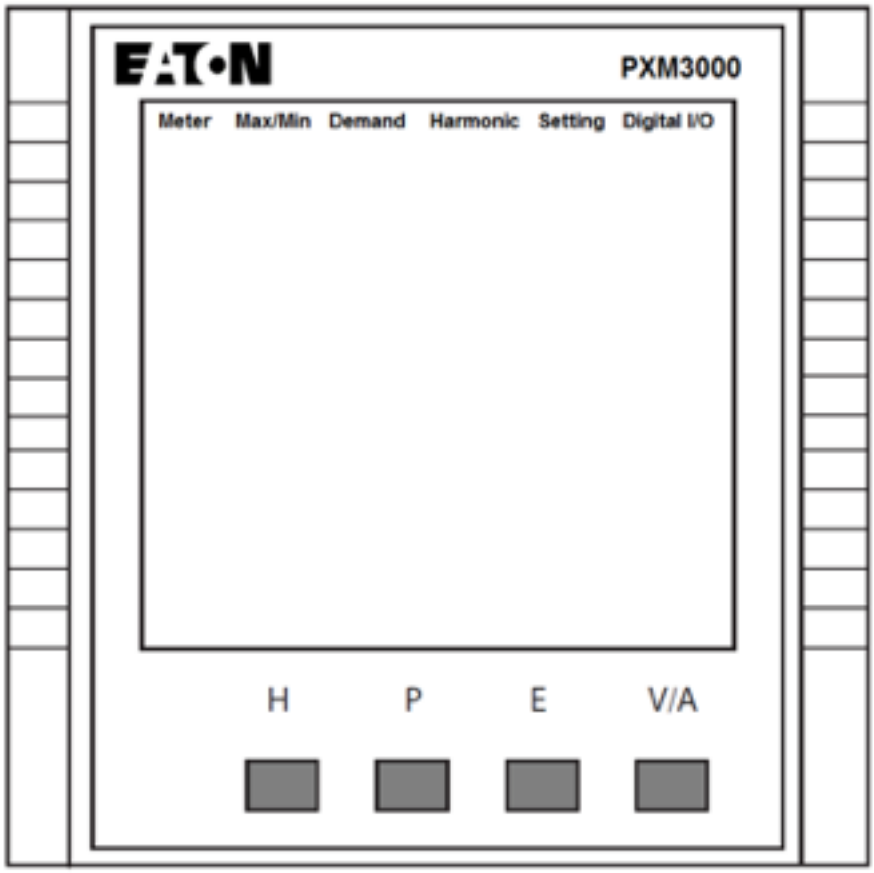
3.2 Metering Data

Pressing H and V/A simultaneously will activate the display mode selection and the cursor will flash. Press P or E to move the cursor right or left. To enter the metering mode, move the cursor to "Meter" then press V/A.

In the metering mode, pressing P and E simultaneously will access the TOU mode.

In metering mode, the meter displays measurements such as voltage, current, power, power factor, phase angle, unbalance, etc.

In the TOU mode, the meter displays the energy, maximum demand, and time in different tariffs.

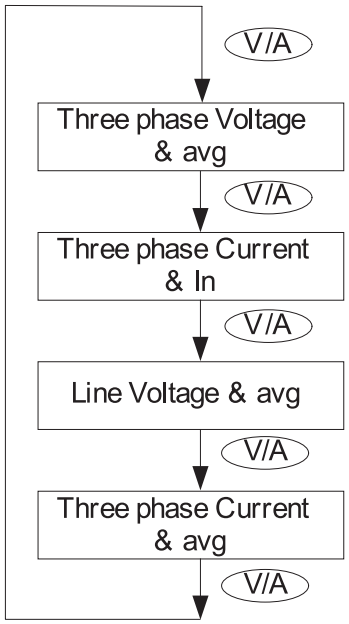


A. Voltage and Current:

Press V/A to read the voltage and current in the metering area. Pressing V/A again will take the user to the next screen in the sequence.

The Figure 26 shows the sequence:

Figure 26. Voltage and Current Sequence.



Notes: When the meter is set to "2LL" or "3LL", there is no phase voltage or neutral current display. Therefore, only the third screen (line voltage & avg) and the fourth screen (three phase current & avg) will be displayed.

When the meter is set to "1LN", only phase A voltage and phase A current will be displayed, without line voltages or other displays.

When the meter is set to "1LL", phase C voltage and phase C current will not be displayed.

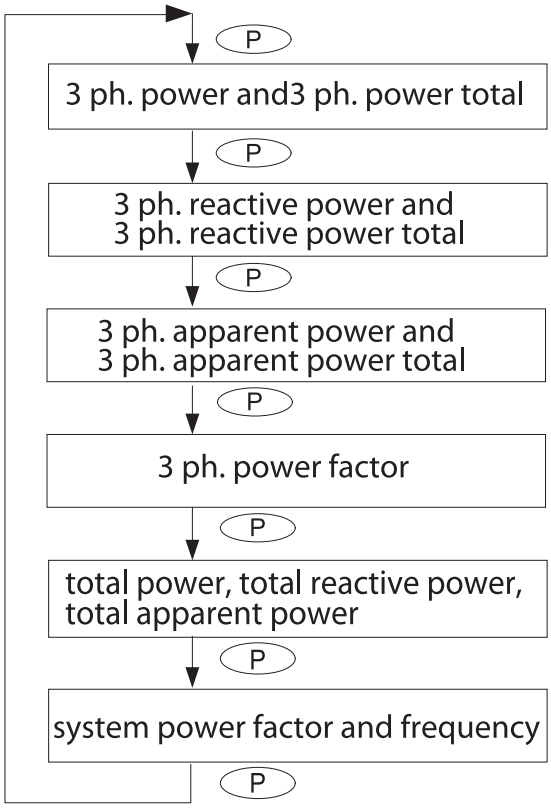
B. Power, Power, Factor and Frequency:

Press P to display power related data.

Pressing P again will take the user to the next screen in the sequence.

The Figure 27 shows the sequence.

Figure 27. Power, Power Factor, and Frequency Sequence.



Notes: When the meter is set to "2LL" or "3LL", only the fifth screen (system power) and the sixth screen (system power factor & frequency) will be displayed.

When the meter is set to "1LN", only phase A power and phase A power factor will be displayed.

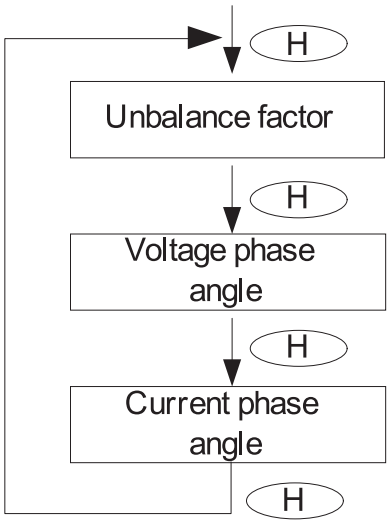
When the meter is set to "1LL", no phase C power and phase C power factor will be displayed.

C. Phase Angles and Unbalance:

Press H to display phase angles and unbalance data. Pressing H again will take the user to the next screen in the sequence.

The Figure 28 shows the sequence:

Figure 28. Phase Angles and Unbalance Sequence.



When using "2LL" or "3LL" wiring setting mode, voltage stands for line to line voltage. Otherwise, voltage stands for line-to-neutral voltage.

When the meter is set to "1LN", there is only phase A current to phase A voltage angle display.

When the meter is set to "1LL", there is no phase C voltage or current to phase A voltage angle factor display.

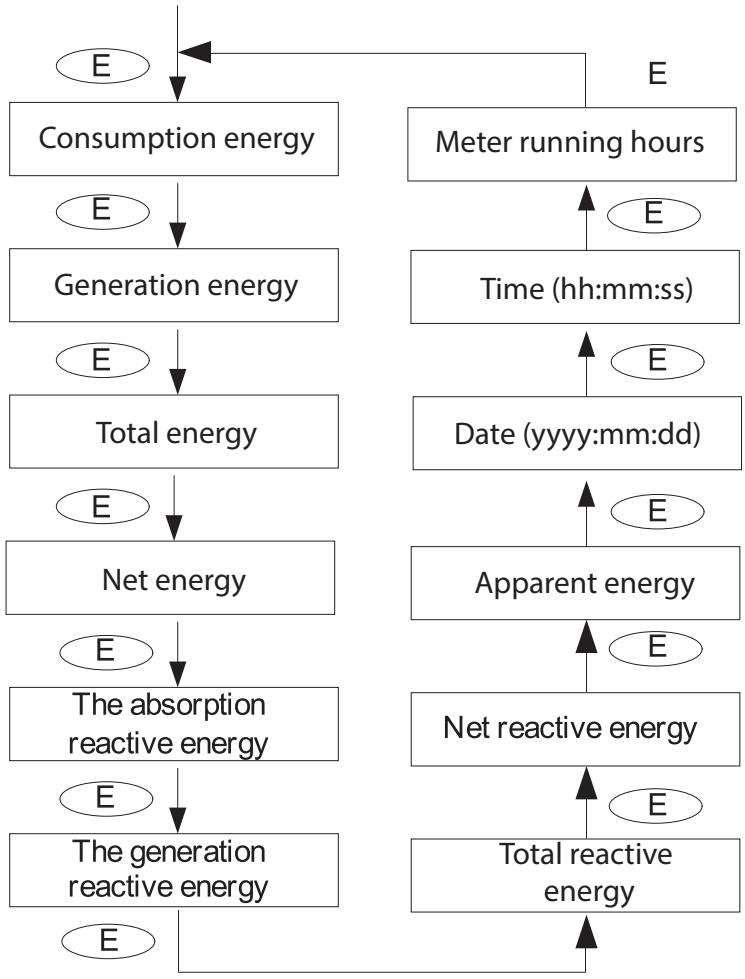
D. Energy:

Press the E key to display energy and real time clock. Pressing E again will take the user to the next screen in the sequence.

The PXM 3000 meter can be set to record primary energy or secondary energy. The unit of energy is kWh for active energy, kvarh for reactive energy, and kVAh for apparent energy. The running time has a resolution of 0.01h. The meter begins accumulating energy upon initial power-up of the unit. The accumulated data is stored in the non-volatile memory. It can be reset via communication or from the front of the meter.

The Figure 29 shows the sequence:

Figure 29. Energy Sequence.



E. TOU display

Press "P" and "E" simultaneously to enter the TOU energy and maximum demand page. Press "E" to display the TOU energy. Press "P" to display the TOU maximum demand. Pressing "P" again from this screen takes users to the TOU maximum demand year, month, and date. Pressing "P" for a second time takes users to the TOU maximum demand hour, minute, and second. Pressing "H" takes users to the tariffs screen. It displays energy under different tariffs in the energy page. It also displays demand under different tariffs in the maximum demand page. Pressing "V/A"will display the different types of energy and maximum demand. Press "P" and"E" simultaneously to exit the current page and return to metering mode.

Figure 30 shows the sequence:

Figure 30. TOU Sequence.

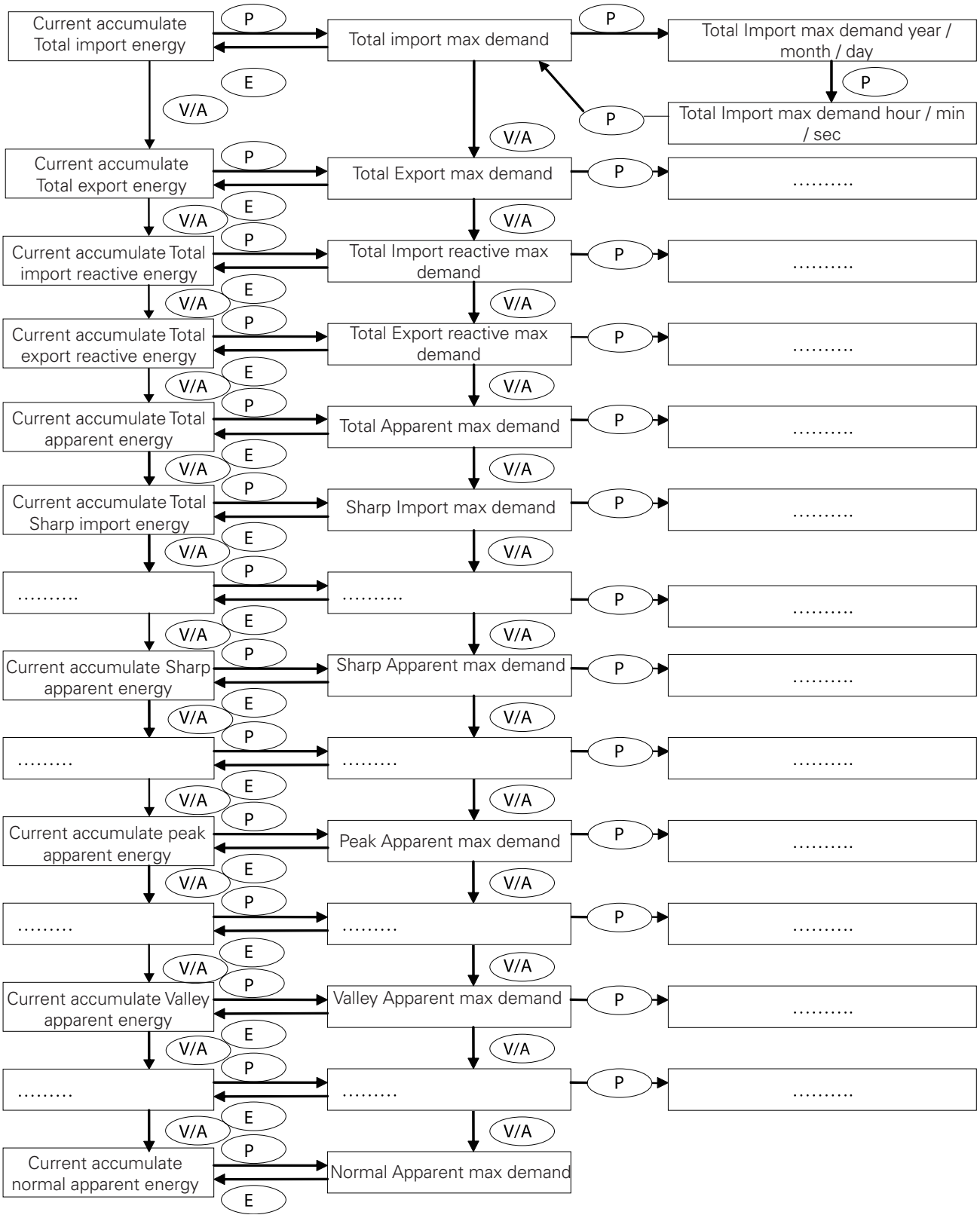
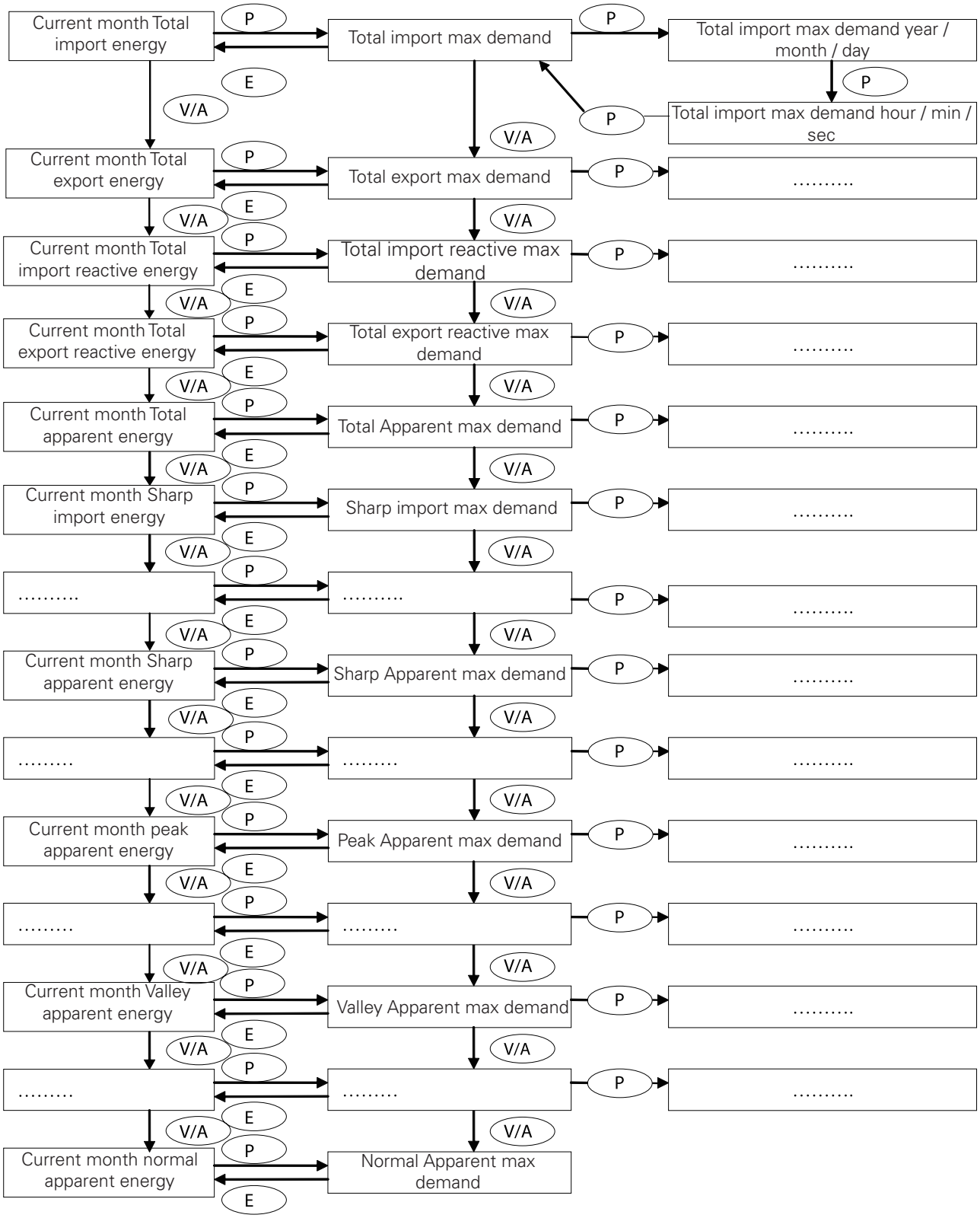


Figure 30. TOU Sequence (Continued).



3.3 Max/Min Data

Pressing H and V/A simultaneously will activate the display mode selection and the cursor will flash. Press P or E to move the cursor right or left. To enter the statistics data mode, scroll the cursor to "Max/Min" then press V/A.

In max/min data mode, the meter displays the maximum values and minimum values for voltage, current, power, power factor, unbalance, demand, THD, etc. The user should note that time stamp for the parameters can be viewed only from the software through communication. No commands are associated with the H key in "Max/Min" display mode.

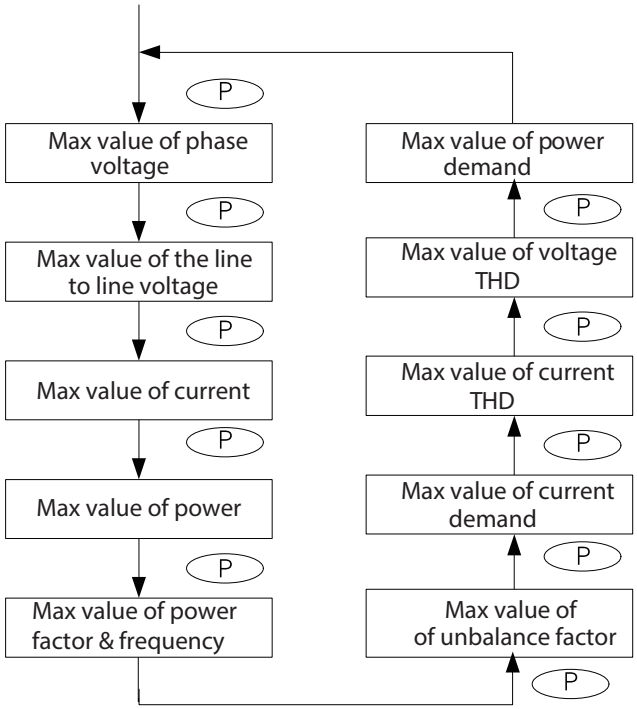
Pressing P will take the user to the next screen in the sequence.

Pressing E will take the user to the previous screen in the sequence.

Pressing V/A will toggle the displayed parameter between maximum and minimum. For example, if the current display is the maximum phase voltage value, when V/A is pressed, the display will show the minimum phase voltage value. If V/A is pressed again, the display will switch back to show the maximum phase voltage value.

Figure 31 shows the sequence:

Figure 31. Max/min Data Sequence.



Notes: The figure shows the rolling sequence when pressing P. The sequence will be reversed when pressing E.

When the meter is set to "2LL" or "3LL", the first screen (max value of phase voltage) will not be displayed.

When the meter is set to "1LL", there are no such displays as phase C voltage,Ubc and Uca line voltage, phase C current, three phase voltage and current unbalance factor, Uc and Ic THD, phase C current demand, etc.

When the meter is set to "1LN", there is only phase A display of phase voltage and current, only Ua and Ia THD display, only demand display of phase A. And there are no such displays as three-phase voltage and current unbalance factor, line voltage, etc.

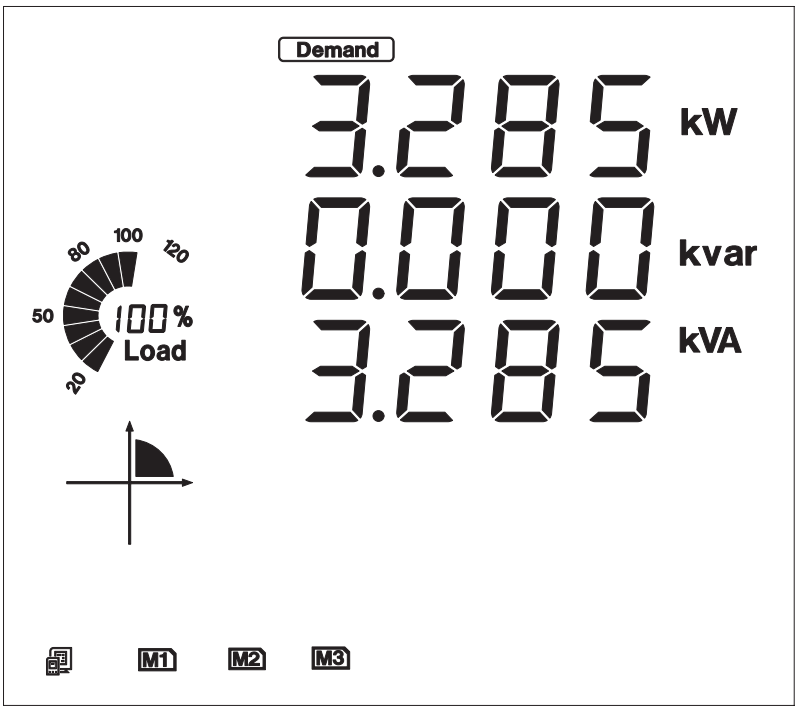
3.4 Demand Data

Pressing H and V/A simultaneously will activate the display mode selection and the cursor will flash. Press P or E to move the cursor right or left. To enter the demand mode, move the cursor to "Demand" then press V/A.

In the demand data mode, the first screen displays the demand of active power, reactive power, and apparent power, and the second screen displays the current demand of phase A, phase B, and phase C. When the meter is set to "1 LL", the current demand for phase C is not available. When the meter is set to "1 LN", phases B and C for current demand display are not available.

As shown in the Figure 32, system active power demand is 3.285 kW, system reactive power demand is 0 kvar, and system apparent power demand is 3.285 kVA.

Figure 32. Demand Data.



3.5 Harmonic Data

Pressing H and V/A simultaneously will activate the display mode selection and the cursor will flash. Press P or E to move the cursor right or left. To enter harmonic mode, move the cursor to "Harmonic" then press V/A.

In the harmonic data mode, the meter displays the harmonic ratio of voltage and current, THD, odd HD, even HD, THFF, CF, and KF.

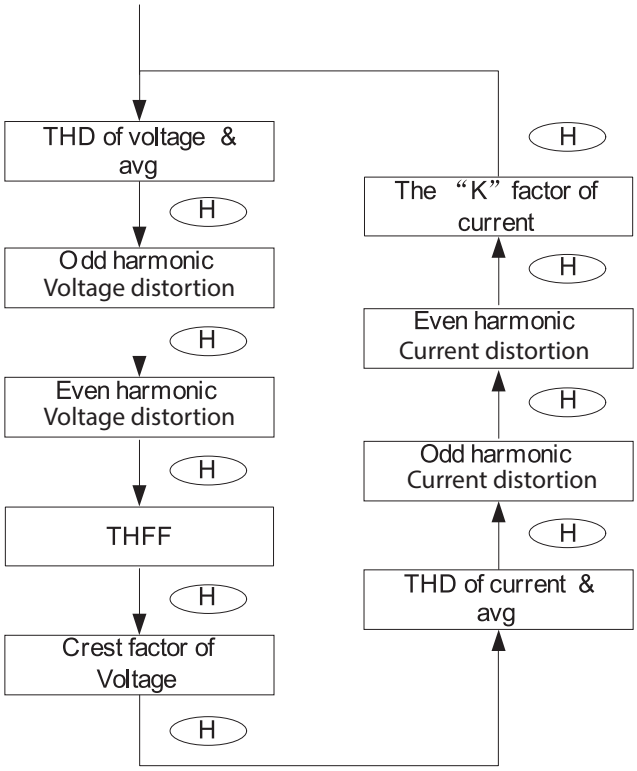
A. Power Quality Data:

Press H to display the power quality data. Pressing H again will take the user to the next screen in the sequence.

The P and E keys have no functionality in this mode.

Press V/A to switch to harmonic ratio data display.

Figure 33. Power Quality Data Sequence.



Notes: When the meter is set to "1LN", there is only phase A display for voltage THD, voltage odd HD, voltage even HD, THFF, voltage crest factor, current THD, current odd HD, current even HD, and current K factor.

When the meter is set to "1LL", there is no phase C display.

B. Harmonic Ratio Data

Press H to switch to power quality data display.

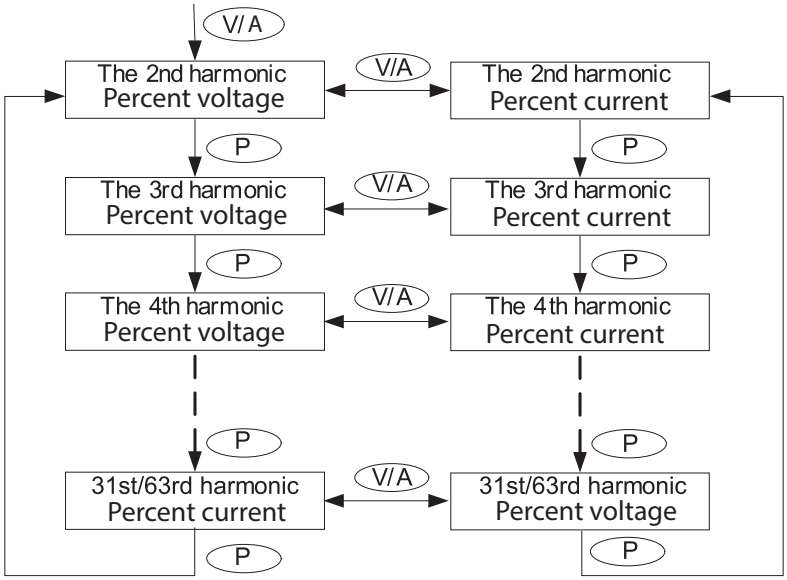
Pressing P increases the harmonic order by one until the user reaches the 63rd harmonic. The user will then be returned to the 2nd harmonic.

Pressing E decreases the harmonic order by 1 until the user reaches the 2nd harmonic. The user will then be returned to the 63rd harmonic.

Press V/A to switch display between voltage harmonics and current harmonics.

Figure 34 shows the sequence:

Figure 34. Individual Harmonic Data.



Notes: The figure shows the rolling sequence when pressing P. If E is pressed, the sequence will reverse.

Harmonic is 2nd ~ 63rd.

When the meter is set to "1LN", there is only phase A display for voltage and current harmonic magnitude.

When the meter is set to "1LL", there is no phase C display for voltage and current harmonic magnitude.

3.6 Expanded I/O Module Data

Pressing H and V/A simultaneously will activate the display mode selection and the cursor will flash. Press P or E to move the cursor right or left. To access data from the expanded I/O modules, move the cursor to "Digital I/O" then press V/A to enter the expanded I/O module data mode.

In the expanded I/O module data mode, the meter displays the data from expanded I/O modules, such as DI status, pulse counter number, relay status, analog input, analog output, etc.

In this mode, the first page is module selection. You can choose to view the available modules that are attached to the meter. If no expanded I/O modules are connected, the screen will display "NO IO".

A. Module Selection:

The H key has no functionality on this screen.

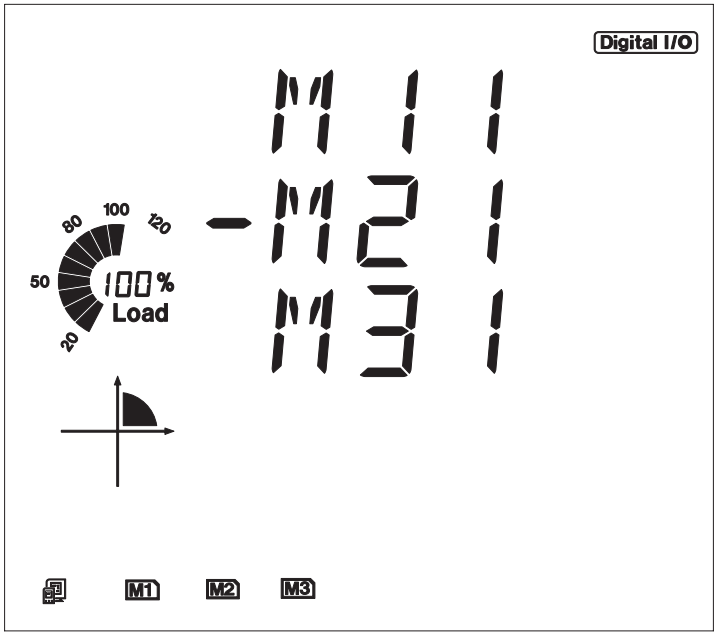
If more than one module is connected, pressing P will allow the user to select the module listed below. When the bottom of the list is reached, the cursor will return to the top.

If more than one module is connected, pressing E will allow the user to select the module listed above. When the top of the list is reached, the cursor will return to the bottom.

Press V/A to select the module and enter the I/O module data selection mode.

As shown in Figure 35, three modules are connected, PXM1K-11, PXM1K-21, PXM1K-31, which are indicated by M11, M21, M31 respectively. The cursor points to M21, which indicates that PXM1K-21 is chosen now.

Figure 35. Three Connected Modules Shown.



B. I/O Module Data Selection:

Press H to return to module selection screen.

Press P to move the cursor downwards; the cursor will return to the top when it reaches the bottom. Please note that there are three parameters for PXM1K-1, three parameters for PXM1K-1, and four parameters for PXM1K-3.

Press E to move the cursor upwards; the cursor will return to the bottom when it reaches the top.

Press V/A to select the parameter and take the user to the data display screen.

C. I/O Module Data Display

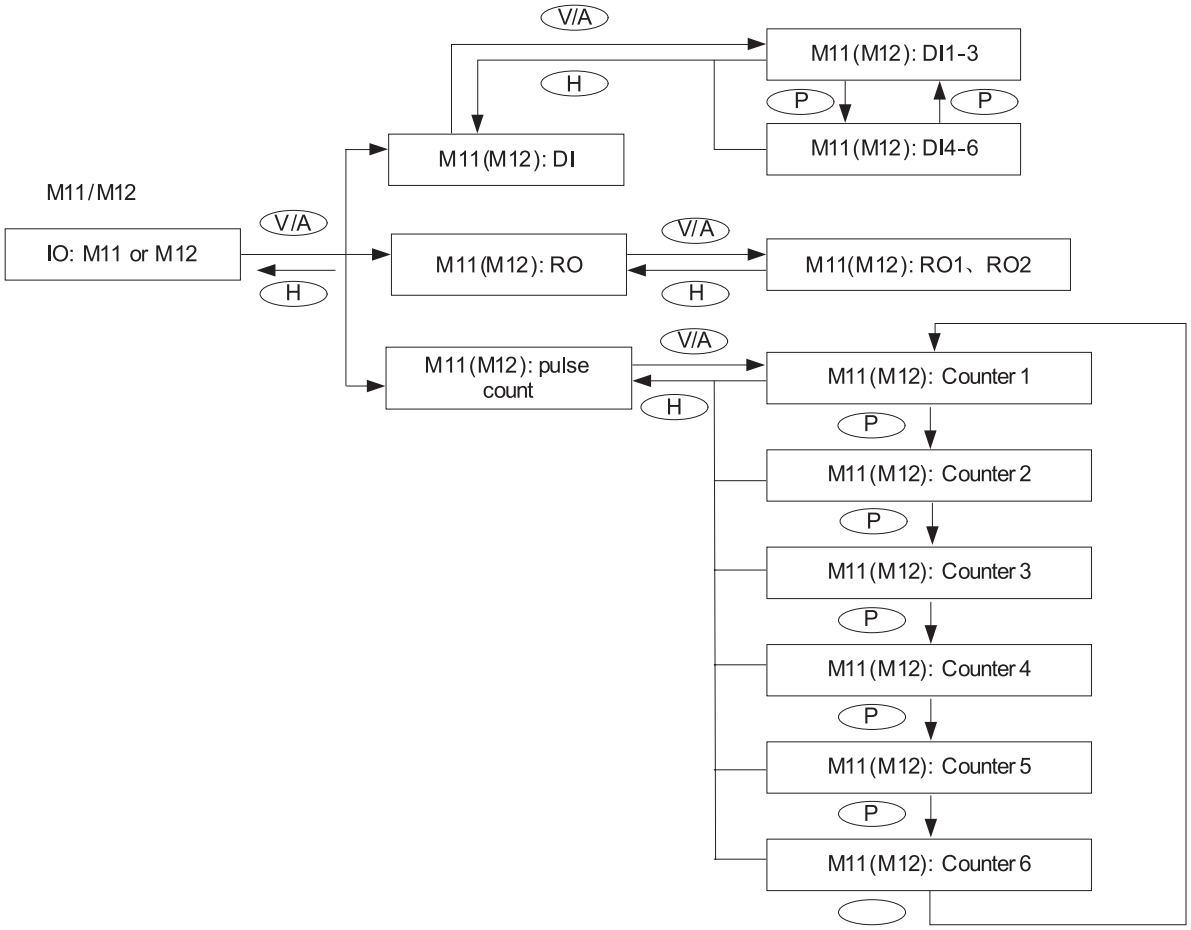
Pressing H returns the user to the I/O module data selection screen.

If more than one page is available, pressing P will take the user to the next screen in the sequence. Pressing E will take the user to the previous screen in the sequence.

The V/A key has no functionality on these screens.

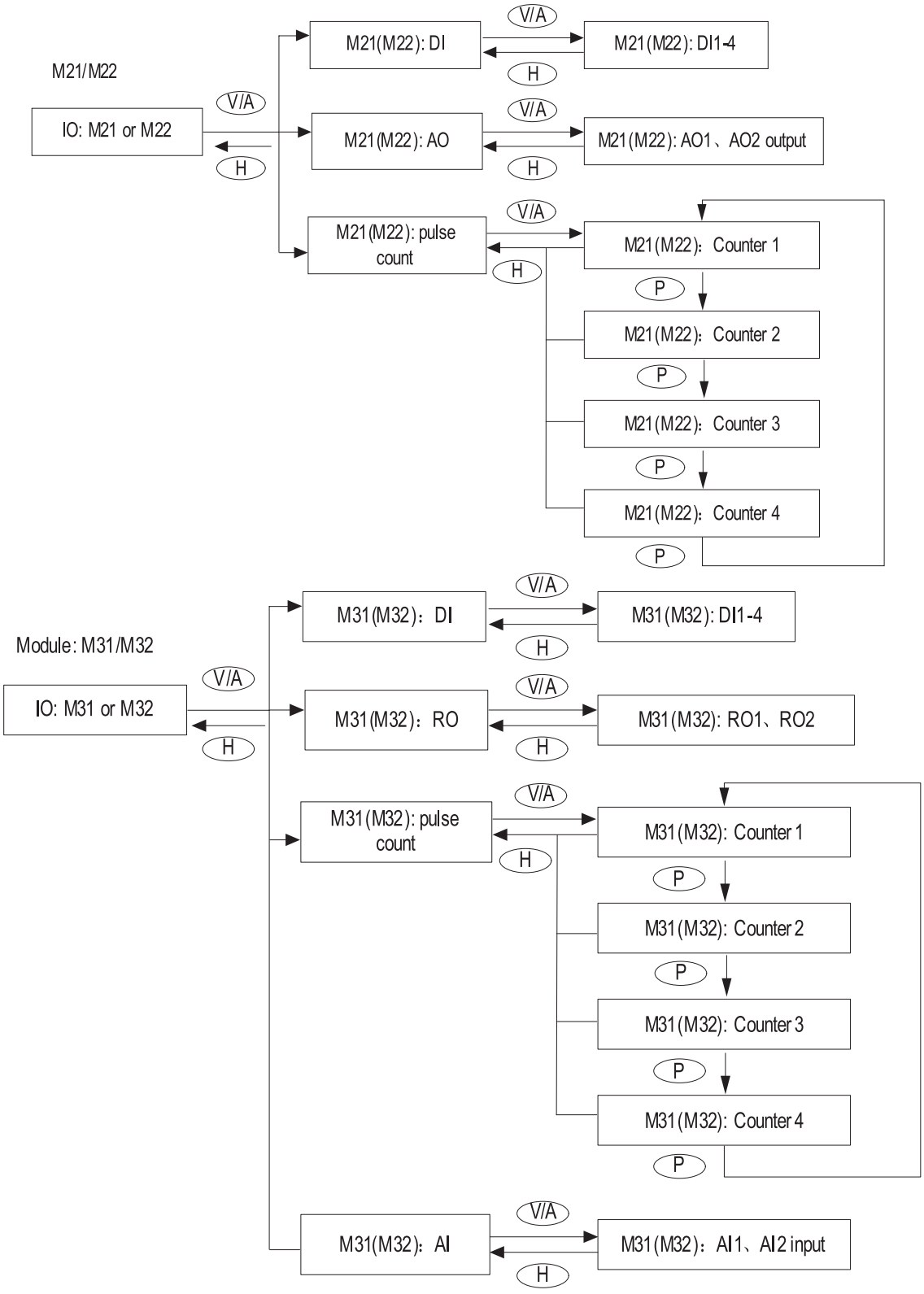
Figure 36 shows the sequence:

Figure 36. I/O Module Data Display Sequence.



Note: The Figure 36 shows the scrolling sequence for using the P key. If using the E key for scrolling the page, the sequence will be reversed.

Figure 36. I/O Module Data Display Sequence (continued).



3.7 Parameter Setting Mode

Pressing H and V/A simultaneously will activate the display mode selection and the cursor will flash. Press P or E to move the cursor right or left. To enter parameter setting mode, move the cursor to "Setting" then press V/A.

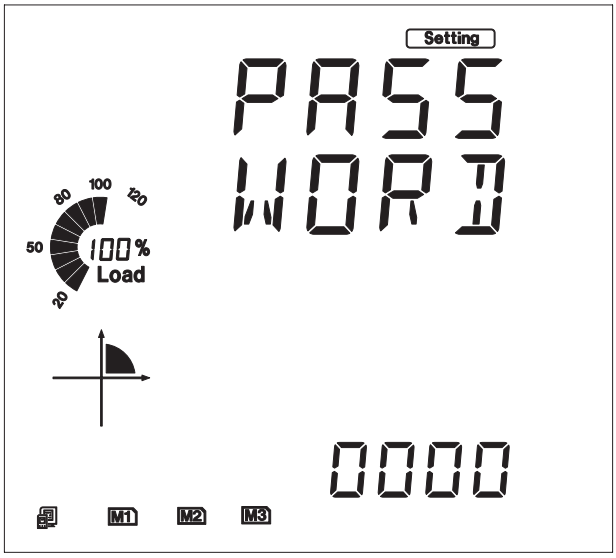
In the parameter setting mode, parameters such as system parameters, expanded I/O module parameters, and alarm parameters can be read and modified.

A. Password Entry

Parameter setting mode is password protected. Before entering the password and getting into the parameter setting mode, the meter's device communication address will display for three seconds. A four digit password (0000 to 9999) is required every time before accessing the parameter setting mode. The default password is 0000. After entering the password, press V/A to go to the parameter selection page. The meter will be still in the password entry page if a wrong password is entered.

Figure 37 shows the password entry page.

Figure 37. Password Entry Page.



To input password:

1. Press H to move the flashing cursor to the next position.
2. Press P to increase the number by 1.
3. Press E to decrease the number by 1.
4. Press V/A to confirm the password.

B. Parameter Selection Mode

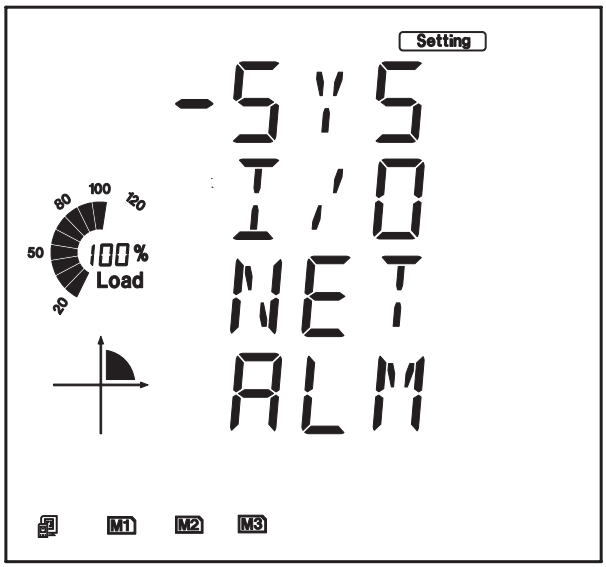
There are three parameters to choose from in the parameter selection manual: system, expanded I/O module, and alarm.

The H key has no functionality in this mode.

Press P to move the cursor downwards. The cursor will return to the top when it reaches the bottom.

Press E to move the cursor upwards. The cursor will return to the bottom when it reaches the top.

Figure 38. Parameter Selection Mode Page.



Press V/A to select and modify the parameter. Figure 38 shows the parameter selection page. "SYS" stands for system parameter, "I/O" stands for expanded I/O module parameter, and "ALM" stands for alarm parameter. As shown in the figure, the cursor points to the "SYS", which means system parameter is selected.

C. System Parameter Setting

Users can select and modify system parameter in the system parameter setting mode.

Key functions for selecting a parameter:

Press H to return to the parameter selection mode.

Pressing P will take the user to the next screen in the sequence.

Pressing E will take the user to the previous screen in the sequence.

Press V/A to modify the selected parameter.

Key functions for modifying the parameter:

Press H to move the flashing cursor to the next position.

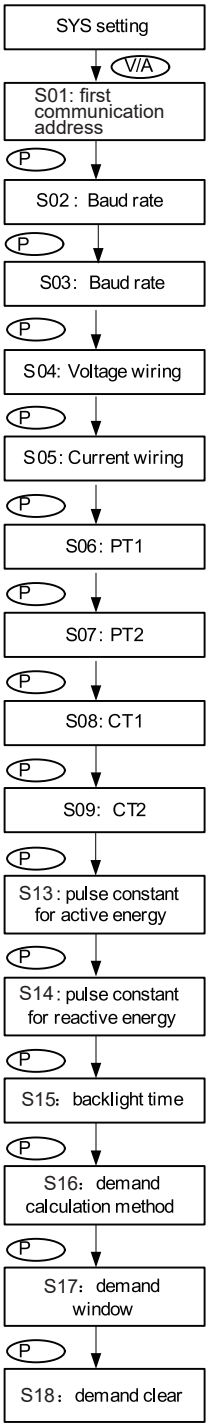
Press P to increase the number by 1.

Press E to decrease the number by 1.

Press V/A to confirm the modification and return to parameter selection mode.

Figure 39 shows the sequence:

Figure 39. System Parameter Setting Sequence.



MODBUS: range is 1~247
DNP3.0: range is 0~65534

Select from 1200, 2400, 4800, 9600, 19200, 38400, 115200 bps

Fixed as 115200 bps

Voltage wiring mode can be "3LN", "2LL", "3LL", "1LN", or "1LL".

Current wiring mode can be "3CT", "1CT", "2CT".

PT1: primary nominal value of PT, range from 50.0V~500,000.0V

PT2: secondary nominal value of PT, range from 50.0V~400.0V
If no PT is used, set PT1 and PT2 as the nominal voltage of the meter.
IF not using analog output, disregard. Only pertinent if using analog output.

CT1: primary value of CT, range from 1A~50000A.

CT2: secondary value of CT, choose from 1A or 5A. When the CT is 333mV voltage type, choose 333. When the CT is Rogowski coil, choose 100. When the CT is mA, choose 80/100/200.

Which is active energy of each pulse from DO. The pulse of each 1kWh energy can be set to 1~60000 pul/kWh (secondary).
Description: the specific setting method, see the address table.

Which is reactive energy of each pulse from DO. The pulse of each 1kvarh energy can be set to 1~60000 pul/kvarh (secondary).
Description: The specific setting method, see the address table.

The "on" time can be set from 0 to 120 minutes. The backlight will always be "on" if the setting value is 0. If it is another value, it means after that period of time, it will go off if no other key has been pressed. It will be "on" whenever you press any key if it is "off".

0: Fixed block, 1: Sliding block, 2: thermal , 3: Rolling block

Range from 1~30 minutes

Yes: clear
No: do not clear

Figure 39. System Parameter Setting Sequence (continued).

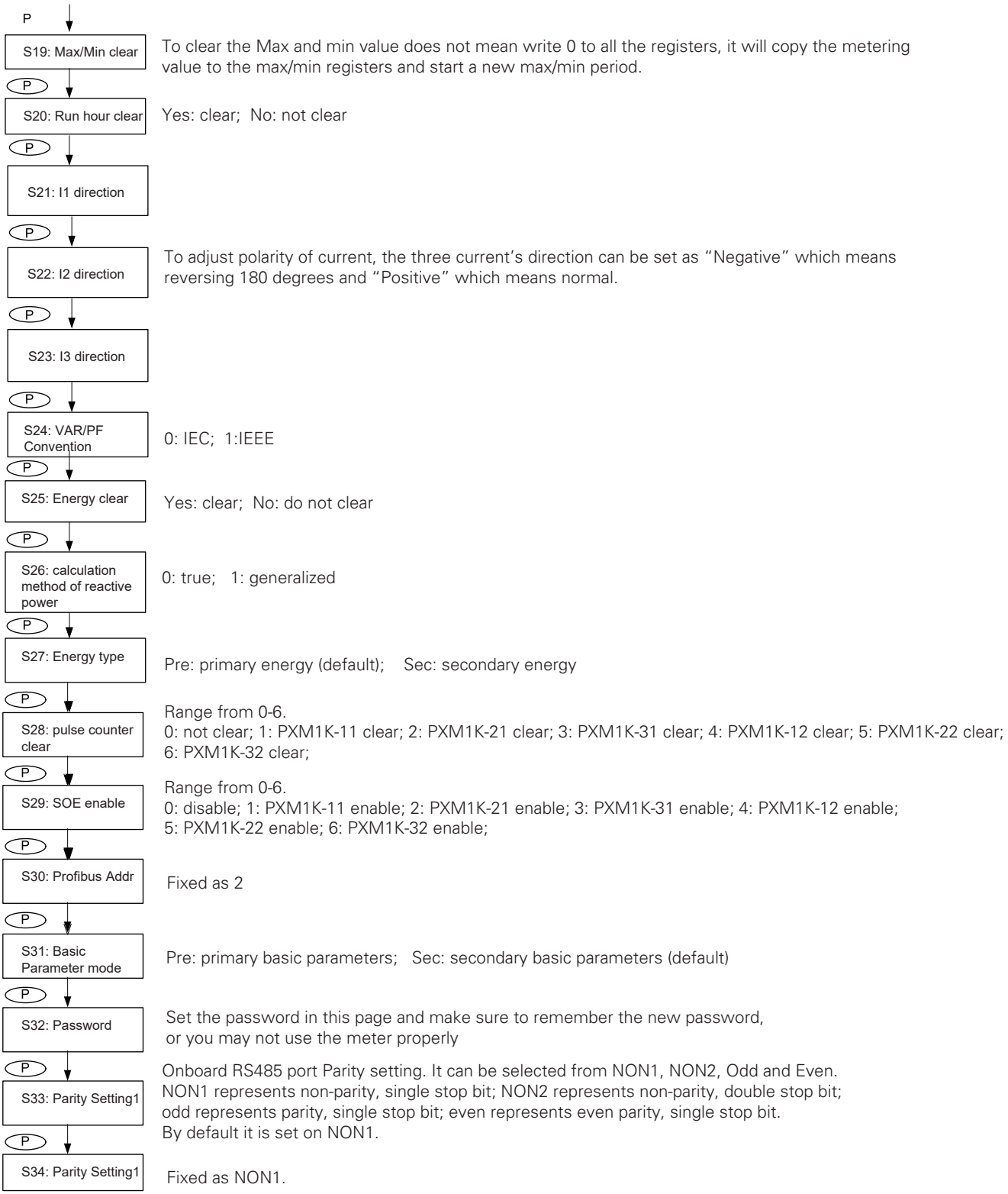
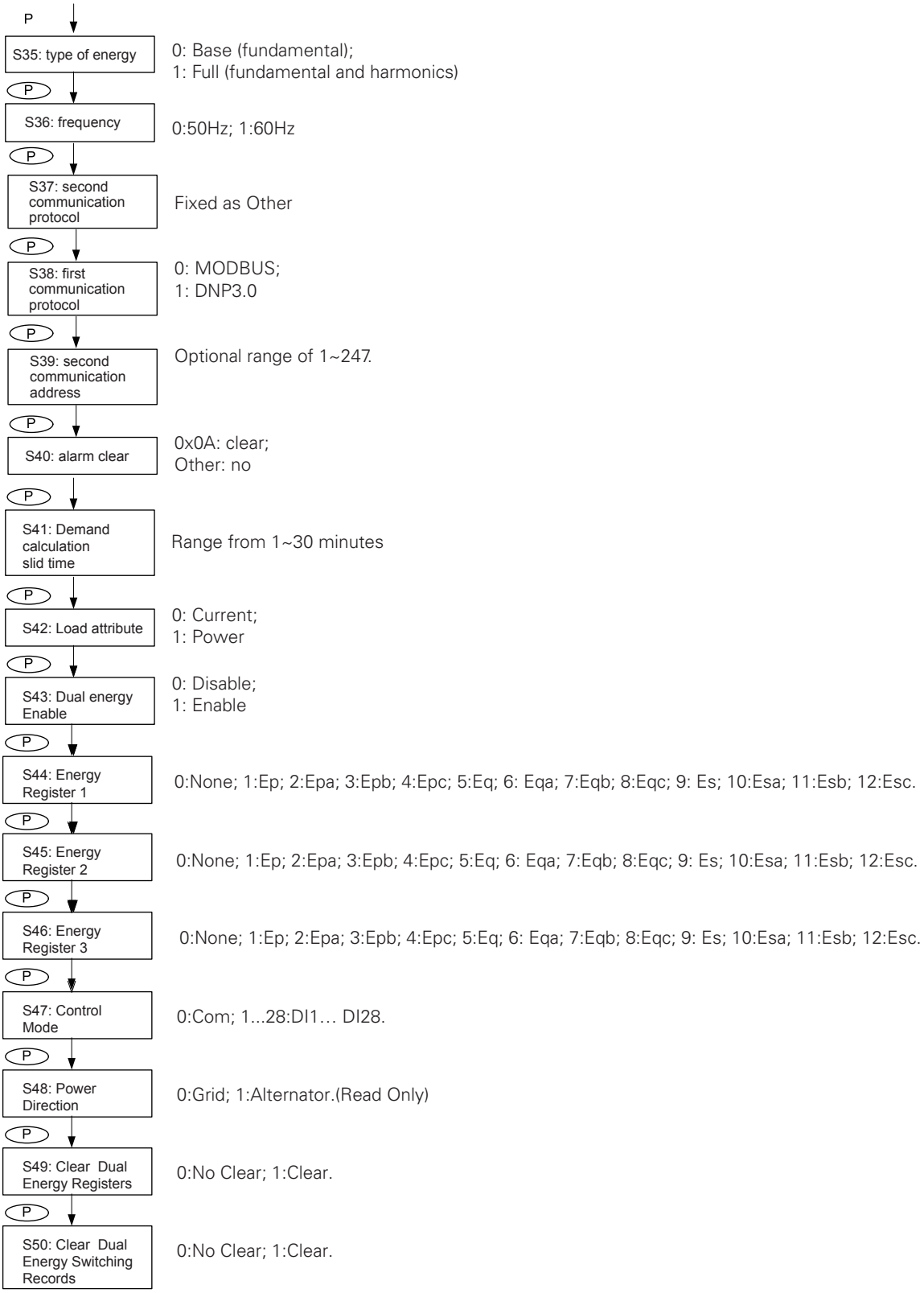


Figure 39. System Parameter Setting Sequence (continued).



Note: The figure shows the sequence for using the P key. If using the E key to change the page, the sequence will reverse.

D. Expanded I/O Module Parameter

In the expanded I/O module parameter mode, the user can choose to view the available modules that are attached to the meter and modify their parameters. If no expanded I/O modules are connected, the screen will display "NO IO". To return to system parameter setting mode main menu, press H (no other keys have functionality on this screen).

Key functions for I/O module selection:

Press H to return to parameter selection mode.

If more than one module is connected, pressing P moves the cursor downwards. The cursor will return to the top after it reaches the bottom. Pressing E will move the cursor back upwards.

Press V/A to select the module and enter the I/O module parameter setting mode.

Key functions for setting the I/O module parameter:

Press H to return to I/O module selection mode.

Pressing P will take the user to the next screen in the sequence.

Pressing E will take the user to the previous screen in the sequence. Press V/A to modify the selected parameter

Key functions for modifying the parameter:

Press H to move the flashing cursor to the next position.

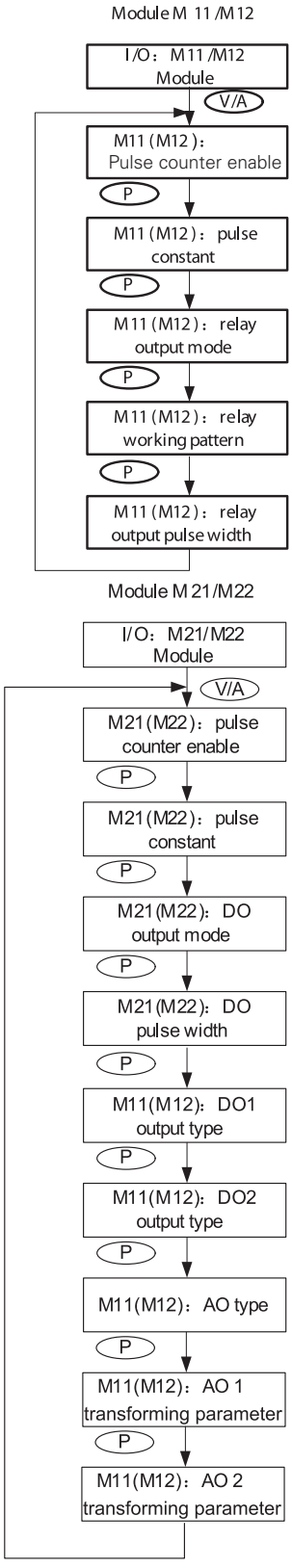
Press P to increase the number by 1.

Press E to decrease the number by 1.

Press V/A to confirm the modification and return to parameter selection mode.

Table 3 shows the sequence:

Table 3. Expanded I/O Module Parameter Sequence.



DI of PXM1K-1 can be used as the pulse counter, each DI function corresponds to one bit of a 6-bit register. The correspondence bit of 0 means that the DI works as the digital status input and correspondence bit of 1 means that the DI works as the pulse counter. For example, if the setting value is 000001, it means that DI1 is set as the pulse counter and other DIs work as digital status inputs.

If the DI works as a pulse counter, when the number of pulses counted by the DI equals to the pulse constant, the pulse counter will increase by one. This means that the actual pulse number equals the number of pulses counted multiplied by the pulse constant.

Relays of PXM1K-1 can be used as alarm output or control output.
ALM: alarm output;
CTRL: control output

When set as control output, relays have two control methods: latch or pulse.
LATCH: latch mode;
PUL: pulse mode

If the relay pulse control method is selected, the relay contact will close for a preset period and open afterwards. The pulse width range is 50~3000 ms.

DI of PXM1K-2 can be used as the pulse counter, each DI function corresponds to one bit of a 4-bit register. The correspondence bit of 0 means that the DI works as the digital status input and correspondence bit of 1 means that the DI works as the pulse counter. For example, if the setting value is 0001, it means that DI1 is set as the pulse counter and other DIs work as digital status inputs.

If the DI works as a pulse counter, when the number of pulses counted by the DI equals to the pulse constant, the pulse counter will increase by one. This means that the actual pulse number equals the number of pulses counted multiplied by the pulse constant.

DO of PXM1K-1 can be used as alarm output or energy pulse output.
ALM: alarm output;
PUL: energy pulse output

Range from 20-1000ms.

Choose output energy type for DO1. Range from 0-4.
0: no output; 1: import active energy; 2: export active energy 3: import reactive energy; 4: export reactive energy

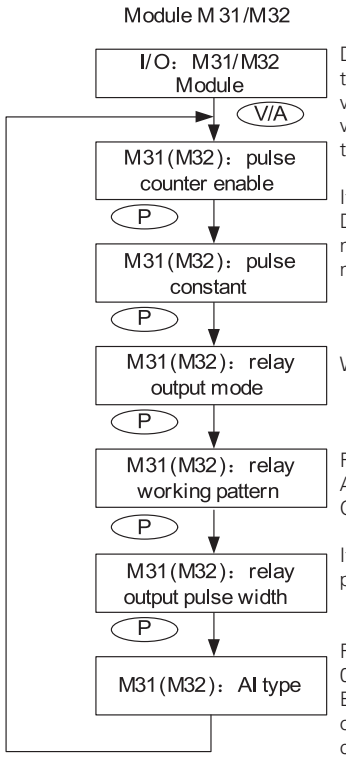
Follow the DO1 setup method to setup DO2.
If DO type is set as alarm output, DO1 and DO2 output type parameters will have no effect.

Range from 0 to 3.
0: 0-20mA; 1: 4~20mA; 2: 0-5V; 3: 1-5V

Be aware that module with current option cannot be set as voltage type (i.e. option 2 and 3 are unavailable); modules with voltage option cannot be set as current type (i.e. option 0 and 1 are unavailable).

For AO1 and AO2 parameter:
Range: 0~29, see Chapter 5 page 99 "AO transforming parameter settings" for more details. When modifying AO parameter, AO input/output values reset default value.

Table 3. Expanded I/O Module Parameter Sequence (continued).



DI of PXM1K-3 can be used as the pulse counter, each DI function corresponds to one bit of a 4-bit register. The correspondence bit of 0 means that the DI works as the digital status input and correspondence bit of 1 means that the DI works as the pulse counter. For example, if the setting value is 0001, it means that DI1 is set as the pulse counter and other DIs work as digital status inputs.

If the DI works as a pulse counter, when the number of pulses counted by the DI equals to the pulse constant, the pulse counter will increase by one. This means that the actual pulse number equals the number of pulses counted multiplied by the pulse constant.

When set as a control output, relays have two control methods: latch or pulse.

Relays of PXM1K-3 can be used as alarm output or control output.
ALM: alarm output;
CTRL: control output.

If relay pulse control method is selected, the relay contact will close for a preset period and open afterwards. The pulse width range is 50~3000ms.

Range from 0 to 3.
0: 0~20mA; 1: 4~20mA; 2: 0~5V; 3: 1~5V
Be aware that module with current option cannot be set as voltage type (i.e. option 2 and 3 are unavailable); modules with voltage option cannot be set as current type (i.e. option 0 and 1 are unavailable).

Note: Table 3 shows the sequence for using the P key. If using the E key to change, the sequence will be reversed.

E. Alarm Parameter

In the alarm parameter mode, the user can view and modify the parameters.

Key functions for finding the alarm parameter:

Press H to return to parameter selection mode.

Pressing P will take the user to the next page in the sequence.

Pressing E will take the user to the previous page in the sequence.

Press V/A to modify the selected parameter.

Key functions for modifying the parameter:

Press H to move the flashing cursor to the next position.

Press P to increase the number by one.

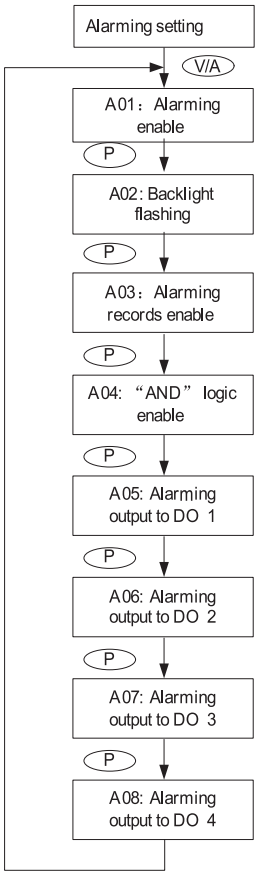
Press E to decrease the number by one.

Press V/A to confirm the modification and return to parameter selection mode.

Figure 40 shows the sequence:

Note: The figure shows the scrolling sequence for using key P. If using E for scrolling the page, the sequence will reverse.

Figure 40. Alarm Parameter Rolling Sequence.



Yes: Alarm enable; No: Alarm disable

It can be selected as cue signal for alarming.
Yes: backlight flashes upon alarm condition; No: no backlight flashing

There are 16 alarm channels available. Each channel is controlled and enabled 1 bit each from a 16-bit register. Bit value of 1 means that the corresponding alarm channel is enabled whereas 0 means that the channel is disabled. The meter will display the value of this 16-bit register in decimal numbers (for different channel combination). For example, 00000 means that all channels are disabled; 00001 means only the first channel is enabled; 65535 means that all channels are enabled. Refer to section 4.4 on page 65 for more details.

"AND" logic relationship can be set among channels. When an "AND" logic is in place, both channels have to be triggered before the meter sends out the alarm signal. The logic can be set according to the predefined rule (refer to section 4.4 for more details). User can set up to 8 logic relationships for alarming. Each logic relationship is controlled and enabled 1 bit each from a 16-bit register (only the lower 8 bits are used). Bit value of 1 means that the corresponding logic relationship is enabled whereas 0 means that the relationship is disabled. The meter will display this 8-bit value in decimal numbers (for different relationship combination). For example, 000 means that all relationships are disabled; 001 means only the first relationship is enabled; 255 means that all relationships are enabled.

When DO1 works in alarming mode, a 16-bit register is used to control which channels are associated with this output. Similar to the alarm channel selection, this 16-bit value is expressed in decimal when reading on the meter front. For example, 00000 means that no alarm channels are associated to this output; 00001 means that alarm channel 1 is associated to DO1; 65535 means that all alarm channels are associated to DO1. Refer to section 4.4 for more details.

If 2 PXM1K-2 modules are attached to the meter, DO1 and DO2 denote to the first and the second DO channel of PXM1K-21; DO3 and DO4 denote to the first and the second DO channel of PXM1K-22 respectively. DO2, DO3 and DO4 use the same setup method as DO1.

3.8 Page Recovery Function

The PXM 3000 meter has a page recovery function. This means that the meter stores the current display page in the non-volatile memory upon power loss and reloads the page when power recovers. If power is lost while in parameter setting mode, the meter will show voltage display when power recovers. If power is lost while in expanded I/O module data mode, but this module is no longer connected when power is recovered, the meter will return to the voltage display page instead.

4. Web Server Interface

4.1 General

4.1.1 Accessing the Web Pages

Launch the PXM 3000 web UI application on a web browser by using the meter IP address, which redirects to login screen below.



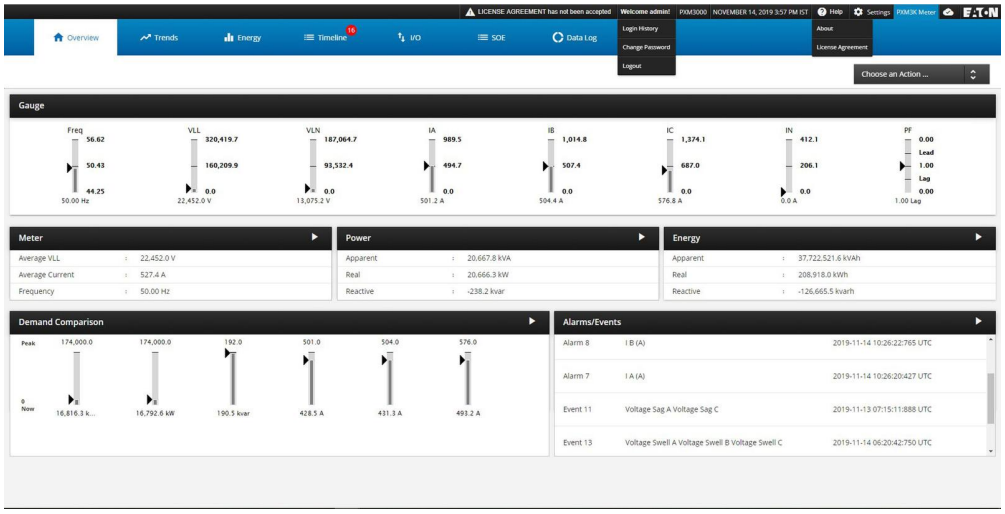
Supported Web Browser and Devices

- Google Chrome (Resolution – 1280 x 1024 & 1920 x 1080)
- I-pad

The default username is admin and the default password is **admin**. Log in using these credentials.

- On successfully login, the user will be redirect to home page.
- Login error scenario:
  - Login failure – Invalid credential;
  - Session limit exceeded – Number of allowed concurrent login sessions is already fulfilled.
  - Invalid Certificate – If valid certificate is not installed, the user will be redirect to download certificate page.

4.1.2 Home Page (Default)



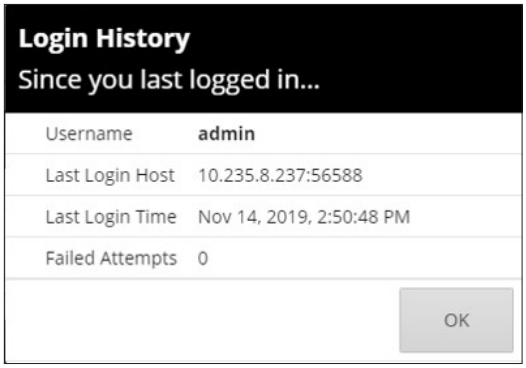
Login History

To see the Login History via the web interface:

1. Access the meter home page. You may be challenged for the admin account username and password.
2. Click on the **Welcome [USER_NAME]** link.
3. Click on the **Login History**.

This will display details shown below:

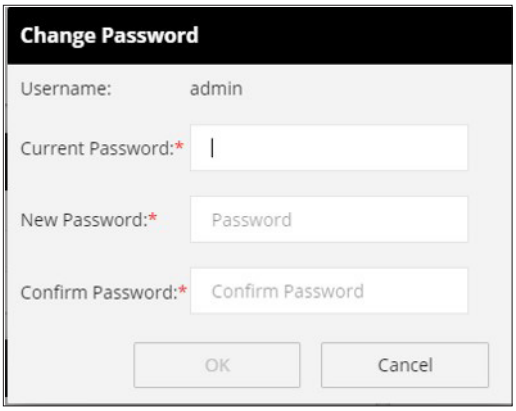
- Username;
- Last Login Host;
- Last Login Time; and
- Failed Attempts.



Change Password:

The PXM 3000 meter has two built-in accounts and an additional password for access to data deletion functions from the meter web UI. You cannot change the name of any of the accounts, but you may change the password.

- The Numeric Meter password account provides access to configuration settings and resets from the meter web UI. This can be configured only from this page. This must be matched with the password policy.
- The user account is for access to the meter web pages only. Someone logged in as user cannot change any settings. The default password is **user.**
- Use the Admin account to change settings or delete data. The default password is **admin**.

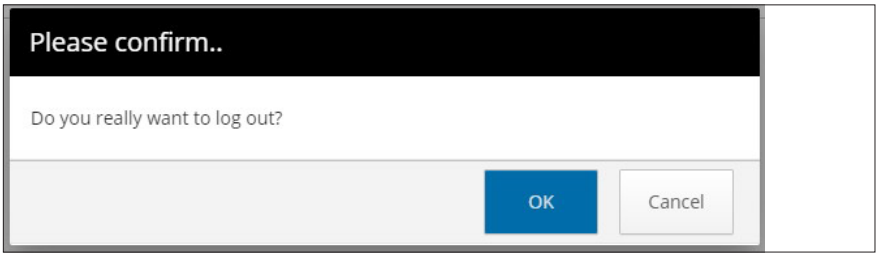


Logout:

To sign-out from the web application -

1. Access the meter home page. You may be challenged for the admin account username and password.
2. Click on the **Welcome [USER_NAME]** link.
3. Click on the **Logout**.

This will prompt a confirmation window.



End User License Agreement (EULA):

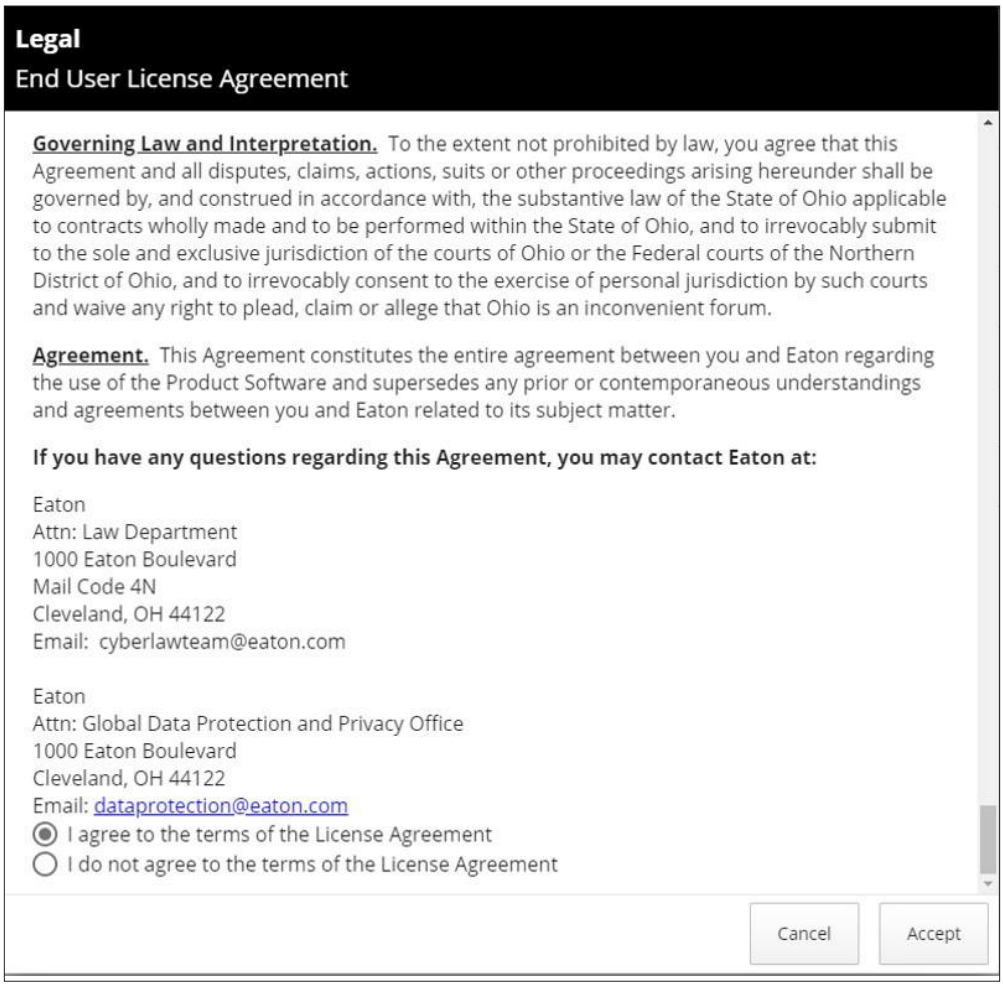
If EULA is not accepted, the user will get an Alert message on top bar stating that it is not accepted



To accept the EULA -

1. Access the Meter home page. You may be challenged for the admin account username and password.
2. Click on the **Help** link from top nav bar.
3. Click on the **License Agreement.**

This will open a popup with the content shown below.



To accept the agreement, the user must scroll down to the pop-up window and select first radio button then click on Accept button. After acceptance pop window closes, the Alert from the top window will be hidden.

4.1.3 SOE

There are 20 groups of records with the same format. The Function code is "03H" for reading. Before gathering SOE records, the selected I/O module must be SOE enabled. If the SOE enabled I/O module is not connected, SOE record logs will not be collected.

Address	Parameter	Code	Range	Data Type	Property
4399J~439fH	First group: time stamp: yyyy:;mm:dd:hh:mm:ss:ms	F3		Word	R
43a0H	first group: DI status	F1		Word	R
43a1H~4438H	2nd to 20th group			Word	R
4439H	I/O module of SOE	F1	0: none; 1: PXM*1K*-11; 2: PXM*1K*-21; 3: PXM*1K*-31; 4: PXM*1K*-12; 5: PXM*1K*-22; 6: PXM*1K*-32	Word	R

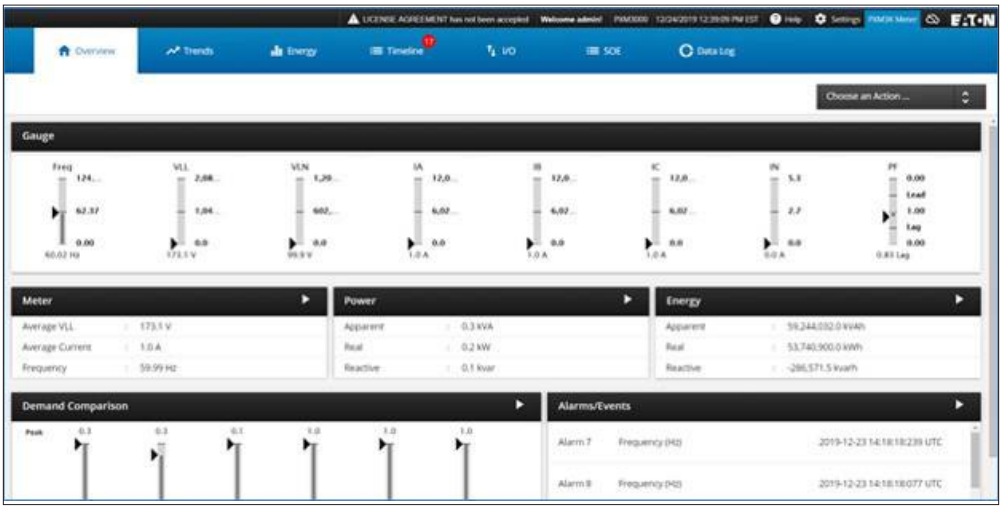
4.2 Web Server Functions

4.2.1 Home Page (A summary of values and navigation to other pages)

4.2.2 Overview

This is the landing page after successful login. The Overview pages show various parameters in the form of values and gauges. The gauges display values along with arrow indication.

Meter, Power, Energy, Demand Comparison, Alarms/Events – by clicking of the arrows at the top-right of each card, the interface navigates to the respective pages.



There is also a "Choose an Action" dropdown to perform the following actions.

- Reset All Min/Max: This action will reset all minimums and maximums for all measured values, including Peak Demands.
- Reset Circuit Gauges: This action will reset the +/-3 Sigma values and mean values for the Home page bar gauges.
- Erase All Trend Data: This action will permanently erase the set of Trend data since 6:30 AM of previous day.
- Erase Power Quality Event Log: This action will permanently erase the full set of Power Quality Event data.
- Erase Alarm Event Log: This action will permanently erase the full set of Alarm Event data.
- Manually Capture Waveform: This action generates a new event with the current waveform [takes around 20 seconds to complete].
- Demand Reset: It resets energy and demand registers from meter.

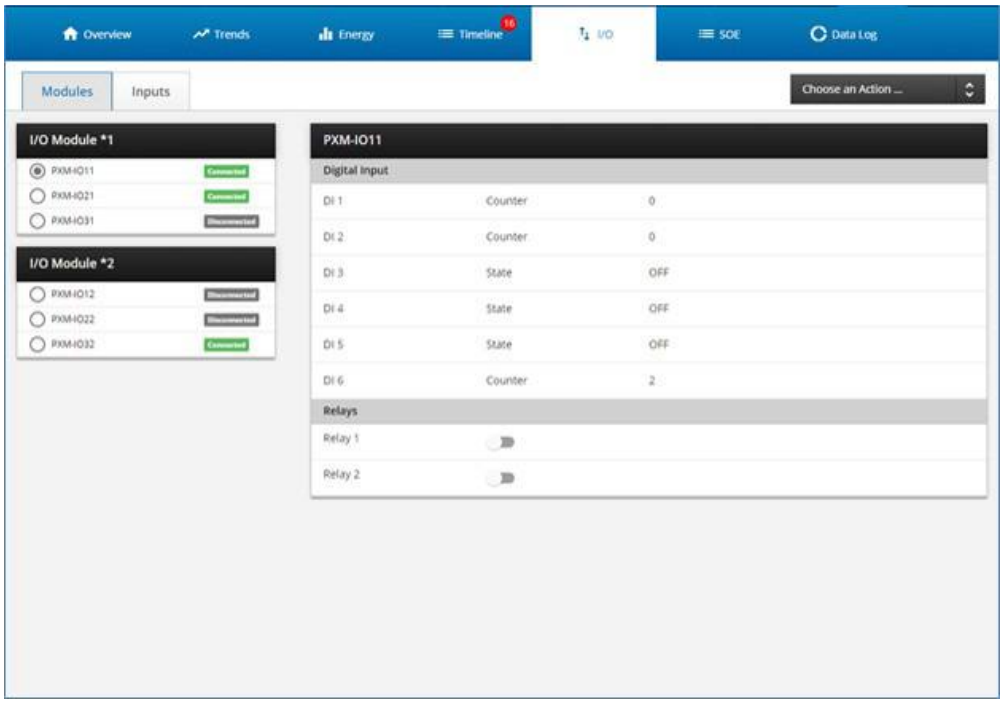
This option will be available on all core pages as per module.

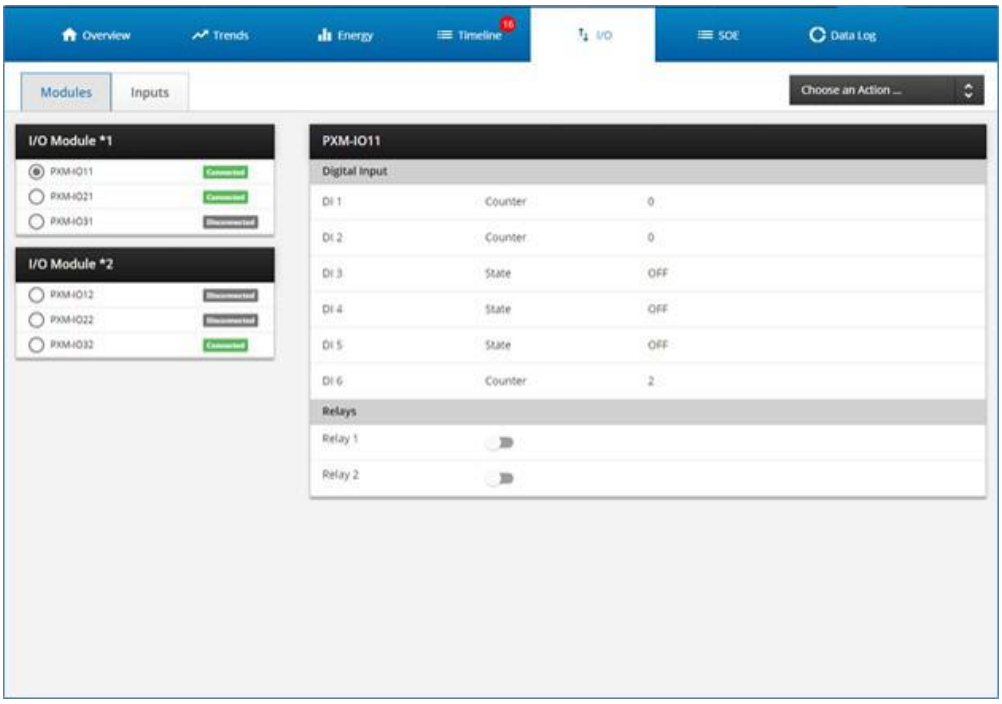
4.2.3 Input/Output (I/O)

I/O module data will be visible only if the corresponding I/O modules are installed. If no I/O modules are installed, the screen will display "No I/O Module Connected".

1. Modules

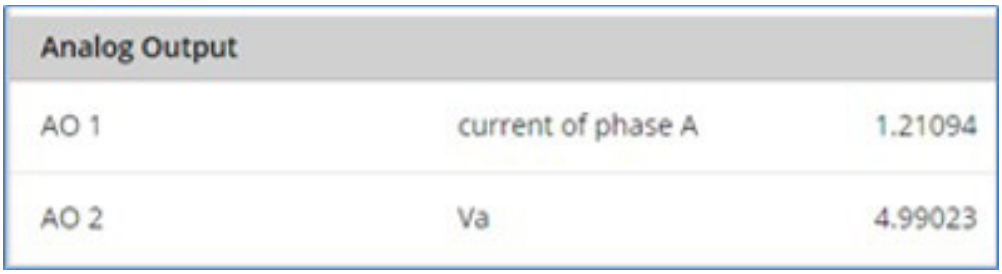
   The user can view I/O Modules *1 and *2, thus resulting in PXM-IO11, PXM-IO21, PXM-IO31, and PXM-IO12, PXM-IO22, PXM-IO32. Module statuses are shown alongside the radio buttons as "Connected" or "Disconnected".





Digital Inputs show DI status as State (ON/OFF) or Counter with values.

Analog Outputs display channels configured with its corresponding value.

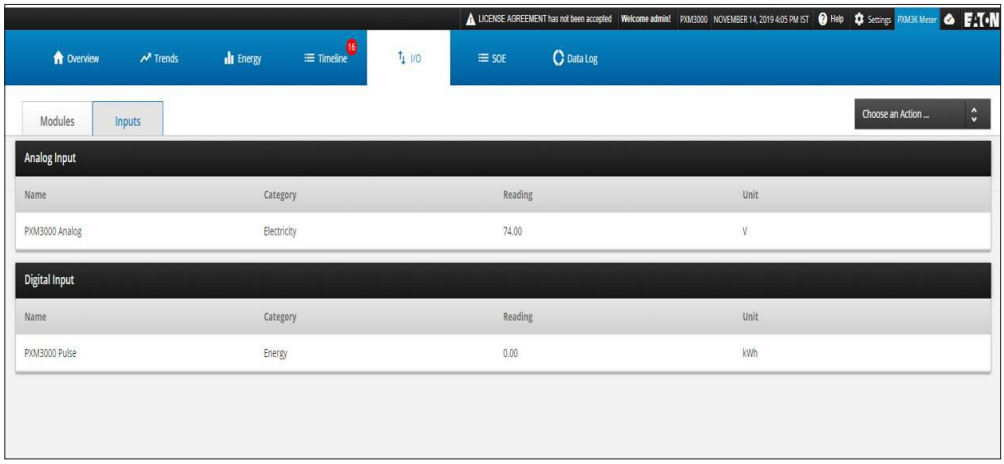


Relays 1 and 2 can be toggled ON or OFF.



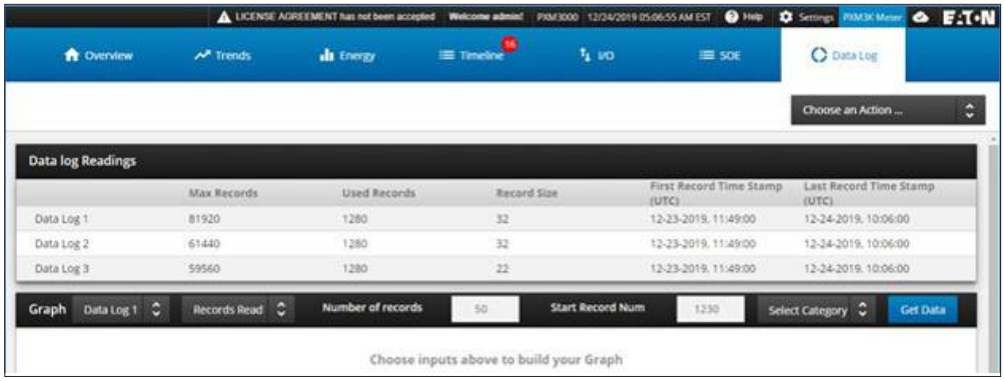
2. Modules

- This tab displays Analog and Digital inputs. As per the I/O settings configured, Name, Category, Reading, and Unit are populated.



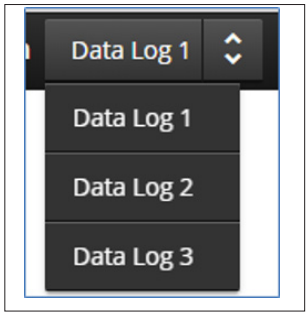
4.2.4 Data Log

Data Log readings are displayed in table form with following data: Data logs categorized into three types, each with different configuration as needed by the user. By default, the user can see details for the three data log types (if they are configured - see data log settings) displaying Max Records, Used Records, Record Size, First Record Time Stamp, and Last Record Time Stamp. The time stamps are displayed in UTC [aka Zulu or Greenwich Mean Time].

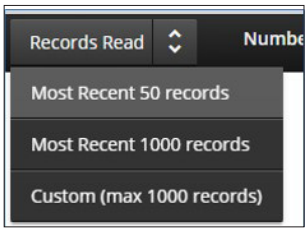


Graphs can be seen by clicking "Get Data". The button is disabled if the required inputs are invalid. The inputs include:

- Data Log Type

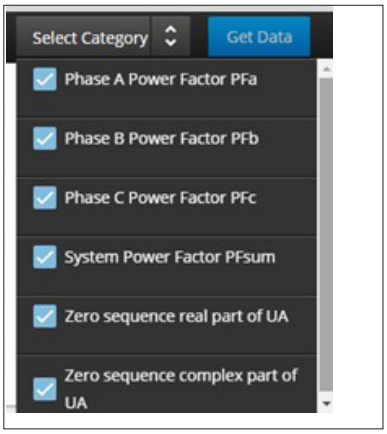


- Records Read



- Number of Records - Accepts positive integer values.
- Start Record Number - Accepts positive integer values.

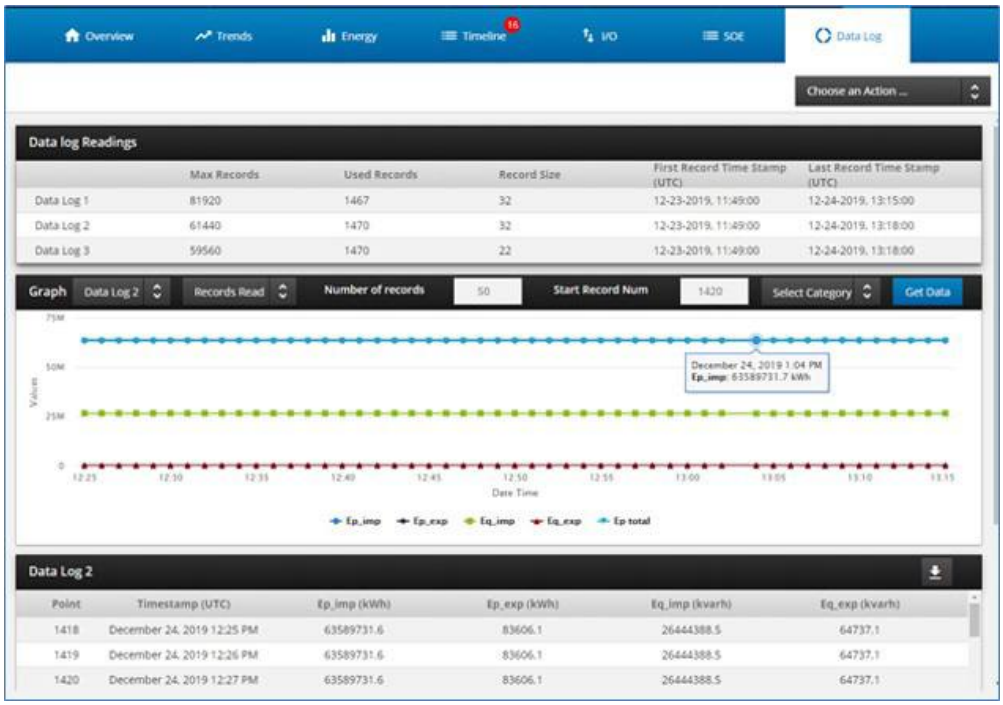
  Note: "Number of Records" + "Start Record Number" should be less than "Used Records"
- Select Category - Maximum of six categories can be selected.



On clicking "Get Data", the following modal pops up.



This shows the progress on the number of records fetched vs. the total requested. The fetching can be cancelled until loading completes. Until then, whatever records are fetched until the point of cancellation will be displayed.



The graph displays selected categories mapped as per their Values vs Date Time. Tooltips are used to display time stamp, category, and value with units when the user hovers over a point. The legends at the bottom of the chart can be used to show/hide them.

Data Log # displays selected data type along with the data contents in tabular format. The Point (record numbers) are mapped with Time stamps (UTC) and all the categories configured for the selected data type. Download icon provided at the top of table can be used to export the tabular data into csv format.

4.2.5 Meter

The Meter page provides voltage, current, and frequency values.

Voltage

The PXM 3000 measures line-to-line and line-to-neutral voltages and calculates rms values every 100 ms. The system average values are calculated for main voltage channels. The Power Xpert Meter maintains a log of all voltage rms measurements, date, and time stamps. This information is also available as a trend graph. The minimum, maximum, and averages are recorded for each interval.

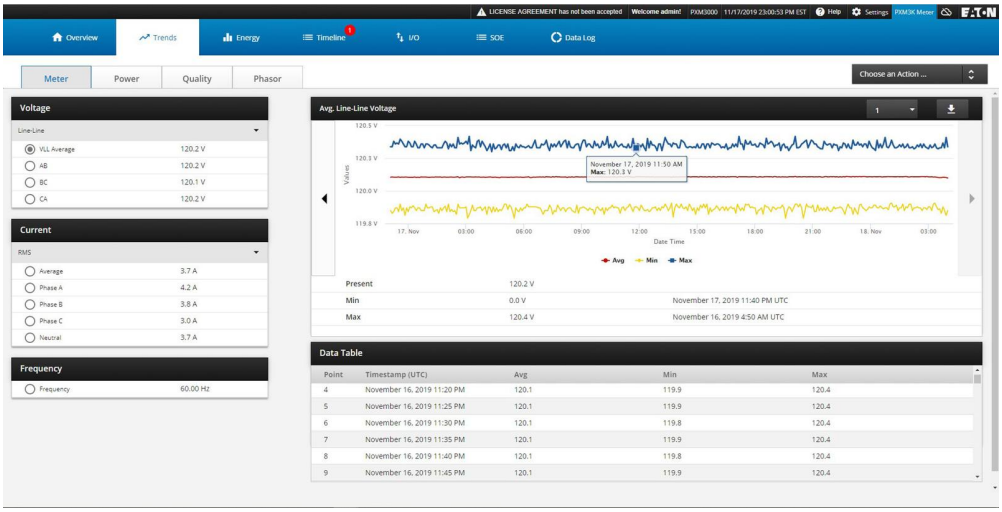
All meter models also store 100,000 global event log entries and 1,000,000 global dead-band log entries. The most recent minimum and maximum values are detected for all the voltage channels and are logged with date and time stamps. Current minimum and maximum values can be cleared by the user at any time through the "Choose an Action" menu. After the current values are cleared, the meter will begin calculating new minimum and maximum values.

Locating Voltage Information

From the Meter page, select **Line-Line, Line-Neutral**, or **Symmetric** from the drop down box beside Voltage on the left side of the page. Click one of the radio buttons below **Voltage** to select a specific voltage measurement:

- **VLL Average, AB, BC, CA** for **Line-Line** or **VLN Average**;
- **AN, BN, CN** for **Line-Neutral**; or
- **Zero-Sequence, Pos-Sequence, Neg-Sequence** (% Voltage and % Current imbalance are also shown) for **Symmetric**.

The page will update to display the selection.

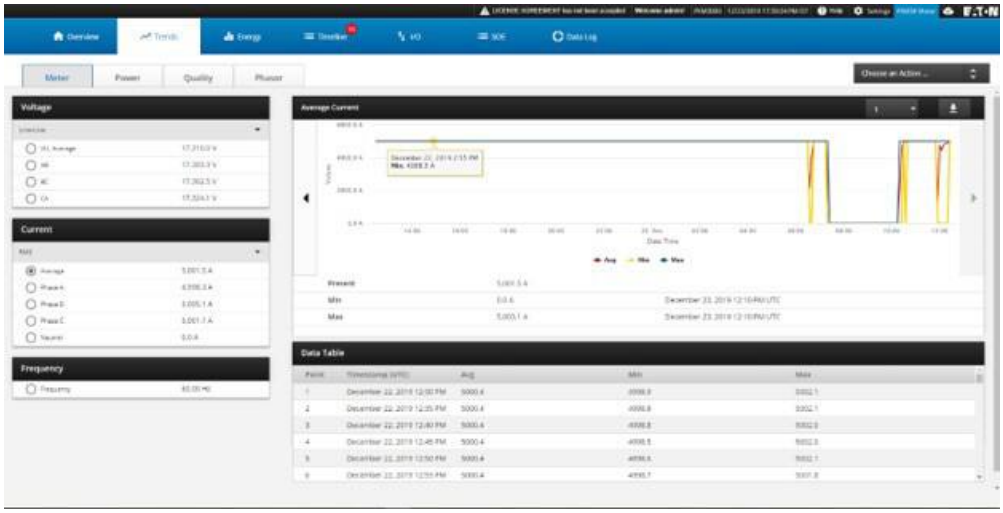


Meter Page: Voltage Display

Current

The PXM 3000 measures per phase and average system current and calculates rms values every 100 ms. The meter will maintain a log of all rms measurements with date and time stamps. The same information is available as a trend.

The most recent minimum and maximum values are detected for all the channels and are logged with date and time stamps. Current minimum and maximum values can be cleared by the user at any time. After the current values are cleared, the meter will start calculating new minimum and maximum values.



Meter Page: Current Display

Locating Current Information

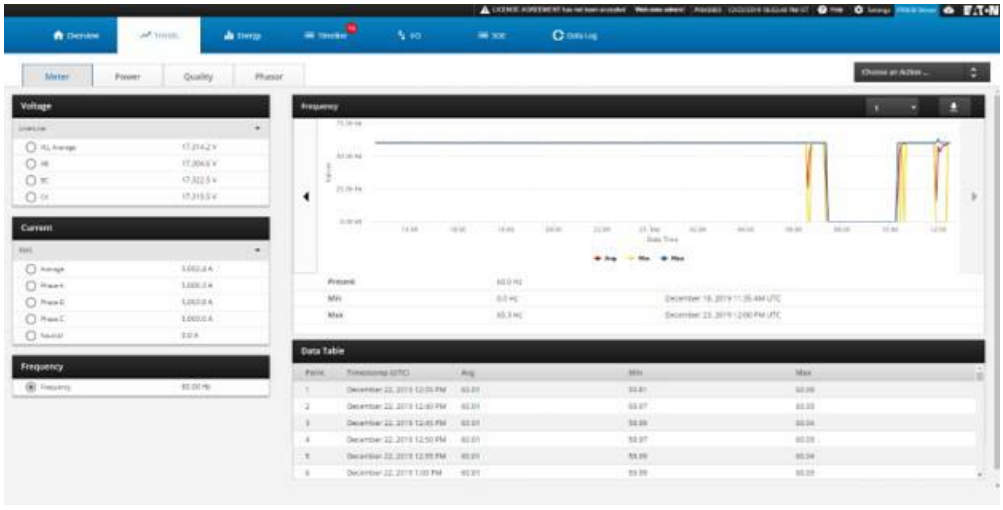
From the **Meter** page, click one of the radio buttons listed below **Current** to select a specific phase or neutral. The page will update to display the selection.

Frequency

Frequency measurements for the meter occur every 100 ms using phase A voltage on V1. In the absence of voltage, there is no frequency measurement. The result is a gap in the trend data for frequency-dependent items. A trend of frequency measurements is maintained with date and time stamps. Minimum and maximum values are detected and saved with date and time stamps of the occurrence.

Locating Frequency Information

From the Meter page, click the Frequency radio button on the left side of the page.



Meter Page: Frequency Display

2.4.6 Power

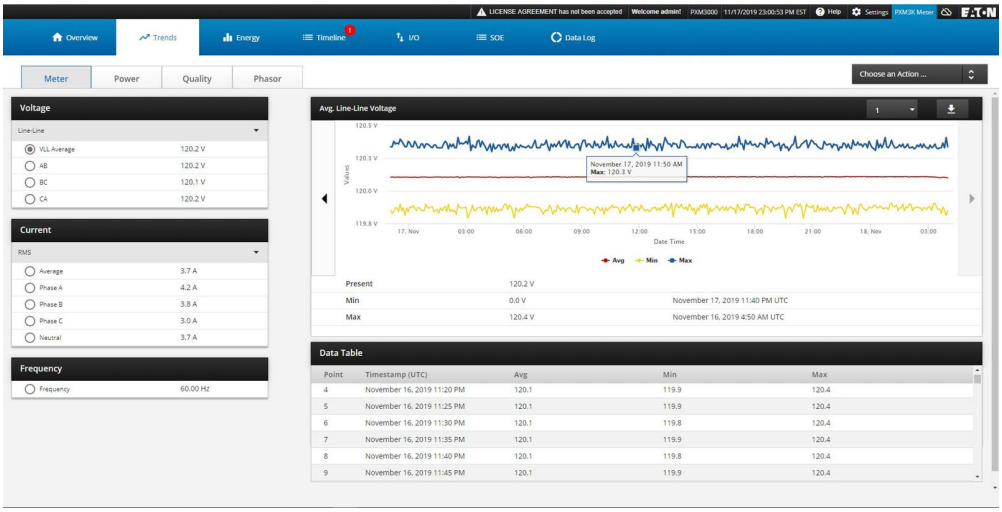
The Power page contains information about the present power consumption and power factor. The meter calculates the following:

- Apparent Power Factor (PF) for System, Phase A, Phase B, and Phase C;
- Apparent Power (kVA) for System, Phase A, Phase B and Phase C;
- Real Power (kW) for System, Phase A, Phase B and Phase C; and
- Reactive Power (kvar) for System, Phase A, Phase B and Phase C.

The meter will log the trend for all of the above measurements, as well as detect and save minimum and maximum values with date and time stamps for all of the above parameters.

Locating Power Information

From the Power page, click one of the radio buttons on the left side of the page that represents the type of power measurement you want to view. The page will update to display a graph with minimum, maximum, and average values as per the selection.



Power Page

4.2.7 Quality

The Quality page contains information about the present power quality for PXM 3000 series meters. These meters calculate the following instantaneous values, averaged over a period of 60 cycles:

- Percent THD of Current for Phases A, B, and C; and
- Percent THD of Voltage for Phases A, B, and C.

Using the drop down box above the graph, you can select to view this information as a trend, with minimum and maximum values, or as harmonics.

Total Harmonic Distortion (THD)

The THD is the ratio, expressed as a percentage, of the rms value of the ac signal after the fundamental component is removed and inter-harmonic components are ignored to the rms value of the fundamental. The formula for THD is provided below. The variables X1 and xn may represent either voltage or current and may be expressed either as rms or peak values, as long as all are expressed in the same fashion.

$$D_x = \frac{\sqrt{\sum_{n=2} x_n^2}}{X_1} \cdot 100\%$$

$$\%THD\ x(t) = 100 * \sqrt{\{x_2^2 + x_3^2 + x_4^2 + \ldots x_n^2\}} / x_1$$

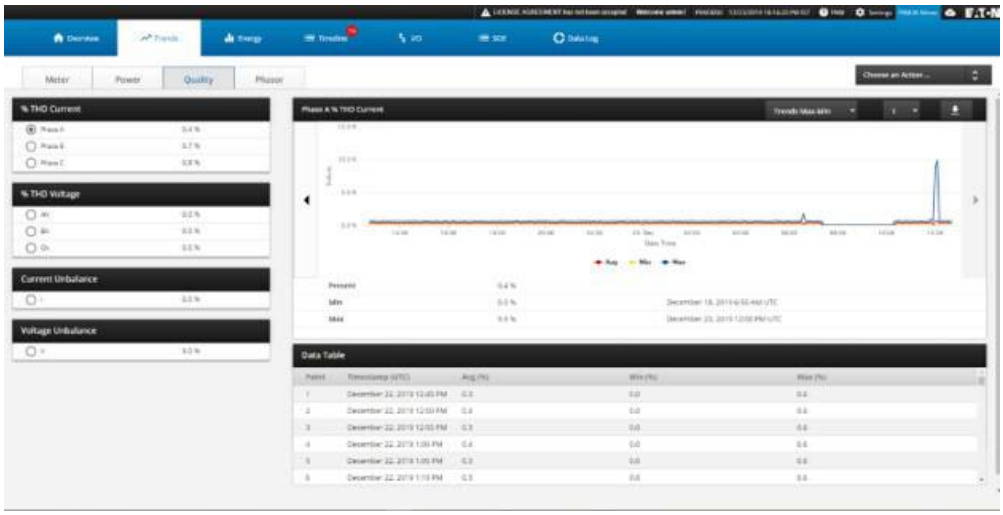
where n is the highest harmonic number used

X1 = fundamental value of current or voltage.

xn = nth harmonic value of current or voltage.

Trend and Min/Max View

Like most other real-time measures, the % THD Current and Voltage measurements in the Power Xpert meter are trended and minimum and maximum values are logged with date and time stamps.



Trend View

Harmonics View

The individual harmonics of the THD may be viewed for up to 63rd harmonics of the fundamental frequency. Select this view from the upper-center drop down list.

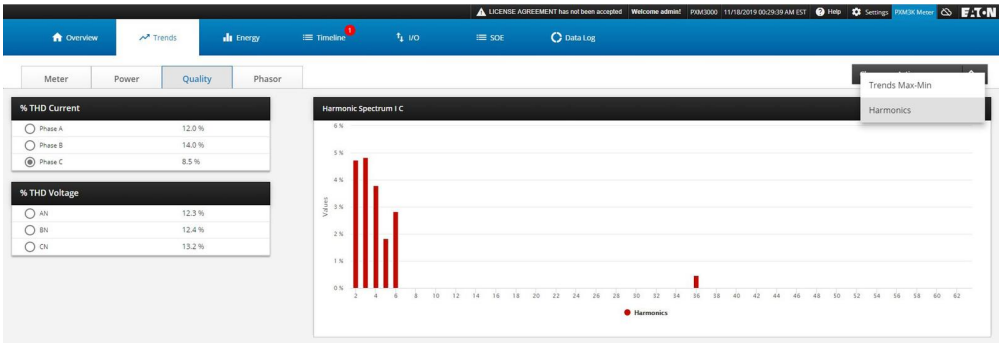
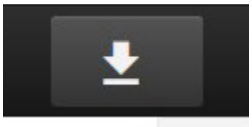
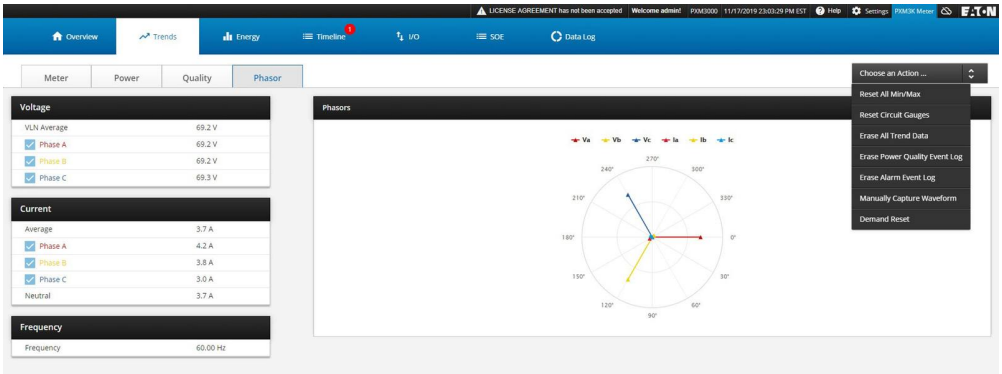


Table of Data Points for Trend Min-Max view, a table below the graph provides the numeric values as well. This table can be saved as a csv file by clicking the download button (pictured to the below).



4.2.8 Phasor

By default, all the options are selected when the Phasor diagram is selected. You can clear an option on the left side of the page to remove the correlating point of the phasor diagram. Phasor is not available if wiring configuration is set to Delta.



4.2.9 Energy

The Power Xpert meter measures and accumulates three kinds of energy values every 60 cycles: Real Power (Watts), Reactive Power (VAR) and Apparent Power (VA). Real Power and Reactive Power take direction into account, and so are recorded for forward, reverse, net and sum.

- Real Power (Watts).
- Reactive Power (VAR) is the product of the voltage, the current, and the sine of the phase angle between them.
- Apparent Power (VA) is the square root of the sum of the squares of the active and reactive powers.

Demand

Demand is the power consumption at the receiving terminal averaged over a specific interval of time. The Power Xpert meter is capable of measuring demand by two methods, Fixed (Block) or Sliding (Rolling), in intervals of 1 to 30 minutes. These measurements are performed for the Total Current, kW Forward/Reverse, Net and Sum; kVAR Lead/Lag, Net and Sum, and kVA.

Fixed and Sliding Demand

Fixed demand (Total Current), which is an average system current over time, can be set to average current over a range of 1 to 30 minutes. This is known as fixed window. For example, setting the current demand window to 15 sets the meter to determine the average current over the past 15 minutes and update the value every 15 minutes.

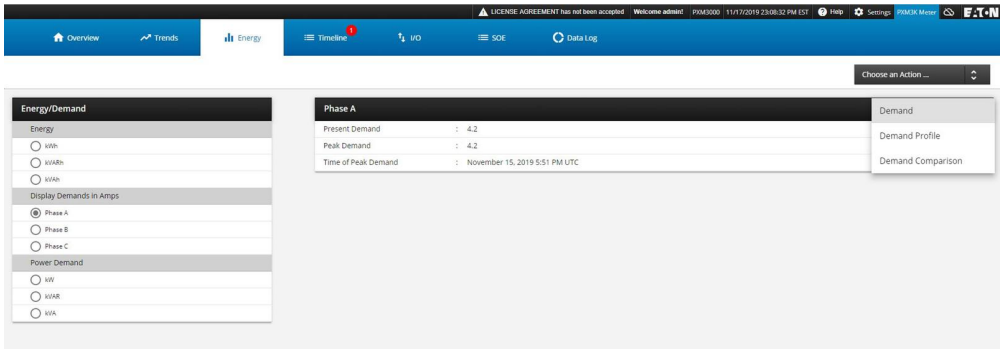
Power demand can be fixed, as described above, or sliding - e.g.: 15-minute average can be obtained that is updated every 5 minutes. To accomplish this, the Subinterval is set to 5 minutes and the Total Interval is set to 15.

Note: The Total Interval is the number of minutes in the average calculation. The subinterval is the number of minutes between updates.

The Power Xpert meter starts the demand window on meter startup.

The Power Xpert logs the Demand Profile data from the following sources:

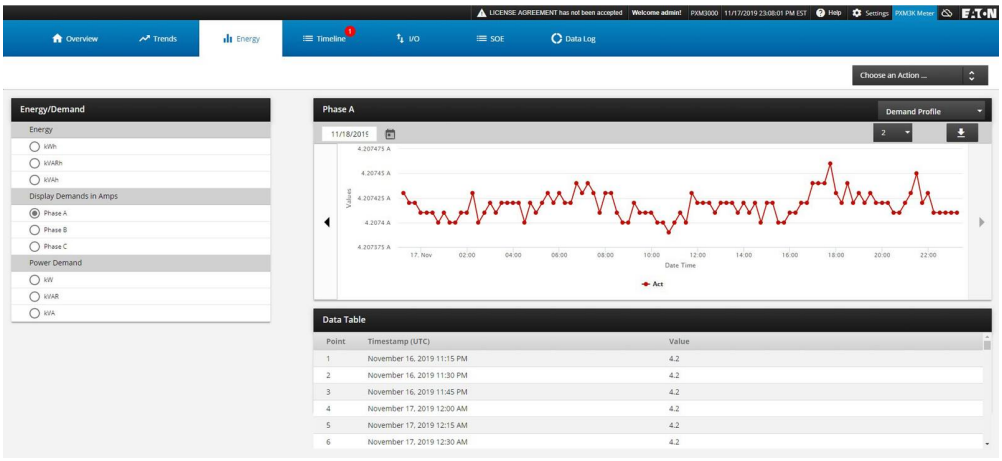
- KVA;
- kW;
- kVAR; and
- Phase Current demand.



Demand Profile

Demand Profile is the magnitude and duration of loads applied in a prescribed time sequence, including the transient and steady-state characteristics of the individual loads.

For the Demand Profile, you can select a date from the drop down list to change the graph to reflect data from the different day

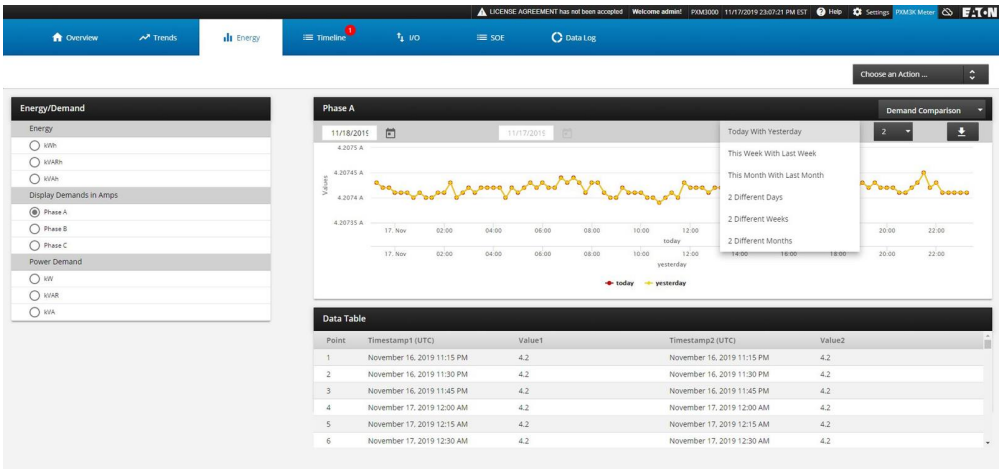


Demand Comparison

The Demand Comparison page is similar to the Demand Profile page but provides a graph you can use to compare two overlaid demand plots from selectable periods of time.

You can select to compare plots between:

- Today and yesterday;
- This week and last week;
- This month and last month;
- Two different days;
- Two different weeks; or
- Two different months.



While viewing the Demand Comparison page, you can also select to compare the information to previous time periods: day-to-day (Demand Comparison only), week-to-week, or month-to-month. If you are comparing two specified time periods, such as two different days, a calendar control appears when you click either date. Click a day in the calendar to select the beginning and ending dates in the comparison.

Locating Demand and Energy Information from the Energy Page

1. Click a radio button on the left side of the page that represents the kind of Energy and Demand you wish to view.
2. From the drop down box near the top of the page, select either Demand, Demand Profile, or Demand Comparison.

The page will update to display the information selected.

The table shown below the graph shows each demand update value along with the time stamp. The Export Data button provides the option to write a comma separated values (.csv) file. The first row in the file contains the column titles.

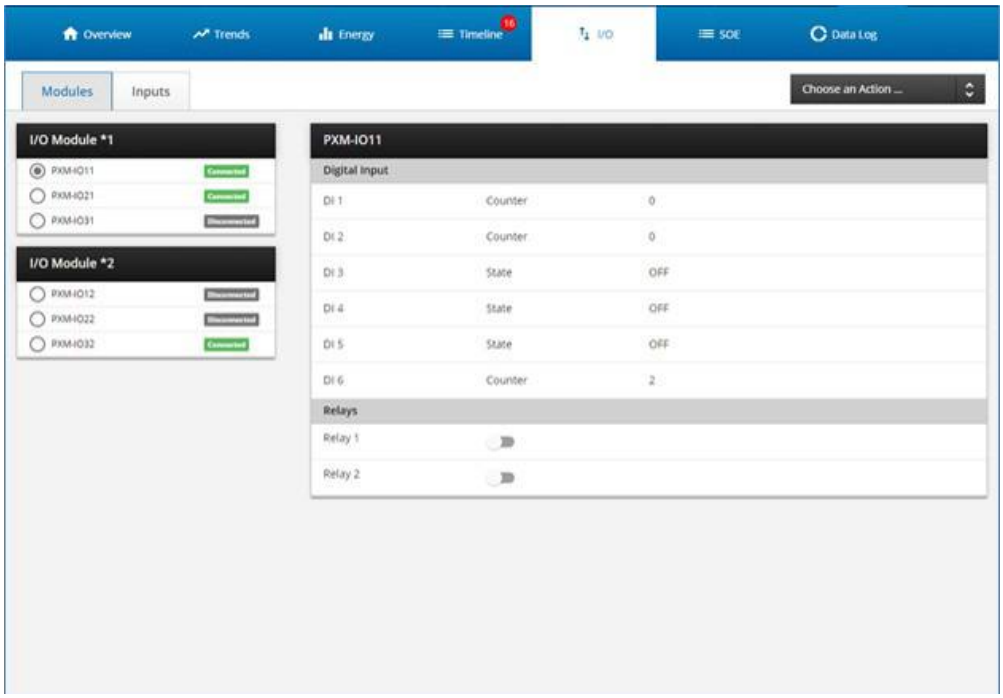
The Log Data display time interval will vary as per selected Zoom level.

Zoom Level	Interval (Minutes)
1	5
2	15
3	30
4	60

4.2.10 I/O

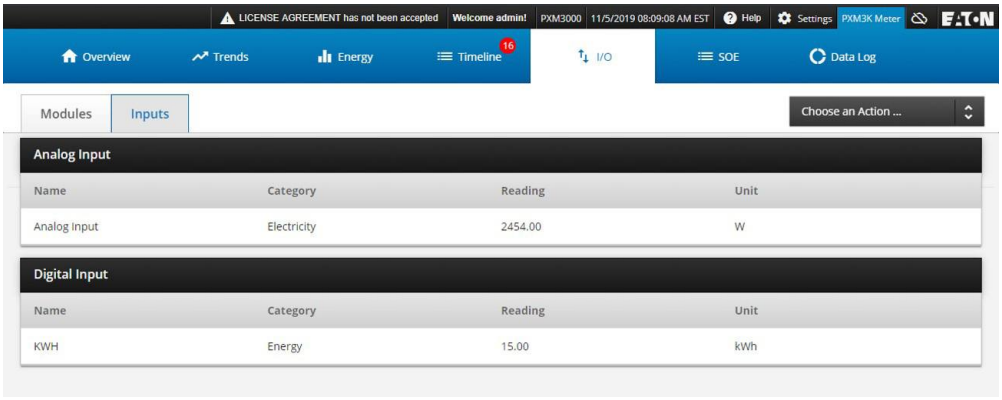
Modules

The I/O -> Modules shows the current status of all of the inputs and outputs from any installed I/O cards.



Inputs

I/O -> Inputs page is the viewing window for Pulse and Analog inputs.



4.2.11 Sequence of Events (SOE)

Before gathering SOE records, the selected I/O module must be SOE enabled. If the SOE is enabled but an I/O module is not connected, SOE record logs will not be collected.

When the digital input circuit is set to detect remote signals, the recording function of SOE can be enabled. Therefore, when the remote signals change, the I/O module can record this information accordingly.

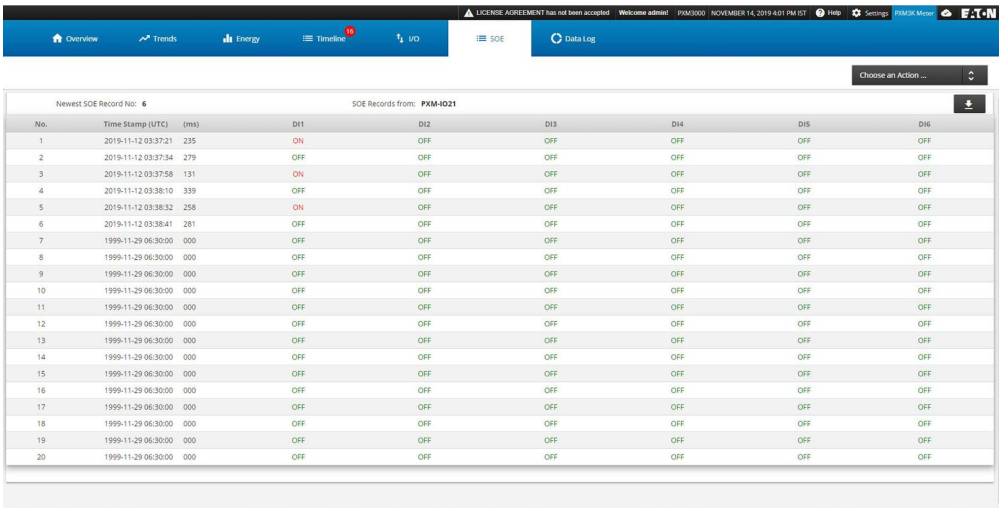
The 20 groups of SOE records are arranged based on time. When more than 20 groups of SOE records are generated, the records will begin at the first one.

When the PXM 3000 meter is powered on, the SOE begins to record immediately. The data in the SOE records will not be lost if the meter is powered off. When the enabled SOE function is changed, the records will be lost.

Only one I/O module can be SOE enabled at one time. If the I/O module is not linked to the PXM 3000 power meter, then there is no need to enable SOE function in the software.

SOE Enable setting is in Setting-> General-> SOE.

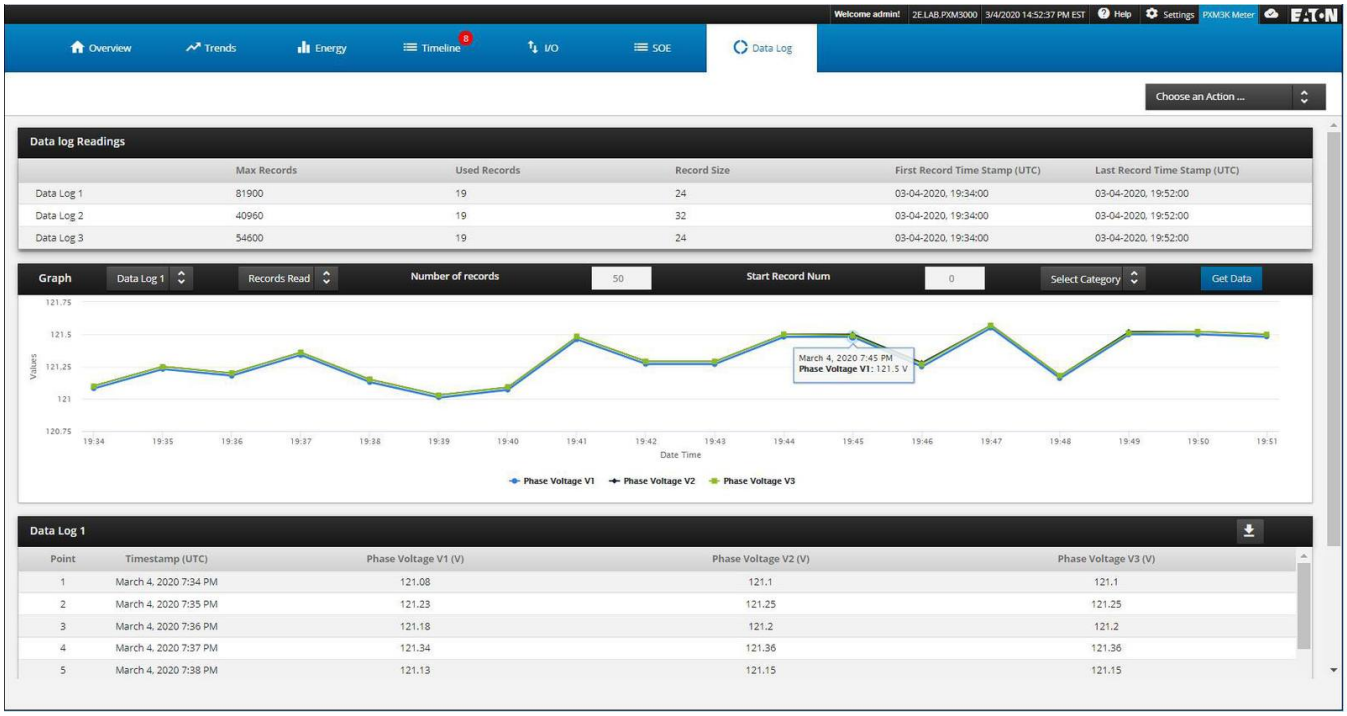
Data of SOE records can be read and viewed in PXM 3000 webpage as shown in the following image.



User can download SOE logs from webpage in csv format.

4.2.12 Data Log

This page provides option for retrieving, plotting, and downloading recorded data log.



The following procedures show how to retrieve, and plot required data.

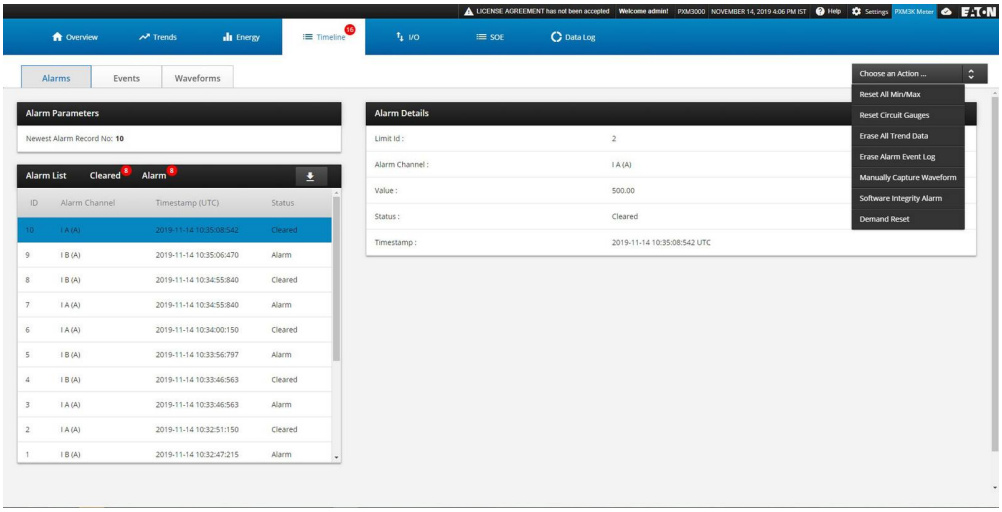
1. Choose the Data Log that is to be retrieved (Data Log 1 / Data Log 2/ Data Log 3).
2. Select the number of records to be retrieved (Most Recent 50 records/ Most Recent 1000 records/ Custom [Max 1000 records]). If Custom (Max 1000 records) is selected:
   a. Enter the number of records to be retrieved; or
   b. Update the starting record number.
3. Select the parameters whose records are to be retrieved.
4. Click "Get Data".

To download the retrieved data, use the Download button.

4.2.13 Alarms

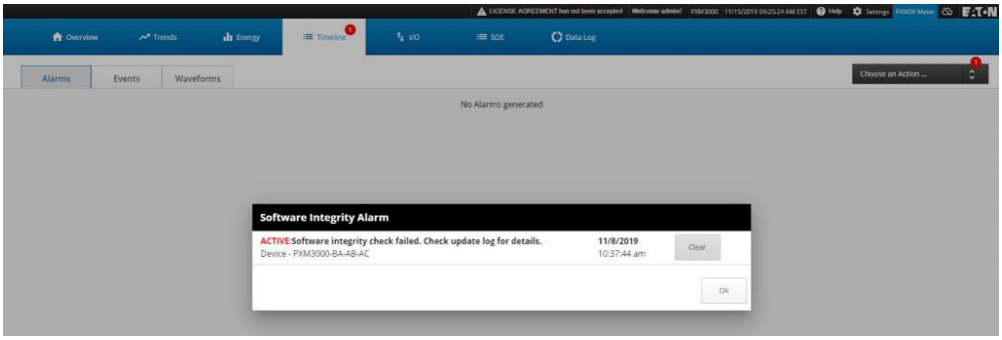
PXM 3000 meter has over/under limit alarming capabilities. When the monitored parameter goes over/under the preset limit and stays at the level over the preset amount of time delay, the over/under limit alarm will be triggered. The over/under limit value and its time stamp will be recorded in the alarming log. The meter can record up to 16 alarming records. When extended I/O modules are attached, digital outputs (DO) and relay outputs (RO) can be triggered upon alarm conditions and used to activate downstream devices such as a beacon light or a buzzer.

Before using the alarming function, alarm conditions such as logic dependency, target setpoint, time delay, etc. must be set correctly. Settings can be accessed and modified from the Alarm menu under the Settings tab.



Alarm details can be viewed and downloaded under Alarms tab on Timeline page. The user can clear the alarms from Choose an Action drop down menu.

In addition to 16 Metering Alarms, 1 additional alarm is for the software integrity check for PXM 3000 meters. The Integrity alarm can be checked under "Choose an Action" tab on Alarms page.



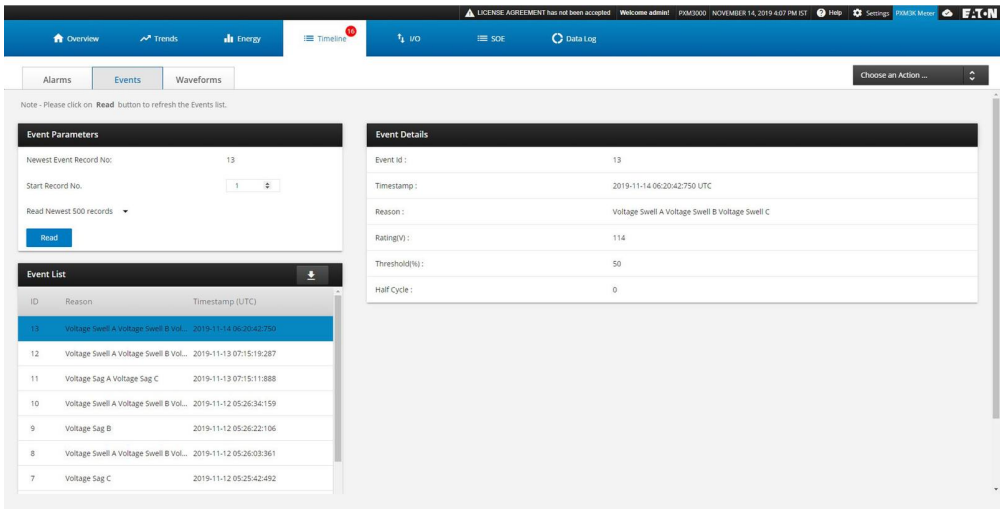
4.2.14 Events

Power quality events can be found under Timeline tab.

1. Power Quality Event Logging

When a power quality event happens, such as voltage sag and swell, etc., the PXM 3000 meter will record the time stamp and the triggering condition of the event. It can save up to 50, 000 power quality events.

The event logging feature can log 50,000 events. If the 50,000 events are full, no more events will be logged even if the triggering condition happens. The user must clear the event log, and then the logging will log the new event. When the log is cleared, the new event will be logged from the first event happening. There will be no data loss after the power is off.



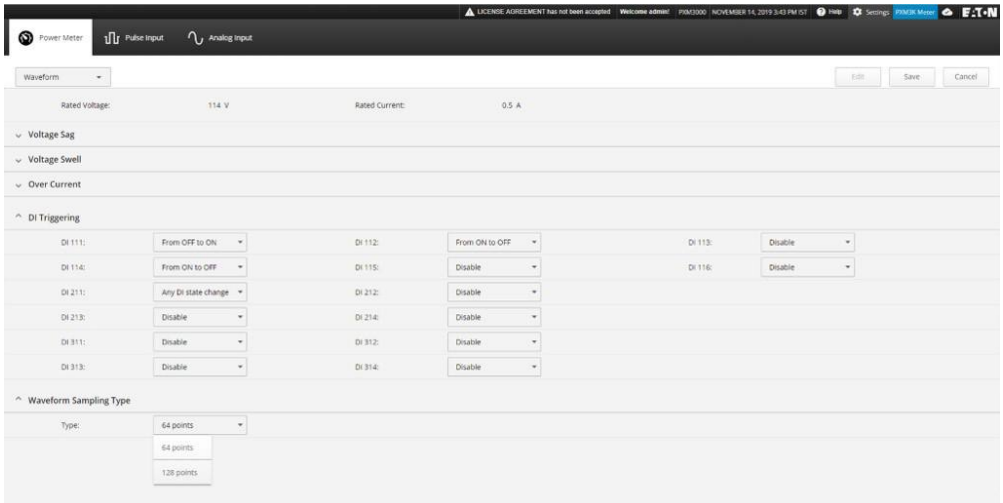
The user can read 10, 50, or 500 events in one request.

The user can download and erase the power quality events from the Choose an Action drop down.

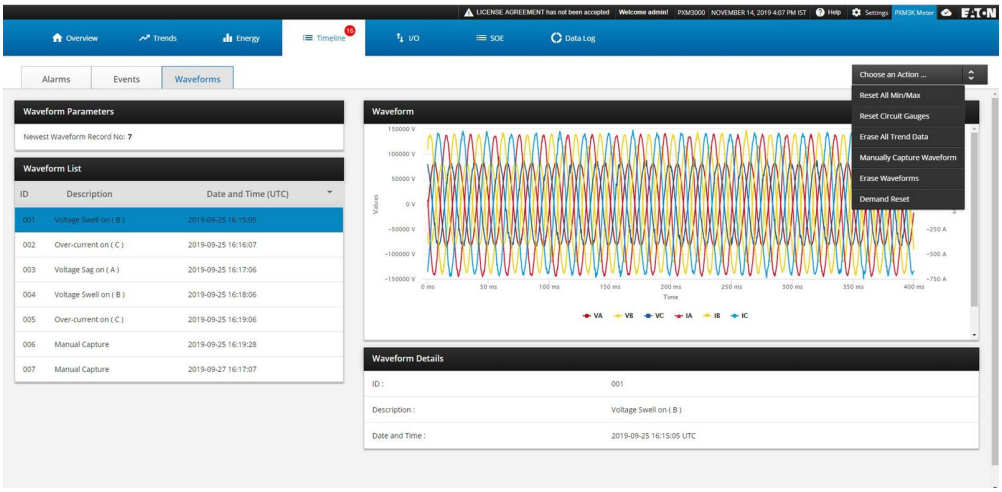
4.2.15 Waveforms

The PXM 3000 can record 200 groups of voltage and current waveform data at a sampling rate of 64 points per cycle or 100 groups of voltage and current waveform at a sampling rate of 128 points per cycle. It provides the captured waveform of ten cycles before and after the triggering point.

The waveform trigger condition can be set from the Setting tab by selecting Waveform from drop down menu on setting page.



The waveform can be read from Waveform tab under Timeline section. The user can download and erase the waveforms from Choose an Action drop down.



Note: Be sure that the configuration software and webpage UI are not accessing the waveform at the same time.

4.2.16 Help

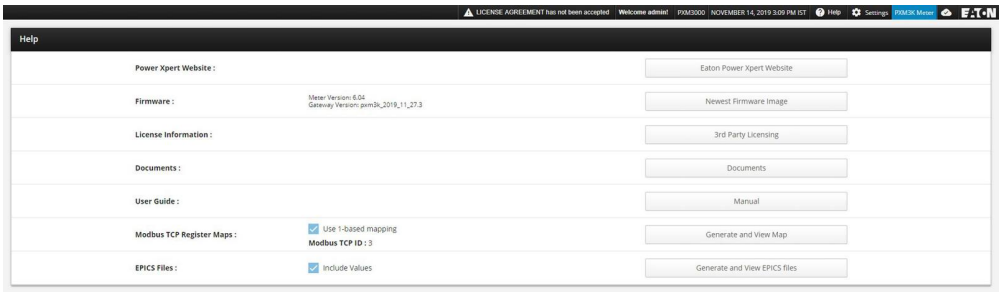
To visit the Help page -

1. Access the meter Home page.
2. Click on the **Help** link from top navigation bar.
3. Click on the **About**.

The Help page not only provides access to the manual, but also lets you:

- Generate a map of the Modbus registers for both the meter and installed I/O cards;
- Access a web page to download the various MIB files;
- Access the Eaton firmware update page;
- Access the Eaton Power Xpert website;
- View licensing information for all 3rd party software used by the Gateway card; and
- Generate the EPICS files for each of the BACnet/IP Devices (meter, gateway, and optional I/O card).

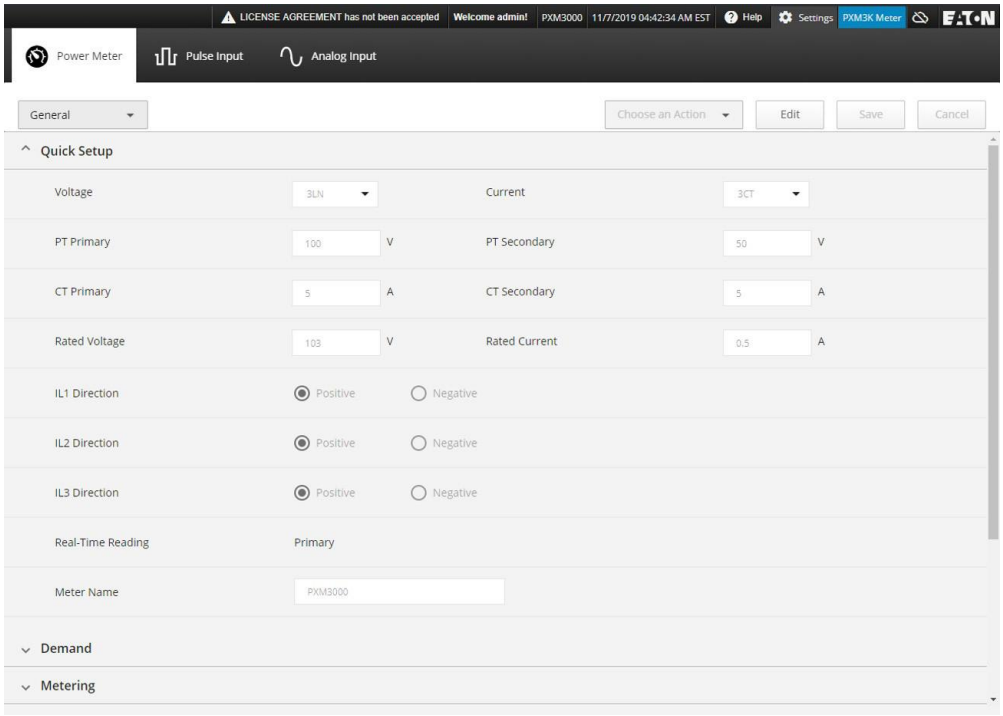
Note: You will be prompted to specify a location on your local file system for the generated TPI file.



4.3 Setup on the Web Server Pages

4.3.1 General

4.3.1.1 Quick Setup



Voltage: Voltage wiring mode can be one of "3LN", "2LL", "3LL", "1LN", or "1LL"

Current: Current wiring mode can be one of "3CT", "1CT", or "2CT".

PT Primary: Primary nominal value of PT, range from 50 V to 500,000 V.

PT Secondary: Secondary nominal value of PT, range from 50 V to 400 V.

CT Primary: Primary nominal value of CT, range from 1 A to 50,000 A.

CT Secondary: Secondary nominal value of CT, can be set to 1 A or 5 A.

Note: If no PT is used, set PT primary and PT Secondary as nominal voltage of the meter.

Rated Voltage: Set rated voltage.

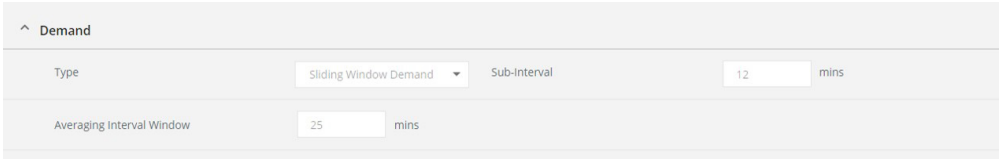
Rated Current: Set rated current.

I1 Direction, I2 Direction, I3 Direction: To adjust polarity of current, the three current directions can be set as "Negative" which means reversing 180 degrees and "Positive" which means normal.

Real Time Reading: Real time reading is always set to Primary.

Meter Name: Meter name can be updated by the user for identification purpose in case of using multiple PXM 3000 meters.

4.3.1.2 Demand

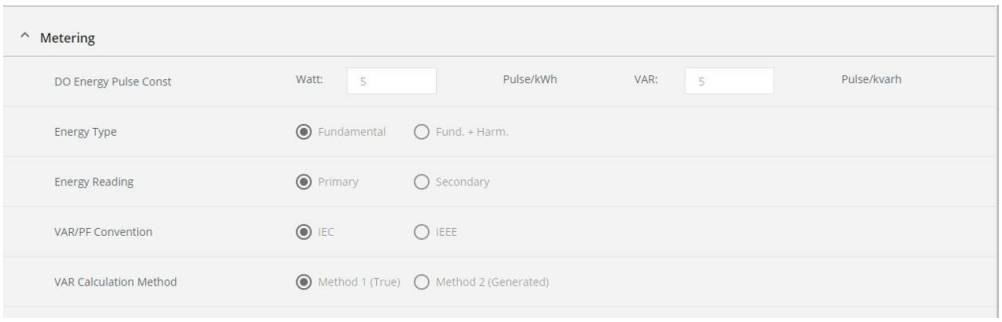


Type: Demand type can be one among "Fixed Demand", "Sliding Window Demand", "Thermal Demand", or "Rolling Demand".

Sub-interval: Range from 1 - 30 Min.

Averaging Interval Window: The width of the demand interval, Range from 1 - 30 Min.

4.3.1.3 Metering



DO Energy Pulse Const: The actual DO pulse number equals the kWh or kvarh measured multiplied by the DO energy pulse constant.

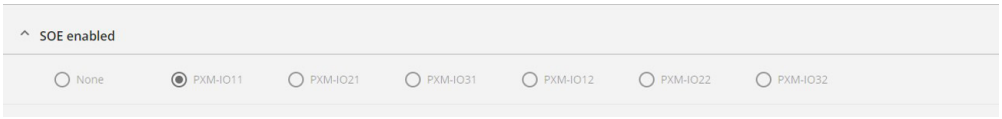
Energy Type: Can be one among "Fundamental" or "Fundamental + Harmonics".

Energy Reading: The PXM 3000 can be set to record primary energy.

VAR/PF Convention: The user can configure VAR/PF convention as IEC or IEEE.

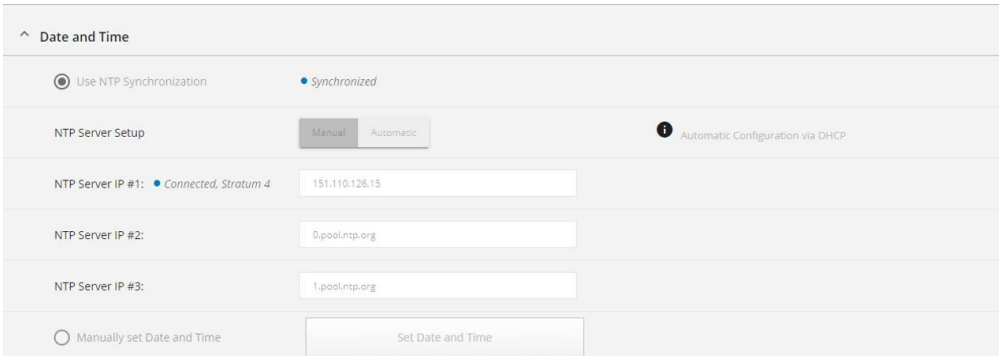
VAR Calculation Method: The PXM 3000 offers two methods for VAR calculation - "True" or "Generated".

4.3.1.4 SOE Enabled



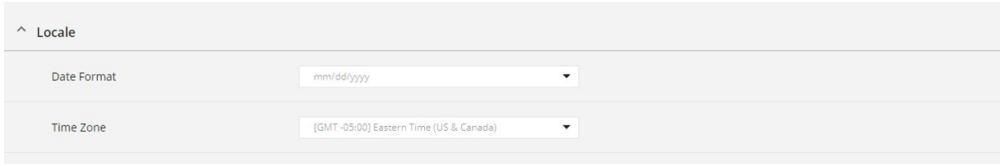
This configuration determines that which I/O module will be SOE enabled.

4.3.1.5 Date and Time



The PXM 3000 uses a real-time clock for time keeping. The meter can also be configured and synchronized using Network Time Protocol (NTP). By using NTP, time will be maintained with firmware, and the real time clock automatically corrected. When selecting NTP as the Time Sync Source, at least one IP address must be entered (as many as three NTP server IP addresses may be entered). This page also allows you to set date and time manually.

4.3.1.6 Locale

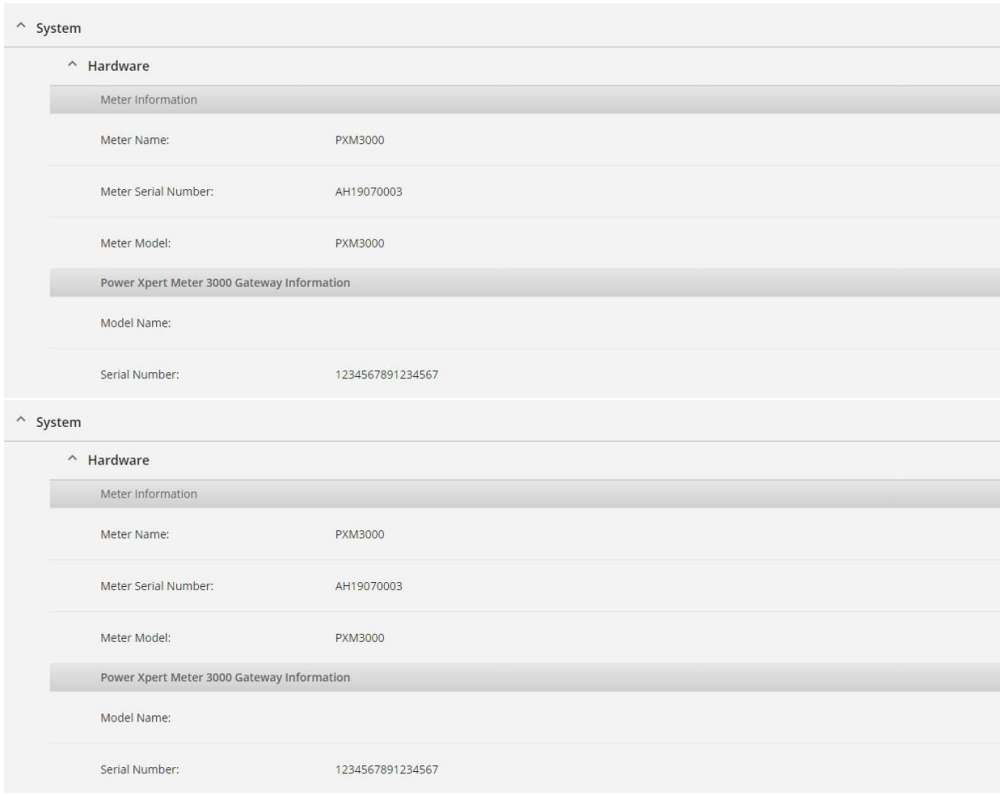


Date format and time zone can be configured.

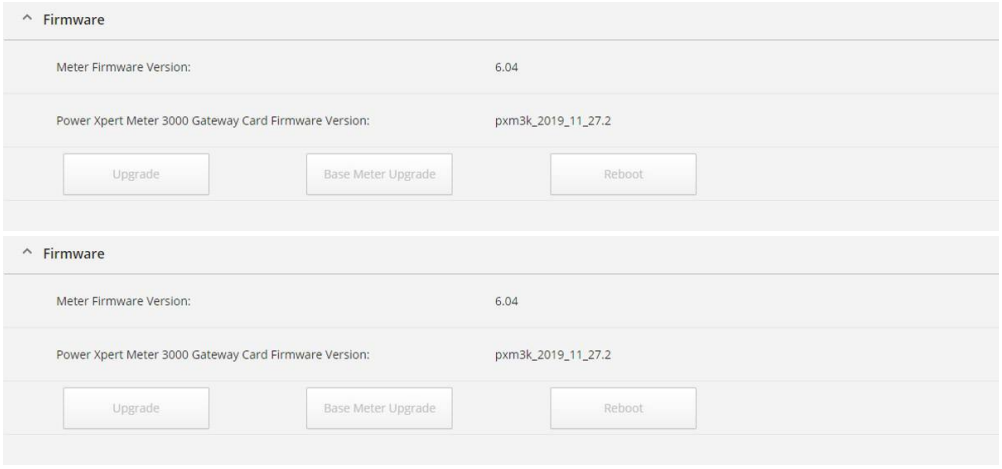
4.3.1.7 System

The System page is divided into two sections: Hardware and System Firmware.

1. Hardware



2. Name and serial number of both the meter and gateway is displayed on the Hardware page. This page also displays the meter model.
3. Firmware



The Firmware page displays both the meter and gateway firmware version. Firmware files can be downloaded from the Eaton website. Click the **Upgrade** button to upload and update the meter gateway card and **Base Meter Upgrade** to update base meter. The gateway card can be rebooted by clicking **Reboot** button.

The "Choose an Action" dropdown has the following actions.

- Save/ Restore Configuration: To download configuration files to your local file system and then uploaded from your local file system to a meter using "Restore Configuration". This is a handy convenience feature, especially if you are setting up multiple meters with either the same or similar settings. You can set up one meter, download the configuration file to your local computer, and then upload that same configuration file to other meters

  To download the current configuration file:

  1. Select the "Save Configuration" drop down within the "Choose an Action" dropdown; then
  2. Click Yes.

  To upload a configuration file to a meter:

  1. Select the "Restore Configuration" drop down within the "Choose an Action" dropdown;
  2. Click Yes;
  3. Select the configuration file; then
  4. Click OK.

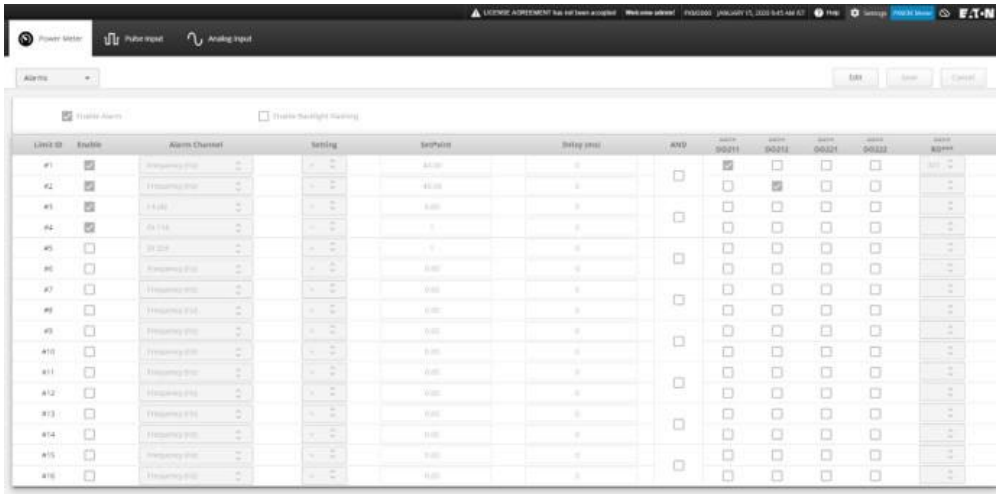
- Factory Reset - Factory reset will:
  - verride the existing network settings and assign the local IP address;
  - Reset the HTTP passwords; and
  - Reset/return the meter to factory defaults.

⚠ IMPORTANT!

When the meter was originally installed, you should have recorded information concerning the IP, subnet mask, and DNS servers. You will need this information to reconfigure the meter.

- Backlight – It will set the time in minutes for which backlight is ON for the meter face plate.

4.3.2 Alarm

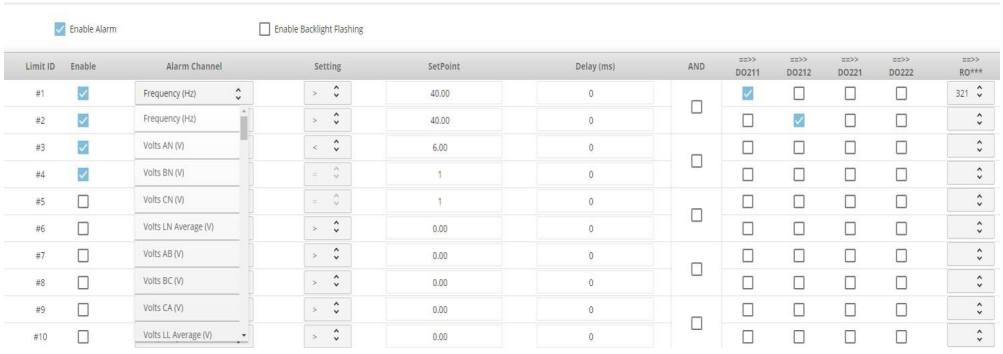


The PXM 3000 meter supports 16 different Alarms. Each channel can be used to set an alarm for 80 different parameters known as Alarm Channels.

The user can click on the "EDIT" tab to set the alarm for different parameters.

On the top bar, "Enable Alarm" checkbox can be used to enable or disable all the alarms.

The "Enable Backlight Flashing" checkbox can be used to flash the meter screen when an alarm occurs.



As shown in the above image, the user can select the different parameters like frequency, voltage, etc. for which the alarm is to be recorded.

Under the setting column, the "greater than, equal to, and less than" condition can be set.

In the "SetPoint" box, the user can enter the value condition at which the alarm should be recorded.

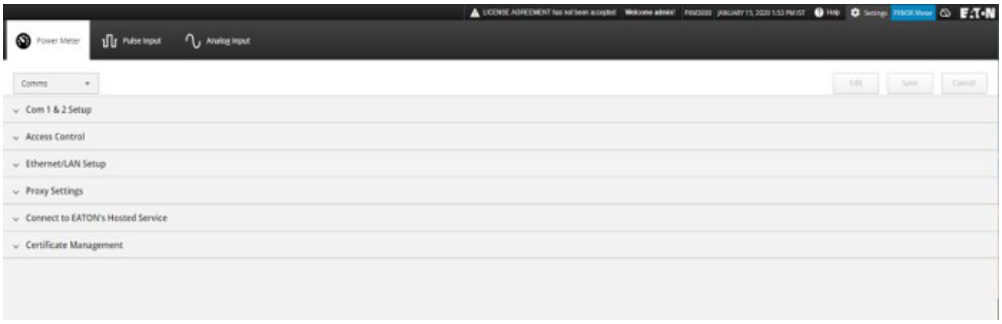
In the "Delay" box, the user can set the delay in milliseconds after which the alarm should be recoded.

Logic "AND" between alarm setting: The 16 alarming records in the meter are divided into 8 pairs. Each pair has two alarm groups. The two groups can be logically "AND" by controlling the logic check box. When two groups are "AND", alarming triggers only if both "AND" conditions are met. If the "AND" logic box is unchecked, the two alarm channels will work independently.

The user can forward the alarm to different DO and relays using the checkbox on alarm setting page.

4.3.3 Comms

The user can view and change the communication related settings by selecting the Comms tab in the dropdown menu on settings page.



Different settings under comms tab are as identified below:

- Com 1 & 2 Setup;
- Access Control;
- Ethernet/LAN Setup;
- Proxy Settings;
- Connect to EATON's Hosted Service; and
- Certificate Management.

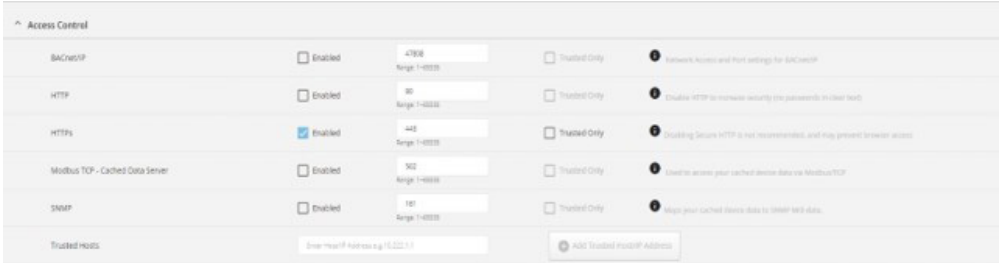
4.3.3.1 Com 1 & 2 Setup



The user can view the Com port settings for the PXM 3000 meter. Com 2 settings cannot be changed. The user can edit only Com 1 settings.

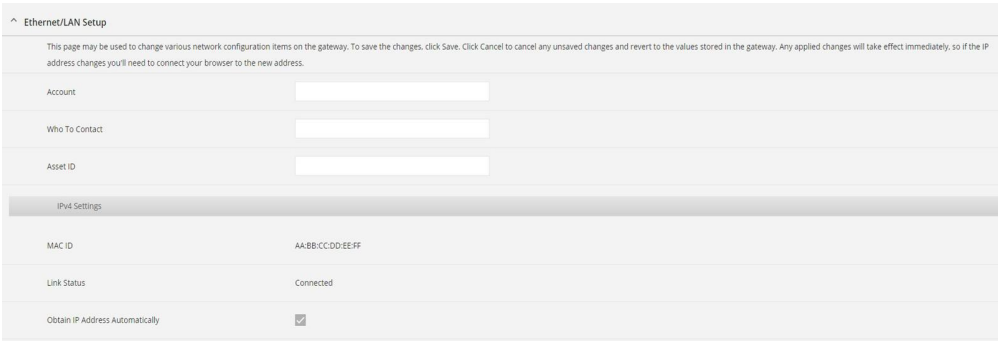
4.3.3.2 Access Control

Under this tab, the user can Enable / Disable different protocol support for and ports for communication.



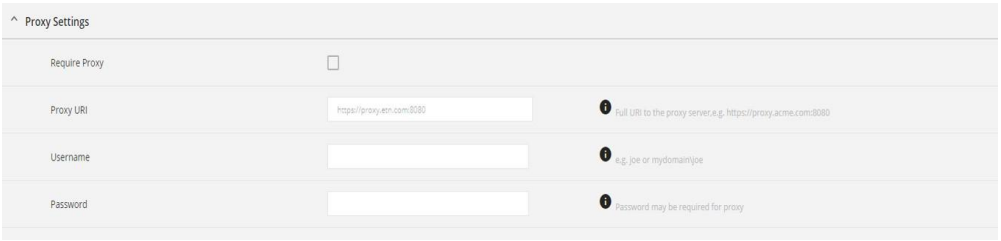
4.3.3.3 Ethernet/LAN Setup

Under this tab, the user can view and edit different Ethernet and LAN settings like DHCP settings or IPv4 settings.



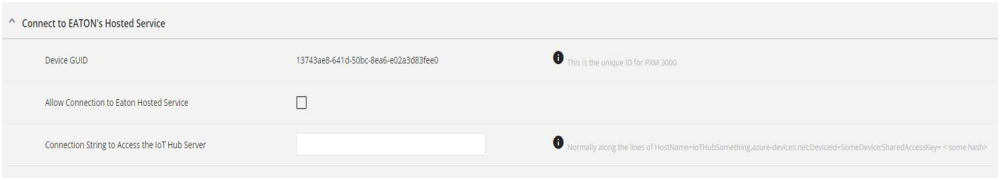
4.3.3.4 Proxy Settings

Under this tab, the user can add the proxy related settings for network.



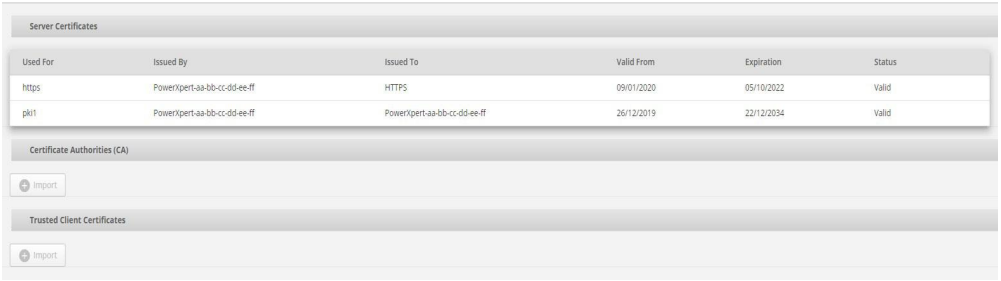
4.3.3.5 Connect to Eaton's Hosted Service

Under this tab, the user can connect the meter to the Eaton Cloud to enable the data logging on the Cloud.



4.3.3.6 Certificate Management

The user can View/Add/Remove different certificates for PXM 3000 meter.

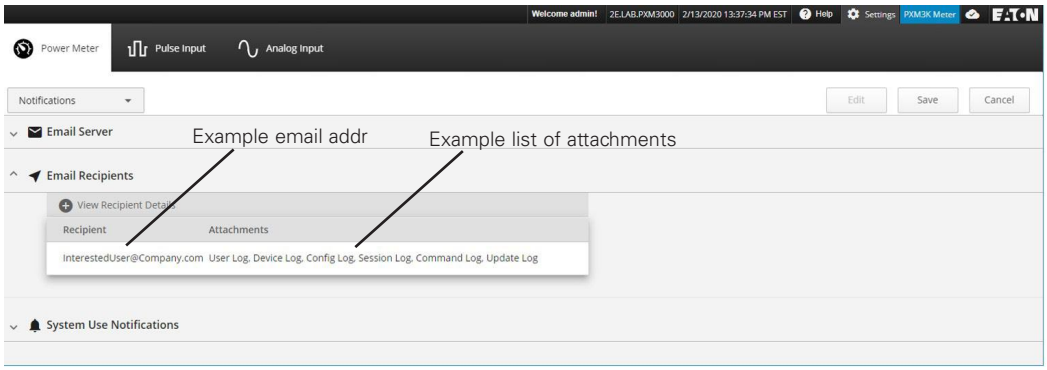


4.3.4 Notification

This page will be used to setup email server configuration, adding email recipients, and system use notification.

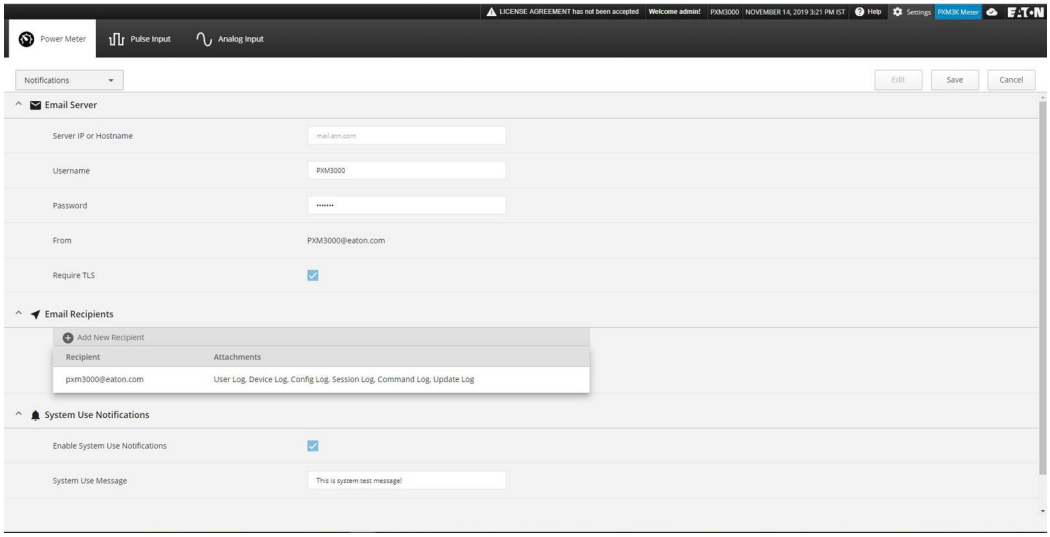
The PXM 3000 series meter can be set to send emails whenever an alarm or event occurs or when an alarm is cleared. It can also be configured to send periodic status emails. Eaton recommends that you periodically examine this list and remove any email addresses that are no longer required. To access the email settings:

1. Access the Email Recipients page and click on View Recipient Details.
2. Click the "+" icon.
3. Make the additions/deletions needed.
4. Click the Save button.



To access the PXM 3000 series meter security settings, follow the procedure below.

1. Access the meter home page. You may be challenged for the admin account username and password.
2. Click the Settings tab.
3. Navigate to the Notifications.
4. Click the Edit button.
5. If you do not have permission to this account, you must enter the admin account username and password.



Email Server

This is a one-time configuration to setup the email server settings. Here we provide a valid SMTP **Server IP or Hostname** details. **Username and Password** are optional. "**From**" is merely text to display for the recipient.

The PXM 3000 meter provides a highly configurable capability to report on meter activity via email. You can configure up to ten separate recipients. If you are unsure as to how to do this, ask your information technologies group or the person in charge of your mail server for assistance.

Email Recipients

Here we can add lists of recipient's email address along with the attachment's options, shown in tabular format.

System Use Notification

Here the user can enable or disable the notification message. If enabled, this message will be displayed before the User Login screen.

4.3.5 User

In this page, you can manage users, roles, sessions time out, concurrent logins, and set password policy.

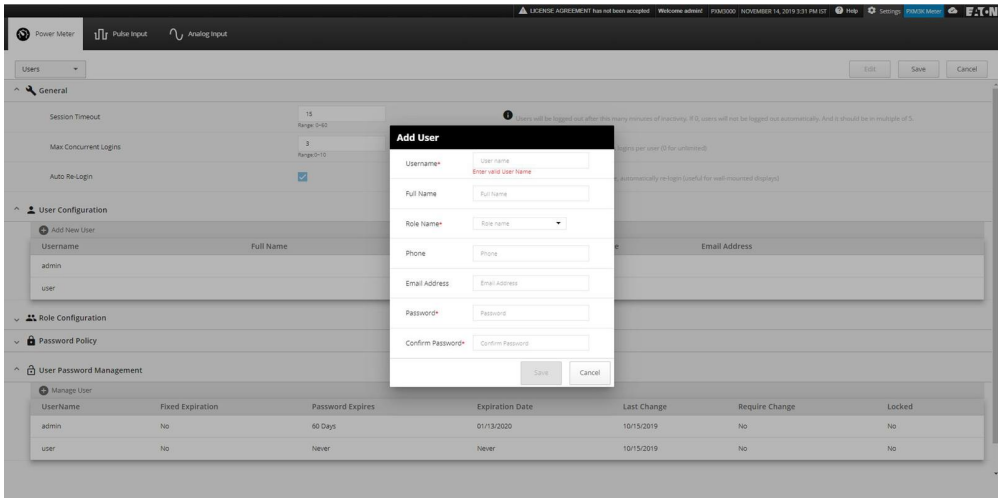
To access the PXM 3000 series meter user settings, follow the procedure below.

1. Access the meter Home page. You may be challenged for the admin account username and password.
2. Click the Settings tab.
3. Navigate to the User.
4. Click the Edit button.
5. If you do not have permission to this account, you must enter the admin account username and password.

Manage User

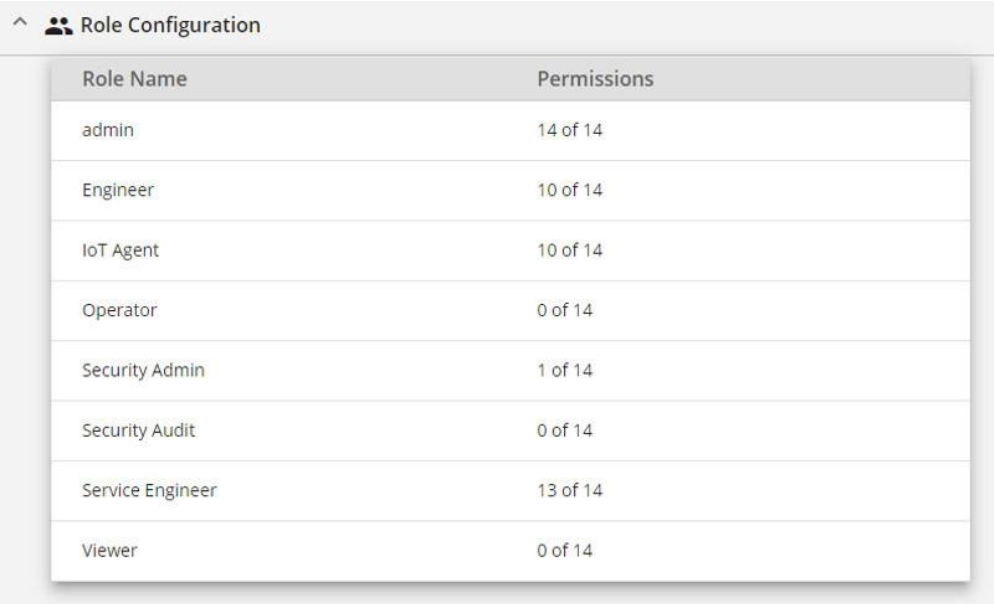
You can view, add, edit, and delete users from the list.

To add a user, you need security permission. After clicking on Add User, a pop-up will be open as shown in below image. Username, Role Name, and Passwords are mandatory. The username should be unique and can be alpha-numeric. Passwords should follow the Password Policy settings. Once a user is created, table should update.



Manage Roles

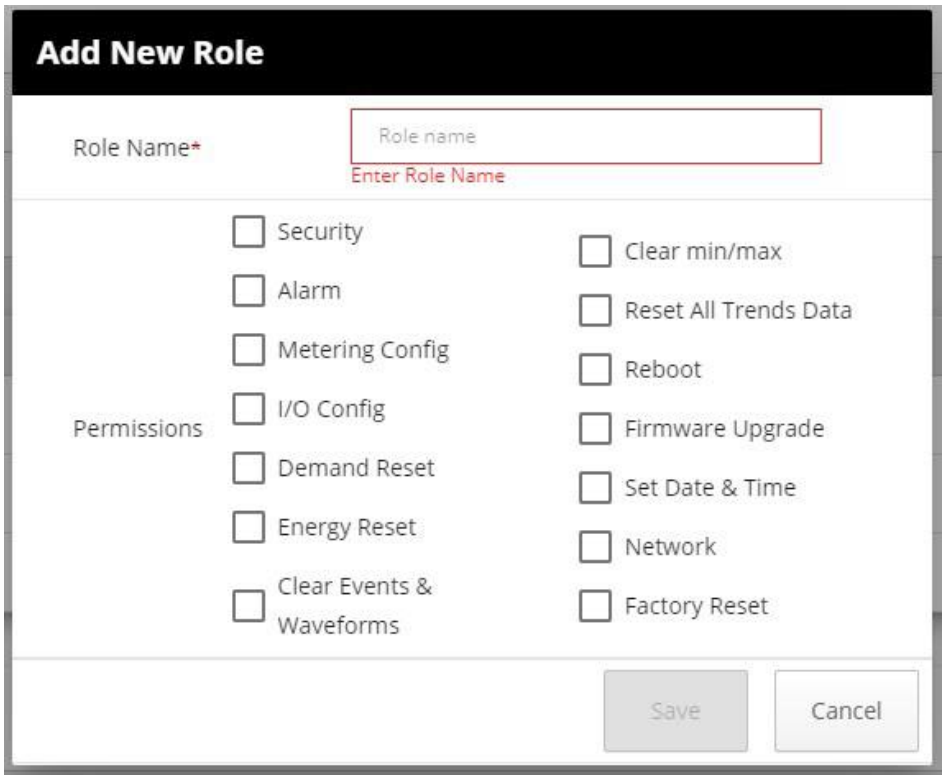
There are eight predefined roles.



Role should be unique, can be alpha-numeric and is a mandatory field. Once the role is created, the table should update. Only a user with appropriate permissions can perform this operation.

Below is a list of permissions:

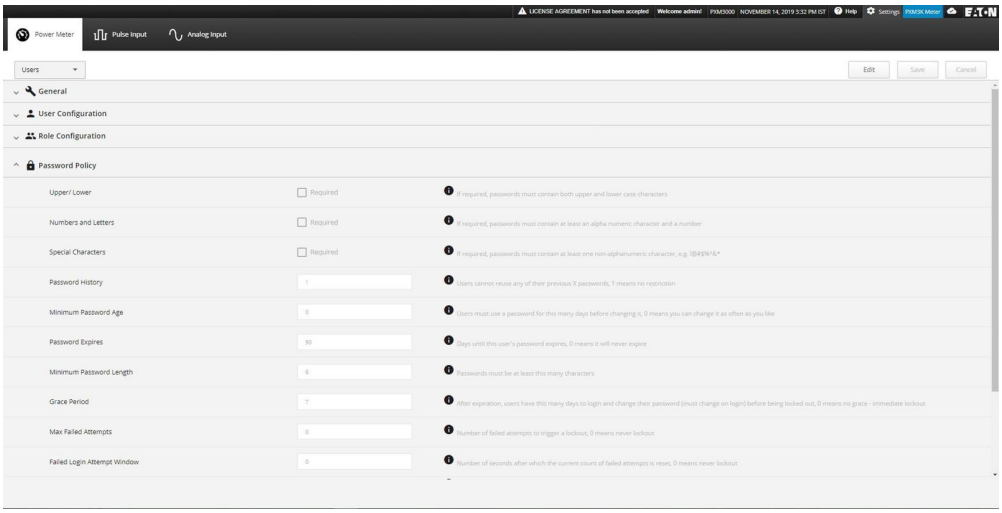
1. Security;
2. Alarm;
3. Metering Config;
4. I/O Config;
5. Demand Reset;
6. Energy Reset;
7. Clear Events and Waveforms;
8. Clear Min/Max;
9. Reset All Trends Data;
10. 1Reboot;
11. Firmware Upgrade;
12. Set Date and Time;
13. Network; and
14. Factory Reset.



Password Policy

Below is list of options for Password Policy, which will be useful while creating new users and change passwords.

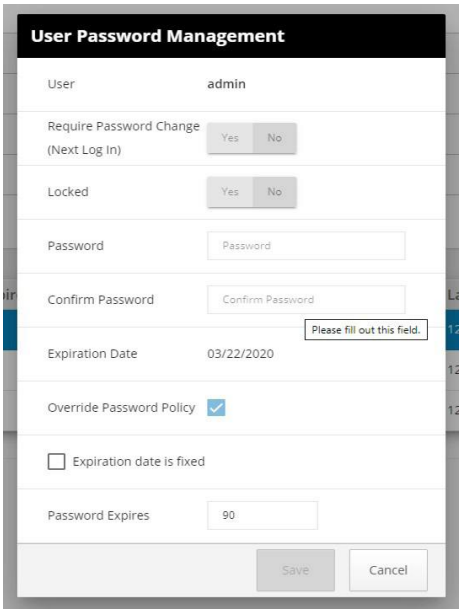
- **Upper/Lower:** If required, passwords must contain both upper and lower-case characters.
- Numbers and Letters: If required, passwords must contain at least an alpha-numeric character and a number.
- **Special Characters:** If required, passwords must contain at least one non-alpha-numeric character (e.g. !@#$%^&*).
- **Password History:** Users cannot reuse any of their previous X passwords - 1 means no restriction [range 1 - 32].
- **Minimum Password Age:** Users must use a password for this many days before changing it -0 means you can change it as often as you like [range 0 - 90].
- **Password Expires:** Days until this user's password expires - 0 means it will never expire [range 0 - 90].
- **Minimum Password Length:** Passwords must be at least this many characters [range 6 - 64].
- **Grace Period:** After expiration, users have this many days to login and change their password (must change on login) before being locked out - 0 means no grace - immediate lockout [range 0 - 4294967295].
- **Max Failed Attempts:** Number of failed attempts to trigger a lockout - 0 means never lockout [range 0 - 30].
- **Failed Login Attempt Window:** Number of seconds after which the current count of failed attempts is reset -, 0 means never lockout [range 0 - 3600].
- **Failed Login Wait:** Lockout due to exceeding Max Failed Attempts. This is the number of seconds until the lock will automatically be removed - 0 means never auto-unlock [range 0 - 3600].



User Password Management

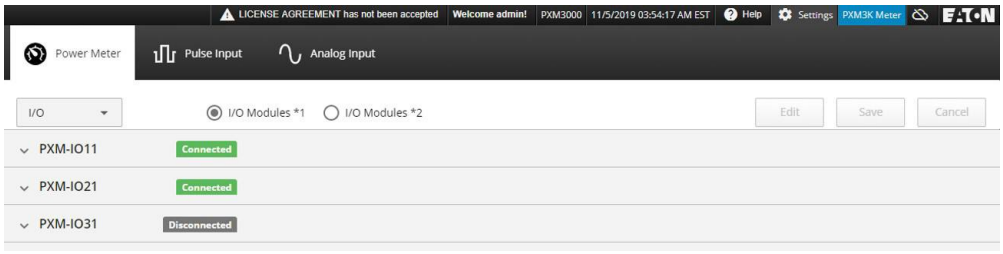


This provides information related to existing users and their password expiration. One can also change the user password expiration, lockout a user, or set a fixed expiration date. If you select the option for a fixed expiration date, a set date is required.



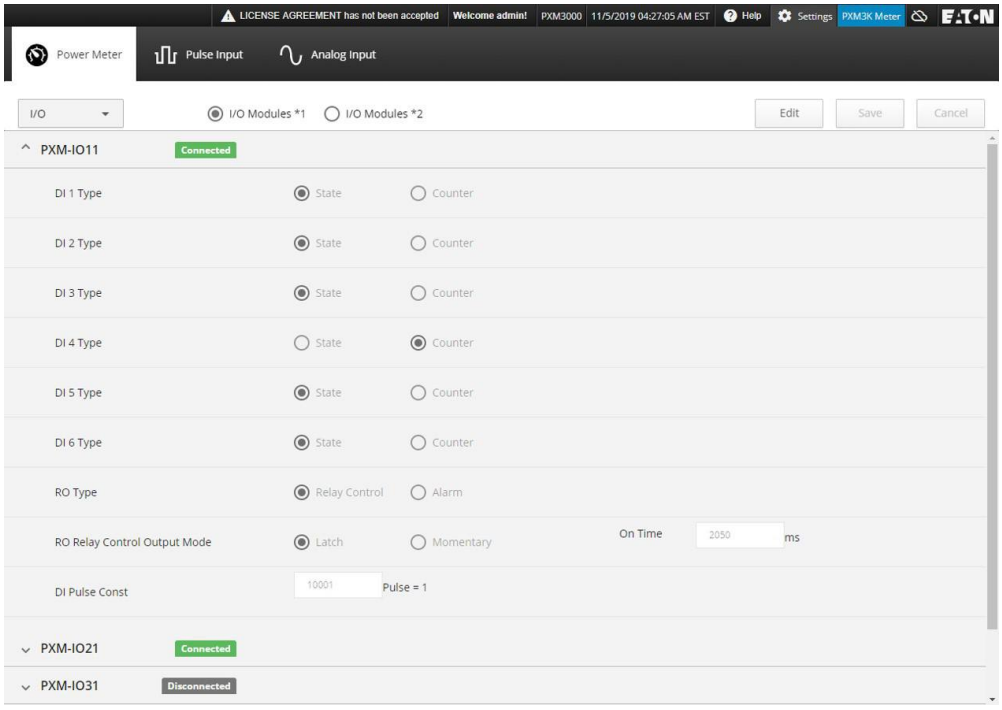
4.3.6 I/O

There are three types of I/O modules supported by PXM 3000: PXM-IO1, PXM-IO2; and PXM-IO3. I/O module setting changes will be made only if the corresponding I/O modules are installed. No changes will be made otherwise. Please note that a maximum of three modules may be attached to the meter. No more than two of the same type I/O modules can be attached. Each type of I/O module also has two modes, logic NO.1 and logic NO.2. Modules with mode logic NO.1 and logic NO.2 will be displayed under I/O Module *1 and I/O Module *2 respectively.



PXM IO1 Page

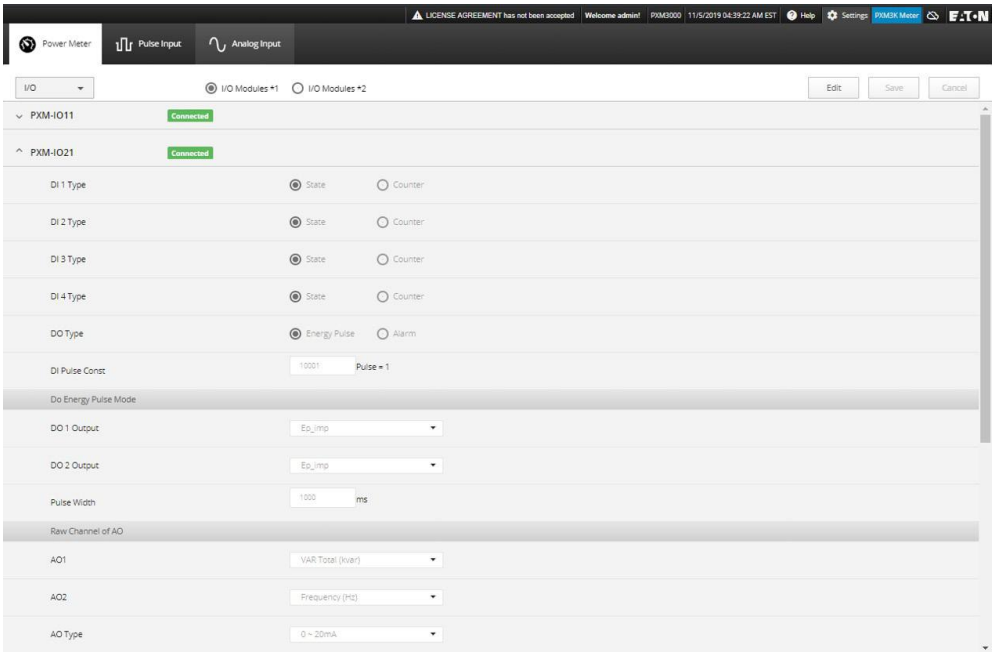
The PXM-IO1 module is composed of six digital inputs (DI), two relay outputs (RO), and a 24 V isolated power supply. Each DI can be used to detect remote signals (state mode) or be used as an input pulse counter. RO can be used for controlling or alarming. Each of the relay outputs work in the same mode. When it operates in controlling mode, there are two output options: latching mode and pulse. When it operates in alarm mode, it has only one latching output mode. If the DI works as a pulse counter, when the number of pulses counted by the DI equals to the pulse constant, the pulse counter will increase by one. This means that the actual pulse number equals the number of pulses counted multiplied by the pulse constant. If relay pulse control method is selected, the relay contact will close for a preset period set as "On Time" and open afterwards.



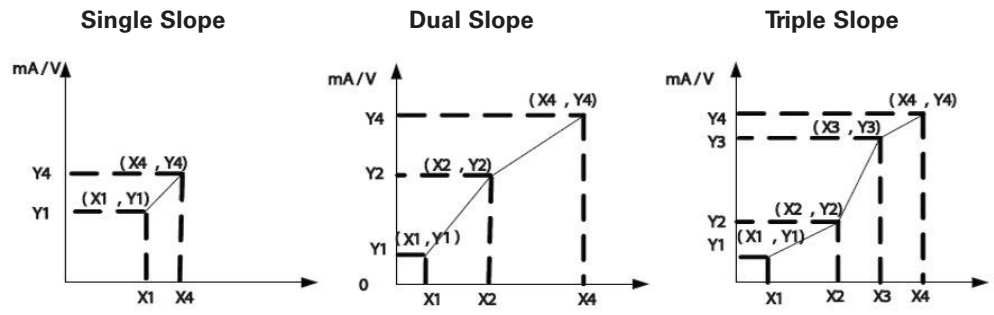
PXM IO2 Page

The PXM-IO2 module is composed of four digital inputs (DI), two analog outputs (AO), and two digital outputs (DO). Each DI can be used to detect remote signals (state mode) or be used as an input pulse counter. If the DI works as a pulse counter, when the number of pulses counted by the DI equals to the pulse constant, the pulse counter will increase by one. This means that the actual pulse number equals the number of pulses counted multiplied by the pulse constant. DOs can be used in alarm mode or energy pulse output mode. Both DOs work in the same mode. When it operates in energy pulse output mode, it can output various types of energy. Choose output energy type for DO1 or DO2 from None, import active energy(Ep_imp), export active energy(Ep_exp), import reactive energy(Eq_imp), and export reactive energy(Eq_exp). If DO type is set as alarm output, DO1 and DO2 output type parameters will have no effect. When outputting energy pulses, pulse width can be set from 20 to 1000 ms. The minimum interval between two pulses is 20 ms.

The AO circuit can convert any of 30 user-selected electrical quantities to an analog voltage or current. The analog output circuit supplies four output modes, including 0 to 20 mA mode, 4 to 2 0mA mode, 0 to 5 V mode, and 1 to 5 V mode. Beware that modules with current option cannot be set as voltage; modules with the voltage option cannot be set as current type.

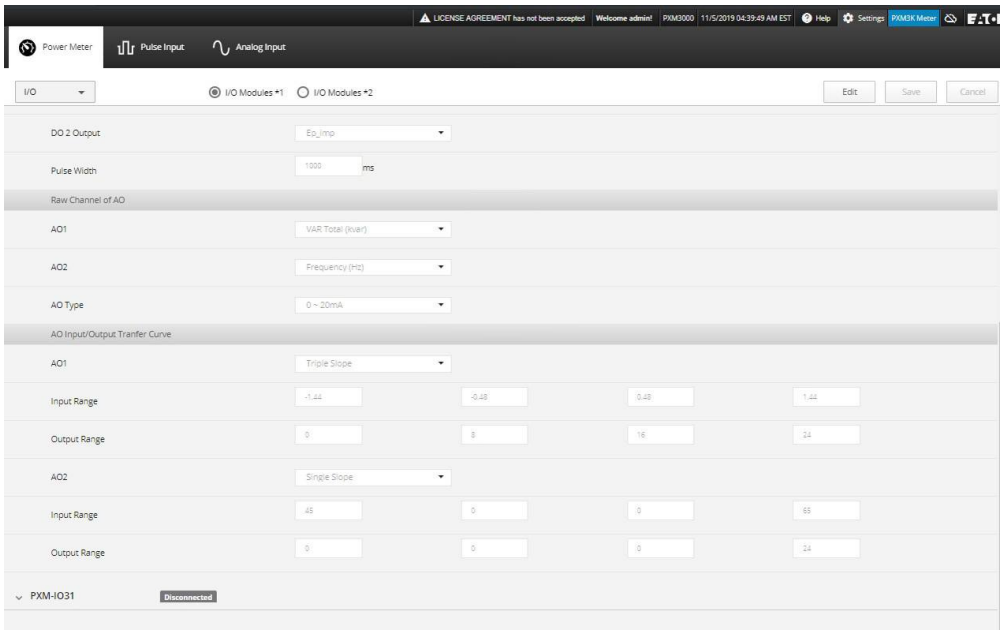


AO Input/Output Transfer Curve:



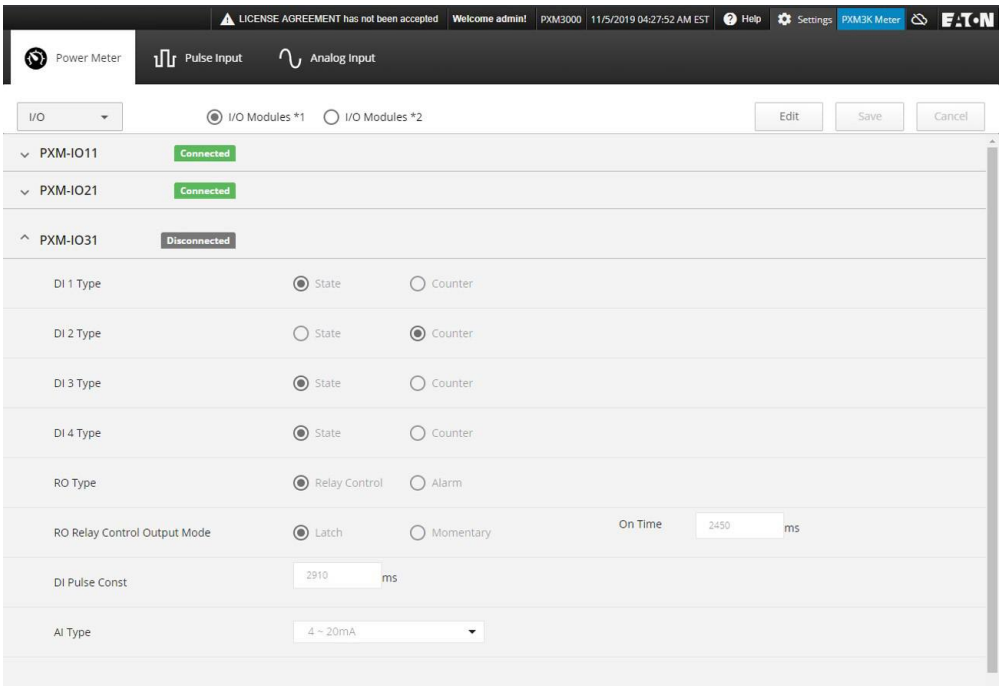
Relationship between analog output and various electrical quantities.

Depending on the application, the user can select Single Slope, Dual Slope, or Triple Slope. Input and the output range should be entered for each selected parameter and should be in the specified range.

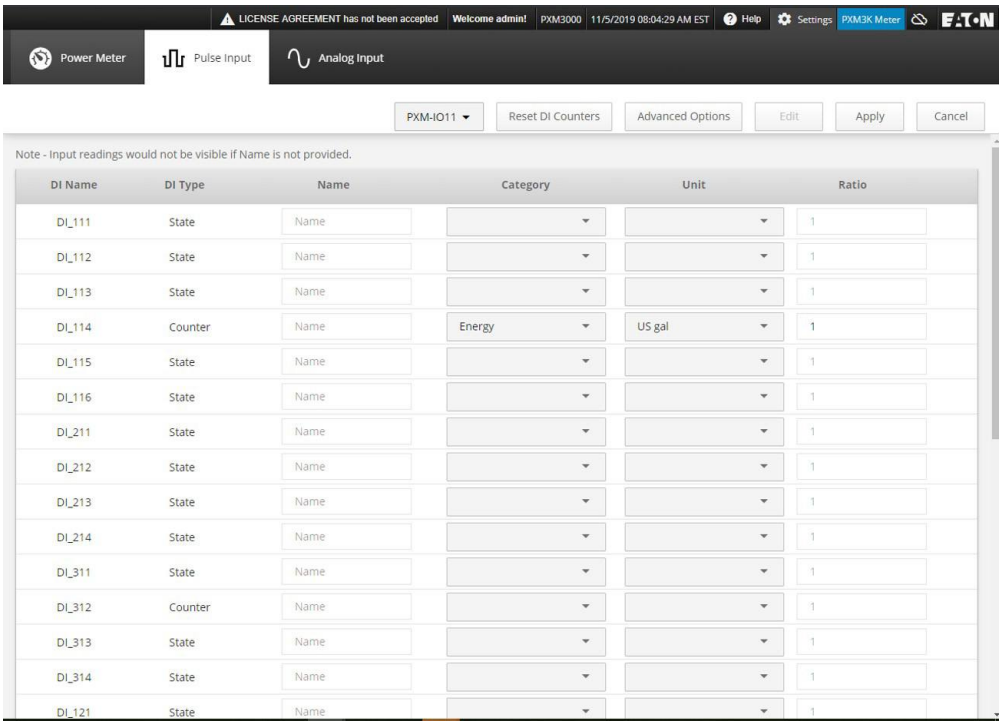


PXM IO3 Page

The PXM-IO3 module is composed of four digital inputs (DI), two relay outputs (RO), and two a Analog inputs (AI). Each digital input can be used to detect remote signals (state mode) or be used as an input pulse counter. RO can be used for controlling or alarming. Each of the relay outputs work in the same mode. When it operates in controlling mode, there are two output options: latching mode and pulse. When it operates in alarm mode, it has only one latching output mode. If the DI works as a pulse counter, when the number of pulses counted by the DI equals to the pulse constant, the pulse counter will increase by one. This means that the actual pulse number equals the number of pulses counted multiplied by the pulse constant. If relay pulse control method is selected, the relay contact will close for a preset period set as "On Time" and open afterwards. AI can detect input analog voltage or analog current. When it detects input analog voltage, the range of voltage is from 0 to 5 V or from 1 to 5 V. When it detects input analog current, the range of current is from 0 to 20 mA or from 4 to 20 mA.

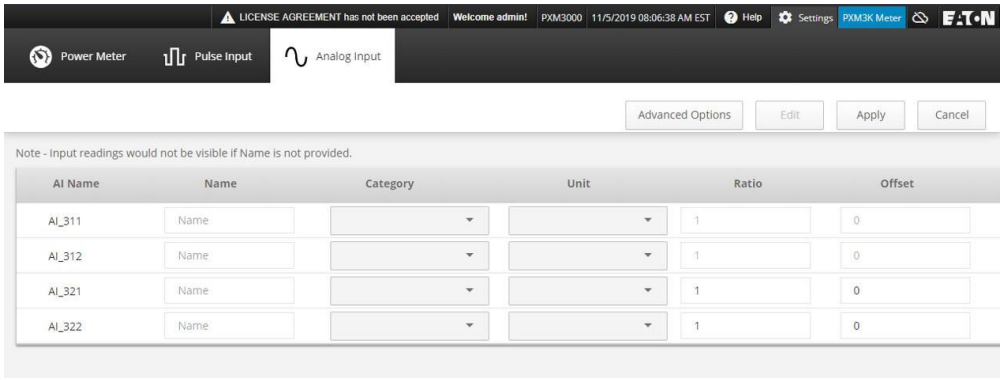


Pulse Input Page



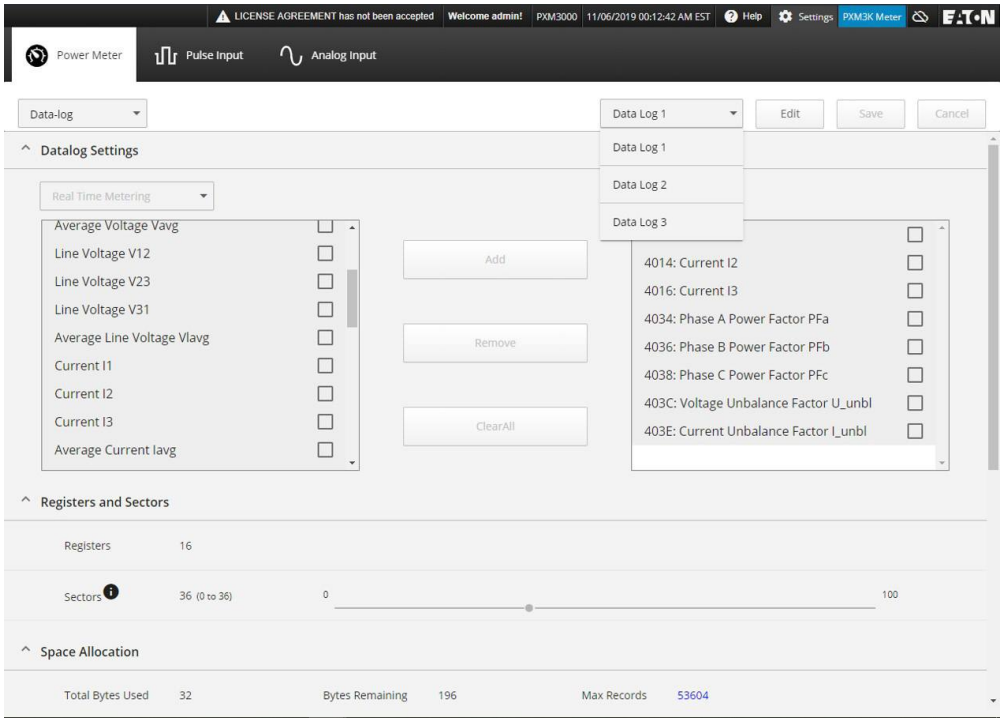
Pulse input can be set only for digital Inputs (DI) that are in counter mode. The user can enter Name, Category, Unit, and Ratio for each digital input. While Name can be any random word, Category and Units should be selected from dropdown menu. The dropdown menu can be updated using Advanced Option. Ratio acts as a multiplication factor for count.

Analog Input Page



Analog Input can be set only for I/O modules connected to the meter. The user can enter Name, Category, Unit, Ratio, and Offset for each analog input. While Name can be any random word, Category and Units should be selected from dropdown menu. The dropdown menu can be updated using Advanced Option. Ratio acts as a multiplication factor and there is an option for providing offset. The value would be calculated as "multiplied with Ratio minus the offset"

4.3.7 Data Log



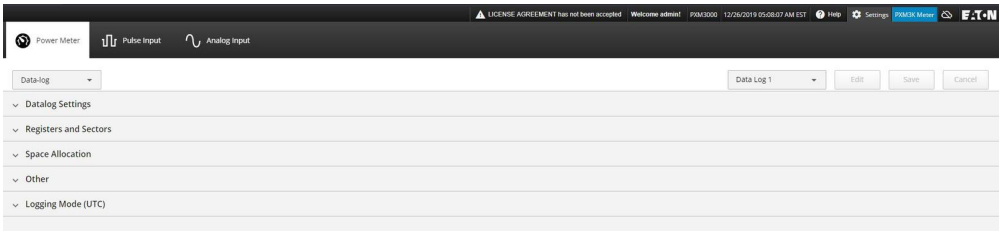
The PXM 3000 has three sets of historical data logs. Each log can be independently programmed with individual settings, meaning that each can be used to monitor different parameters. You can program up to 117 parameters per log. You also can allocate available system resources among the three logs, to increase or decrease the size of the individual historical logs. The total size is no more than 100 sectors that has 64k bytes.

Having three sets of historical logs provides you with the option of programming each log with unique parameters. For example, you can program Historical Log 1 to record measured values parameters such as frequency, voltage, or Current; Log 2 to record energy values parameters; and Log 3 to record power quality parameters.

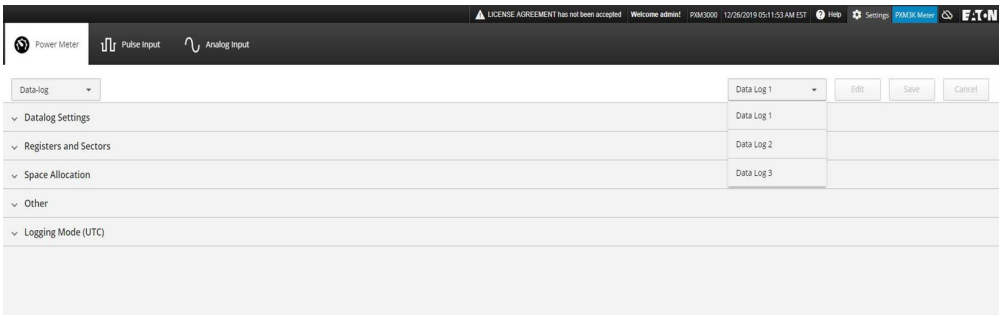
In order to generate historical logs for the selected parameters, users should program the meter so that selected parameters from the corresponding Modbus registers can be copied to the historical log record. As some parameters occupy two registers, the programmable settings for the historical logs contain a list of descriptors. Each descriptor lists the number of Modbus registers for the specified parameter. By combining these two lists, the historical log record can be interpreted.

Data log settings can be found on the Setting page under the Power Meter tab.

Select the Data-log from the drop down menu.

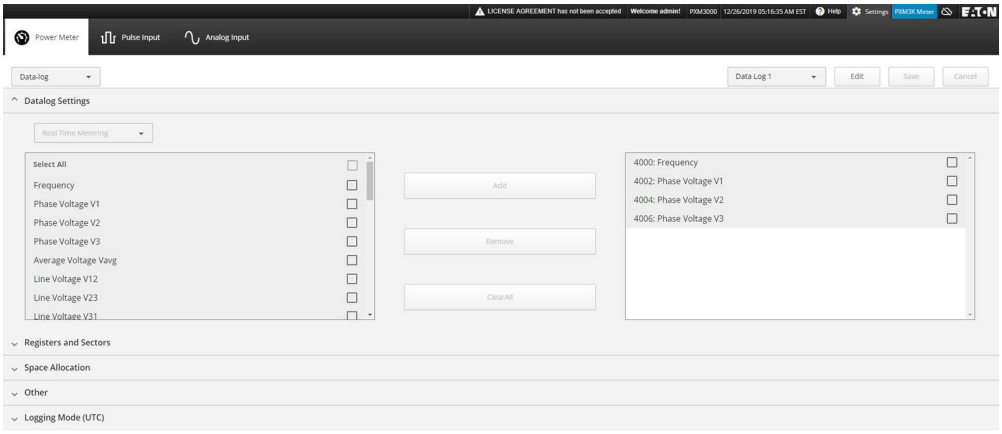


PXM 3000 supports three data log options which can be selected from the drop down menu located at the right side of the above-mentioned page.



Under the Datalog Setting tab, the user can select different parameters for logging.

The left side is the list of all the parameters that the meter supports for logging. The user can Add selected parameters to the right-side list, which the meter will use to start logging. The user can also Remove or Clear the list using the specified buttons.



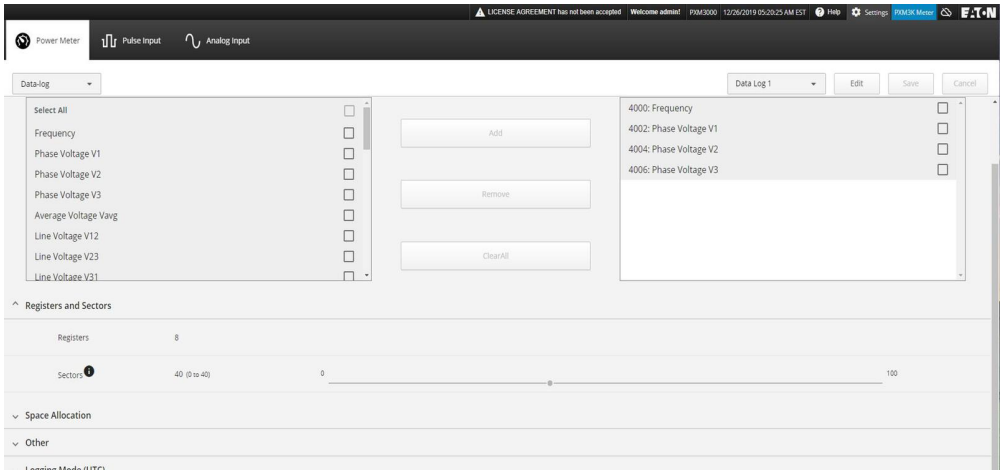
Max Records: The maximum number of records the log can hold given the record size and sector allocation.

Used Records: The number of records stored in the log. This number will equal the Max Records when the log has filled. This value will be set to 1 when the log is reset.

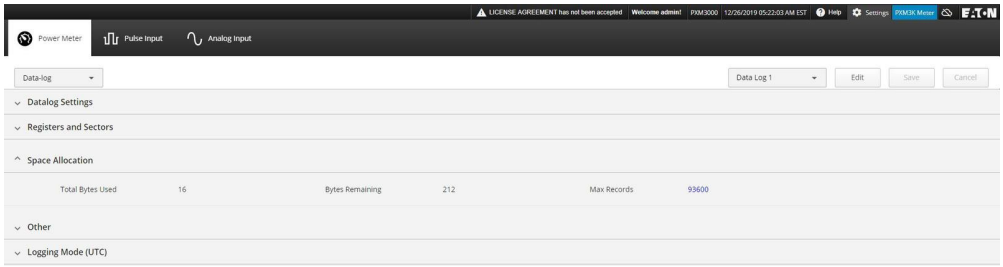
Record Size: The number of bytes in this record, including the time stamp.

The record's format in the meter is: record number (4 bytes) + time stamp (6 bytes) + [data1~dataN] (2 Nbytes) + CRC (2 bytes).

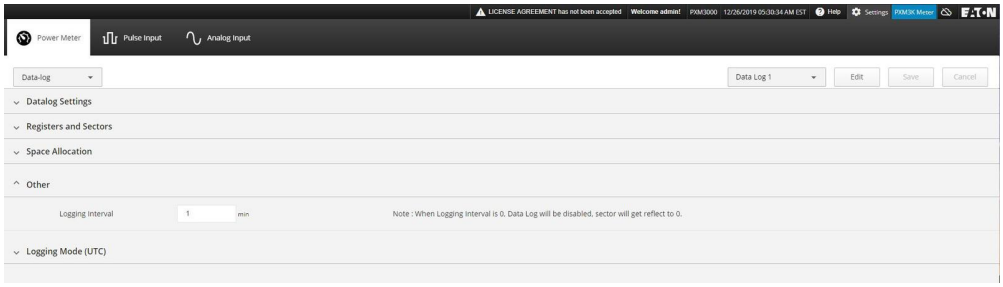
After selecting the number of parameters to log, the user can view the total registers used and sectors allocated for the memory under the Register and Sector tab.



Under the "Space Allocation" tab, the user can see the number of bytes used by currently selected parameters and the number of bytes remaining. It also shows the information regarding the Max Records that meter can log.

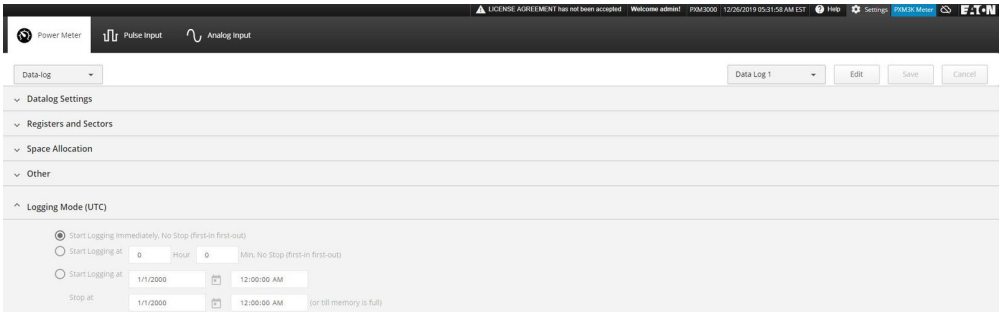


Under "Other" tab, the user can select the interval at which user wants to log the parameter.



Under "Logging Mode (UTC)", the user can select the time from which to log the data (e.g. immediate logging, or start and end time can be specified).

Note: Time is always in UTC.



Notes:

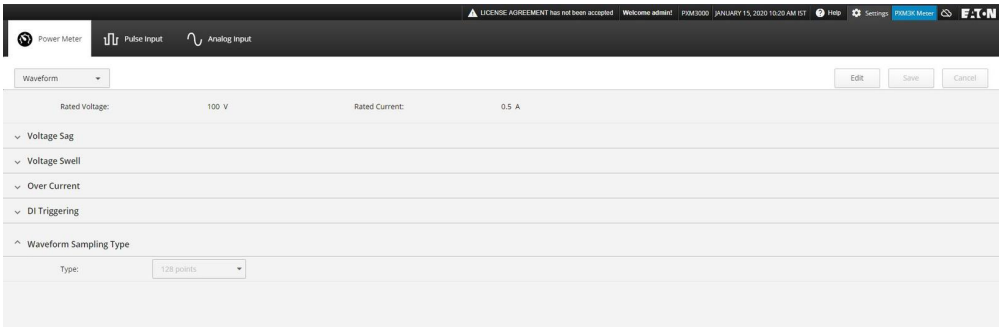
- If the memory of the historical data log is full, the meter will erase the first sector (64 KB). The next sector (2nd sector) becomes the first sector and the data from the erased sector will not be recoverable. Therefore, the user should save the entire log before memory is full to retain all data.
- This screen also shows the registers in the logs, the total bytes used, and the bytes remaining for this historical log. These fields are updated as you make selections on the screen. The total number of bytes available per log record is approximately 234.

4.3.8 Waveform

The user can select Waveform setting under the "Waveform" tab on Settings page.

PXM 3000 can record up to 100 (128 samples/cycle) and 200 (64 samples/cycle) waveforms.

It provides the captured waveform of 20 cycles before and after the triggering point for 128 sample/cycle waveforms, and it provides 10 cycles before and after the triggering point for 64 sample/cycle waveforms.



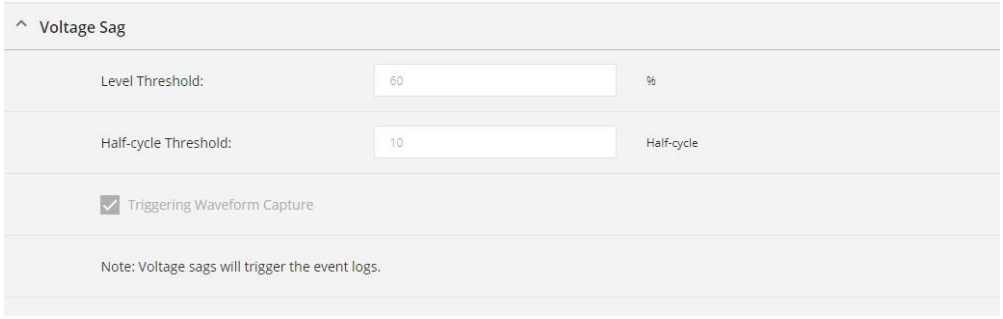
Under "Waveform Sampling Type", the user can select waveform sampling resolution of 128 or 64 bit. If waveform resolution is changed, all the previously captured waveforms are erased.

Waveform triggering conditions are as mentioned below:

- Voltage Sag;
- Voltage Swell;
- Over current; and
- DI Triggering.

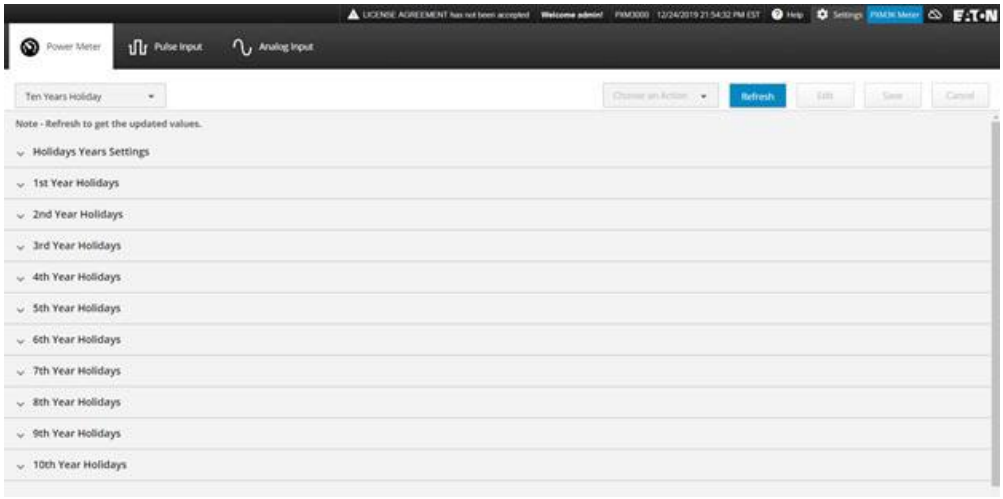
Threshold for each condition can be provided in the corresponding "Level Threshold" box.

Waveform will only be captured when user enables the "Triggering Waveform Capture" checkbox.



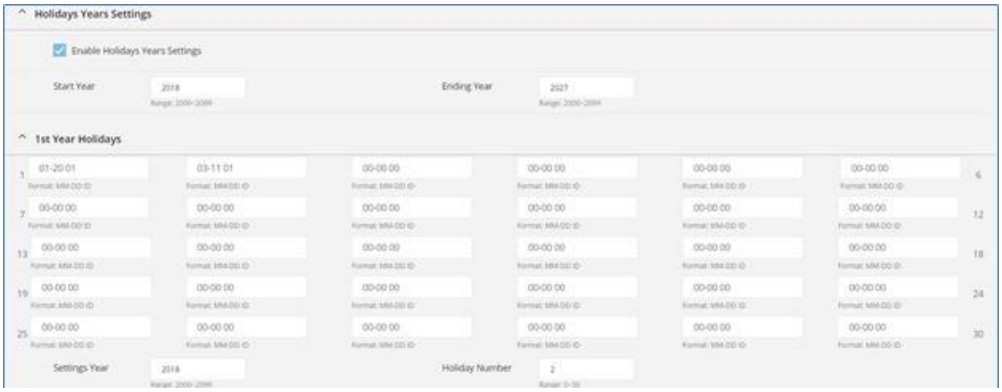
4.3.9 Ten Years Holiday

Users can preset holidays of the next decade via this feature.



When Ten-year Holiday is enabled, if the current year of the meter falls into the Ten-year Holiday setting, it automatically loads the Ten-year Holiday settings into the current TOU settings. If the current year of the meter does not fall into the Ten-year Holiday setting, it remains the current TOU settings.

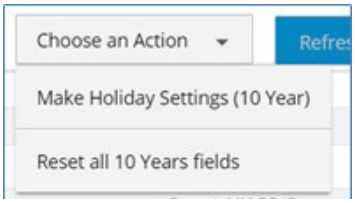
On web page user can find inputs for "Holiday Years Setting" and "1st Year Holidays" to "10th Year Holidays". User must enter Edit mode to input required settings.



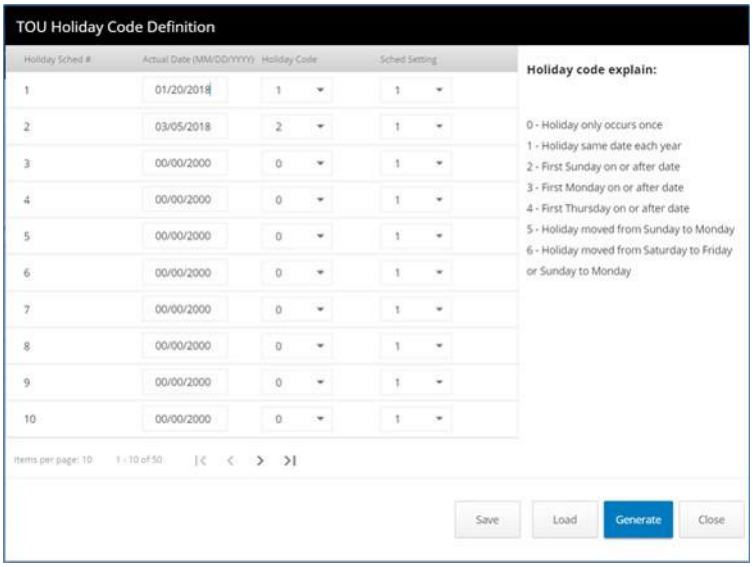
Holiday Years Settings allows the user to enable/disable holidays. The user can enter start and end year while the scope of Holiday-Setting being less than 10. The valid range for years is kept as between 2000 ~ 2099. Any years outside this range would be considered as invalid and user will not be able to save the changes.

1st to 10th Year Holidays each take 30 holidays in "MM-DD ID" format (00-00 00 being default) with no field kept blank, Settings Year - stating the year of configuration, Holiday Number (i.e.: total holidays within that year) which can range from 0 to 30.

Choose an Action dropdown at the top allows the user to select either of the two options:



1. **Make Holiday Settings:** This opens a modal providing an automated functionality so that the holiday calendar can be generated. Holiday codes from 0 - 6 are explained at the right side of modal. Schedule setting range from 0 - 14. Total 50 entries can be added, and this configuration can be saved (use **Save**) or uploaded (use **Load**). On clicking **Generate**, the settings come into effect and the calendar is filled accordingly.



2. **Reset all 10 Years fields:** This button click resets all 10 Years Holidays fields to default values. It will not affect Holidays Years Settings.

4.3.10 Time of Use (TOU)

The user can customize the TOU calendar (including its tariffs, seasons, schedules, and segments) according to different applications. To make sure that the TOU calendar is setup correctly, the meter will check the TOU settings according to the predefined rules (see below for "TOU setting format requirement" for details). The TOU function will be disabled if the TOU calendar is set up incorrectly. If no errors are found in the calendar and the TOU function is enabled, TOU energy accumulation will begin.

TOU setting format requirement:

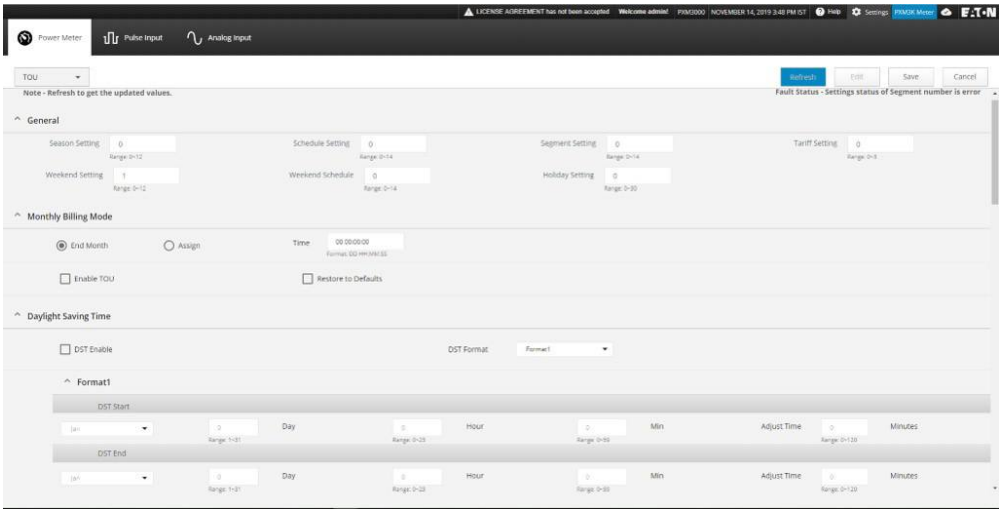
1. Season setting parameter: The calendar year will be divided up into different seasons depending on the season setting parameter. The parameter can be selected from any integer between 1 to 12. The user must enter the correct value for the season setting parameter in accordance to the TOU season table. If the season setting parameter is set as two, the first two slots of the TOU season table must be set, otherwise it will be considered as an invalid input (TOU function will be disabled).
2. TOU season format: Enter the start date into the TOU season table slot following this format "MM-DD ID" - MM stands for the month, DD stands for the day, and ID stands for the TOU schedule ID (available from 01 to 14). The dates should be organized so that they are in sequence according to the calendar year (the earlier date comes first, and the later date comes last). For example, if three seasons are selected, the date parameters are January 1, June 6, and September 7, and TOU schedule 02, 01, 03 will be used respectively: The first TOU season table slot shall enter 01-01 02, the second slot shall enter 06-06 01, and the third slot shall enter 09-07 03. Entering 01-01 02 for the first slot, 09-07 03 for the second slot and 06-06 01 for the third slot is considered invalid.
3. Schedule setting parameter: The number of available TOU schedules depends on the schedule setting parameter. The parameter can be selected from any integer between 1 to 14. This parameter determines the number of TOU schedules available for the TOU calendar setting. A maximum of 14 TOU schedules (from TOU Schedule #1 to TOU Schedule #14) can be used.
4. Segment setting parameter: Each TOU schedule consists of various timing segments. The number of segments depends on the segment setting parameter setup. The parameter can be selected from any integer between 1 to 14 (inclusively). The user must enter the correct value for the segment setting parameter in accordance to the TOU schedule table. If the segment setting parameter is set as three, the first three slots of the TOU schedule table must be set, otherwise, it will be considered as an invalid input (TOU function will be disabled).
5. TOU schedule format: Each TOU schedule represents a 24-hour cycle. Similar to the TOU season format, enter the start time into the TOU schedule table slot following this format "HH:MM ID" - HH stands for hour (in 24-hour format), MM stands for minutes, and ID stands for tariffs (available from 00 to 03). The time should be organized according to the hour sequence. For example, if three segments are selected, timing parameters are 01:00, 15:30, 22:45, the order of the three segments should be one of the following: 01:00, 15:30, 22:45, or 15:30, 22:45, 01:00 or 22:45, 01:00, 15:30. Entering time information in a wrong sequence (for example, entering 15:30, 01:00, 22:45) is considered as an invalid operation, TOU function will be disabled.
6. Tariff setting parameter: This parameter corresponds to the number of tariffs available for the TOU calendar and can be selected from any integer from 0 to 3. The four tariffs: sharp, peak, valley, and normal are represented by four integers: 0, 1, 2, and 3 respectively. If the tariff setting parameter is set to three, all of the four tariffs will be available for the TOU calendar; if the parameter is set to one, only the first two tariffs (sharp and peak) will be available.
7. Holiday setting parameter: This parameter can be set from any integer between 1 and 30, meaning a maximum of 30 holidays can be programmed to the TOU calendar. If the holiday setting parameter is set as three, the first three slots of the holiday schedule must be set, otherwise it will be considered as an invalid input (TOU function will be disabled).

Note: The user can either customize the TOU calendar factory settings or use the default factory settings. The user can reset the TOU calendar to its default value either via communication or from the meter front.

8. Holiday schedule: The holiday schedule uses the same format as the TOU seasons "MM-DD ID". The user can select which TOU schedule to be used for the holiday. The dates of the holiday schedule do not need to be organized in a sequential order (i.e. the first slot can be January 1, the second slot can be December 26, and the third slot can be December 25).

9. Daylight saving time (DST): Daylight saving time can be enabled in one of two formats: the fixed date option, or a fixed day of one of the weeks in the month (also named as the non-fixed date option). If you choose a fixed date option, you set the format according to a fixed date, for the daylight savings time switch: the format month / day / hour / minute / adjust time (in minutes). If you choose non-fixed date option, DST will be implemented by which day of which week, whose setting format is month/which day (i. e. Tuesday)/which week (i. e. 1st week)/hour/minute/adjust time (in minutes).

   By using the function, you can cause the instrument to automatically switch to and from daylight saving time. When the clock starts to run to daylight saving time, the meter will automatically adjust the clock to a time period in advance, while the clock is running to the end of daylight savings time, meter will automatically adjust the clock pushed back to a time period, as shown in below screenshot.



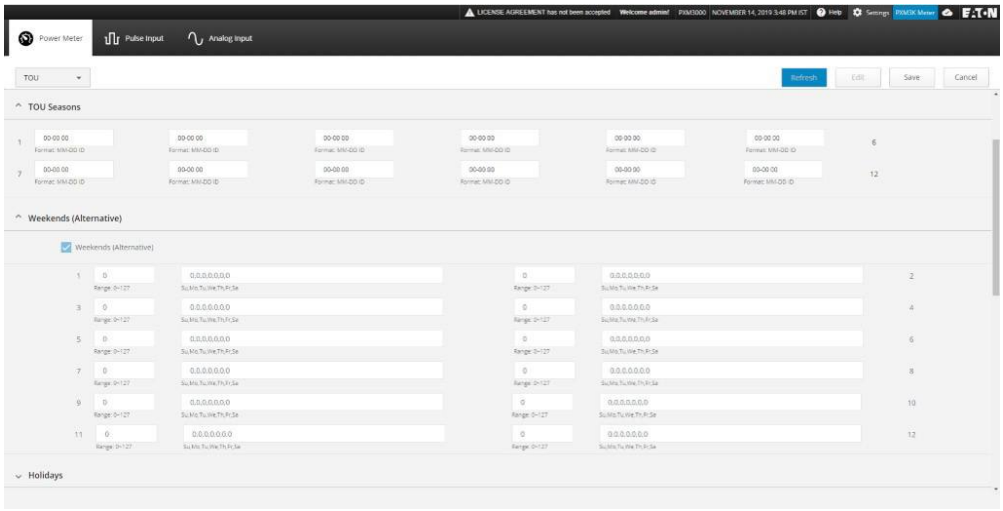
10. Weekend Schedule: Weekend Setting (bit0 : Sunday; bit1 ~ bit6: Monday to Saturday; bit 0 means not effective, bit 1 means effective). For example, when the Weekend Setting bit0 is 1, it means Sunday is effective. When the Weekend Setting bit1 is 1, it means Monday is effective. For example, if a user wants to set Saturday and Sunday effective, he should put 65 (1000001) into the Weekend Setting field. When the meter clock is within the preset Weekend Schedule, the energy will accumulate under the tariff that corresponds to the schedule.

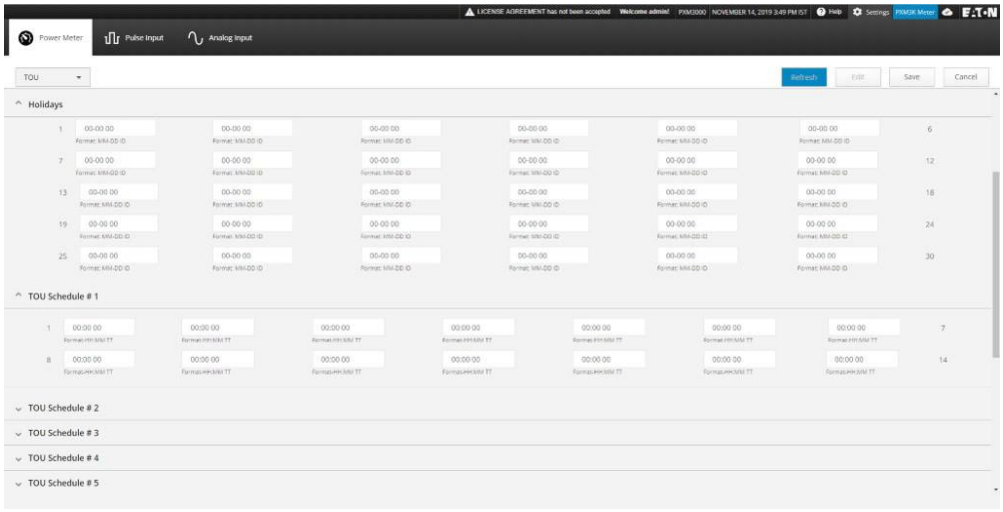
Note: Holiday schedule has the highest priority among all the schedules. Weekend schedule's priority is followed by Holiday schedule. When Holiday schedule is not enabled, Weekend schedule has the highest priority, overriding the normal (weekday) schedule.

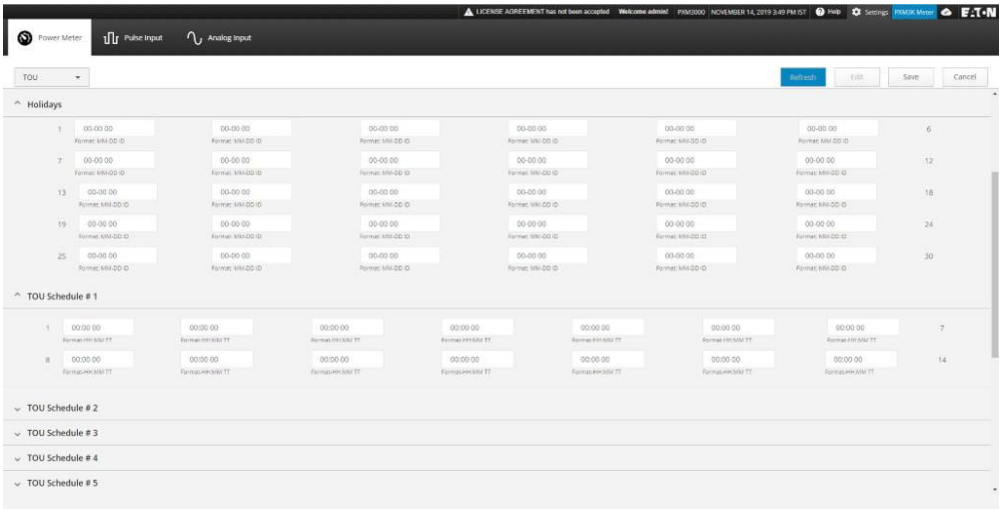
There are two ways of automatic resetting of current month TOU.

1. End of Month: This is the default method. All values from Current Accumulation Month TOU will be copied over to Prior Accumulate Month TOU at the very beginning of each month (the first day of each month at time 00:00:00). Current Accumulation Month TOU will continue to accumulate. All values from Current Month TOU will be copied over to Prior Month TOU at the very beginning of each month (the first day of each month at time 00:00:00) be cleared and reset to 0.

2. Assigned Clock: The user can select when the values from Current Accumulation Month TOU or Current Month TOU would be copied over to Prior Accumulation Month TOU or Prior Month TOU. The user can set the time in the following format: DD HH:MM:SS"- DD stands for day, HH stands for hour, MM stands for minute, SS stands for second. Similar to the previous method, once Current Accumulation Month TOU is transferred to Prior Accumulation Month TOU, Current Accumulation Month TOU will continue to accumulate. Once Current Month TOU is transferred to Prior Month TOU, Current Month TOU will be cleared and reset to 0.



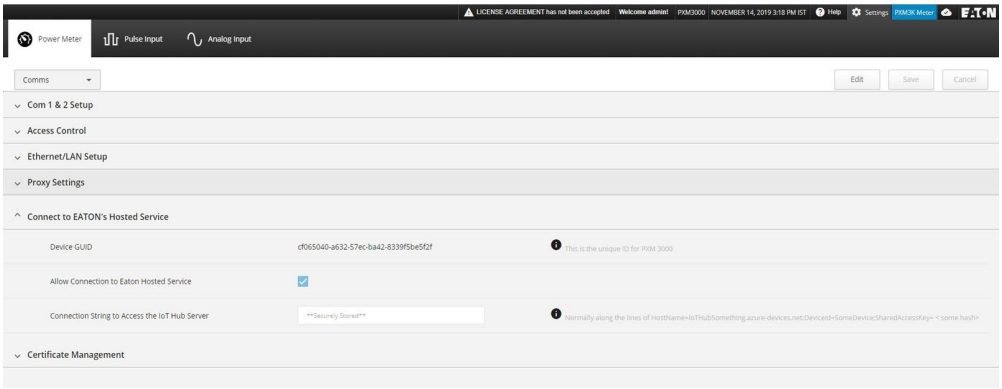




4.3.11 IoT Cloud

The user can configure IoT connection from here.

- Check/uncheck “Allow Connection to Eaton Hosted Service” checkbox to connect/disconnect IoT connection.
- Specify the connection string to access the IoT Hub Server:
- Normally along the lines of HostName=IoTHubSomething.azure-devices.net;DeviceId=SomeDevice;SharedAccessKey= < some hash>.
- After successful connection, the Device GUID will be generated automatically and populated here.
- The connection status (listed below) to IoT hub server is shown on top-right corner with cloud icons:
  - Connected;
  - Not Connected; or
  - Connecting.



4.4 Securing the Power Xpert Meter

4.4.1 Inbound/Outbound TCP/IP Ports

Access to the PXM 3000 meter is restricted by ports and can be restricted by trusted hosts. The following lists the ports open by default on the meter.

Port	Protocol
20 (TCP)	FTP Control
21 (TCP)	FTP Data
80 (TCP)	HTTP
161 (UDP)	SNMP
162 (UDP)	SNMP Trap
443 (TCP)	HTTPS
502 (TCP)	Modbus TCP
1900/7014 (TCP)	UPnP
47808 (UDP)	BACnet IP

4.4.2 Trusted Hosts

Trusted host restrictions can be applied to SNMP, Modbus TCP, and FTP. However, trusted host restrictions are not applied to ping, HTTP, HTTPS, or other protocols.

4.4.3 Hardening Security

Eaton's recommendations for hardening the meter, in decreasing order of importance, are:

- Change the Admin password from the default;
- Change the User password from the default;
- Change the Meter password from the default; and
- Require HTTPS only.
- Change the SNMP v1 and v3 user names and passwords from their defaults (public, private).
- Set the trusted host list for SNMP and Modbus TCP.
- Disable SNMP v1 if not needed. Use SNMP v3 only if you can.
- Disable SNMP v3 if not needed.
- Remove all the SNMP trap recipients if traps are not needed.
- Remove all email recipients if email is not needed.
- Change the default port for Modbus TCP.
- Change the default port for HTTP and HTTPS.

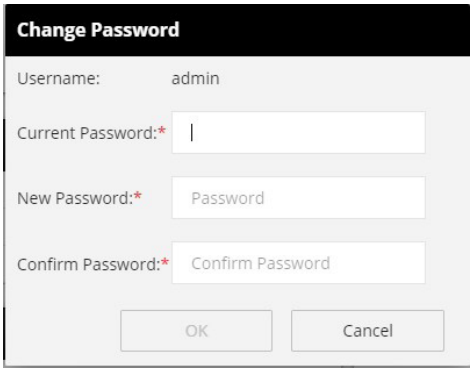
4.4.4 Passwords and Accounts

The PXM 3000 series meter provides two user accounts for network access: a User account and an Admin account. The User account provides read-only access to all pages in the web interface while the admin account provides that plus the ability to modify any of the meter settings. Meter users should only be given access through the User account. The Admin account should be reserved for use by only those people who are authorized to change meter settings. The web UI account provides administrative access as well, and you can use it to change any meter settings directly from the web UI. The password for this account is also mentioned in the following table.

⚠ IMPORTANT!

You should change the passwords of all accounts during the commissioning process. When creating new network passwords, use a combination of letters (mixed case), numbers, and special characters.

Default Passwords are "admin".



To change the passwords via the web interface:

1. Access the meter home page. You may be challenged for the admin account username and password.
2. Click on the **Welcome [USER_NAME]** link.
3. Click on **Change Password.**
4. You must enter the new password twice, and both entries must match to change the new password along with the current password. Also, the password must be set as per the Password Policy setup in the Setting->User->Password Policy page.
5. Click on Ok.

4.4.5 Recommended Meter Access Control Settings

The following are "best practice" recommendations for Power Xpert Meter security settings. You should adopt these to harden the security of the meter.

To access the PXM 3000 series security settings:

1. Access the meter home page. You may be challenged for the admin account username and password.
2. Click the Settings tab.
3. Navigate to the Comms > Access Control page.
4. Click the Edit button.
5. If you did not log in earlier, you must enter the admin account username and password. The following comprise Eaton's recommendations for Access Control best practices:

Trusted IPs/Hostnames Separated by Semi-colons (;)

If you are using SNMP, Modbus TCP, BACnet/IP, or FTP, you should specify the IP addresses or host names for one or more computers that are to be allowed access to the meter and then enable the trusted host feature for that protocol. This provides an added measure of security. You should also audit this list periodically and ensure any unused hostnames are removed.

Note: SNMP and Modbus/TCP-IP access are restricted to trusted hostnames by default. Therefore, an empty list of IP addresses/hostnames means that meter will not respond to SNMP or Modbus/TCP-IP.

SNMP Access Restricted by IP/hostname:

Restricting SNMP access to a set of trusted hosts is a best practice. As this is enabled by default, Eaton strongly recommends that you leave this enabled.

Modbus/TCP-IP Access Restricted by IP/hostname:

Restricting Modbus access to a set of trusted hosts is a best practice. As this is enabled by default, Eaton strongly recommends that you leave this enabled.

Modbus-TCP Port:

This is set to 502 by default. Changing this provides an added measure of security.

BACnet/IP Port:

This is set to 47808 by default. Changing this provides an added measure of security.

HTTP Port:

This is set to 80 by default. Changing this provides an added measure of security. Should you change this, users must append the port number (after a colon) to the meter URL.

HTTPS Port:

This is set to 443 by default. Changing this provides an added measure of security. Should you change this, users must append the port number (after a colon) to the meter URL.

4.4.6 SNMP Hardening

As noted in the previous section, you can specify a set of trusted hosts for SNMP. On the SNMP setup page, you can disable one or both versions of the SNMP protocol (the older version 1 or the more secure version 3). You should disable any version you are not using. We recommend using version 3 only if possible as this is more secure. You can also set passphrases for both the user and admin accounts for both versions of the SNMP protocol. You can specify a set of trap recipients by either IP address or host name. If you are not using traps, clear these fields.

To Access the PXM 3000 Series Meter Security Settings:

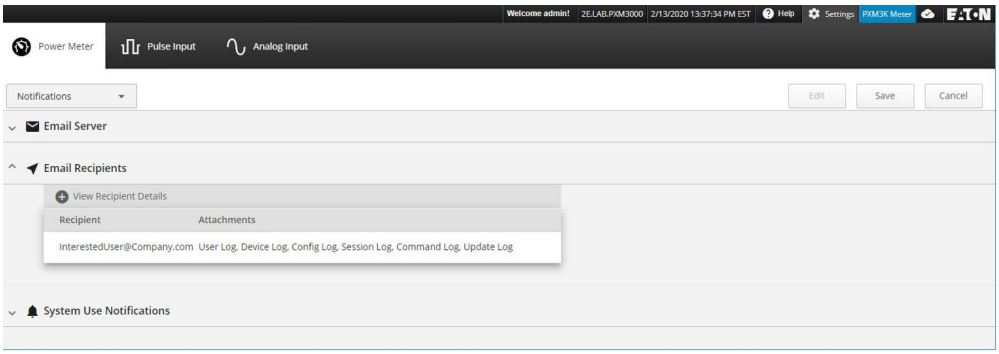
1. Access the meter home page. You may be challenged for the admin account username and password.
2. Click the Settings tab.
3. Select the Comms tab from dropdown menu.
4. Select Access Control menu.
5. Click the Edit button.

4.4.7 Email Security

This page will be used to setup Email server configuration, adding email recipients, and system use notification.

The PXM 3000 series meter can be set to send emails whenever an alarm or event occurs, or when an alarm is cleared. It can also be configured to send periodic status emails. Eaton recommends that you should periodically examine this list and remove any email addresses that are no longer required. To access the Email settings:

1. Access the Email Recipients page and click on View Recipient Details.
2. Click the "+" icon.
3. Make the additions/deletions needed.
4. Click the Save button.



To access the PXM 3000 series meter security settings:

1. Access the meter Home page. You may be challenged for the admin account username and password.
2. Click the Settings tab.
3. Navigate to the Notifications.
4. Click the Edit button.
5. If you do not have permission to this account, you must enter the admin account username and password.

4.4.8 Enabling Root Certificate Authority

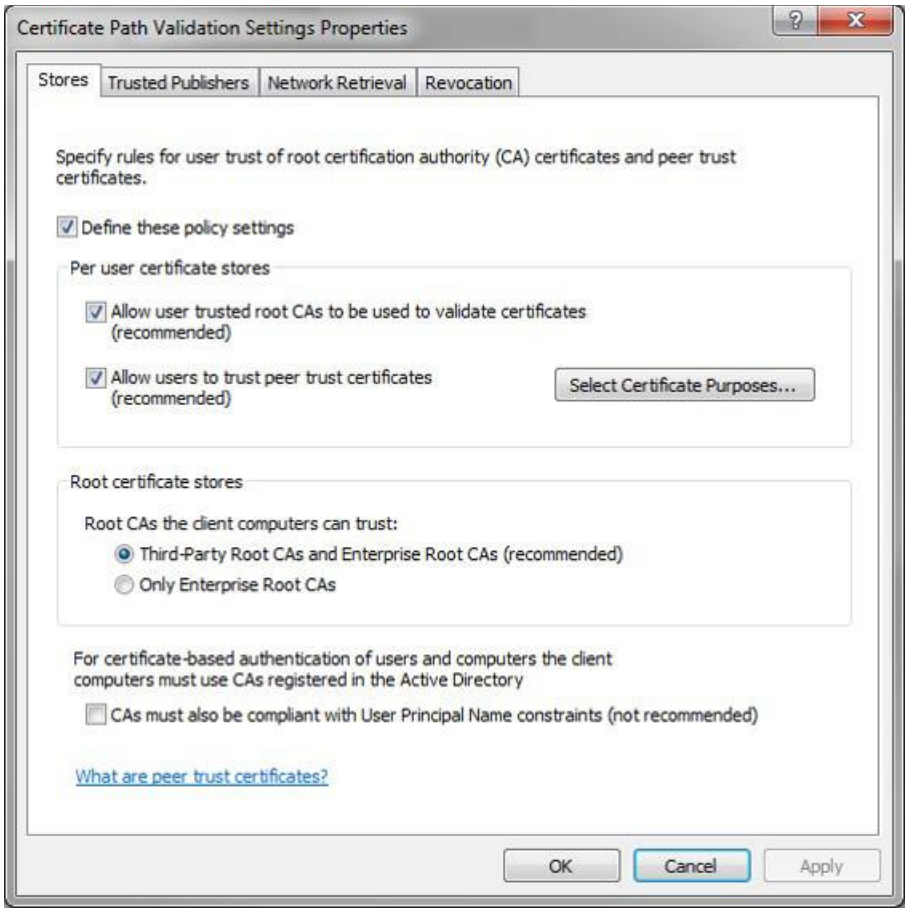
For added security, you should always disable HTTP access and enable HTTPS access for the gateway. To use HTTPS to connect to the PXM 3000, you'll need to:

1. Set either the local machine policy to allow users to manage certificates or, if multiple people in your organization will access the PXM 3000, set a group policy. You will need assistance from your IT organization to set a group policy.
2. Download the certificate file from the PXM 3000.
3. Install the certificate in the Trusted Root Certification Authorities store. This is not the default store that the certificate installation wizard will choose, which is why you need permission to manage certificates.

Both enabling user management for certificates and installing a certificate require administrative privileges for the PC. If you do not have such privileges, you will need to contact your IT organization for assistance before proceeding.

The process for either enabling this on a local machine or setting a group policy is outlined in the following Microsoft Technet Article: https://technet.microsoft.com/en-us/library/Cc754841.aspx .

Essentially, you are going to enable the users to allow trusted root CAs to be used to validate certificates and to trust peer trust certificates. You will do this through MMC. If you are changing the local policy, you will have policy settings for certificate stores set as is shown in the following figure.



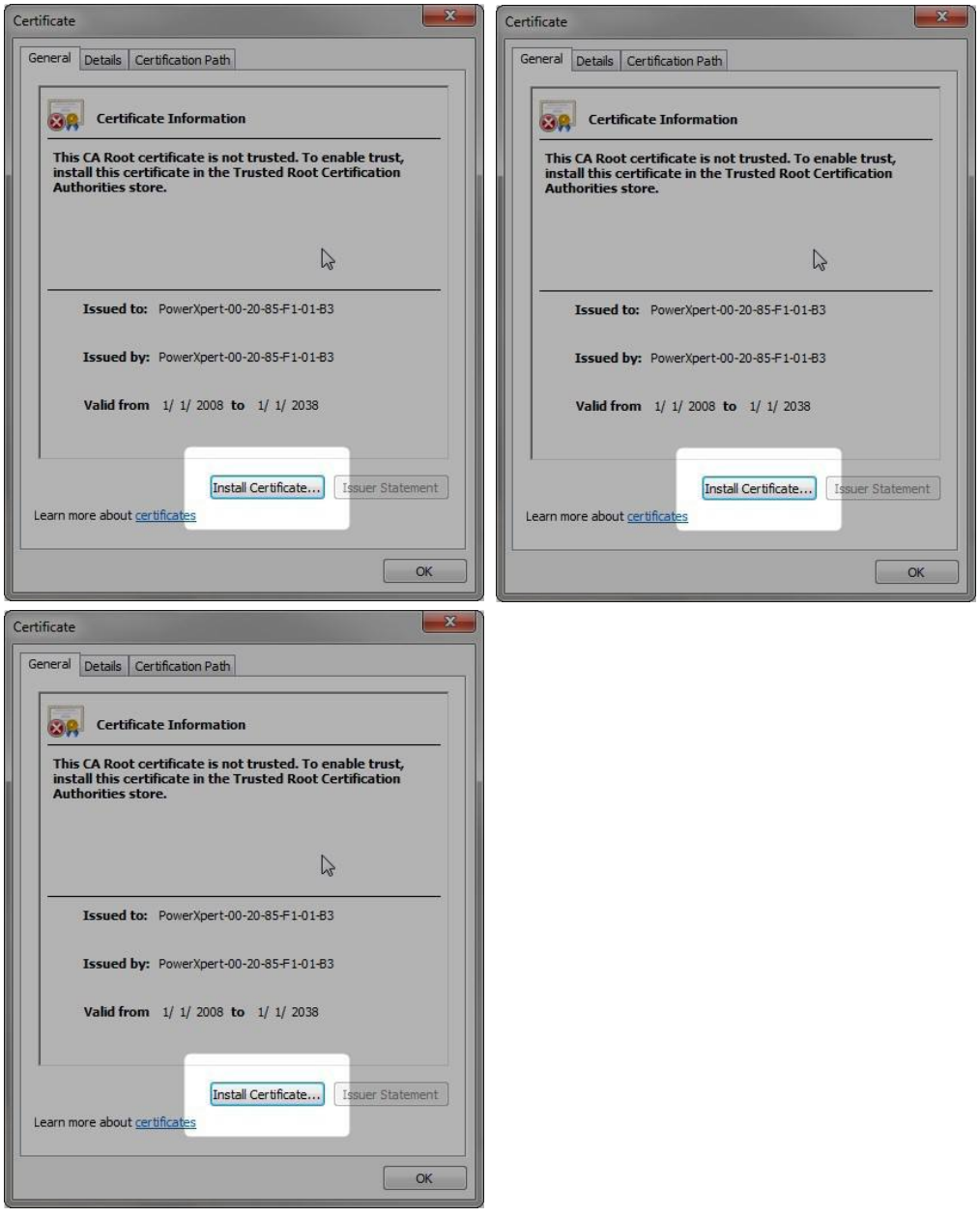
Downloading the Certificate File from the gateway:

1. Point either Google Chrome or Microsoft Internet Explorer to the IP address of the gateway followed by /ca.html. For example: http://192.168.1.1/ca.html.
2. Click the Root CA Certificate link. The browser will download the certificate.

Note: The certificate uses SHA-256 as its cryptographic hash function to avoid incompatibility problems with various browsers.

4.4.9 Installing the Certificate

1. Double-click the certificate file. This will launch the certificate installation wizard.
2. Click Install Certificate.

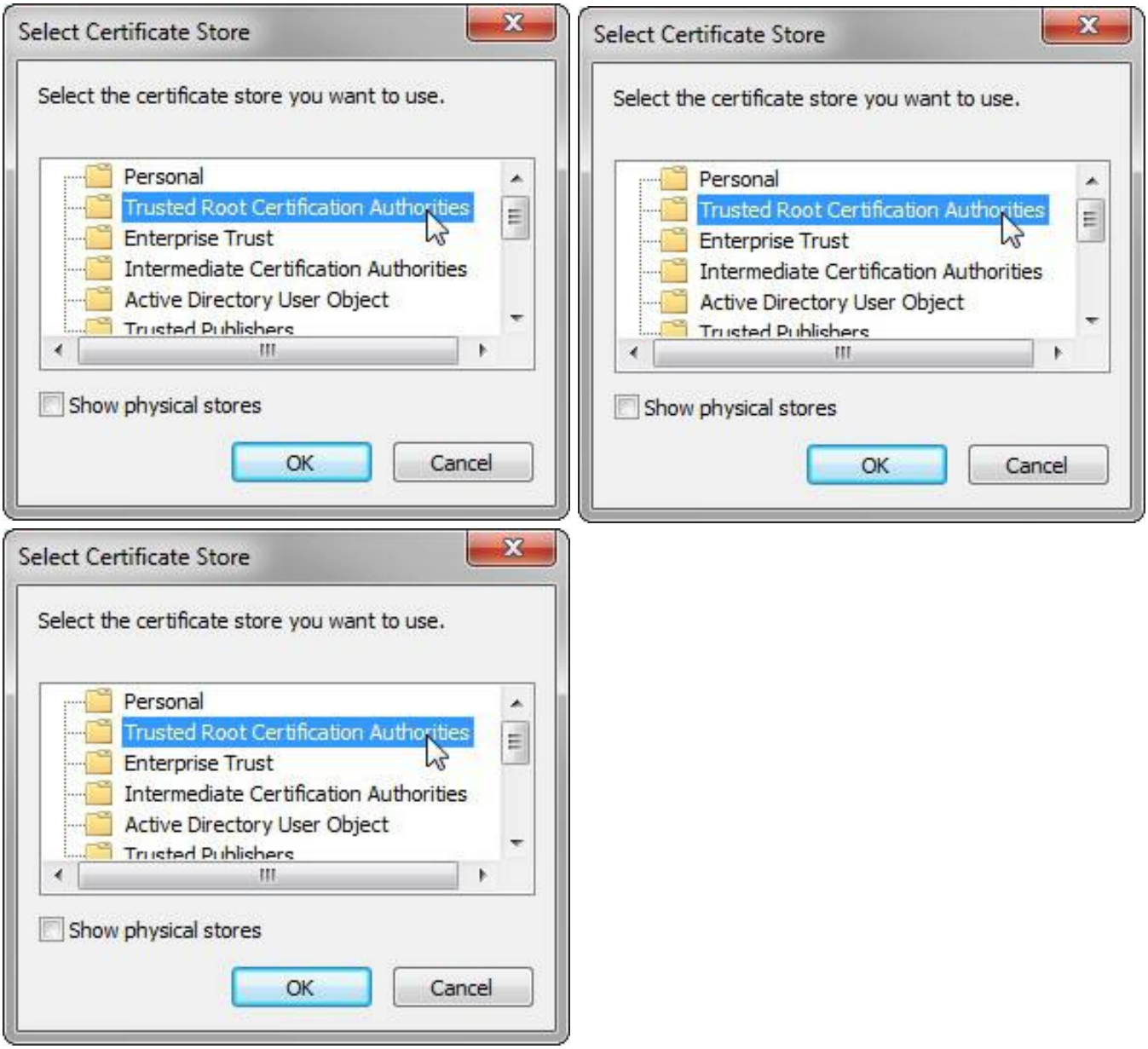


3. On the Welcome dialog box, click Next.
4. Select "Place all certificates in the following store" and then click Browse.

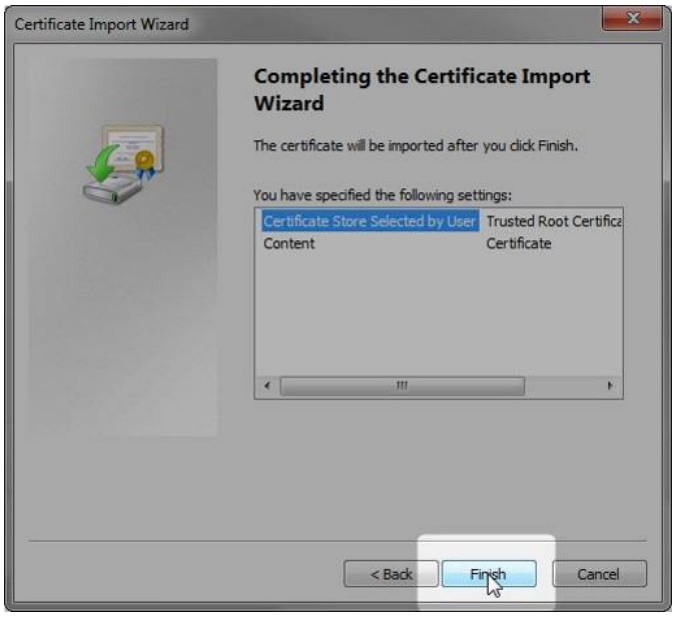


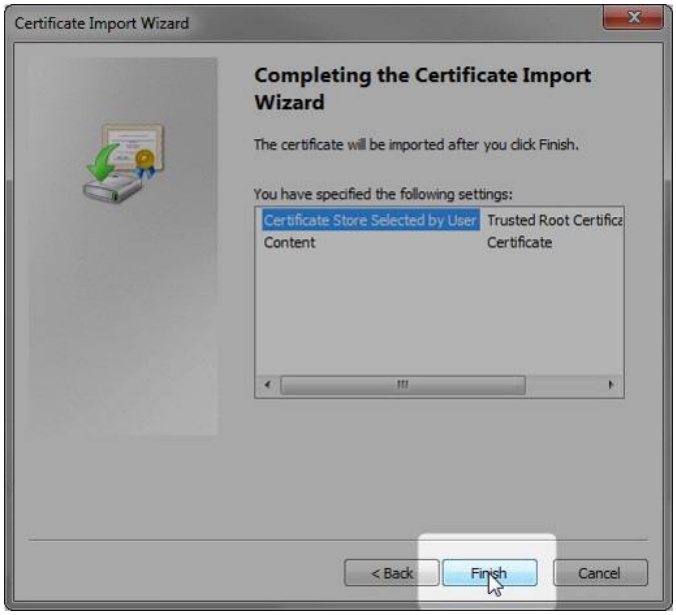


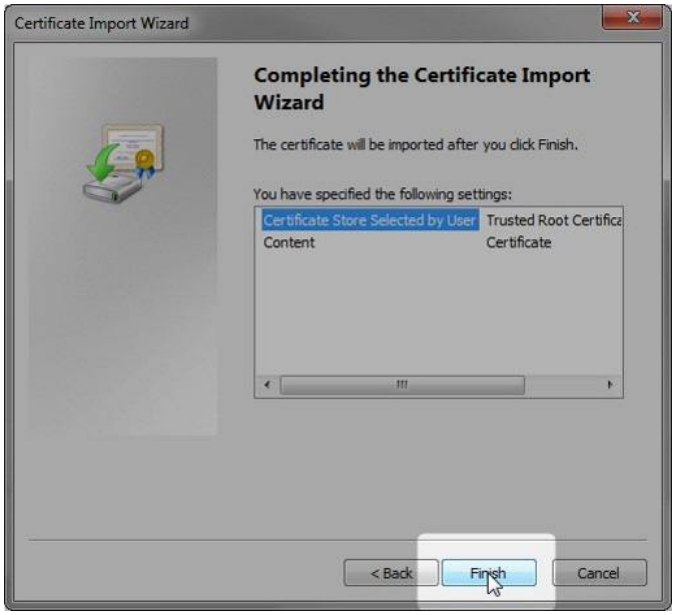
5. Select "Trusted Root Certification Authorities" from the list, then click OK.



6. On the Completing the Certificate Import Wizard dialog box, then click Finish.







7. You should see an alert box stating that the import was successful. Click OK.

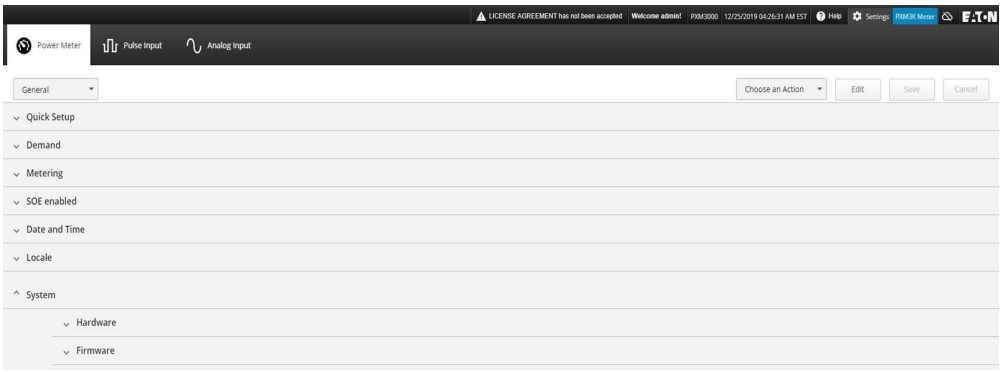


Browser Specific Notes:

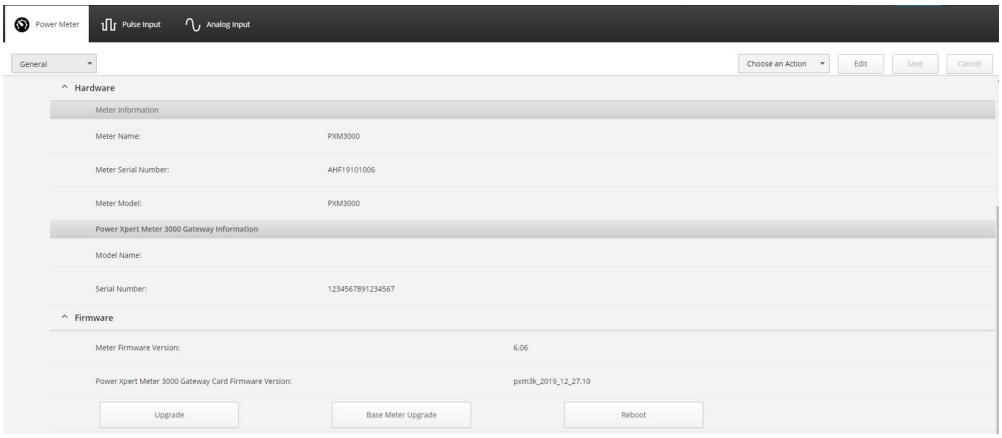
- If you use HTTPS with Internet Explorer 10, you must enable TLS version 1.2 support in the browser. The PXM 3000 does not support SSL version 3.
- Restart your browser after loading the certificate to avoid any problems caused by the browser caching data.

4.4.10 Firmware Update

PXM 3000 Web UI provides user friendly method to update the firmware for Gateway Card.



Under the System menu, there are two options one for the Hardware and one for the Firmware.



Under the Hardware section, the user will find the Hardware related information like Meter Model, Serial Number, Name, for both the meter and gateway card.

Under Firmware section, the user will be able to view Meter Firmware version and Gateway Firmware version.

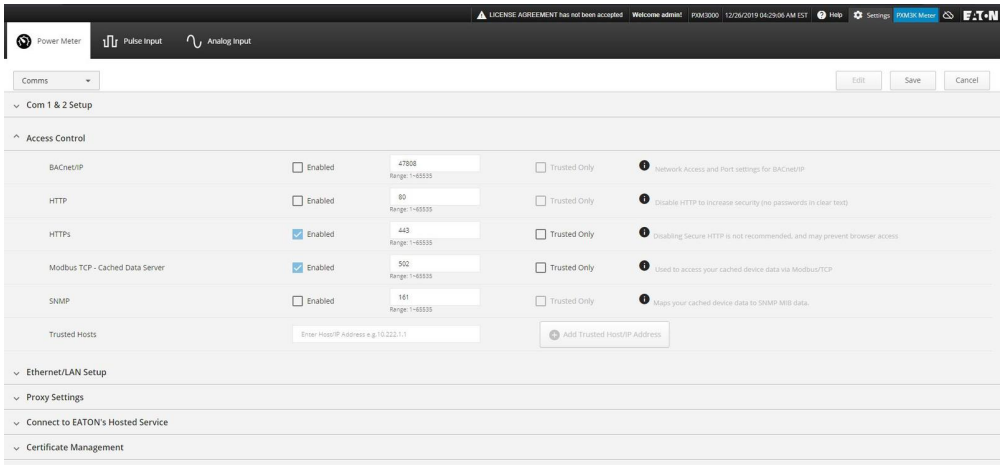
The user can upgrade PXM 3000 Gateway and PXM 3000 Base-meter firmware from this webpage. The user can also reboot gateway by clicking Reboot button. Select appropriate firmware file to upgrade the meter.

4.4.11 Modbus TCP

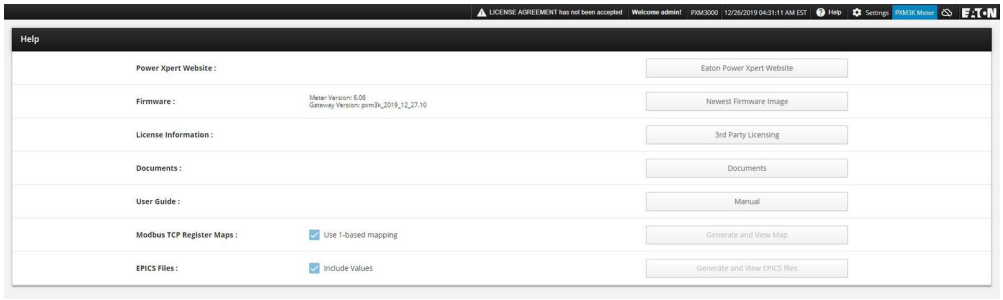
The PXM 3000 meter provides support for the MODBUS TCP connectivity.

The user will have to enable MODBUS TCP in settings page.

On the **Settings** page, select the **Comms** tab from dropdown menu. Under the Comms tab, select the Access Control tab. The user will have to enable the **Modbus TCP - Cached Data** Server option.



After enabling the MODBUS TCP, the user can go to the Help Page About section and click on "Generate and View Map" for Modbus register Map.



Note: The PXM 3000 Modbus TCP only supports register Read Commands, not Register Write Commands.

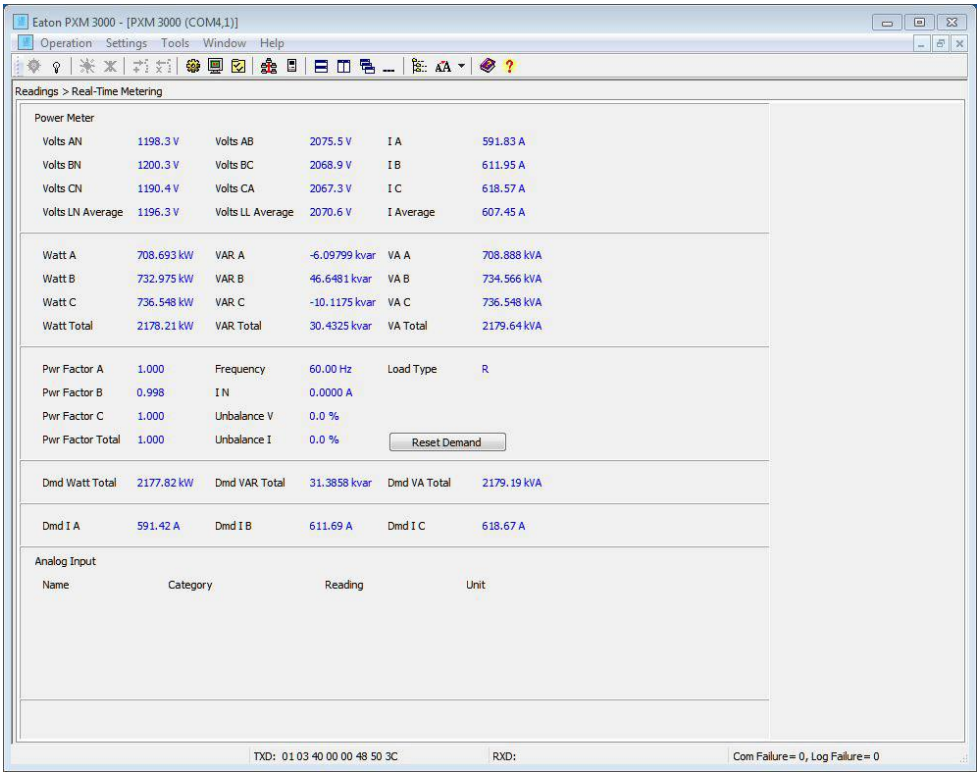
5. Detailed Functions and Software

The PXM 3000 meter contains advanced metering tools and is able to measure a multitude of power, energy, and power quality parameters. Some advanced functions may not be accessible directly from the front of the meter; therefore, every meter comes with a software that helps access the information. This chapter introduces these functions and the software.

5.1. Basic Analog Measurements

The PXM 3000 meter can measure voltage, current, power, frequency, power factor, demand, etc., with high accuracy via the software shown below.

Figure 41. Real-time Metering.



Demand:

This meter has the capability to calculate several types of demand: total active power demand, total reactive power demand, total apparent power demand, phase A current demand, phase B current demand, and phase C current demand. When demand is reset, demand memory registers are set as 0.

The user can configure the demand calculation mode to sliding window or thermal. Figure 47 shows how it works.

When using the sliding window interval mode, the user defines an interval between 1 and 30 minutes (which is the period of calculation). The demand is updated once a minute as the window slides.

Thermal demand mode calculates the demand based on a thermal response (mimicking a thermal demand meter). The user defines the period for the calculation and the demand is updated at the end of each period.

Energy:

This meter measures and accumulates energy in different directions (import and export). For real-time energy monitoring, it accumulates energy in kWh, kvarh, and kVAh continuous (since its last reset).

Calculation Mode

1. The user can select different energy calculation, fundamental-based or true RMS based. This selection can be made from the front of the meter or via communication. Fundamental-based calculations are used to accumulate energy without taking harmonics into consideration while true RMS based calculations are used to accumulate energy including fundamental and harmonics.

Note: When fundamental-based calculation mode is selected, PF calculation will be based on the fundamental wave.

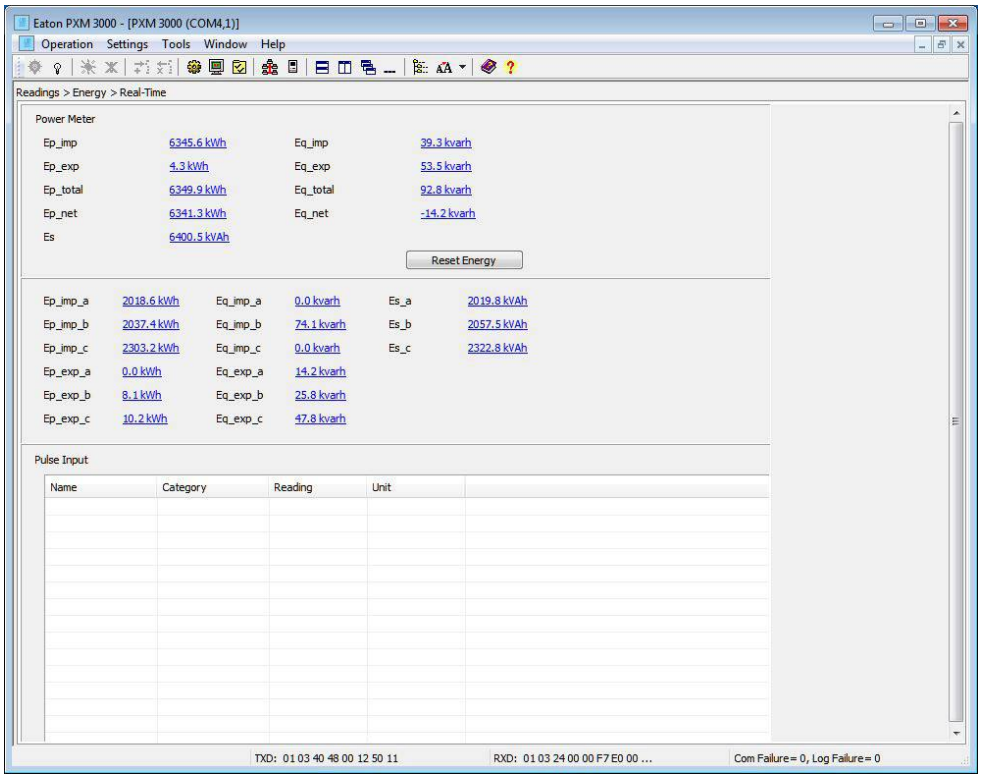
Note: When 400 Hz is the selected type, only true RMS calculations are supported.

2. There are two ways to calculate reactive energy (power):
   Mode 0: Real reactive energy $Q = \sqrt{S^2 - P^2 - D^2}$
   Mode 1: General reactive energy $Q = \sqrt{S^2 - P^2}$

3. The user can choose primary energy or secondary energy, either by pressing keys on the front of the meter or via communication as shown in Figure 47.

Note: PXM 3000 is able to display either primary energy or secondary energy on the LCD screen. However, it is only able to send out pulses based on secondary energy via the PXM1K-2 module.

Figure 42. Energy and Power Quality Parameters.



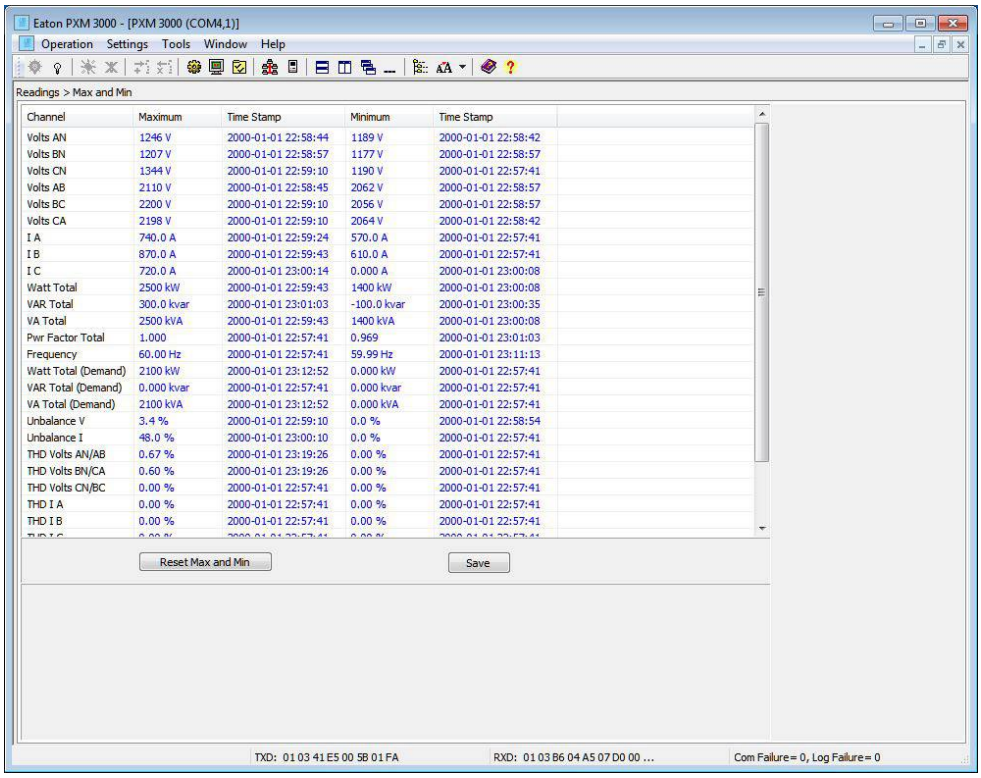
Current Direction Adjustment:

Under normal circumstances, current flows from input terminal 1 to terminal 2 (i.e. from I11 to I12 for phase A current). However, current may flow in the opposite direction due to incorrect wiring. In this scenario, users may reverse the polarity of the current by 180 degrees by setting the current direction to "negative." By default, the current direction is set as "positive". Refer to Figure 47 for more details.

5.2 Max/Min

The PXM 3000 meter logs maximum and minimum value statistics for phase/line voltages, current, power, reactive power, apparent power, power factor, frequency, demand, unbalance factor, and THD as well as the time they occur. All data is stored in non-volatile memory so that max/min information can be preserved even when the meter is powered off. All maximum and minimum data can be accessed via communication or from the front of the meter, but time stamps can only be accessed via communication. Max/min can be cleared either via communication or from the front of the meter.

Figure 43. Max/Min.



5.3 Harmonics and Power Quality Analysis

1. Harmonics:

The PXM 3000 meter can measure and analyze THD, calculate 2nd to 63rd order harmonics, even HD, odd HD, crest factor, THFF, K factor etc. They are shown in Figure 42.

Note: When 400 Hz type is selected, only 2nd~15th harmonics are supported.

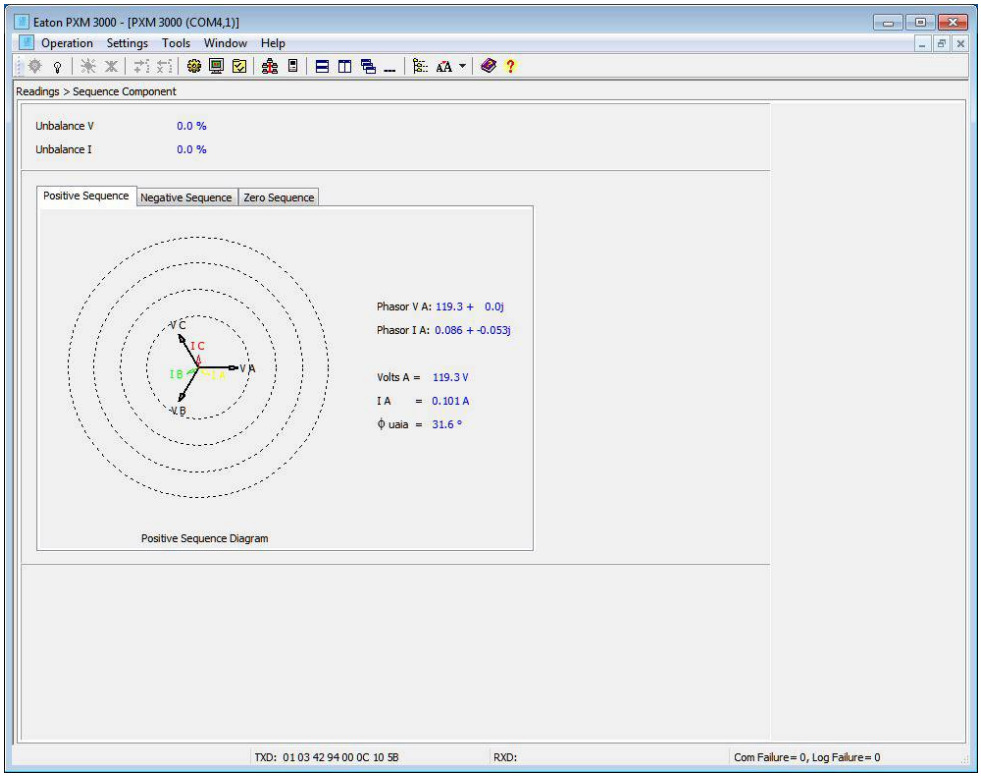
2. Phase Angle:

The term "phase angle" refers to angle of voltage/current parameters with respect to Phase A voltage. Angles ranges from 0 to 360 degrees. This function is to help users find the relationship between all input signals and avoid incorrect wiring. When the mode is set to "2 LL" or "3 LL", it gives the phase angles of u23, u31, i1, i2, i3 corresponding to u12. When it is set to "3 LN", it gives the phase angles of u2, u3, i1, i2, and i3 corresponding to u1. When it is set to "1 LL", it gives the phase angles of u2, i1, i2, corresponding to u1. They are shown in Figure 44.

3. Sequence Component and Unbalance Analysis:

The PXM 3000 meter is able to perform sequence analysis on the input signal. It looks at the positive sequence, negative sequence, and zero sequence of the fundamental frequency and performs unbalance analysis for voltage and current. Sequence components are shown in Figure 44, unbalance of voltage and current are shown in Figure 41.

Figure 44. Sequence Component and Phase Angle.

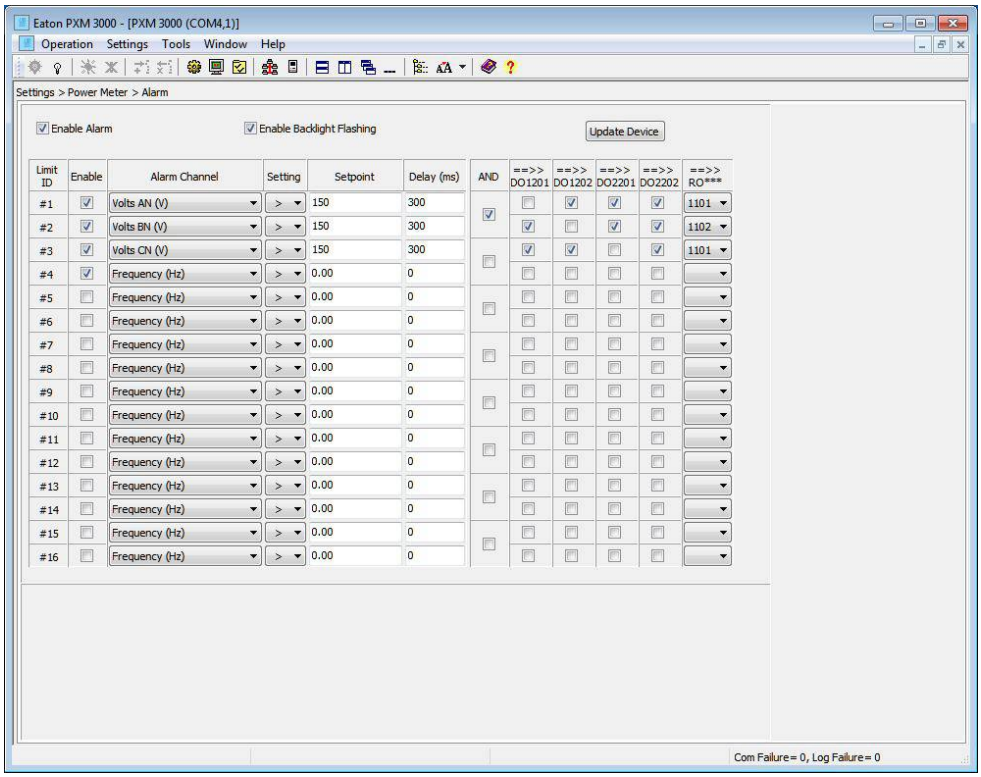


5.4 Out-of-limits Alarms

The PXM 3000 meter has configurable alarms for out-of-limits conditions. The alarm is triggered when the monitored parameter exceeds the set limit for some duration (delay, given in ms). The event is recorded in the alarms log with the parameter value and time stamp. The meter stores up to 16 triggers. When extended I/O modules are used, alarms can be used to trigger digital outputs (DO) and relay outputs (RO) in order to activate downstream devices such as a beacon light or a buzzer.

Before alarms can be used, alarm conditions such as logic dependency, target setpoint, time delay, etc., must be set correctly. Settings can be accessed and modified from the software via communication connection as shown in Figure 45.

Figure 45. Alarm Setting.



1. Single Alarms Settings Groups:

Table 4 indicates the first group of settings. There are 16 groups in total with the same format.

Table 4. First Group of Alarming Settings.

Address	Parameter	Range	Property
104eH	First group: parameter code	0~79	R/W
104fH	First group: comparison mode	1: larger, 2:equal, 3:smaller	R/W
1050H	First group: setpoint value	Related with parameters	R/W
1051H	First group: delay time	0~3000(*10ms)	R/W
1052H	First group: output to relay	0: none, 1-8: related relay	R/W

Parameter code: select the target parameter for alarm monitoring. For example: 0-frequency, 44-AI4 sampling data.

Comparison mode: set alarm condition 1: greater than, 2: equal to, 3: less than. For example, if you choose the target parameter to be "frequency", condition to be "greater than" and setpoint to be "50", an alarm will be triggered when the frequency is greater than 50 Hz.

Note: Setpoint value is the same as the actual value of the selected parameter.

Delay time: If the alarm condition lasts for the configured time period, the alarm signal will be triggered. The delay range is from 0 to 3000 (unit: 10 ms). When it is set to 0, there is no delay. An alarm will be triggered as soon as the alarm condition is met. If it is set to 20, there will be a 200 ms (20 x 10 ms) delay.

Output to relay: 0-alarm signal will not be sent to RO. If it is set as 1 and PXM1K-11 is connected, it will output to RO1 when the alarm is triggered. RO1 will be turned off when the alarm output to RO1 is cleared. Relays RO2 - RO8 work in the same way.

Note: If an alarm is sent to RO, it can only work in "latch" mode.

Note: If the parameter code is set with a value between 51 and 79, please refer to Chapter 6 for detailed description of parameter and usage.

In addition to configuring an individual alarm, users must also configure global alarm settings in order to use alarms

2. Global Settings:

Register addresses for global alarm settings are from 1046H~104dH.

"Global alarming enable" enables alarms across the meter. Alarms are enabled when this register is set to 1, and disabled when set to 0.

Setting "Alarming flash enable" to 1 will cause the backlight to flash when an alarm is triggered.

The "Alarming channel enable setting" is a 16-bit register that controls the different alarm groups. There are a total of 16 groups, each is controlled by one bit of the register. An alarm channel is activated when the corresponding bit is set to 1.

Logical AND between pairs of alarm triggers: The 16 alarm triggers can be considered in 8 pairs (with two alarm groups in each). If the logical AND check box is selected, the two triggering conditions are considered together. An alarm will only occur of both conditions are met. If the logical AND check box for a pair remains unchecked, the two alarm channels will work independently.

The 8 "AND" logic pairs are arranged as follows: 1st, 2nd channel form Pair 1; 3rd, 4th channel form Pair 2; 5th, 6th channel form Pair 3; 7th, 8th channel form Pair 4; 9th, 10th channel form Pair 5; 11th, 12th channel form Pair 6; 13th, 14th channel form Pair 7; 15th, 16th channel form Pair 8.

This functionality is controlled by the lower 8 bits of this 16-bit register with each bit corresponding to a pair. "1" means this function is enabled and "0" means disabled.

"Alarm output to DO1 setting": "Digital output mode" allows DO1 to be used as an alarming output. This is controlled by a 16 bit register. Each bit in the register corresponds to one of the 16 groups. When the appropriate I/O module is connected and alarms are enabled, the corresponding bit will be set to 1. When the alarm is triggered, the signal will be sent to DO1 until the alarm is cleared. Setting the corresponding bit to 0 disables this functionality. Outputs DO2-DO4 work the same way.

3. Setting Example:

Here is an example showing how to apply the logic "AND" function for a pair of alarm channels.

The conditions are as follows: I1 greater than 180 A, delay 5s for the 1st alarm channel; U1 less than 9,980 V, delay 10 s for the 2nd alarm channel. No alarm signals will be sent to outputs. The CT has a primary value of I1 is 200 A, and CT2 is 5 A. The PT ratio for U1 is 10000:100. The following shows how all the related registers are to be set.

Table 5. Register Settings.

Trigger Threshold	I1 (current)	Greater than 180 A	Delay 5s for the 1st alarm channel	Output none
	U1 (voltage)	Less than 9,980 V	Delay 10 s for the 2nd alarm channel	Output none
CT Primary	CT1	200A		
CT Secondary	CT2	5A		
PT Primary	PT1	10000		
PT Secondary	PT2	100		
Secondary Value	Rx			

Settings of first group:

- "Parameter code (104eH)" is set to 9, which represents I1.
- "Comparison mode (104fH)" is set to 1, which represents "greater than".
- "Setpoint value (1050H)" is set to 4500 since the 4.5 is represented as an integer . Rx= (I/(CT1/CT2))*1000.
- "Delay time (1051H)" is set to 500, so the actual delay time is 500*10 ms = 5 s.
- "Output to relay (1052H)" is set to 0, because there is no output to RO.

Settings of second group:

- "Parameter code (1053H)" is set to 1, which represents U1.
- "Comparison mode (1054H)" is set to 3, which represents "smaller than".
- "Setpoint value (1055H)" is set to 998, since the 9980 is represented as an integer . Rx= (U/(PT1/PT2))*10.
- "Delay time (1056H)" is set to 1000, so the actual delay time is 1000*10 ms = 10 s.
- "Output to relay (1057H)" is set to 0, because there is no output to RO.

Global settings:

- "Alarming channel enable setting (1048H)" set to 0003H to enable the first and the second channel.
- "Logic "AND" between alarming setting (1049H)" set to 0001H to enable logic "AND" in Pair 1.
- "Alarming output to DO1 setting (104aH)" set to 0, since no output to DO1.
- "Alarming output to DO2 setting (104bH)" set to 0.
- "Alarming output to DO3 setting (104cH)" set to 0.
- "Alarming output to DO4 setting (104dH)" set to 0.
- "Alarming flash enable (1047H)" set to 0 to disable backlight flashing when alarming occurs.
- "Global alarming enable (1046H)" set to 1 to enable over/under limit alarming.

4. Records of Alarming Event:

The PXM 3000 meter has built in alarm logging capabilities. A total of 16 entries can be recorded. The record sequence of these entries does not depend on the sequence of the 16 alarm channels. The meter begins logging alarm status starting from the first record location to the last one. Alarm logs are recorded in a "FIFO" fashion which means the latest event will overwrite the oldest record. When out-of-limit parameters return to normal, its value and time stamp will be recorded as well. Therefore, users can determine the out-of-limit duration by checking the time difference.

Table 6 shows the first group of records. Other groups of records have the same format.

Table 6. Alarming Status of the 1st Group of Records.

Address	Parameter	Range
42a9H	First group: alarming status	0~65535
42aaH	First group: parameter code	0~79
42abH	First group: over/under limit or reset value	Related with parameters
42acH~42b2H	First group: occur time: yyyy:mm:dd:hh:mm:ss:ms	Time

"Alarm status" indicates information of the current alarm status. It is a 16-bit unsigned integer. The Parameter code is stored in the higher 8 bits. Bit1 indicates whether logic "AND" is enabled or not. 1 means enabled and 0 means not enabled. Bit0 indicates whether the alarm has occurred or recovered. 1 means occurred and 0 means recovered. Undefined bits are 0.

"Parameter code" specifies the monitored parameter.

"Value" shows the recorded value of the selected parameter when an alarm is triggered and when it recovers.

"Time" indicates the time stamp with the precision in milliseconds (ms).

An alarm event will set bit0 of the "system status (102eH)" to be 1. When the software sends clear alarm command, bit0 of "system status (102eH)" will be set to 0.

Alarm group number (1032H): the range is 0~16, 0 is no alarm record, and 1~16 identify the records from newest alarm to oldest. The alarm group number can be saved during meter power off.

Here is an example:

Figure 46. Alarming Records.

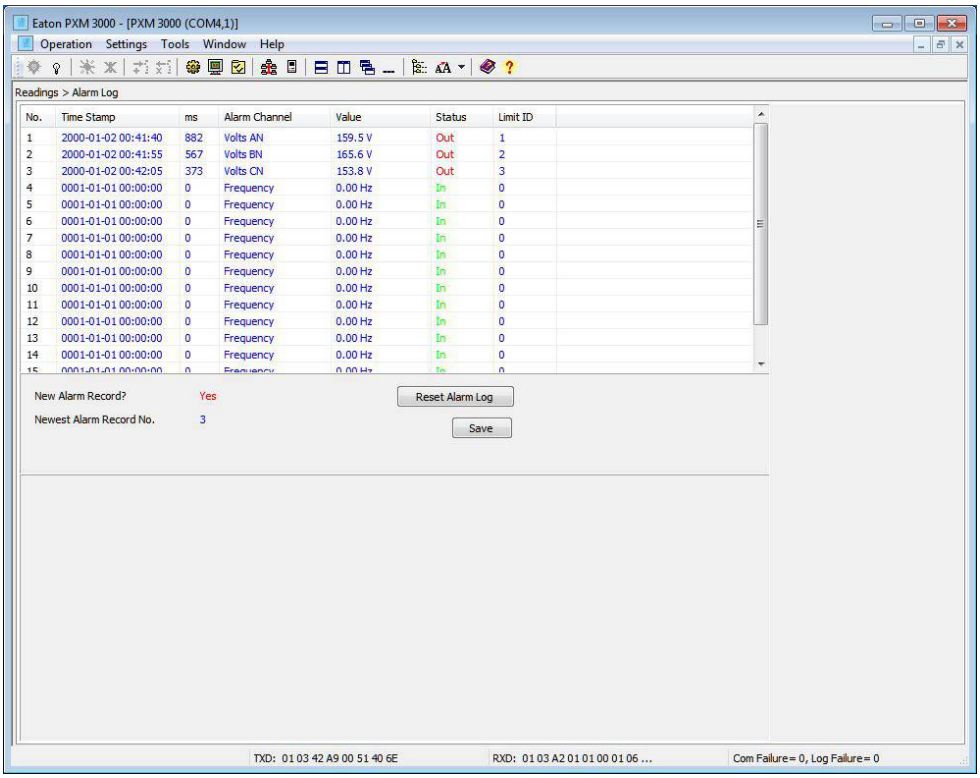
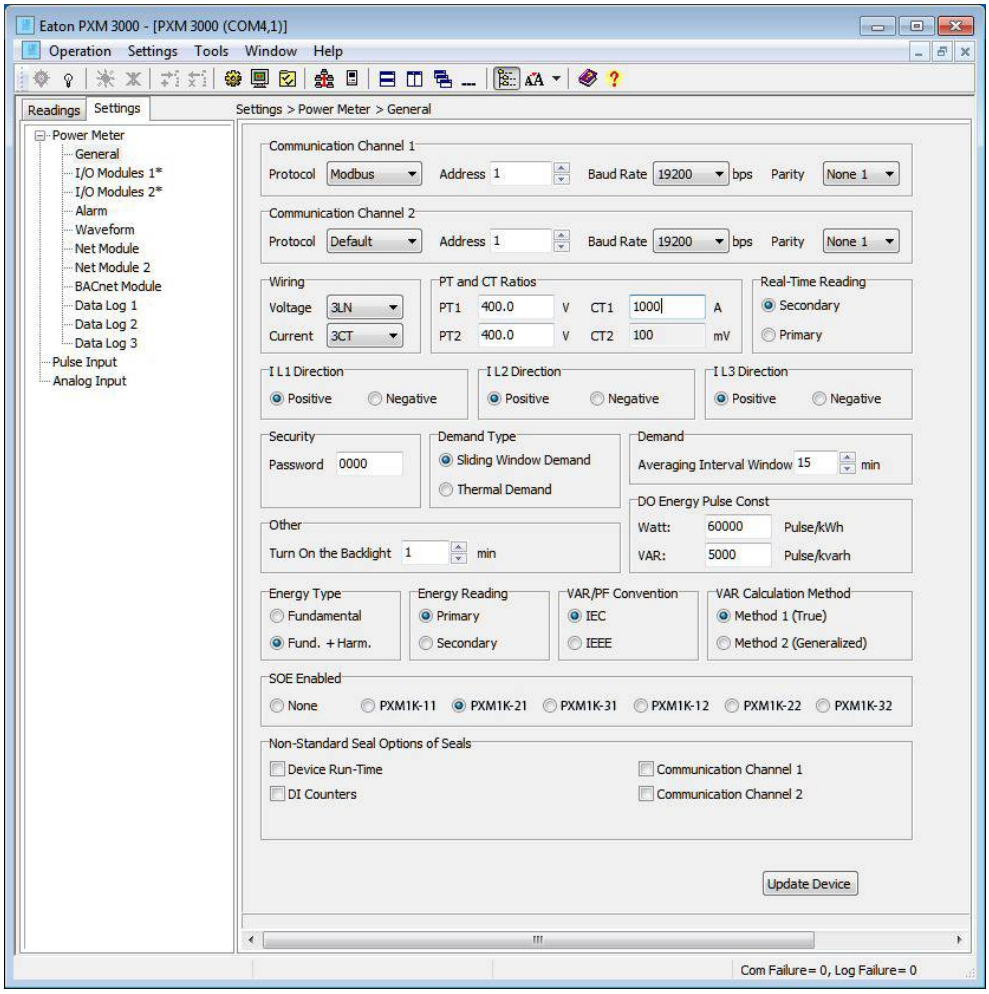


Figure 47. Basic Settings.



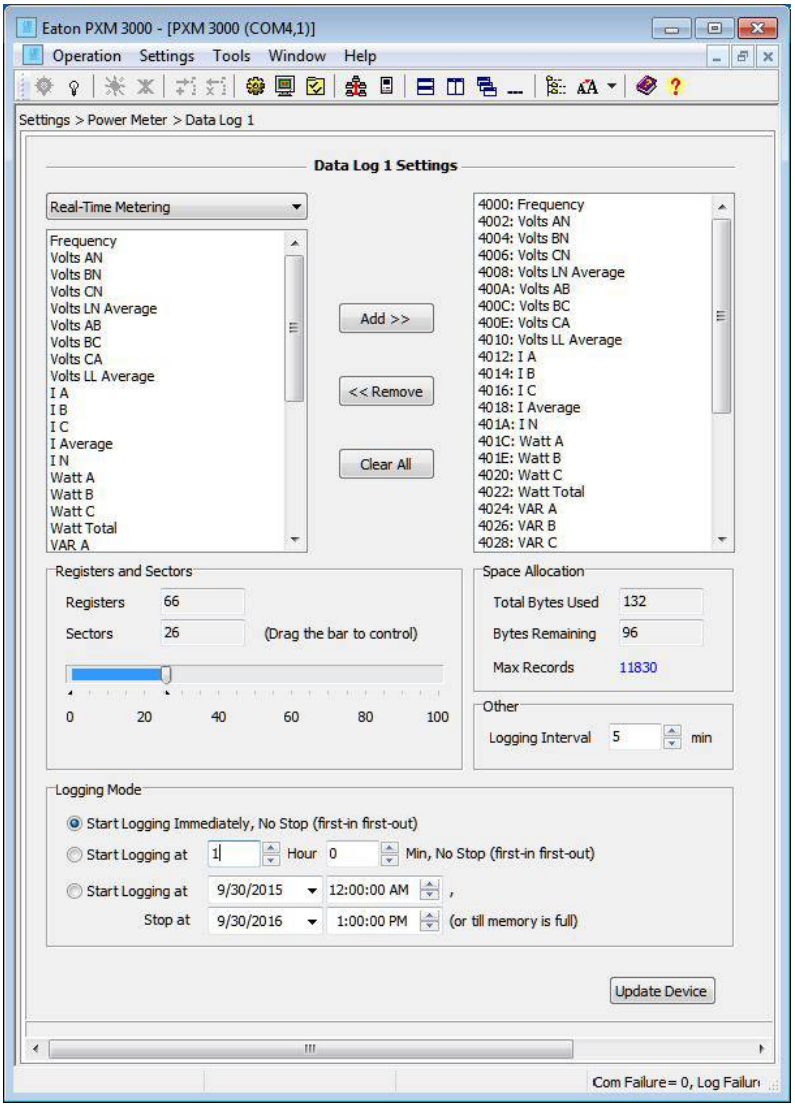
5.5 Data Logging

The PXM 3000 meters provide data logging that records the data at a set interval. This meter has four megabytes of memory which gives it extensive data-logging capabilities. It has a real-time clock that allows logs to be time-stamped when log events are created.

1. Data Log Settings:

The PXM 3000 meters have three sets of historical data logs. Each log can be independently programmed with individual settings, meaning that each can be used to monitor different parameters. The user can program up to 117 parameters per log. The user also has the ability to allocate available system resources among the three logs, to increase or decrease the size of the individual historical logs. The total size is no more than 63 sectors that have 64 k-bytes. The data log 1 setting is shown in Figure 48.

Figure 48. Data Log 1 Setting.



Having three sets of historical logs provides the user with the option of programming each log with unique parameters. For example, you can program historical log 1 to record measured values parameters (for example, frequency, voltage, current), log 2 to record energy values parameters, and log 3 to record power quality parameters. Historical log parameters can be selected from the following 13 groups:

- Real-Time Metering (Frequency; Instantaneous Voltage; Instantaneous Current; Total and Per Phase Power and Power Factor; Neutral Current ; Unbalance V/I; load type; Current demand; and Per Phase/ Total Power demand)
- Energy (Ep_imp; Ep_exp; Ep_total; Ep_net; Eq_imp; Eq_exp; Eq_total; Eq_net; Es; Epa_imp; Epa_exp; Epb_imp; Epb_exp; Epc_imp; Epc_exp; Eqa_imp; Eqa_exp; Eqb_imp; Eqb_exp; Eqc_imp; Eqc_exp; Esa; Esb; Esc).
- THD Volts AN/AB(THD, 2nd -63rd Harmonic Magnitudes, ODD, EVEN, CF and THFF of Volts AN/AB).
- THD Volts BN/BC (THD,2nd -63rd Harmonic Magnitudes,ODD,EVEN,CF and THFF of Volts BN/BC).

- THD Volts CN/CA (THD, Average THD, 2nd-63rd Harmonic Magnitudes, ODD, EVEN, CF and THFF of Volts CN/CA).
- THD IA (THD, 2nd -63rd Harmonic Magnitudes, ODD, EVEN, KF of IA).
- THD IB (THD, 2nd -63rd Harmonic Magnitudes, ODD, EVEN, KF of IB).
- THD IC (THD, average THD, 2nd -63rd Harmonic Magnitudes, ODD, EVEN, KF of IC).
- Sequence Component (positive, negative, and zero sequence).
- Phase Angles(the angle between U1 and other voltage and current parameters).
- DI Counter (the DI numbers of the IO modules).
- AO/AI Raw Value (the AO output register values and the AI sample register values).
- AO/AI Value(the AO output values and the AI sample values)

The following procedures show how to select and store parameters in historical log 1. The Group field determines the items that are available for selection.

1. Select a Group. The possible selections are: Real-Time Metering, Energy, THD Volts AN/AB, THD Volts BN/BC , THD Volts CN/CA , THD IA , THD IB , THD IC, Sequence Component, Phase Angles, DI Counter, AO/AI Raw Value, and AO/AI Value.
2. Select items for your log:
   A. Highlight the parameter(s) you want to log into the meter's memory.
   B. Click Add to add the parameter to the Selected Parameter Area.
   C. To remove parameter(s), highlight them in the Selected Parameter Area and click Remove.
3. Set the logging interval (in minutes). Interval can be set from 0 - 1444 minutes according to different applications.

   The logging interval determines when the meter takes a snapshot. When the interval is set as 0, the set of historical data log is disabled.
4. There are 100 sectors in total for the 3 historical data logs. The user can assign a different sector size to each log according to different applications (as long as the total sector sizes of the 3 logs do not exceed 100).
5. Three Modes of historical log:

   Mode 1: The historical log can record, even if date and time are not set. The log is first-in first-out, so earlier entries are replaces as the log fills.

   Mode 2: With date and time set, the historical log can record for the duration. Recording will stop once the buffer fills.

   Mode 3: With the hour and minute set, the historical log can record, starting at that set time. This log is also first-in first-out.

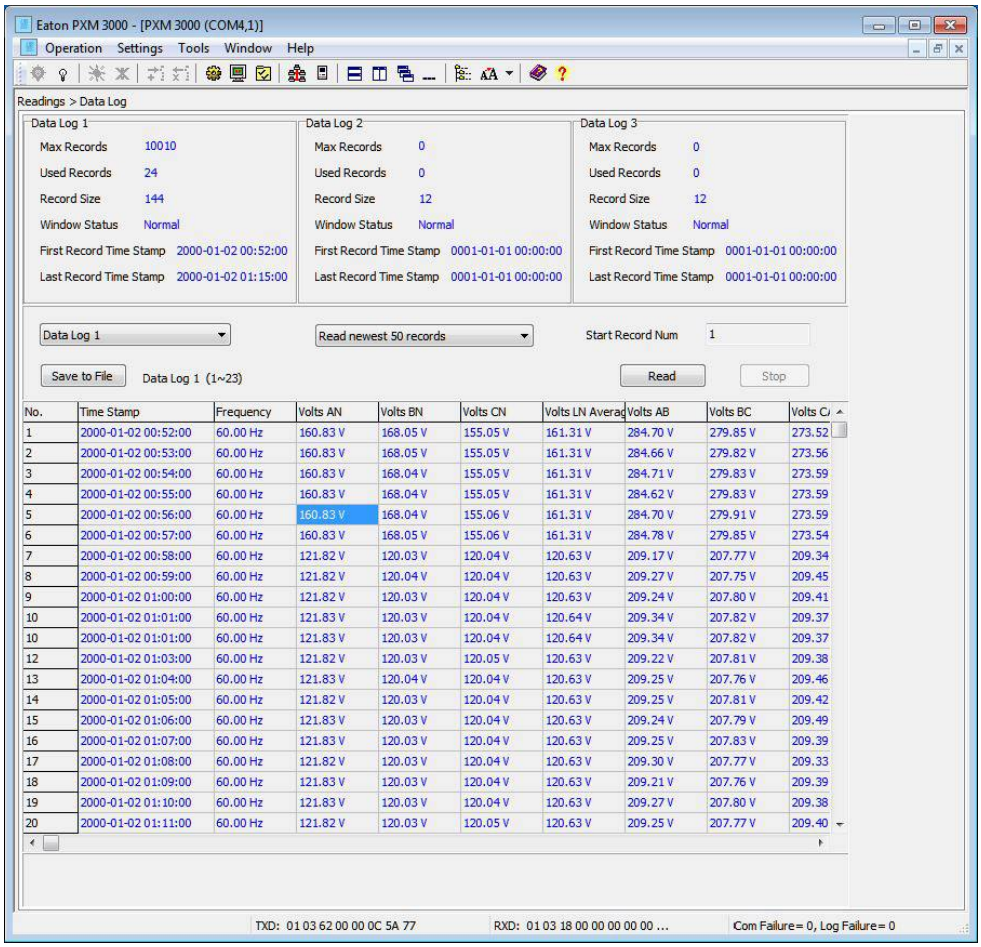
Note: If the memory of the historical data log is full, the meter will erase the first sector in which the memory size is 65536 bytes (64kb). The following sector (the second sector) will become the first sector and the data from the erased sector will not be recoverable. Therefore, the user should save the whole log before memory is full to maintain all the data.

Note: There are two display fields at the bottom of the data log setting screen. They show the registers in the logs, the total bytes used, and the bytes remaining for this historical log. These fields are updated as you make selections on the screen. The total number of bytes available per log record is approximately 234.

2. Retrieving Logs:

There are two ways of retrieving the logs: "read one window" and "read all". The retrieval screen is shown in Figure 49.

Figure 49. Retrieval Screen.



The "read one window" method allows the user to access and read a specific log location at an offset from the first log. The "window record num" is the maximum number of record entries the software can read at a time. It is calculated by:

246 / record size.

The larger this number is, the faster data can be retrieved. "Log type" is used to retrieve the logs the user wants to retrieve. For example, log type 0 is data logging 1, log type 1 is data logging 2, and log type 2 is data logging 3.

The "read all" method accesses and reads the historical data log automatically. The offset increases automatically until all the logs are retrieved.

The data logs contents are shown on the bottom section of the screen.

5.6 Time of Use (TOU)

The user can assign up to four different tariffs (sharp, peak, valley, and normal) to different time periods within a day depending on the billing requirements. The meter will calculate and accumulate energy to different tariffs according to the meter's internal clock timing and TOU settings.

The user can set a maximum of 12 TOU seasons. Each season can be assigned to a TOU schedule (a maximum of 14 TOU schedules are available). Each schedule can be divided up into 14 segments (in which each segment can have its own tariff). The user can customize the TOU calendar (including its tariffs, seasons, schedules, and segments) according to different applications. To make sure that the TOU calendar is setup correctly, the meter will check the TOU settings according to the predefined rules (see below for "TOU Setting Format Requirements" for details). The TOU function will be disabled if the TOU calendar is configured incorrectly. If no errors are found in the calendar and the TOU function is enabled, TOU energy accumulation will begin.

TOU Setting Format Requirements:

1. Season setting parameter: The calendar year will be divided into different seasons depending on the season setting parameter. The parameter can be selected from any integer between 1 and 12. The user must enter the correct value for the season setting parameter in accordance with the TOU season table. If the season setting parameter is set as 2, the first 2 slots of the TOU season table must be set. Otherwise it will be considered as an invalid input (TOU function will be disabled).
2. TOU season format: Enter the start date into the TOU season table slot following this format "MM-DD ID" (MM stands for the month; DD stands for the day; and ID stands for the TOU schedule ID [available from 01 to 14]). The dates should be organized so that they are in sequence according to the calendar year (the earlier date comes first and the later date comes last). For example, if 3 seasons are selected, the date parameters are January 1, June 6, and September 7, and TOU schedule 02, 01, 03 will be used. The first TOU season table slot will enter 01-01 02, the second slot will enter 06-06 01, and the third slot will enter 09-07 03. Entering 01-01 02 for the first slot, 09-07 03 for the second slot, and 06-06 01 for the third slot is considered invalid.
3. Schedule setting parameter: The number of available TOU schedules depends on the schedule setting parameter. The parameter can be selected from any integer between 1 and 14. This parameter determines the number of TOU schedules available for the TOU calendar setting. A maximum of 14 TOU schedules (from TOU Schedule #1 to TOU Schedule #14) can be used.
4. Segment setting parameter: Each TOU schedule consists of various timing segments. The number of segments depends on the segment setting parameter configured. The parameter can be selected from any integer between 1 and 14 (inclusively). The user must enter the correct value for the segment setting parameter in accordance to the TOU schedule table. If the segment setting parameter is set as 3, the first 3 slots of the TOU schedule table must be set. Otherwise, it will be considered as an invalid input (TOU function will be disabled).
5. TOU schedule format: Each TOU schedule represents a 24 hour cycle. Similar to the TOU season format, enter the start time into the TOU schedule table slot following this format "HH:MM ID" (HH stands for hour [in 24 hr format]; MM stands for minutes; and ID stands for tariffs [available from 00 to 03]). The time should be organized according to the hour sequence. For example, if 3 segments are selected, timing parameters are 01:00, 15:30, 22:45, the order of the 3 segments should be one of the following: 01:00, 15:30, 22:45 or 15:30, 22:45, 01:00, or 22:45, 01:00, 15:30. Entering time information in a wrong sequence (for example, entering 15:30, 01:00, 22:45) is considered as an invalid operation and the TOU function will be disabled.
6. Tariff setting parameter: This parameter corresponds to the number of tariffs available for the TOU calendar and can be selected from any integer from 0 to 3. The four tariffs: sharp, peak, valley, and normal are represented by four integers: 0, 1, 2, and 3 respectively. If the tariff setting parameter is set to 3, all of the 4 tariffs will be available for the TOU calendar. If the parameter is set to 1, only the first 2 tariffs (sharp and peak) will be available.

7. Holiday setting parameter: This parameter can be set from any integer between 1 and 30, meaning a maximum of 30 holidays can be programmed to the TOU calendar. If the holiday setting parameter is set as 3, the first 3 slots of the holiday schedule must be set. Otherwise it will be considered as an invalid input (TOU function will be disabled).

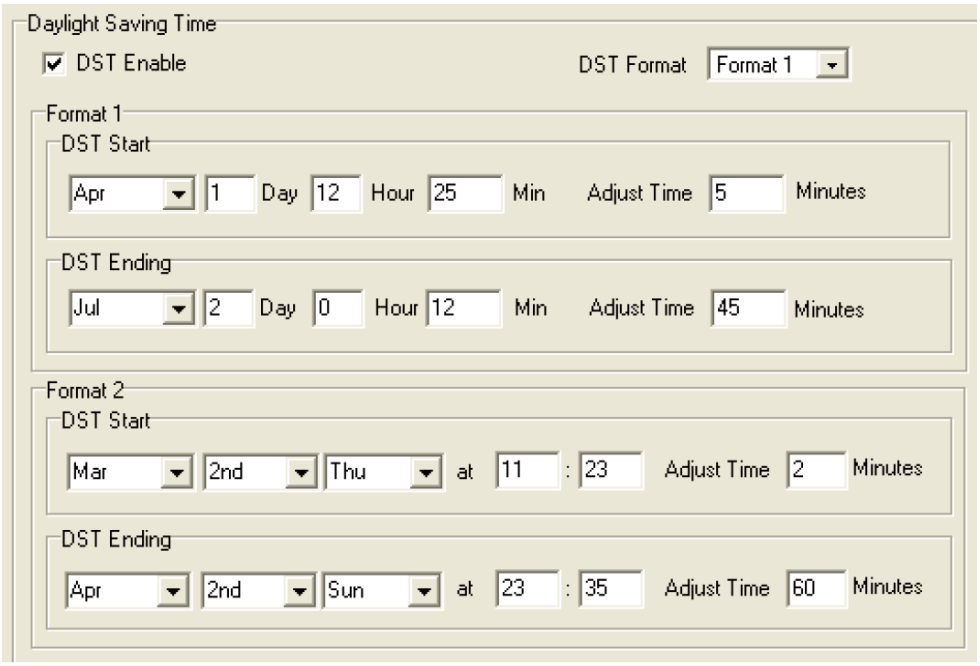
Note: User can either customize the TOU calendar factory settings or use the default factory settings. The user can reset the TOU calendar to its default value either via communication or from the front of the meter.

8. Holiday schedule: The holiday schedule uses the same format as the TOU seasons "MM-DD ID". The user can select which TOU schedule to be used for the holiday. The dates of the holiday schedule do not need to be organized in a sequential order (i.e.: the first slot can be January 1, the second slot can be December 26 and the third slot can be December 25).

9. Daylight saving time (DST): Daylight saving time can be enabled in one of two formats: 1. the fixed date option, or 2. a fixed day of one of the weeks in the month (also named as the non-fixed date option). If the user choose a fixed date option for the DST switch, set the format according to a fixed date. The format is month / day / hour / minute / adjust time (in minutes). If the user selects the non-fixed date option, DST will be implemented by which day of which week. The format is month/which day (i. e. Tuesday)/which week (i. e. 1st week)/hour/minute/adjust time (in minutes)

   By using this function, the user can cause the instrument to automatically switch to and from DST. When the clock starts to run to daylight saving time, the meter will automatically adjust the clock to a time period in advance. While the clock is running to the end of daylight saving time, the meter will automatically adjust the clock, pushing it back to a time period, as shown in Figure 50.

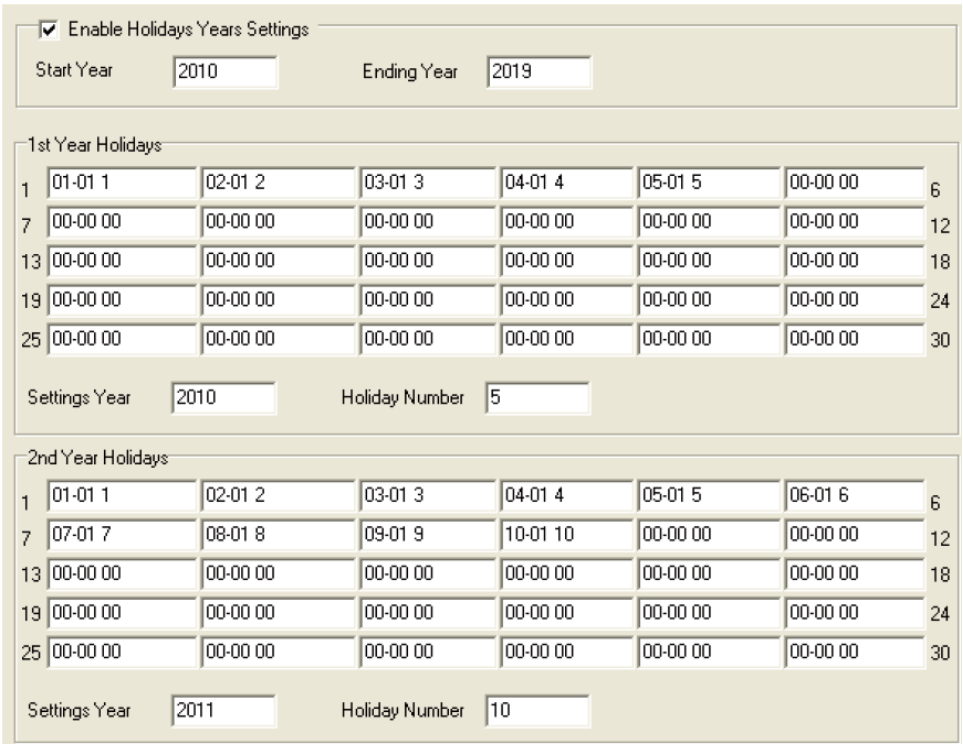
Figure 50. Daylight Saving Time Setting Interface.



10. Ten-Year Holiday setting: Users can preset holidays of the next decade via the meter software. The holiday format is month/day/year; holiday code; holiday schedule. After the format setup, click on "Make Holiday Settings (10 year)", then a holiday table for the next decade will be generated.

    Holiday Auto Switch: When Ten-Year Holiday is enabled, if the current year of the meter falls into the Ten-Year Holiday setting, it automatically loads the Ten-Year Holiday settings into the current TOU settings. If the current year of the meter does not fall into the Ten-Year Holiday setting, it retains the current TOU settings.

Figure 51. Ten-Year Holiday Settings.



11. Weekend Schedule: Weekend Setting (bit0 : Sunday; bit1 ~ bit6: Monday to Saturday; bit 0 means not effective, bit 1 means effective). For example, when the Weekend Setting bit0 is 1, it means Sunday is effective. When the Weekend Setting bit1 is 1, it means Monday is effective. If a user wants to set Saturday and Sunday effective, he should put 65 (1000001) into the Weekend Setting field. When the meter clock is within the preset Weekend schedule, the energy will accumulate under the tariff that corresponds to the schedule.

Note: The Holiday schedule has the highest priority among all the schedules. Weekend schedule's priority is followed by Holiday schedule. When Holiday schedule is not enabled, Weekend schedule has the highest priority, overriding the normal (weekday) schedule.

The PXM 3000 can record maximum power and current demand under different tariffs, as well as the time stamp of the maximum value. It can also clear the maximum demand under different tariffs.

Except normal energy parameter readings, the PXM 3000 has four separate logs: 1. Current Accumulation Month TOU; 2. Prior Accumulation Month TOU; 3. Current Month TOU; and 4. Prior Month TOU. When setup appropriately and when TOU is enabled, energy will be accumulated in a month-to-month basis. The current energy usage will be stored under Current Month TOU (or Current Accumulation Month TOU) and is divided up into different tar-

iffs. When the next month (or counting period) starts, all Current Month TOU values will be moved to Prior Month TOU (or Prior Accumulation Month TOU).

There are two ways of automatic resetting of current month TOU.

1. End of Month: This is the default method. All values from Current Accumulation Month TOU will be copied over to Prior Accumulate Month TOU at the very beginning of each month (the first day of each month at time 00:00:00). Current Accumulation Month TOU will continue to accumulate. All values from Current Month TOU will be copied over to Prior Month TOU at the beginning of each month (the first day of each month at time 00:00:00) will be cleared and reset to 0.
2. Assigned Clock: The user can select when the values from Current Accumulation Month TOU or Current Month TOU would be copied over to Prior Accumulation Month TOU or Prior Month TOU. The user can set the time in the following format: DD HH:MM:SS": DD stands for day; HH stands for hour; MM stands for minute; and SS stands for second. Similar to the previous method, once the Current Accumulation Month TOU is transferred to Prior Accumulation Month TOU, Current Accumulation Month TOU will continue to accumulate. Once the Current Month TOU is transferred to Prior Month TOU, Current Month TOU will be cleared and reset to 0.

5.7 Power Quality Event Logging and Waveform Capture

Power Quality Event Logging:

When a power quality event happens, such as voltage sag and swell, the PXM 3000 will record the event timestamp and the triggering condition. It can save up to 50,000 events.

1. Event Logging Data Format
   Timestamp (4 words) + Triggering Condition (1 word) + Rated Value (1 word) + Threshold (1 word) + Half Cycle Count (1 word)

   Each event has 8 words in total.

   Event Time: W1: Year-High Byte; Month-Low Byte; W2: Day-High Byte, Hour-Low Byte; W3: Minute-High Byte; Second-Low Byte; W4: Millisecond Triggering Condition; W5: Voltage Sags or Voltage Swells; 0: Logging Disabled;

   Bit0: 1 – u1 voltage sag, 0 – no u1 voltage sag;

   Bit1: 1 – u2 voltage sag, 0 – no u2 voltage sag;

   Bit2: 1 – u3 voltage sag, 0 – no u3 voltage sag;

   Bit3: 1 – u1 voltage swell, 0 –u1 no voltage swell;

   Bit4: 1 – u2 voltage swell, 0 –u2 no voltage swell;

   Bit5: 1 – u3 voltage swell, 0 –u3 no voltage swell;

   Rated Value: W6 - voltage rated value; threshold:

   W7 - threshold for voltage sag and swell, (expressed in secondary volts when PT's are used).

   Half cycle count: W8 (voltage swell: 0; voltage sag: 4-200).

2. Logging Events

   The event logging feature can log 50,000 events. If the 50,000 events are full, no more events will be logged even if the triggering condition happens. The user must clear the event log, and then the logging feature will log the new event. When the log is cleared, the new event will be logged from the first event that happens. There will be no data loss after the power is off.

3. Event Logging Triggering Conditions

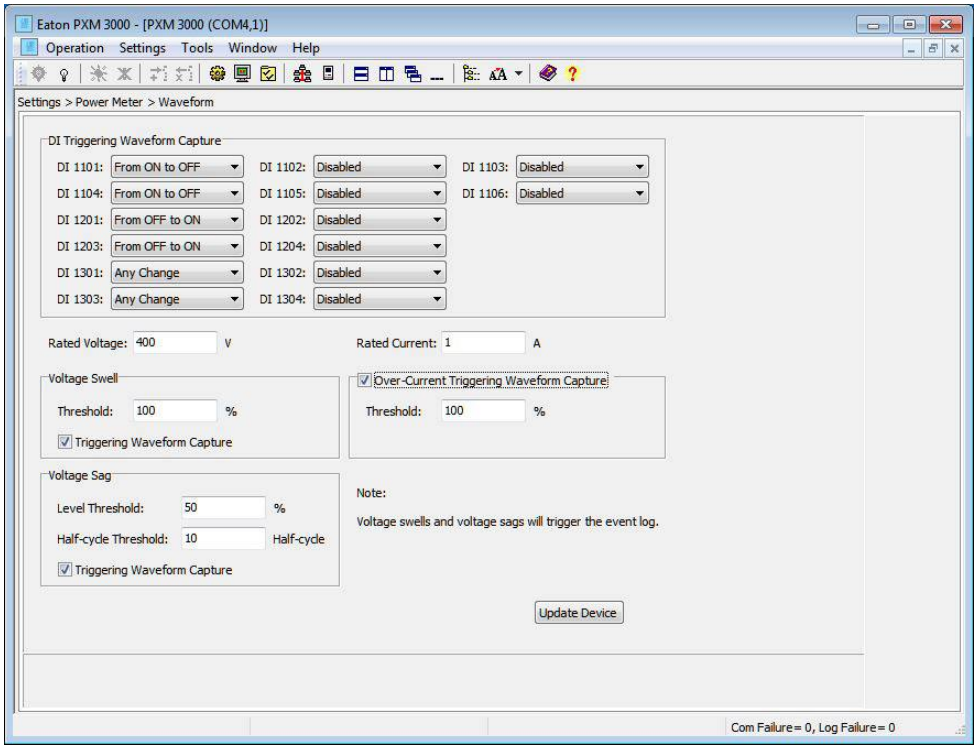
   A. Voltage Sag
   When any phase of the three-phase voltage is lower than the set value (voltage

rated value x threshold %), there will be a voltage sag event. When one phase voltage sag happens, the other phase will not respond to voltage sag event logging. A new voltage sag event will be responded to only when all of the phase voltages are restored to normal.

B. Voltage Swell
When any phase of the three-phase voltage is higher than the set value (voltage rated value x threshold %), there will be a voltage swell event. When a voltage swell happens on one phase, the other phase will not respond to the voltage swell event logging. A new voltage swell event will be responded to only when all of the phase voltages are restored to normal.

Note: Figure 52 demonstrates how to set the parameters for Power Quality Event Logging and Waveform Capture. In the parameter settings, voltage sag and voltage swell share the same voltage rated value. The parameters for event logging includes: voltage rated value, voltage sag threshold, voltage sag half cycle count, and voltage swell threshold. Those parameters also fit the voltage sag waveform capture. The other triggering conditions for waveform capture can be set when necessary. When the waveform capture triggering by voltage sag and voltage swell is enabled, the corresponding event log and waveform will be recorded when voltage sag or voltage swell happens.

Figure 52. Event Triggering Conditions.



4. Event Log Retrieve

When a new event log starts, the newest event number address (0X8CFDH) contains the newest event number. When the log is being retrieved, the starting event log number (0X8CFEH) and the event quantity for each retrieve (0X8CFF) must be set correctly. It must be ensured that the starting number of event log should equal or be smaller than the newest log number. When setup is correct, reading registers 0X8D00H-0X8D4FH will acquire the event log data. Each time, a maximum of ten logged events can be retrieved. The event log retrieve page is shown in Figure 53. The Modbus register address of the event log is shown in the Table 7 (see details in Chapter 6).

Figure 53. Event Log Retrieve Page.

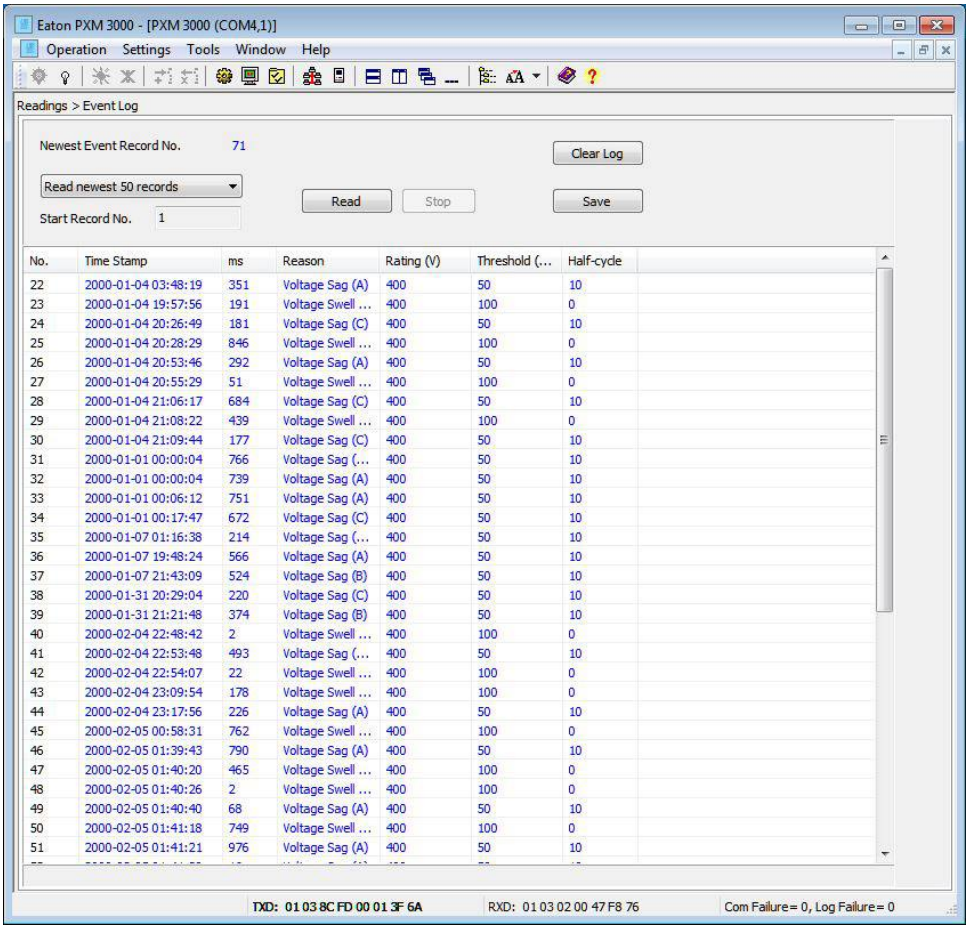


Table 7. Modbus Register Address of the Event Log.

8CFDH	The newest event number	word	R	Range: 1~50000; 0: No event
8CFEH	The starting event log number	word	R/W	Range: 1-50000 **Note**: smaller than or equal to the newest event number.
8CFFH	The event quantity of each time retrieve	word	R/W	1-10

Waveform Capture:

Function description PXM3000 supports waveform recording function. The number of sampling points per week supports 64/128 points. Record up to 200 groups (64 points / cycle) or 100 groups (128 points / cycle). It provides the captured waveform of 10 cycles before and after the triggering point (including U1, U2, U3, I1, I2, I3). At the same time, trigger conditions can be set, providing real-time data for user power quality analysis.

1. Waveform Capture Data Format

Frequency	Voltage/bit	Current/bit	Timestamp	Triggering Condition	Kept Data Storage	Before triggering point 10 wave for ms date	After triggering point 10 wave for ms date
2 Word	2 Word	2 Word	7 Word	9 Word	48 Word	N * 6 * 10 Word	N * 6 * 10 Word

Frequency: Float data, high byte first;

Voltage/bit: Float data, high byte first;

Current/bit: Float data, high byte first;

Timestamp: Year (W1), Month (W2), Day (W3), Hour (W4), Minute (W5), Second(W6), Millisecond(W7)

Triggering Condition:

W8 - Manual triggering (0: disable; 1: enable);

W9 - PXM1K-11 DI triggering (bit1 bit0: DI1[bit3 bit2: DI2] bit5 bit4: DI3; bit7 bit6: DI4; bit9 bit8: DI5; bit11 bit10: DI6);

W10 - PXM1K-21 DI triggering (bit1 bit0: DI7; bit3 bit2: DI8; bit5 bit4: DI9; bit7 bit6: DI10);

W11 - PXM1K-31 DI triggering (bit1 bit0: DI11; bit3 bit2: DI12; bit5 bit4: DI13; bit7 bit6: DI14);

(Two bits meaning: 00: No DI triggering; 01: DI triggering from OFF to ON; 10: DI triggering From On to OFF)

W12 - Voltage sag triggering (0: disabled;

Bit0: 1 - u1 voltage sag waveform, 0 - no u1 waveform;

Bit1: 1 - u2 voltage sag waveform, 0 - no u2 waveform;

Bit2: 1 - u3 voltage sag waveform, 0 - no u3 waveform)

W13 - Voltage swell triggering (0: disabled;

Bit0: 1 - u1 voltage swell waveform, 0 - no u1 waveform;

Bit1: 1 - u2 voltage swell waveform, 0 - no u2 waveform;

Bit2: 1 - u3 voltage swell waveform, 0 - no u3 waveform;)

W14 - Over-current triggering (0: disabled;

Bit0: 1 - I1 voltage over-current waveform, 0 - no I1 waveform;

Bit1: 1 - I2 voltage over-current waveform, 0 - no I2 waveform;

Bit2: 1 - I3 voltage over-current waveform, 0 - no I3 waveform;)

W15, W16 - 0 (Reserved)

Waveform Order:

Before triggering point: 10 U1 cycles, 10 U2 cycles, 10 U3 cycles, 10 I1 cycles, 10 I2 cycles, 10 I3 cycles.

After triggering point: 10 U1 cycles, 10 U2 cycles, 10 U3 cycles, 10 I1 cycles, 10 I2 cycles, 10 I3 cycles.

Note: When 64 points / cycle sampling is selected, N = 64; when 128 points / cycle sampling is selected, N = 128.

2. Waveform Capture Group

Waveform Capture can log up to maximum recording group of waveform data. When the recording group data is full, it does not respond to any waveform triggering condition. Only when all the waveform data is reset / emptied, waveform capturing function will be normal. When the waveform data is emptied, new waveform data starts from the 1st group. The waveform data will not be lost when the power is off.

Note: since the amount of each waveform group data is large, it takes more time to write into the flash memory. Therefore, Waveform Capture only responds to one triggering condition at one time. During the process of writing data into the flash memory, it does not respond to new triggering condition. After 99 the process of memory writing, it will respond to new waveform triggering condition.

3. Waveform Capture Group

Waveform Capture can log up to 100 groups of waveform data. When the 100 group data is full, it does not respond to any waveform triggering condition. Only when all the waveform data is reset / emptied, will the waveform capturing function return to normal. When the waveform data is emptied, new waveform data starts from the first group. The waveform data will not be lost when the power is off. Waveform Capture only responds to one triggering condition at one time.

4. Waveform Capture Triggering Condition

A. Manual Triggering

Manually trigger one group waveform capture.

B. DI Triggering

DI Triggering must fulfill the following two conditions at the same time.

- IO modules with the logical address of 1 (PXM1K-11, PXM1K-21, PXM1K-31).
- DI channel type is set as "State".

The Modbus address assigns two bits for the DI channel. When they are set as "00", it means DI Triggering Disabled. "01" means DI Triggering will take place when DI state changes from OFF to ON. "10" means DI Triggering will take place when the DI state changes from ON to OFF. "11"means DI Triggering will take place when the DI state has any change.

C. Voltage Sag Triggering

As mentioned in Voltage Sag Event Logging, when the Voltage Sag Triggering Waveform is enabled, both event logging and waveform capture will take place at the same time once a voltage sag occurs.

D. Voltage Swell Triggering

As mentioned in Voltage Swell Event Logging, when Voltage Swell Triggering Waveform is enabled, both event logging and waveform capture will take place at the same time once a voltage swell occurs.

E. Over-current Triggering

When Over-current Triggering is enabled, if any phase of the three-phase current is higher than the set value (rated value x threshold %), the waveform capture will take place. If one phase is over-current, any other phase over-current cannot trigger the waveform capture. Waveform capture will respond only when all of the phase currents are restored to normal.

5. Waveform Capture Retrieve

Because of large quantity of saved waveform, waveform retrieving window use 64 addresses to make retrieving data easier, which keeps consistent with recording points of one period. There are two retrieving methods, one retrieve waveform record reasons, another retrieve all data of one group waveform. Two methods are shown below.

First retrieving method is only retrieving waveform record reasons.

When 0x8F01H (Waveform Group Number) is set to 0, waveform record reason can be retrieved from data retrieving window, by changing Waveform Group Number for Retrieving (8F00H). Waveform record reason is 22 bytes and data window are 64 bytes, so that is the reason why each retrieving access include 2 group records.

For example, if the total number of Waveform Group Number for Retrieving is 19, the by this method, only through 10 times retrieving, all 20 groups waveform can be retrieved. Before retrieving, user should write 0 to 8F01H, and 1 to 8F00H, while at first time retrieving addresses from 8F00H to 8F43H, these window data are the reasons of waveform group 1, 2. After retrieving, Waveform Group Number for Retrieving (8F00H) will automatically update to 3. While at second time retrieving addresses from 8F00 to 8F43, these window data are the reasons of waveform group 3, 4,. Waveform Group Number for Retrieving (8F00H) will stop increasing after the value reaches 19.

Note: Unless user retrieves all data from 8F00H to 8F43H in one time, Waveform Group Number for Retrieving (8F00H) will automatically add 2, otherwise Waveform Group Number for Retrieving (8F00H) will keep present value.

Second retrieving method is retrieving all waveforms data. When 0x8F01H (Waveform Group Number) is set to 1~121(64 points / cycle)or(128 points / cycle), each time only one group data of each number will be retrieved, then Waveform Group Number for Retrieving (8F01H) will automatically add 1 after retrieving. At next time, new Waveform Group Number will be retrieved. Waveform Group Number will stop increasing after value reach 121.

Note: The range of 0x8F01H is 0~121(64 points/cycle), within setting range, windows (8F04H~8F43H) are corresponding to waveform data.

0: Only retrieving waveform record reasons

1: Time and reasons of waveforms

2~11: Each waveform number of ten waveforms before U1 waveform is triggered

12~21: Each waveform number of ten waveforms before U2 waveform is triggered

22~31: Each waveform number of ten waveforms before U3 waveform is triggered

32~41: Each waveform number of ten waveforms before I1 waveform is triggered

42~51: Each waveform number of ten waveforms before I2 waveform is triggered

52~61: Each waveform number of ten waveforms before I3 waveform is triggered

62~71: Each waveform number of ten waveforms after U1 waveform is triggered

72~81: Each waveform number of ten waveforms after U2 waveform is triggered

82~91: Each waveform number of ten waveforms after U3 waveform is triggered

92~101: Each waveform number of ten waveforms after I1 waveform is triggered

102~111: Each waveform number of ten waveforms after I2 waveform is triggered

112~121: Each waveform number of ten waveforms after I3 waveform is triggered

Note: The range of 0x8F01H is 0~241(128 points/cycle), within setting range, windows (8F04H~8F43H) are corresponding to waveform data.

0: Only retrieving waveform record reasons

1: Time and reasons of waveforms

2~21: Each waveform number of ten waveforms before U1 waveform is triggered

22~41: Each waveform number of ten waveforms before U2 waveform is triggered

42~61: Each waveform number of ten waveforms before U3 waveform is triggered

62~81: Each waveform number of ten waveforms before I1 waveform is triggered

82~101: Each waveform number of ten waveforms before I2 waveform is triggered

102~121: Each waveform number of ten waveforms before I3 waveform is triggered

122~141: Each waveform number of ten waveforms after U1 waveform is triggered

142~161: Each waveform number of ten waveforms after U2 waveform is triggered

162~181: Each waveform number of ten waveforms after U3 waveform is triggered

182~201: Each waveform number of ten waveforms after I1 waveform is triggered

202~221: Each waveform number of ten waveforms after I2 waveform is triggered

222~241: Each waveform number of ten waveforms after I3 waveform is triggered

Note: Unless the user retrieves all data from 8F00H to 8F43H at one time, Waveform Group Number for Retrieving (8F01H) will automatically add 1, or Waveform Group Number for Retrieving (8F01H) will keep its present value if only partial data is retrieved.

Note:

Table 8. Waveform Group Number Information.

8F00H	Waveform Group Number for Retrieving	1~100 or 1-200 **Note:** When the value is smaller than or equal to the newest waveform record group number, this value is valid.
8F01H	Waveform Group Number	Waveform number; 0~121
8F02H	Waveform Record Window Status	0x0BH: window data is valid. 0xFF: window data is invalid. 0xAA: waveform record memory is clearing (data is invalid)
8F03H	Newest Waveform Group Number	1~100; 0~no record
8F04H - 8F43H	Waveform Record Data Retrieving Window	-32768~ 32767

Relationship between Voltage or Current Waveform Value and Real Value

The relationship between the voltage waveform value and real value is:

Real Value (Unit: V) = Waveform Value / 37.59105.

The relationship between the current waveform value and real value is:

A. 5A, 1A: Real Value (Unit: A) = Waveform Value/1683.153;

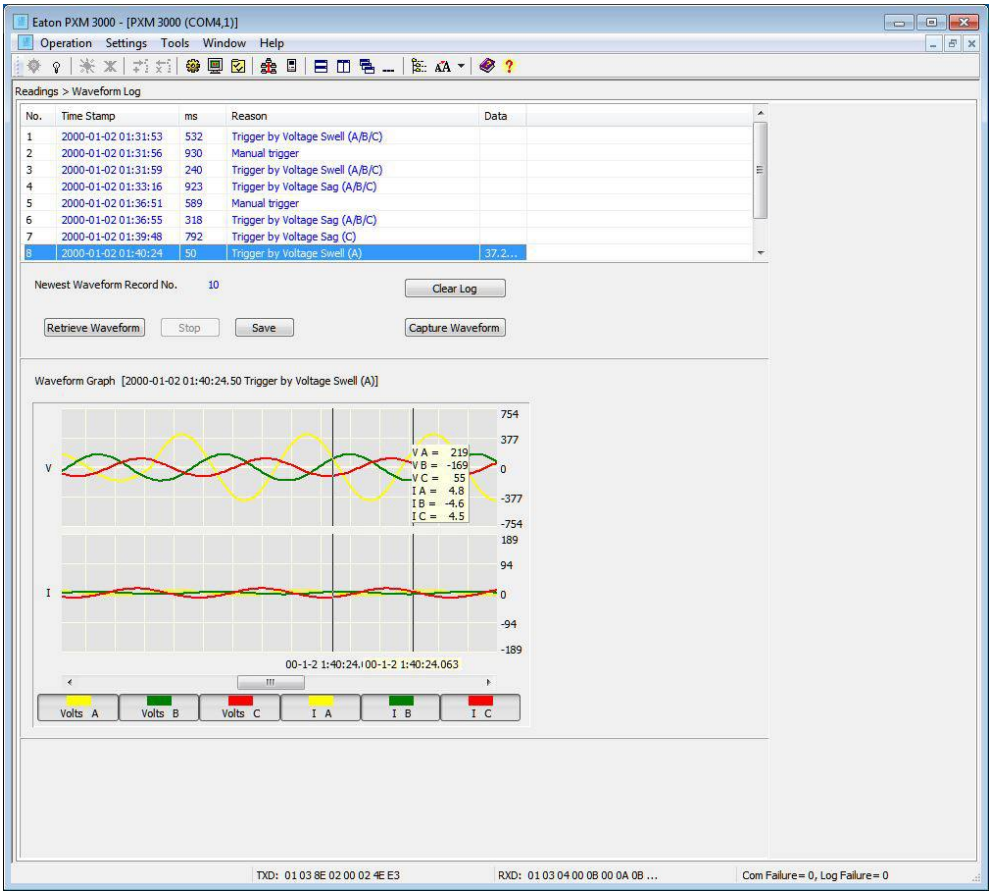
B. 333 mV: Real Value (Unit: A) = Waveform Value/K (firmware above 3.21, K = 14427.15; other: K = 15869.87);

C. 100 mV Rogowski coil: Real Value (Unit: A) = Waveform Value/K (firmware above 3.21, K = 20291.1; firmware 3.20, K = 22068.8, other: K = 15869.87);

D. mA CT: Real Value (Unit: A) = Waveform Value/K (80 mA CT: K = 7414.289; 100 mA: K = 9267.440; 200 mA: K = 18514.68); The voltage and current value obtained from the waveform are the PT or CT secondary side value.

The waveform capture retrieve page is shown in Figure 54.

Figure 54. Waveform Capture Retrieve Page.



Note: When selecting 400 Hz type, there is no support event logging and wave capture.

5.8 Seal Function

Panels with the Seal function, which has a sealed key control, are different from the panels without a seal. When the seal is in Opened status, the functions are the same as normal meters. But when the Seal function is in Sealed status, some functions of the meters, which includes parameters blocked by seal and optional parameters, will be blocked. These parameters can still be accessible by the local display and via communications the same way they could be accessed before. But, in Seal, these parameters cannot be modified by the local display and via communications in the way they were before.

Addresses for the Seal function are 101EH and 101FH.

Address 101EH is for the parameters blocked by Seal, which can be configured by the user. These setting will be valid only when the Seal function is in sealed status.

Address 101FH is for the status of Seal Function. When the panel is a normal one, or the seal panel is in Invalid Sealed status, this address will show that the Seal is open. When the Seal is valid, this address will show the sealed status and corresponding parameters will be blocked.

Table 9. Seal Function Address Status.

101EH	Sealed Nonstandard Parameters Selection	Bit0: 1st communication parameters Bit1: 2nd communication parameters Bit2: run time clear Bit3: DI pulse count Bit4:TOU 1: valid of corresponding selection 0: invalid
101FH	Seal Status	0x0A: Seal sealed; Other: Seal opened.

Parameters blocked by the Seal function:

As long as Seal is in the Sealed status, the parameters below must be blocked, no matter what the value of address 101EH.

Table 10. Seal Parameters Status.

System Parameters Setting:

Address	Parameter	Keys	Communication
1003H	Voltage input wiring type	√	√
1004H	Current input wiring type	√	√
1005H	PT1 (High 16 bit)	√	√
1006H	PT1 (Low 16 bit)	√	√
1007H	PT2	√	√
1008H	CT1	√	√
1009H	CT2	√	√
100AH	kWh pulse constant	√	√
100BH	Kvarh pulse constant	√	√
100DH	Demand slid window time	√	√
100EH	Demand calculating mode	√	√
100FH	Clear demand memory	√	√
1012H	Current I1 direction	√	√
1013H	Current I2 direction	√	√
1014H	Current I3 direction	√	√
1015H	VAR/PF convention	√	√
1016H	Energy clear	√	√
1017H	Energy calculating mode	√	√
1018H	Reactive power measuring mode	√	√
1019H	Energy display mode	√	√
101DH	Basic parameter mode	√	√
101EH	Sealed nonstandard parameters selection	-	√

Note: "√" means these addresses will be blocked for keys and communication, and"-" means this function is unavailable.

Table 10. Seal Parameters Status (continued).

Energy:

Address	Parameter	Keys	Communication
4048H--4049H	Energy IMP	-	√
404AH--404BH	Energy EXP	-	√
404CH--404DH	Reactive Energy IMP	-	√
404EH--404FH	Reactive Energy EXP	-	√
4050H--4051H	Energy TOTAL	-	√
4052H--4053H	Energy NET	-	√
4054H--4055H	Reactive Energy TOTAL	-	√
4056H--4057H	Reactive Energy NET	-	√
4058H--4059H	Apparent Energy	-	√
4620H~4621H	Phase A Energy IMP	-	√
4622H~4623H	Phase A Energy EXP	-	√
4624H~4625H	Phase B Energy IMP	-	√
4626H~4627H	Phase B Energy EXP	-	√
4628H~4629H	Phase C Energy IMP	-	√
462AH~462BH	Phase C Energy EXP	-	√
462CH~462DH	Phase A Reactive Energy IMP	-	√
462EH~462FH	Phase A Reactive Energy EXP	-	√
4630H~4631H	Phase B Reactive Energy IMP	-	√
4632H~4633H	Phase B Reactive Energy EXP	-	√
4634H~4635H	Phase C Reactive Energy IMP	-	√
4636H~4637H	Phase C Reactive Energy EXP	-	√
4638H~4639H	Phase A Apparent Energy	-	√
463AH~463BH	Phase B Apparent Energy	-	√
463CH~463DH	Phase C Apparent Energy	-	√

DO:

Address	Parameter	Keys	Communication
10A5H	Working mode of DO 1 and 2	√	√
10A6H	DO pulse width	√	√
10A7H	DO1 output	√	√
10A8H	DO2 output	√	√
10B7H	Working mode of DO 3 and 4	√	√
10B8H	DO pulse width	√	√
10B9H	DO3 output	√	√
10BAH	DO4 output	√	√

Sealed Nonstandard Parameters:

Table 11. Sealed Nonstandard Parameters

1. When bit 0 of address 101EH is valid, parameters about 1st communication should be blocked.

Address	Parameter	Keys	Communication
0FFEH	Communication Protocol 1	√	√
0FFFH	Parity Setting 1	√	√
1000H	Password	√	√
1001H	Communication address 1	√	√
1002H	Baud rate 1	√	√

2. When bit 1 of address 101EH is valid, parameters about 2nd communication should be blocked.

Address	Parameter	Keys	Communication
102FH	Baud rate 2	√	√
1030H	Parity Setting 2	√	√
1031H	Communication address 2	√	√

3. When bit 2 of address 101EH is valid, parameters below should be blocked.

Address	Parameter	Keys	Communication
1011H	Run time clear	√	√

4. When bit 3 of address 101EH is valid, parameters below should be blocked.

Address	Parameter	Keys	Communication
101CH	Pulse counter clear	√	√
109eH	DI1-6 type	√	√
109fH	DI pulse constant	√	√
10a3H	DI7-10 type	√	√
10a4H	DI pulse constant	√	√
10aaH	DI11-14 type	√	√
10abH	DI pulse constant	√	√
10b0H	DI15-20 type	√	√
10b1H	DI pulse constant	√	√
10b5H	DI21-24 type	√	√
10b6H	DI pulse constant	√	√
10bcH	DI25-28 type	√	√
10bdH	DI pulse constant	√	√

Table 11. Sealed Nonstandard Parameters (continued).

5. When bit 4 of address 101EH is valid, parameters below should be blocked.

Address	Parameter	Keys	Communication
103fH	Week	-	√
1040H	Year	-	√
1041H	Month	-	√
1042H	Day	-	√
1043H	Hour	-	√
1044H	Minute	-	√
1045H	Second	-	√

Note: Standard time is valid within ±5 minutes, it will be invalid if not in this range.

Address	Parameter	Keys	Communication
TOU			
1039H	Ten-year download setting enable	-	√
103aH	Sharp demand clear	-	√
103bH	Peak demand clear	-	√
103cH	Valley demand clear	-	√
103dH	Normal demand clear	-	√
103eH	Total demand clear	-	√
Current and last month TOU energy			
7200H-7263H	Current and last month TOU energy	-	√
DST setting			
7700H-7717H	DST setting	-	√
Season setting			
7800H-780EH	Basis parameter of TOU	-	√
7820H-7AECH	Season setting	-	√
Ten-year holiday setting			
7B00H-7E97H	Ten-Year Holiday setting	-	√

6. Bi-directional Energy Register Function

PXM3000 series meters have a bi-directional energy measuring function. It has three sets of bi-directional energy registers, which control the direction of energy accumulation through DI input and communication, and realize the function of measuring the energy of different power inputs. The two power input names are "grid" and "generator".

Bi-directional power parameter can be selected arbitrarily from the following parameters: System Energy, Phase A Energy, Phase B Energy, Phase C Energy, System Reactive Energy, Phase A Reactive Energy, Phase B Reactive Energy, Phase C Reactive Energy, System Apparent Energy, Phase A Apparent energy, Phase B Apparent energy, and Phase C Apparent energy.

If you want to use the bi-directional energy function, you must set the parameters correctly, otherwise this function will not be enabled normally. First, enable the bi-directional energy function. Secondly, set the energy power of the three bi-directional energy registers. Then, select the control input that controls the direction of the bi-directional energy. Finally, the direction of electric energy accumulation is switched by the control input to realize the measurement of electric energy of different input sources. At the same time, it also supports 20 event records for the energy accumulation direction switching. Of course, the event records are covered by cycles.

1) Instructions for controlling bi-directional energy accumulation direction

Users can choose the communication control or DI1 ~ DI28 control as the control input of bi-directional energy. When communication control is selected as the control input, write 0 to the 1685H address (direction flag), which means switching to the direction of "grid", write 1 to the 1685H address, which means switching to the "generator" direction. When DI1 ~ DI28 are selected as the control input, the direction of the bi-directional energy will be switched in sequence each time the corresponding DI is input once.

The following table shows the corresponding relationship between DI1 ~ DI28 and DI number on the IO module.

Table 12. DI Control Input Number According to IO Module Name

IO module name	DI Number	Bi-directional energy DI control input number
PXM1K-011	DI1 ~ DI6	DI1 ~ DI6
PXM1K-021	DI1 ~ DI4	DI7 ~ DI10
PXM1K-031	DI1 ~ DI4	DI11 ~ DI4
PXM1K-012	DI1 ~ DI6	DI5 ~ DI20
PXM1K-022	DI1 ~ DI4	DI21 ~ DI24
PXM1K-032	DI1 ~ DI4	DI25 ~ DI28

6. Extended Modules

6.1 IO Modules

6.1.1 The Purpose of IO Modules

The standard PXM 3000 meter base does not have any built-in IO functions. However, with the addition of the extended modules, multiple IO options can be added. These functions include digital input, pulse counter, relay output, analog output, analog input, etc.

There are three types of IO modules, PXM1K-1, PXM1K-2, and PXM1K-3. Please note that a maximum of three modules may be attached to the meter. If a communication module is used, it must be installed on the back of the meter first before any IO modules are attached. No more than two of the same type IO modules can be attached. According to the difference in communication with PXM 3000 meter, each type of IO module also has two modes, logic NO. 1 and logic NO. 2. This means, two of each type of IO module can be linked to the PXM 3000 meter simultaneously (one being logic NO. 1 and the other being logic NO. 2).

The PXM1K-1 module is composed of:

Six Digital Inputs (DI) - Each DI can be used to detect remote signals, or be used as an input pulse counter. When it is used to detect remote signals, it also can enable sequence of events (SOE), recording the event and time of the event.

Two Relay Outputs (RO) - Can be used for controlling or alarms. Each of the ROs work in the same mode. When it operates in controlling mode, there are two output options; latching and pulse. When it operates in alarm mode, it has only one latching output mode.

24 V Isolated Power Supply - Used as an auxiliary power supply for digital inputs.

The PXM1K-2 module is composed of:

Four Digital Inputs (DI) - Each DI can be used to detect remote signals, or be used as an input pulse counter. When it is used to detect remote signals, it can also enable a sequence of events (SOE), recording the events and time of the events.

Two Analog Outputs (AO) - Can output analog voltage or analog current. When it outputs analog voltage, the range of voltage is from 0 to 5 V or from 1 to 5 V. When it outputs analog current, the range of current is from 0 to 20 mA or from 4 to 20 mA.

Two Digital Outputs (DO) - Can be used in alarm mode or energy pulse output mode. Both of the DOs work in the same mode. When the DOs operate in the energy pulse output mode, they can output various types of energy.

The PXM1K-3 module is composed of:

Four Digital Inputs (DI) - Each DI can be used to detect remote signals, or be used as an input pulse counter. When a DI is used to detect remote signals, it can also enable a sequence of events (SOE), recording the events and time of the events.

Two Relay Outputs (RO) - Can be used for controlling or alarms. Each of the ROs work in the same mode. When a RO operates in the controlling mode, there are two output options: latching mode and pulse . When it operates in alarm mode, it has only one latching output mode.

Two Analog Inputs (AI) - Can detect input analog voltage or analog current. When an AI detects input analog voltage, the range of voltage is from 0 to 5 V or from 1 to 5 V. When it detects input analog current, the range of current is from 0 to 20 mA or from 4 to 20 mA.

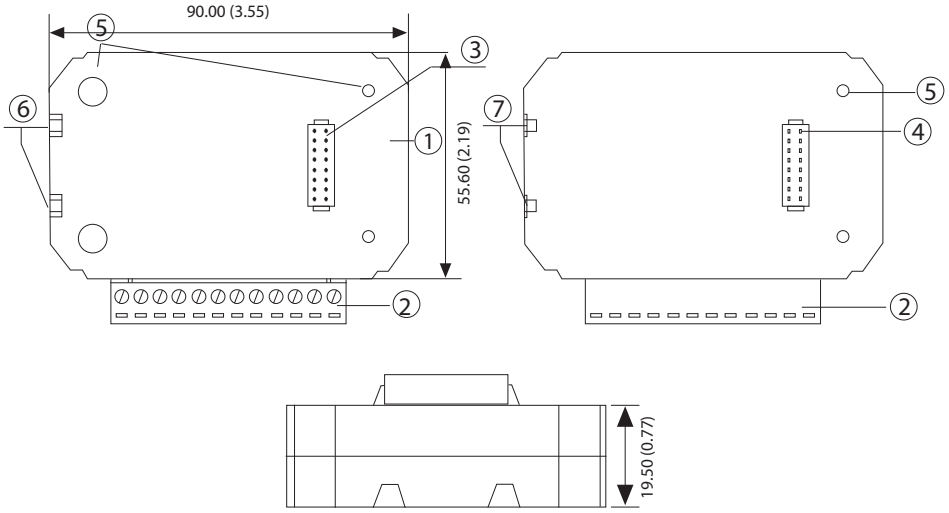
6.1.2 List of Functions of IO Modules

Table 13. Functions of IO Modules.

Functions	PXM1K-1	PXM1K-2	PXM1K-3
Detection of remote signals	•	•	•
Recording of SOE	•	•	•
Counting of input pulses	•	•	•
Output remote controlling by relay	•		•
Output alarm by relay	•		•
Output alarm by digital output		•	
Output power pulses by digital output		•	
Analog output		•	
Analog input			•
24 V isolated voltage output	•		

6.1.3 Appearance and Dimensions

Figure 55. Dimensions



1	Enclosure	5	Installation Screw
2	Wiring Terminals	6	Counterpart of Clip
3	Linking Pins	7	Installation Clip
4	Linking Socket		

6.1.4 Installation Method

Environment

Please verify the installation environment meets the requirements listed as follows:

Temperature:

Operation: -25°C to 70°C (-13°F to 158°F)

Storage: -40°C to 85°C (-40°F to 185°F)

Humidity:

5% to 95% non-condensing.

Location:

The PXM 3000 meter and IO modules should be installed in a dry and dust free environment avoiding heat, radiation, and high electrical noise sources.

Installation Method:

The IO modules are linked to the meter and to each other with link pins. The maximum number of extended modules linked to a PXM 3000 meter, including IO module, RS-485 module, is three. The communication modules must be installed first. No other module can be installed before them.

1. Insert the installation clips into the module to be linked to the PXM 3000 meter. Press the IO module lightly into the meter until a solid link is established.
2. Tighten the installation screws. Torque is 0.2 nm ± 0.02 (1.77 lb-in ± 0.177)
3. Install other IO modules in the same way.

Note: Install the IO Modules carefully to avoid damage;

⚠ WARNING

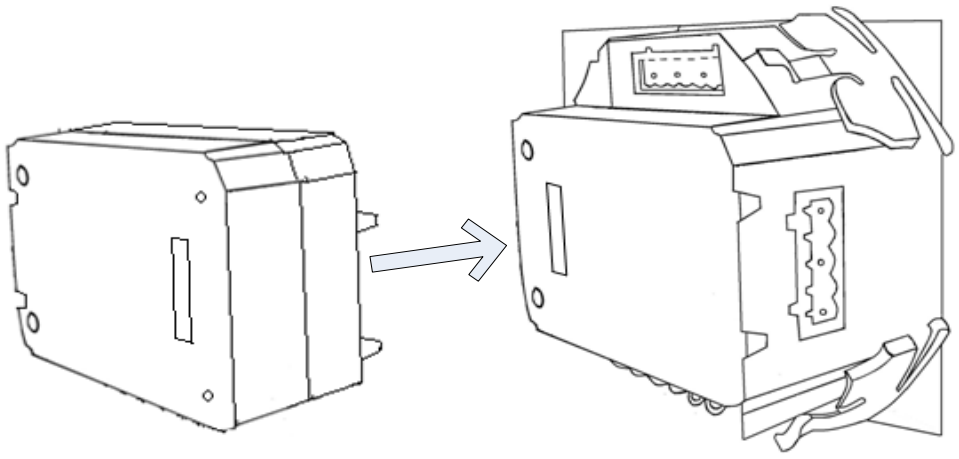
Do not attempt to install or perform maintenance on equipment while it is energized. Death, severe personal injury, or substantial property damage can result from contact with energized equipment. Always verify that no voltage is present. Before proceeding with the task, and always follow generally accepted safety procedures.

⚠ WARNING

Switch main power off and wait five (5) minutes before making any connection or disconnection on the device. Danger of explosion!

Eaton is not liable for the misapplication or misinstallation of its products.

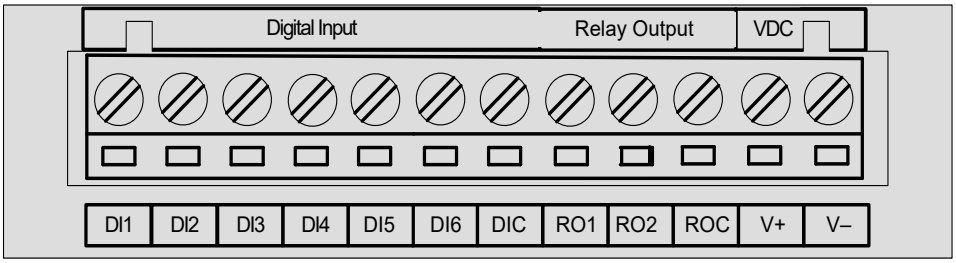
Figure 56. Installation of the IO Modules.



6.1.5 Wiring the IO Modules:

Terminal strips of PXM1K-1 module:

Figure 57. Terminal Strips of PXM1K-1 Module.



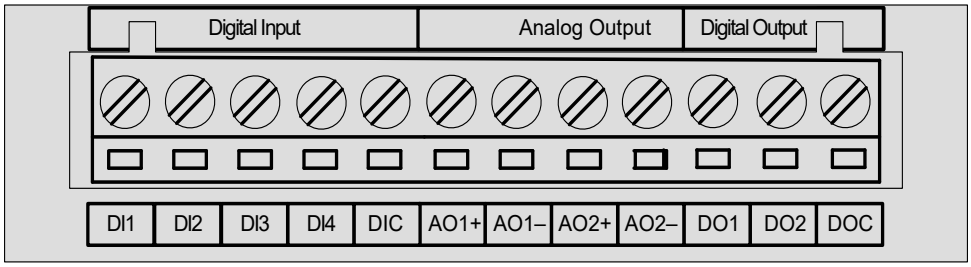
DI1 to DIC: digital input terminals, where DIC is the common terminal for DI1 to DI6 circuits.

RO1 to ROC: relay output terminals, where ROC is the common terminal for RO1 and RO2 circuits.

V24+ and V24-: auxiliary voltage supply terminals.

Terminal Strips of PXM1K-2 Module:

Figure 58. Terminal Strips of PXM1K-2 Module.

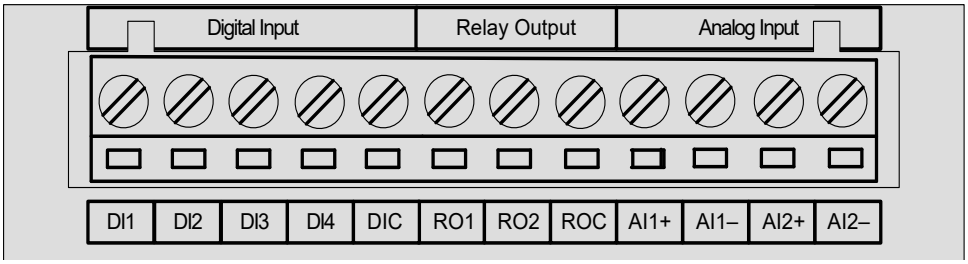


DI1 to DIC: digital input terminals, where DIC is the common terminal for DI1 to DI4 circuits.

AO1+, AO1-, AO2+, AO2-: analog output terminals.

DO1 to DOC: digital output terminals, where DOC is the common terminals for the DO1 to DO2 circuits.

Figure 59. Terminals Strips of PXM1K-3 module:



DI1 to DIC: digital input terminals, where DIC is the common terminal for the DI1 to DI4 circuits.

RO1 to ROC: relay output terminals, where ROC is the common terminal for the RO1 and RO2 circuits.

AI1+, AI1-, AI2+, AI2-: analog input terminals.

Sequence of DI, RO, DO, AO, AI in IO Modules (according to the logical order in the communication address table of the main body):

DI Sequence: PXM1K-11 (PXM1K-1 module in logic NO.1): DI1-6

PXM1K-21 (PXM1K-2 module in logic NO.1): DI7-10

PXM1K-31 (PXM1K-3 module in logic NO.1): DI11-14

PXM1K-12 (PXM1K-1 module in logic NO.2): DI15-20

PXM1K-22 (PXM1K-2 module in logic NO.2): DI21-24

PXM1K-32 (PXM1K-3 module in logic NO.2): DI25-28

RO Sequence: PXM1K-11 (PXM1K-1 module in logic NO.1): RO1-2

PXM1K-31 (PXM1K-3 module in logic NO.1): RO3-4

PXM1K-12 (PXM1K-1 module in logic NO.2): RO5-6

PXM1K-32 (PXM1K-3 module in logic NO.2): RO7-8

DO Sequence: PXM1K-21 (PXM1K-2 module in logic NO.1): DO1-2

PXM1K-22 (PXM1K-2 module in logic NO.2): DO3-4

AO Sequence: PXM1K-21 (PXM1K-2 module in logic NO.1): AO1-2

PXM1K-22 (PXM1K-2 module in logic NO.2): AO3-4

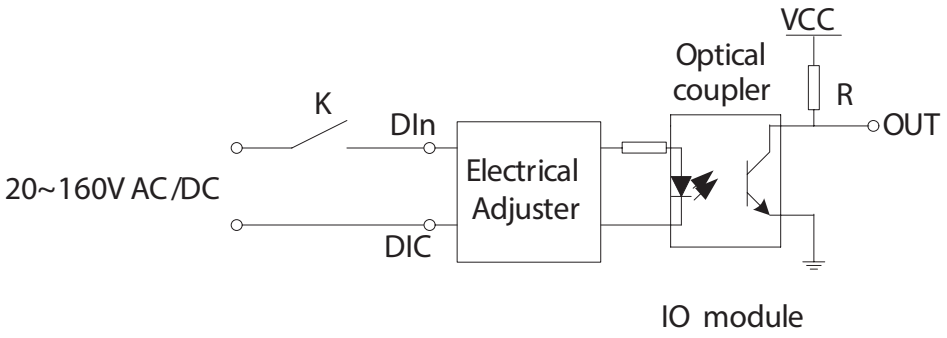
AI Sequence: PXM1K-31 (PXM1K-3 module logic NO.1): AI1-2

PXM1K-32 (PXM1K-3 module in logic NO.2): AI3-4

Wiring of Digital Input Circuit:

There are six digital input circuits, four digital input circuits and four digital input circuits in PXM1K-1, PXM1K-2, and PXM1K-3 modules respectively. The digital input circuit can be used to detect remote signals, or be used as an input pulse counter.

Figure 60. Schematic Diagram of Digital Input Circuit.



The circuit drawing of the digital input is simplified as shown in Figure 60. When K is switched off, OUT is in high state. When K is switched on, OUT is in low state.

The external power supply for the digital input is 20-160 Vac/Vdc. The maximum current in the loop line is 2 mA.

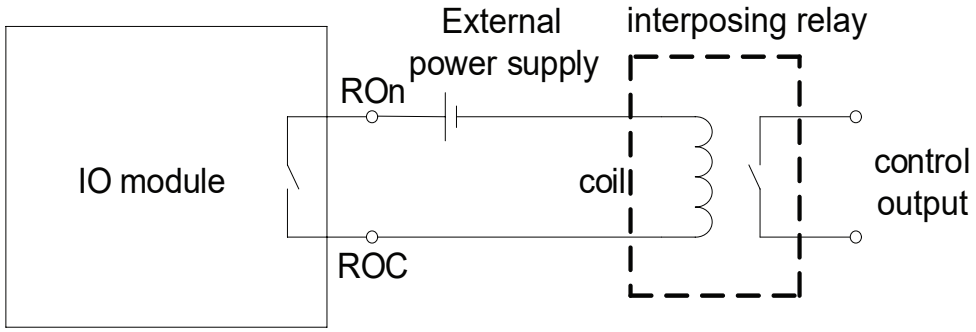
The wire selected for digital input should be between AWG 22~16 or 0.5~ 1.3 mm^2.

Wiring of Relay Output Circuit:

There are two relay output circuits in PXM1K-1 and PXM1K-3 modules respectively. The relay output circuit can work in a controlling state or an alarm state. When it operates in a controlling state, it has two optional output modes: latching and pulse. When it operates in an alarm state, it has only one latching output mode.

The relay type is a mechanical Form A contact with 3 A/250 Vac or 3 A/30 Vdc. An interposing relay is recommended in the output circuit as in Figure 61.

Figure 61. Schematic Diagram of Relay Output Circuit.



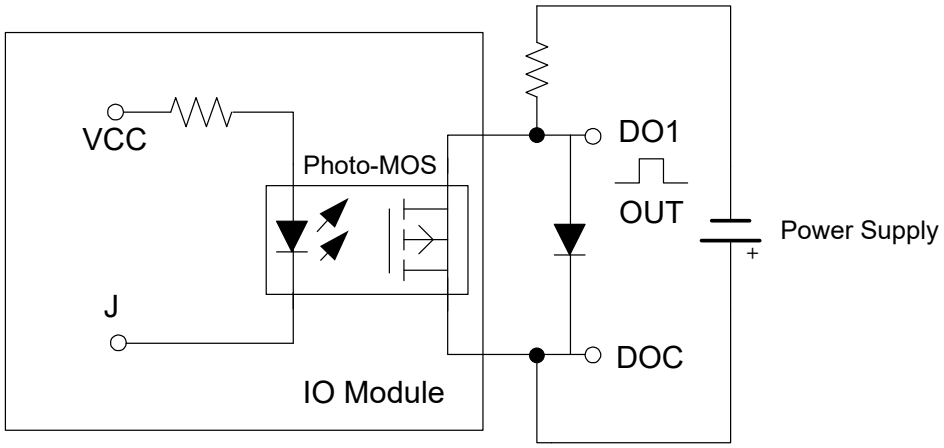
The wire selected for the relay output should be between AWG 22~16 or 0.5~1.3 mm^2.

Wiring of Digital Output Circuit:

There are two digital output circuits in the PXM1K-2 module. The digital output circuit can work in an alarm state or work in an energy pulse output state.

Digital output circuit form is Photo-MOS. The simplified circuit is shown in Figure 62.

Figure 62. Schematic Diagram of Digital Output Circuit 1.

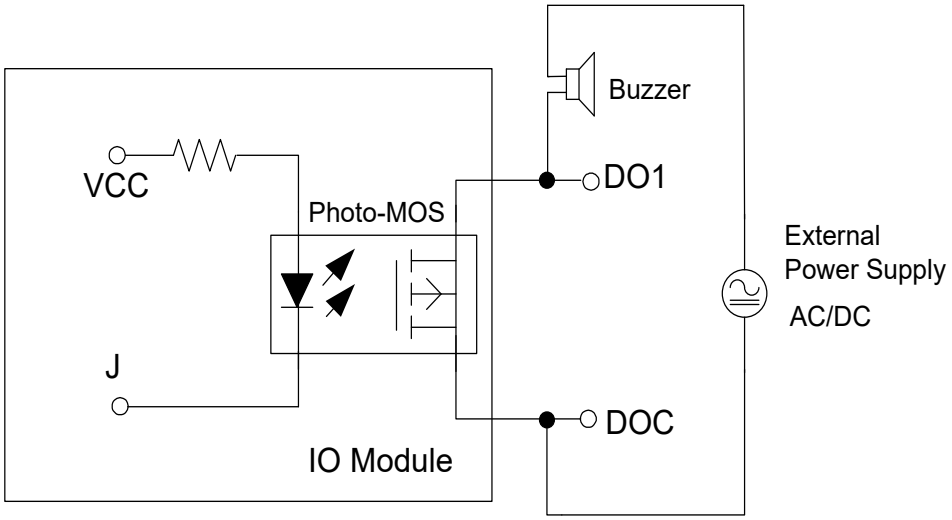


When J is in the low state as shown in Figure 62, OUT is in low state. When J is in the high state, OUT is in high state. OUT can therefore output pulse signals under the control of J.

The max output voltage and current of the digital output circuit are 250 V and 100 mA respectively.

Another drawing of the alarming output with buzzer is shown in Figure 63.

Figure 63. Schematic Diagram of Digital Output Circuit 2.



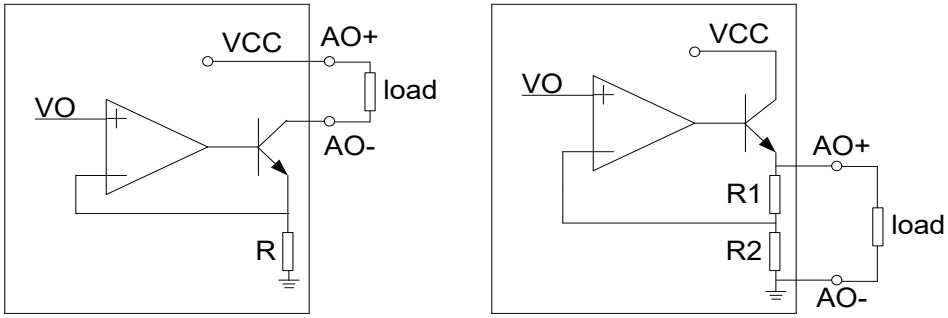
The wire selected for the digital output circuit should be between AWG22~16 or 0.5~1.3 mm^2.

Wiring of Analog Output Circuit:

There are two analog output circuits in PXM1K-2 modules. The terminals of the analog output circuits are AO1+, AO1-, and AO2+, AO2-. The analog output circuit can convert any one of 30 electrical quantities, which is selected by user, to analog voltage or current. The analog output circuit supplies four output modes, including 0 to 20 mA mode, 4 to 20 mA mode, 0 to 5 V mode, and 1 to 5 V mode.

The simplified circuit is as shown in Figure 64.

Figure 64. Schematic Diagram of the Analog Output Circuit.



The Load Capability of the Analog Output Circuit:

0 to 20 mA mode: the max load resistance is 500 Ω.

4 to 20 mA mode: the max load resistance is 500 Ω.

0 to 5 V mode: the max load current is 20 mA.

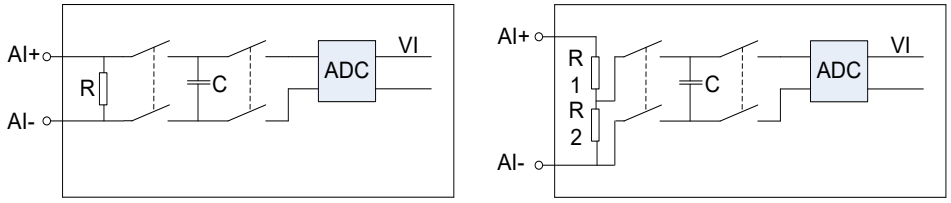
1 to 5 V mode: the max load current is 20 mA.

Wiring of the Analog Input Circuit:

There are two analog input circuits in PXM1K-3 modules. The terminals of analog input circuit are AI1+, AI1- and AI2+, AI2-. The analog input circuit supplies four input modes, including 0 to 20 mA mode, 4 to 20 mA mode, 0 to 5 V mode, and 1 to 5 V mode.

The simplified circuit is as shown in Figure 65.

Figure 65. Schematic Diagram of the Analog Input Circuit.



Current Analog Input

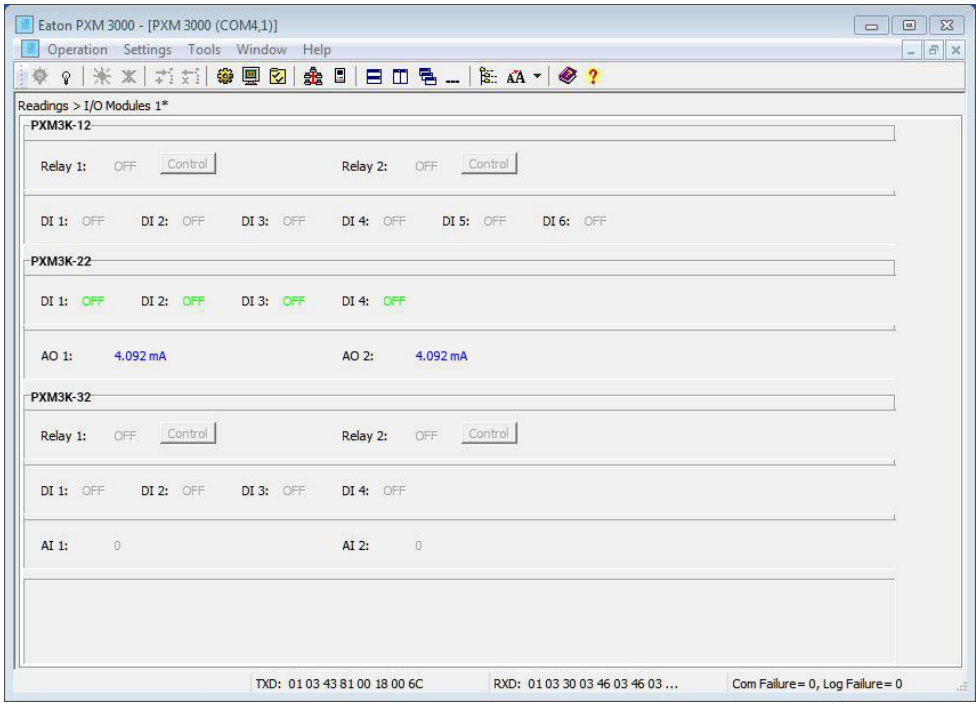
Voltage Analog Input

24 V Isolated Power Supply:

To simplify and make it more convenient for the end user, there is a DI auxiliary power supply provided in PXM1K-1 module. The voltage of the DI auxiliary power supply is 24 Vdc (1W). This power supply can NOT be used for other purposes.

Figure 66 shows the function of IO modules, which is displayed in the utility software as follows, where PXM1K-12 (PXM1K-1 module in logic NO.2), PXM1K-22 (PXM1K-2 module in logic NO.2), and PXM1K-32 (PXM1K-3 module in logic NO.2) are linked to PXM 3000 meter.

Figure 66. Functions of the IO Modules.



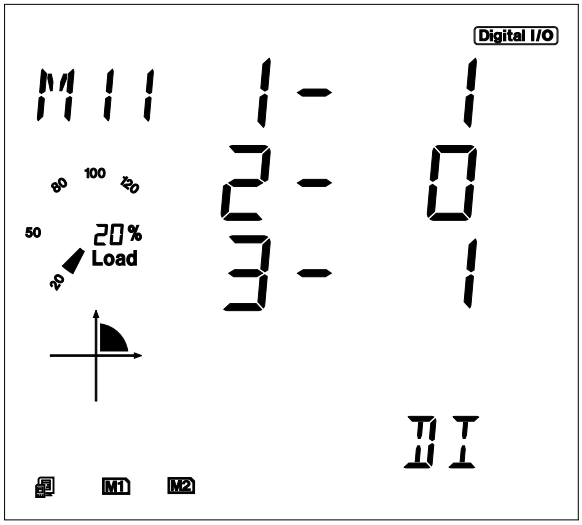
6.1.6 Detection of Remote Signals

The digital input circuit can be set to detect remote signals.

A. Detection of Remote Signals

When the digital input circuit detects a qualified voltage input, it will show "1" on screen and "ON" in the software. Otherwise, it will show "0" on screen and "OFF" in the software.

Figure 67. Showing the DI State on the Screen.



B. Record of SOE

When the digital input circuit is set to detect remote signals, the recording function of SOE can be enabled. Therefore, when the remote signals change, the IO module can record this information accordingly.

SOE Record: Including "4399H to 4439H" address registers. "4399H to 4438H" address registers record 20 groups of SOE records. "4439H" records the IO module which generates the SOE records. For example, if register "4439H" is 1, the 20 groups of SOE records are all generated by PXM1K-11 (PXM1K-1 module in logic NO.1).

The 20 groups of SOE records are arranged based on time. When more than 20 groups of SOE records are generated, the records will begin at the first group.

When the PXM 3000 meter is powered on, the SOE begins to record. The data in the SOE records will not be lost if the meter is powered off. When the enabled SOE function is changed, the records will be lost.

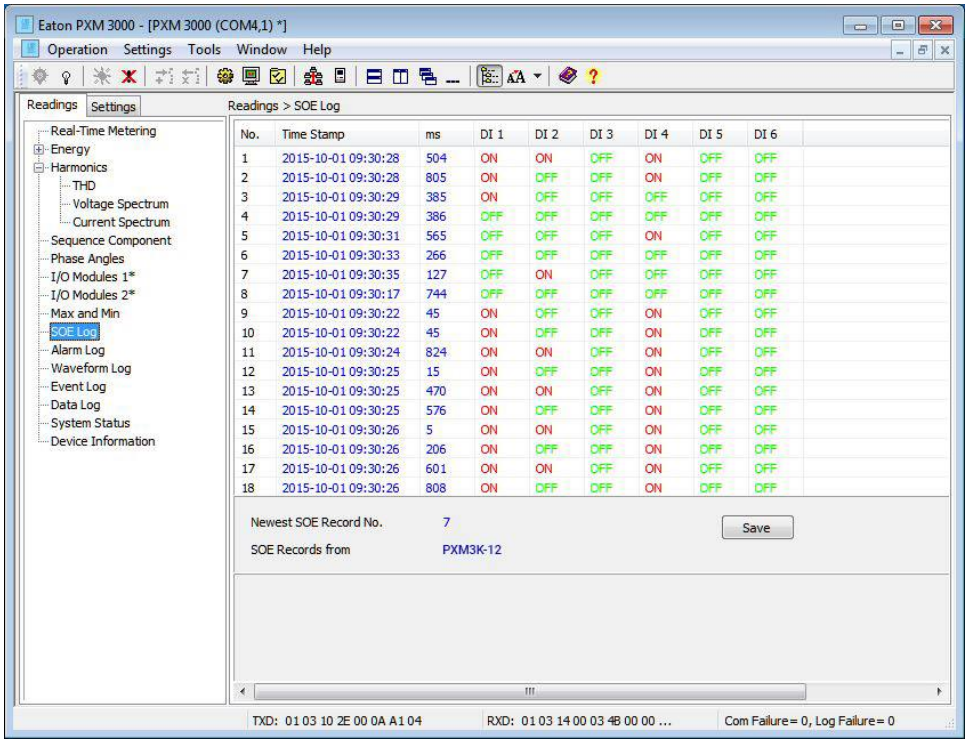
All groups of SOE records are in the same format. Take the first group of SOE records for example. The "4399H to 439fH" registers record the time information, including year, month, day, hour, minute, second, and millisecond. "43a0H" register records the state information, which is an unsigned integer, where bit 0 records DI1 state, bit 1 records DI2 state, and so on. For example, if "43a0H" is "1", it means that DI1 is "1", and others are all "0".

Note: If one of digital input circuits is set to be a pulse counter when the IO module is SOE enabled, then the counterpart bit of "43a0H" register will always be "0".

Data of SOE records can only be read by the utility software. It cannot be read on screen.

Figure 68 shows the data information of SOE records of PXM1K-12 (PXM1K-1 module in logic NO.2) read by the utility software.

Figure 68. Data Information of SOE Records Read by the Utility Software.



C. Parameter Setting of Detection of Remote Signals

Take the parameter setting of PXM1K-11 (PXM1K-1 module in logic NO.1) for example.

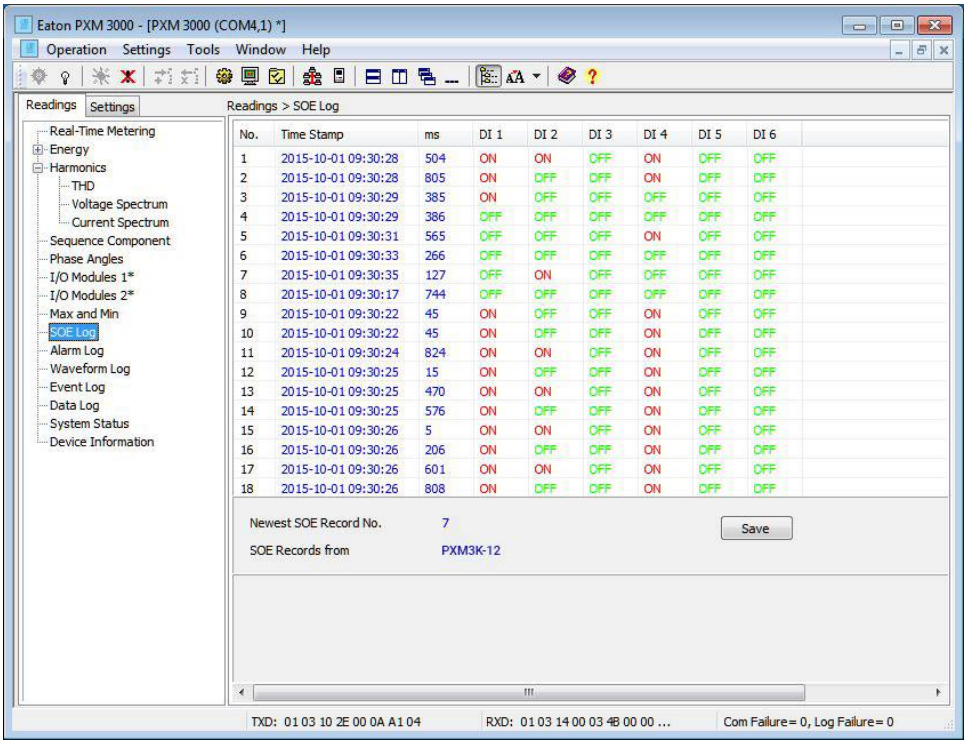
"109eH" register: This register is an unsigned integer, where bit0 determines DI1's working mode, bit1 determines DI2's working mode, and so on. If the bit is "1", then the DI circuit is set to be a pulse counter. Otherwise, the DI circuit is set to detect remote signals. Figure 69 shows the parameter setting of the digital input circuits.

"101bH" register: This register is an unsigned integer. It determines which IO module will be SOE enabled. If the register is "0", then any IO module is SOE disabled. If the register is "1", then PXM1K-11 (PXM1K-1 module in logic NO.1) is SOE enabled. If the register is "2", then PXM1K-21 (PXM1K-2 module in logic NO.1) is SOE enabled. If the register is "3", then PXM1K-31 (PXM1K-3 module in logic NO.1) is SOE enabled. If the register is "4", then PXM1K-12 (PXM1K-1 module in logic NO.2) is SOE enabled. If the register is "5", then PXM1K-22 (PXM1K-2 module in logic NO.2) is SOE enabled. If the register is "6", then PXM1K-32 (PXM1K-3 module in logic NO.2) is SOE enabled.

Only one IO module can be SOE enabled at a time. If the IO module is not linked to the PXM 3000 meter, then there is no need to enable the SOE function in the utility software.

Figure 69 shows the parameters setting of IO module's SOE function.

Figure 69. Parameter Settings of the IO Module's SOE Function.



6.1.7 Pulse Counter

The digital input circuit can also be set to count pulses.

Recorded number of pulses: including "4349H to 4380H" address. The "4349H to 4380H" registers record 28 groups of the individual number of pulses. This includes six groups of records for PXM1K-11 (PXM1K-1 module in logic NO.1), four groups of records for PXM1K-21 (PXM1K-2 module in logic NO.1), four groups of records for PXM1K-31 (PXM1K-3 module in logic NO.1), six groups of records for PXM1K-12 (PXM1K-1 module in logic NO.2), 4 groups of records for PXM1K-22 (PXM1K-2 module in logic NO.2), and four groups of records for PXM1K-32 (PXM1K-3 module in logic NO.2) in sequence. One group of records is an unsigned long integer, for example, "4349H to 434aH" registers record the number of pulses for the DI1 circuit of PXM1K-11 (PXM1K-1 module in logic NO.1).

Figure 70 shows the recorded number of pulses read on the screen.

Figure 70. Recorded Number of Pulses Read on the Screen.

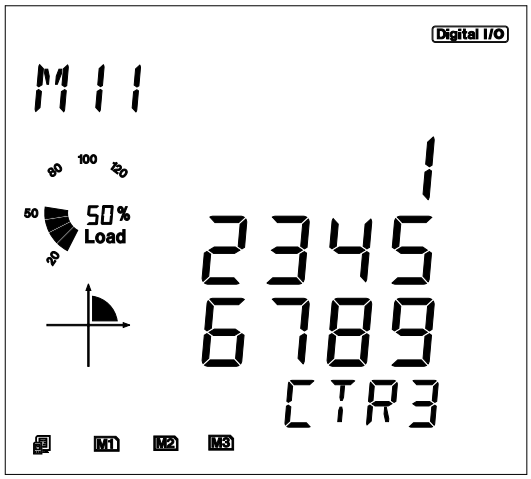
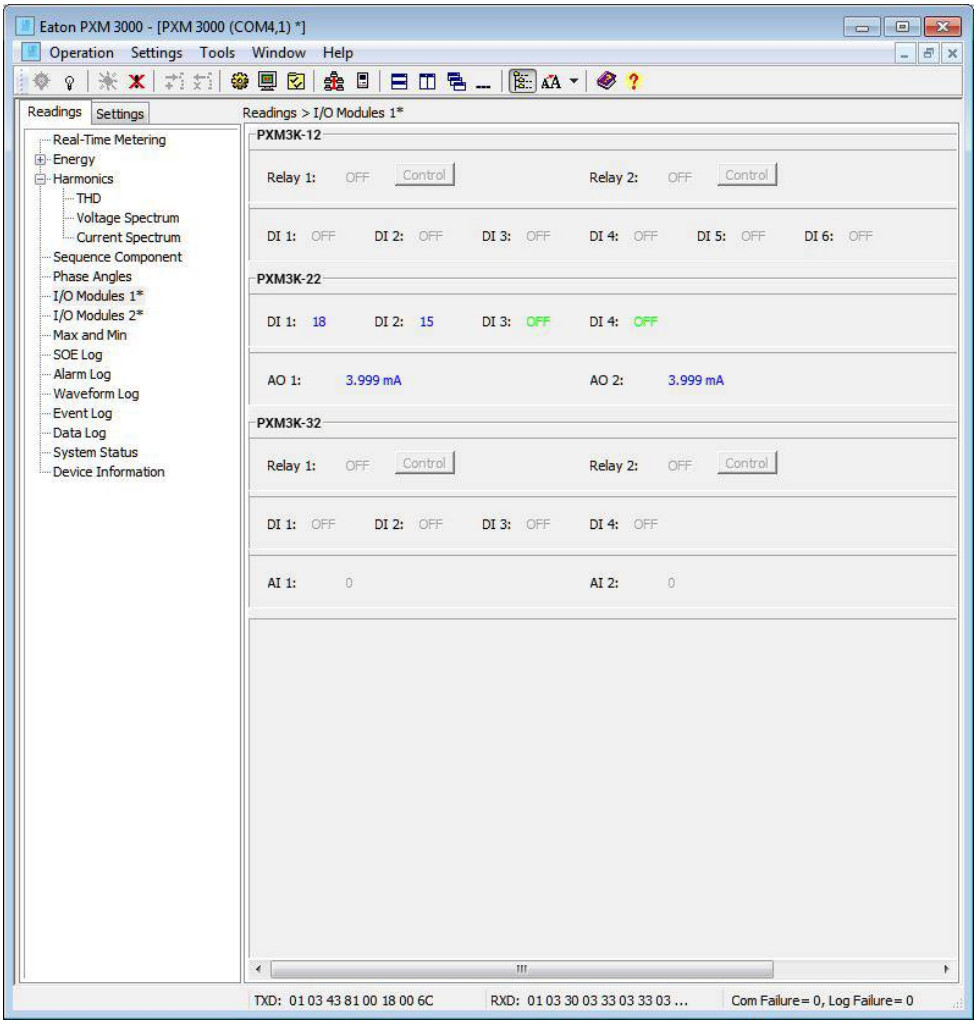


Figure 71 shows the recorded number of pulses read by the utility software.

Figure 71. Recorded Number of Pulses Read by the Utility Software.



Parameter Settings for Counting Input Pulses:

Take PXM1K-11 (PXM1K-1 module in logic NO.1) for example.

1. "109eH" register: if the bit is set as "1", the counterpart digital input circuit is set to be a counter of input pulses.
2. "109fH" register: this register is an unsigned integer. If this register is A , and the digital input circuit is set to be a pulse counter, then the real number of pulses counted by this DI circuit will be as follows:

   Real number of pulses = A × Recorded number of pulses.

For example, if A = 20, the recorded number of pulses counted by DI1 circuit of PXM1K-11 is 100 (4349H to 434aH registers), then the real number of pulses is 20 × 100 = 2,000.

The parameter setting is shown in Figure 69.

6.1.8 Relay Output

Relays in IO modules can work in two different modes, one is the controlling mode, and the other is the alarm mode. For the controlling mode, relays can be switched on and off directly. For the alarm mode, the action of the relays is controlled by whether the alarm has occurred or not.

There are two mode selections for relay output: one is latching and the other is pulse. For the latching mode, the relay can be used to output two states (On or Off). For the pulse mode, the output of the relay changes from Off to On for a period of time and then goes Off. The period can be set from 50 to 3,000 ms.

Note: When the relay is working in alarm mode, the default output mode is latching mode.

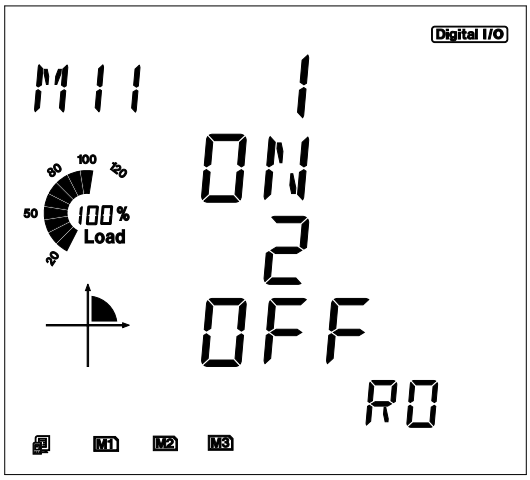
A. Display of Relay State

If the relay state is "ON", it means that relay is switched On. If relay state is "OFF", it means that relay is switched Off.

Figure 72 shows the status of relays read on the screen.

Figure 68 shows the status of relays read by the utility software.

Figure 72. Status of the Relays Read on Screen.



B. Parameter Setting

Take PXM1K-11 (PXM1K-1 module in logic NO.1) for example.

"RO working mode (10a0H)" register: This register determines the working mode of the relays. If the register is "0", then RO1 and RO2 will work in the controlling mode. If the register is "1", then RO1 and RO2 will work in the alarm mode.

"RO output mode (10a1H)" register: This register determines the output mode of the relays. If the register is "0", then RO1 and RO2 will work in the latching output mode. If the register is "1", then RO1 and RO2 will work in the pulse output mode.

"RO pulse width (10a2H)" register: When the relays are working in the pulse mode, this register determines the period of time which can be set from 50 to 3,000 ms. For example, if this register is "100", the relay (RO1 or RO2) will be switched on for 100 ms after receiving the "ON" instruction and then be switched off.

The parameter setting is shown in Figure 69.

6.1.9 Digital Output

There are two mode selections for the digital output circuit: one is the alarm mode and the other is the energy output mode. For the alarm mode, the action of the digital output circuit is controlled by whether the alarm is triggered or not. For the energy output mode, the digital output circuits can output various types of energy, such as import active energy, export active energy, import reactive energy, and export reactive energy. When outputting energy pulses, the pulse width can be set from 20 to 1000 ms. The minimum interval between two pulses is 20 ms.

Parameter Setting:

Take PXM1K-21 (PXM1K-2 module in logic NO.1) for example.

"DO working mode (10a5H)" register: This register determines the working mode of the DO circuits. If the register is "0", then DO1 and DO2 will work in the energy output mode. If the register is "1", then DO1 and DO2 will work in the alarm mode.

"DO pulse width (10a6H)" register: When the DO circuits work in the energy output mode, this register determines the width of energy pulses.

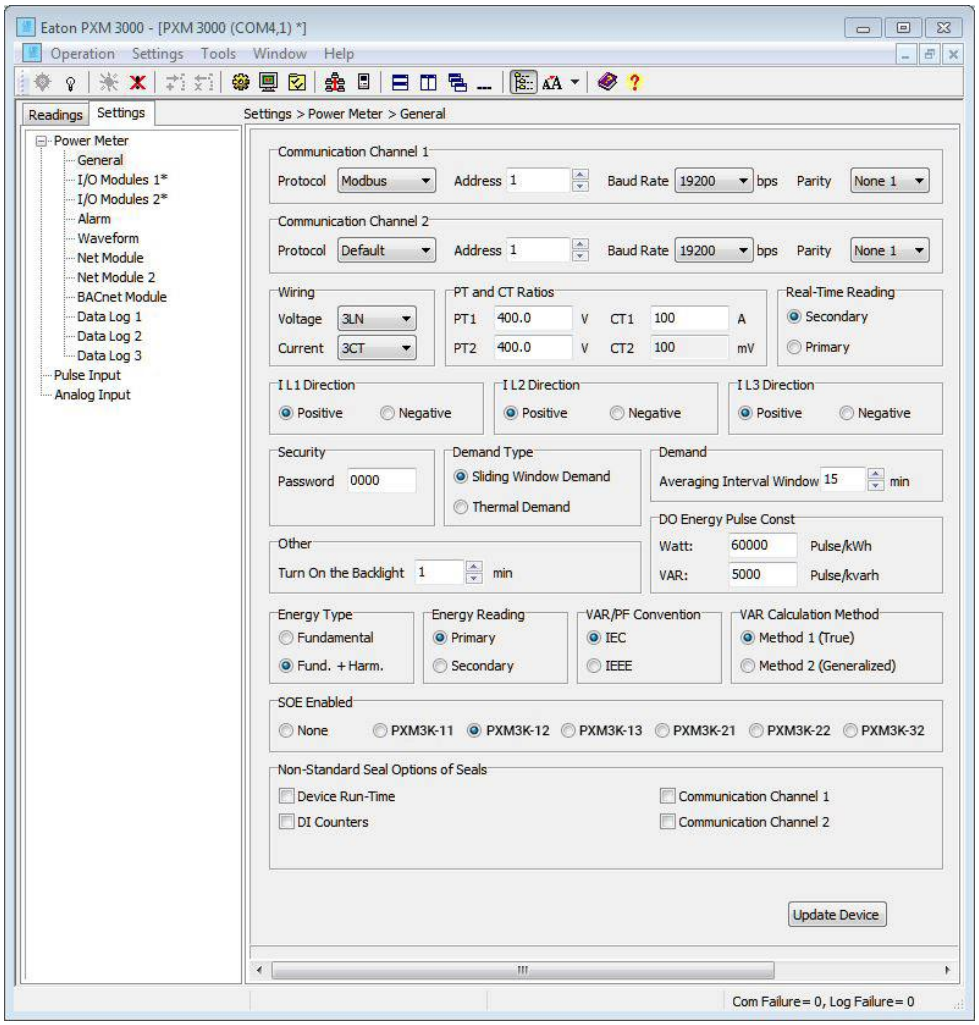
"DO1 output type (10a7H)" register: When the DO circuits work in the energy output mode, this register determines the energy output type for DO1. If this register is "0", DO1 outputs nothing. If this register is "1", DO1 outputs import active energy. If this register is "2", DO1 outputs export active energy. If this register is "3", DO1 outputs import reactive energy. If this register is "4", DO1 outputs export reactive energy.

"DO2 output type (10a8H)" register: When DO circuits work in the energy output mode, this register determines the energy output type for DO2. The value of this register is defined as the same as "DO1 output type" register.

The "DO1 output type" register and "DO2 output type" register can be set to the same value or different values.

The parameter setting is shown in Figure 73.

Figure 73. Parameter Setting of the DO Energy Pulse Constant.



Note: In Figure 73 the DO Energy Pulse Constants are based on the secondary reading; refer to calculator for more information.

6.1.10 Analog Output

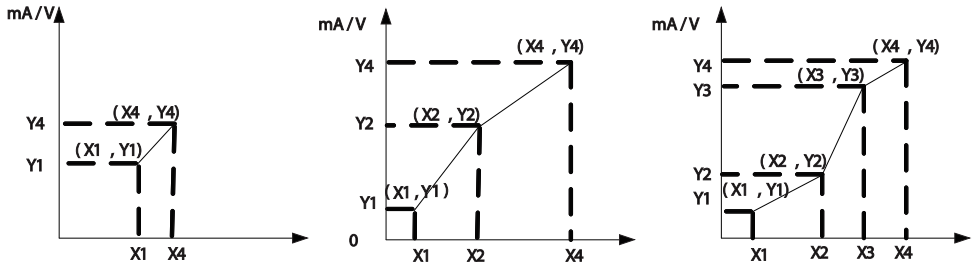
1. Analog Output Relationship with Electrical Quantities

The analog output (AO) circuit can convert anyone of 30 electrical quantities selected by the user (reference Chapter 6) to analog voltage or current.

The analog output circuit supplies four output modes including: 0 to 20 mA mode, 4 to 20 mA mode, 0 to 5 V mode, and 1 to 5 V mode.

Figure 74 shows the relationship between analog output and various electrical quantities.

Figure 74. Relationship Between Analog Output and Various Electrical Quantities.



Note: The following section introduces how the AO function works.

Addresses about the AO function are from 10D0H to 10F3H, which include three groups of parameters such as gradient number, AO following value range, and AO output range.

A. AO Gradient Number Selection of input/output transfer curve.

When the number is 1, which includes (X1, Y1) and (X4, Y4), only the following parameters should be set:

- AO following value range setting start point;
- AO following value range setting end point
- AO1 output range setting start point; and
- AO1 output range setting end point.

When the number is 2, which includes (X1, Y1), (X2, Y2), and (X4, Y4), only the following parameters should be set:

- AO following value range setting start point;
- AO1 following value range setting point 2;
- AO following value range setting end point;
- AO1 output range setting start point;
- AO1 output range setting point 2; and
- AO1 output range setting end point.

When the number is 3, which includes (X1, Y1), (X2, Y2) , (X3, Y3), and (X4, Y4), only the following parameters should be set:

- AO following value range setting start point;
- AO1 following value range setting point 2;
- AO1 following value range setting point 3; and
- AO following value range setting end point.

At the same time, the AO1 output range setting start point, AO1 output range setting point 2, AO1 output range setting point 3, and AO1 output range setting end point should be set.

B. Following the value range setting:

The AO following value range setting start point (X1), AO1 following value range setting point 2 (X2), AO1 following value range setting point 3 (X3), and AO following value range setting end point (X4) are increasing value, while they should be within range of AO following value. Otherwise, the function of the AO will be affected.

- Frequency: When selecting 50 Hz or 60 Hz type, frequency range is 45 Hz ~ 65 Hz, real setting value is 4,500 ~ 6,500. When selecting 400 Hz type, the frequency range is 300 Hz ~ 500 Hz, real setting value is 30,000~50,000.
- Phase voltage V1, V2, V3, and average phase voltage: 0~480 V, real setting value is 0~4,800.
- Line voltage V12, V23, V31, and average line voltage: 0~831 V, real setting value is 0~8,310.
- Current I1, I2, I3, and average current: 0~10 A, real setting value is 0~10,000.
- Power Pa, Pb, and Pc: -4,800~4,800 W, real setting value is -4,800~4,800.
- System power: -14,400~14,400,W, real setting value is -14,400~14,400.
- Reactive power Qa, Qb, and Qc: -4,800~4,800 Var, real setting value is- 4,800~4,800.
- System reactive power: -14,400~14,400 Var, real setting value is -14,400 ~ 14,400
- Apparent power Sa, Sb, and Sc: 0~4,800 VA, real setting value is 0~4,800.
- System apparent power: 0~14,400 VA, real setting value is 0~14,400.
- Power factor PFa, PFb, PFc, and System power factor: -1~1, real setting value is -1,000~1,000.

C. AO output range setting:

The AO output value range setting start point (Y1), AO1 output value range setting point 2 (Y2), AO1 output value range setting point 3 (Y3), and AO output value range setting end point (Y4) are increasing value, while they should be within range of the AO output value.

When the AO type is 0 ~ 20 mA, the corresponding range is 0 ~ 24 mA, the setting value range is 0 ~ 4,915, and the relationship is:

mA = setting value x 20/4,096.

When the AO type is 4 ~ 20 mA, the corresponding range is 4 ~ 24 mA, the setting value range is 819 ~ 4,915, and the relationship is:

mA = setting value x 20/4,096.

When the AO type is 0 ~ 5 V, the corresponding range is 0 ~ 6 V, the setting value range is 0 ~ 4,915, and the relationship is:

V = setting value x 5/4,096.

When the AO type is 1 ~ 5 V, the corresponding range is 1 ~ 6 V, the setting value range is 819 ~ 4,915, and the relationship is

V = setting value x 5/4,096.

Note: If the voltage input wiring of the meter is 2LL or 3LL, then the analog outputs relative to phase voltage, neutral current, phase active/reactive/apparent power, and phase power factor will always be 0.

Note: The maximum of analog output is 1.2 times the range.

2. Display of Analog Output

Value of the analog output is displayed in hex on the screen. The relationship between the displayed value and real value of analog output is:

$$\textit{Real value} = \frac{\textit{Displayed Value}}{4096} \times \textit{20 mA (current output mode)}$$

or

$$\textit{Real value} = \frac{\textit{Displayed Value}}{4096} \times \textit{5 V (voltage output mode)}$$

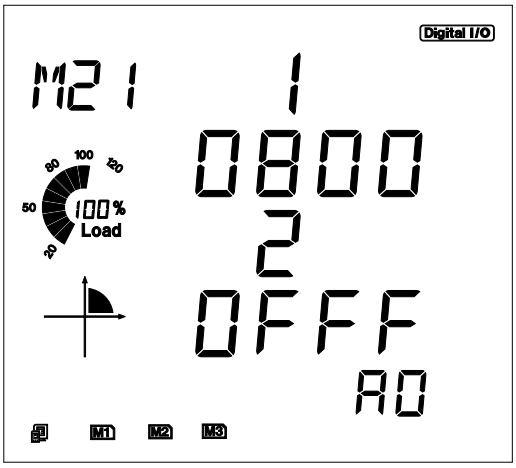
As shown in Figure 75, the displayed value of AO1 is 0 x 0800, so the real value of AO1 is:

(0 x 0800/4096) × 5 V

or

(0 x 800/4096) × 20 mA.

Figure 75. AO Value Read on the Screen.



3. Parameter Setting

Take the PXM1K-21 (PXM1K-2 module in logic NO.1) for example.

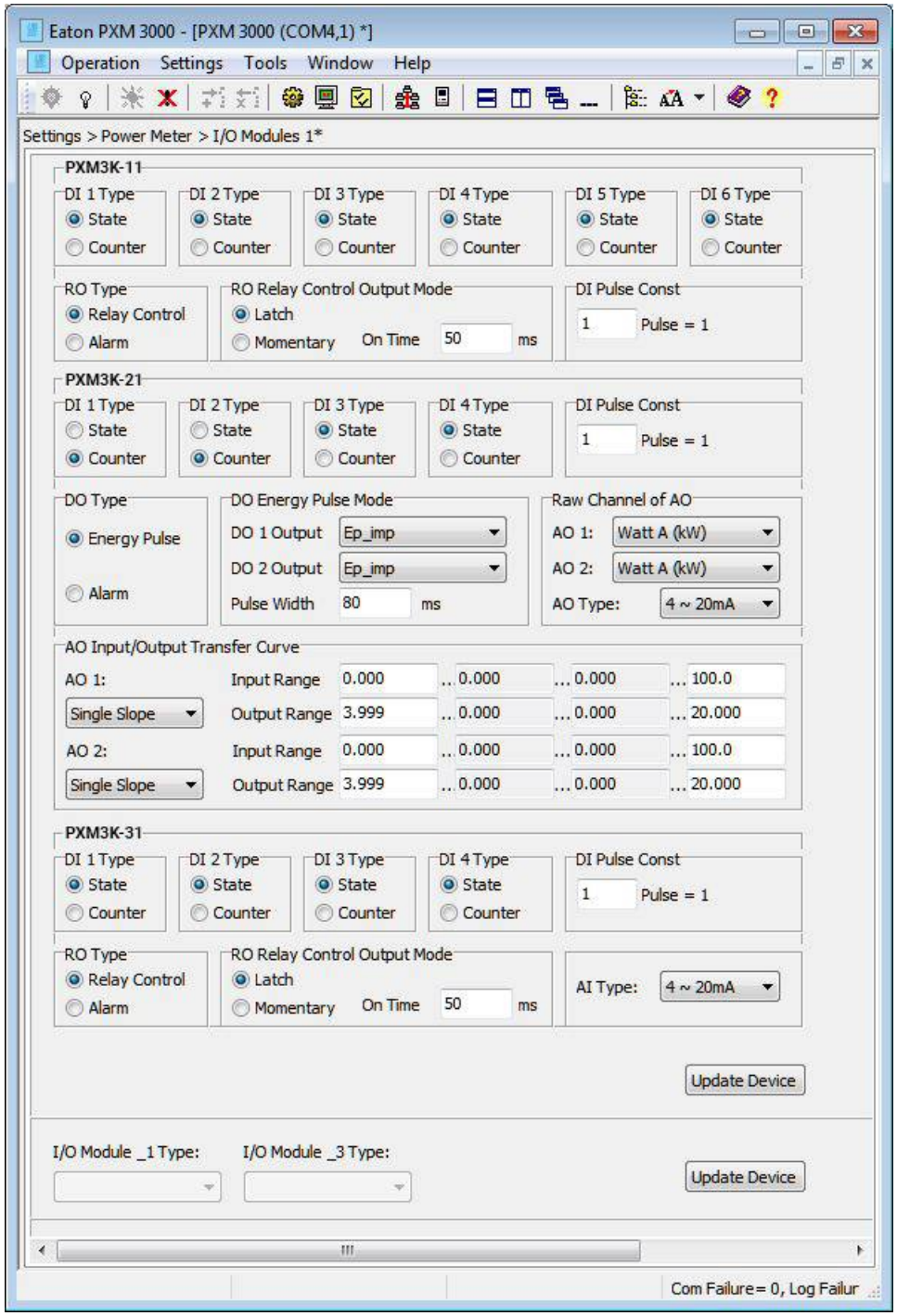
"Electrical quantities mapped to AO1 (10c2H)" register: This register determines to which electrical quantity AO1 should be mapped (see Chapter 6 for explanation). For example, if this register is "0", then AO1 is mapped to "Frequency".

"Electrical quantities mapped to AO2 (10c3H)" register: This register determines to which electrical quantity the AO2 should be mapped. The value of this register is defined as the same as "Electrical quantities mapped to AO1 (10c2H)" register.

"Electrical quantities mapped to the AO1 (10c2H)" register and "Electrical quantities mapped to AO2 (10c3H)" register can be set to the same value .

The parameter setting is shown in Figure 76:

Figure 76. Parameter Setting of the IO Modules.



6.1.11 Analog Input

Analog input circuits supply four types of input modes including: 0 to 20 mA mode, 4 to 20 mA mode, 0 to 5 V mode, and 1 to 5 V mode.

Figure 77 shows the relationship between the AI value and the input analog value.

AI value ranges from 0 to 4,095 without any unit. The AI value is displayed in hex on screen.

Figure 78 shows the AI value read on screen.

Figure 77. Relationship Between the AI Value and the Input Analog Value.

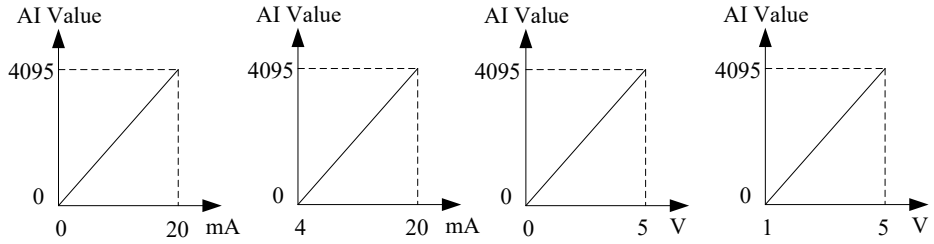
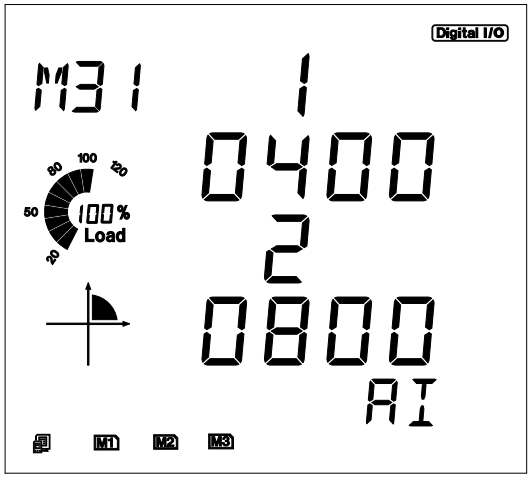


Figure 78 shows the AI value read on screen.

Figure 78. The AI Value Read on the Screen.



7. Communication

This chapter will mainly discuss how to handle the meter via the communication port using the utility software. It is highly recommended that previous chapters be read before moving onto this chapter, a familiarity with Modbus™ would also be helpful.

7.1 Modbus Protocol Introduction

Modbus RTU protocol is used for communication in PXM 3000 series meters. Data format and error check methods are defined in the Modbus protocol. The half duplex query and respond mode is adopted in the Modbus protocol. There is only one master device in the communication network. The others are slave devices, waiting for the query of the master.

Transmission Mode

Transmission Mode, which is compatible with Modbus RTU mode, defines the data structure within a frame and the rules are used to transmit the data. This can be found in Table 14.

Framing

Table 14. Data Frame Format.

Address	Function	Data	Check
8-Bits	8-Bits	N×8-Bits	16-Bits
	Coding System	8-bit binary	
	Start bit	1	
	Data bits	8	
	Parity	no parity, odd parity, even parity	
	Stop bit	1 or 2	
	Error checking	CRC check	

Address Field

The address field of a message frame contains eight bits. Valid slave device addresses are in the range of 0 ~ 247 decimal. A master addresses a slave by placing the slave address in the address field of the message. When the slave sends its response, it places its own address in this address field of the response to let the master know which slave is responding.

Function Code Field

The function code field of a message frame contains eight bits. Valid codes are in the range of 1 ~ 255 decimal. When a message is sent from a master to a slave device, the function code field tells the slave what kind of action to perform.

Table 15. Function Codes.

Code	Meaning	Action
01	Read Relay Output Status	Obtain current status of the Relay Output.
02	Read Digital Input (DI) Status	Obtain current status of the Digital Input.
03	Read Data	Obtain current binary value from one or more registers.
05	Control Relay Output	Force the relay state to "ON" or "OFF".
16	Preset Multiple-Register	Place specific binary values into a series of consecutive Multiple-Registers.

Data Field

The data field is constructed using sets of two hexadecimal digits, in the range of 00 to FF hexadecimal. The data field of messages sent from a master to slave devices contain additional information which the slave must use to take the action defined by the function code. This can include items such as register addresses, the quantity of items to be handled, and the count of actual data bytes in the field. For example, if the master requests a slave to read a group of holding registers (function code 03), the data field specifies the starting register and how many registers are to be read. If the master writes to a group of registers in the slave (function code 10 hexadecimal), the data field specifies the starting register, how many registers to write, the count of data bytes to follow in the data field, and the data to be written into the registers.

If no error occurs, the data field of a response from a slave to a master contains the data requested. If an error occurs, the field contains an exception code that the master application can use to determine the next action to be taken. The data field can be nonexistent (of zero length) in certain kinds of messages.

Error Check Field

Every message includes an error checking field which is based on the Cyclical Redundancy Check (CRC) method. The CRC field checks the contents of the entire message. It is applied regardless of any parity check method used for the individual characters of the message. The CRC field is two bytes long, containing a 16-bit binary value. The CRC value is calculated by the transmitting device and is appended to the message.

The receiving device recalculates the CRC value during reception of the message and compares the calculated value to the actual value it received in the CRC field. If the two values are not equal, an error will be reported. The CRC calculation is first started by preloading the whole 16-bit register to 1's. The process begins by applying successive 8-bit bytes of the message to the current contents of the register. Only the eight bits of data in each character are used for generating the CRC. Start and stop bits, and the parity bit, do not apply to the CRC. When generating the CRC, each 8-bit character is exclusive ORed (connected in logical OR fashion) with the register contents. The result is shifted towards the least significant bit (LSB), with a zero filled into the most significant bit (MSB) position. The LSB is extracted and examined. If the LSB is equal to 1, the register is exclusive ORed with a preset, fixed value. If the LSB equals to 0, no action will be taken. This process is repeated until eight shifts have been performed. After the last (eighth) shift, the next 8-bit byte is exclusive ORed with the register's current value, and the process repeats for eight more shifts as described above. After all the bytes of the message have been applied, the final contents of the register, which should exchange the high-byte and the low-byte, is the CRC value. When the CRC is appended to the message, the low-order byte is appended first, followed by the high-order byte.

7.2 Communication Format

Explanation of Frame

Table 16. Explanation of Frame.

Addr	Fun	Data Start Reg HI	Data Start Reg LO	Data #of Regs HI	Data #of Regs LO	CRC 16 HI	CRC 16 LO
06H	03H	00H	00H	00H	21H	84H	65H

Addr: Address of slave device

Fun: Function code

Data Start Reg HI: Start register address high byte

Data Start Reg LO: Start register address low byte

Data # of Reg HI: Number of register high byte

Data # of Reg LO: Number of register low byte

CRC16 HI: CRC high byte

CRC16 LO: CRC low byte

1. Reading Relay Status

Function Code 01

This function code is used to read the status of the relay in the meter.

1 = On 0 = Off

Relay 1's address is 0000H, Relay 2's address is 0001H, and so on.

The following query is to read the relay status for the meter with communication address 17.

Query

Table 17. Read the Status of Relay 1 and Relay 2 Query Message.

Addr	Fun	Data Start Reg HI	Data Start Reg LO	Relay #of Regs HI	Relay #of Regs LO	CRC 16 HI	CRC 16 LO
11H	01H	00H	00H	00H	02H	BFH	5BH

Response

The PXM 3000 meter's response includes the meter address, function code, quantity of data byte, the data, and error checking. An example response to read the status of Relay 1 and Relay 2 is shown as Table 18. The status of Relay 1 and Relay 2 are responding to the last 2 bits of the data.

Relay 1: bit0 Relay 2: bit1

Table 18. Relay Status Response.

Address	Function Code	Byte Count	Data	CRC High	CRC Low		
11H	01H	01H	02H	D4H	89H		
The content of the data is:							
7	6	5	4	3	2	1	0
0	0	0	0	0	0	1	0
MSB						LSB	

Relay 1 = OFF (LSB), Relay 2=ON (Left to LSB)

2. Read Status of DI

Function Code 02

1 = On 0 = Off

DI1's address is 0000H, DI2's address is 0001H, and so on.

The following query is to read the Status of four DIs of the PXM 3000 meter with communication address 17.

Query

Table 19. Read Four DIs Query Message.

Addr	Fun	DI Start Addr HI	DI Start Addr LO	DI Num HI	DI Num LO	CRC 16 HI	CRC 16 LO
11H	02H	00H	00H	00H	04H	7BH	59H

Response

The PXM 3000 meter's response includes the meter address, function code, quantity of data characters, the actual data characters, and error checking. An example response to read the status of four DIs are shown in Table 20. The DI status corresponds to the last four bits of the data.

DI1: bit0; DI2: bit1; DI3: bit2; DI4: bit3.

Table 20. Read Status of DI.

Address	Function Code	Byte Count	Data	CRC High	CRC Low		
11H	02H	01H	03H	E5H	49H		
The content of the data is:							
7	6	5	4	3	2	1	0
0	0	0	0	0	0	1	1
MSB						LSB	

DI1 = On, DI2 = On, DI3 = Off, DI4 = Off.

3. Read Data (Function Code 03)

Query

This function allows the master to obtain the measurement results from the PXM 3000 meter. Table 21 is an example of reading the measured data (F, V1, and V2) from slave device number 17. The data address of F is 4000H, 4001H; V1's address is 4002H, 4003; and V2's address is 4004H, 4005H.

Table 21. Read F, V1, V2 Query Message.

Addr	Fun	Data Start Addr HI	Data Start Addr LO	Data # of Regs HI	Data # of Regs LO	CRC 16 Regs HI	CRC 16 Regs LO
11H	03H	40H	00H	00H	06H	D2H	98H

Response

The PXM 3000 meter's response includes the meter address, function code, quantity of data bytes, data, and error checking. An example response to read F, V1, and V2 (F = 42480000H [50.00 Hz], V1 = 42C7CCCDH [99.9 V], V2 = 42C83333H [100.1 V]) is shown:

Table 22. Read F, V1, and V2 Message.

Addr	Fun	Byte Count	Data1 HI	Data1 LO	Data 2 HI	Data2 LO	Data3 HI	Data3 LO	Data4 HI	Data4 LO
11H	3H	0CH	42H	48H	00H	00H	42H	C7H	CCH	CDH

Data5 HI	Data5 LO	Data 6 HI	Data6 LO	CRC16 HI	CRC16 LO
42H	C8H	33H	33H	CAH	7FH

4. Control Relay (Function Code 05)

Query

This message forces a relay to either turn "ON" or "OFF". Any relay that exists within the PXM 3000 meter can be forced to either "ON" or "OFF" status. The data value FF00H will set the relay "ON" and the value 0000H will turn it "OFF". All other values are invalid and will not affect that relay.

The example below is a request to the PXM 3000 meter with the address of 17 to turn on Relay 1.

Table 23. Control Relay Query Message.

Addr	Fun	DO Addr HI	DO Addr LO	Value HI	Value LO	CRC 16 HI	CRC 16 LO
11H	05H	00H	00H	FFH	00H	8EH	AAH

Response

The normal response to the command request is to retransmit the message as received after the relay status has been altered.

Table 24. Control Relay Response Message.

Addr	Fun	Relay Addr HI	Relay AddrLO	Value HI	Value LO	CRC HI	CRC LO
11H	05H	00H	00H	FFH	00H	8EH	AAH

5. Preset / Reset Multi-Register (Function Code 16)

Query

Function 16 allows the user to modify the contents of a multi-register. Some registers of the PXM 3000 meter can have their contents changed by this message. The example below is a request to a PXM 3000 meter with the address of 17 to preset Ep_imp as "17807783.3KWh", while its HEX value is 0A9D4089H. The Ep_imp data address is 4048H and 4049H.

Table 25. Preset Multi-register Query Message.

Addr	Fun	Data Start Reg HI	Data Start Reg LO	Data # of Reg HI	Data # of Reg LO	Byte Count
11H	10H	40H	48H	00H	02H	04H

Value HI	Value LO	Value HI	Value LO	CRC HI	CRC LO
0AH	9DH	40H	89H	F1H	6AH

Response

The normal response to a preset multi-register request includes the PXM 3000 meter's address, function code, data start register, the number of registers, and error checking.

Table 26. Preset Multi-register Response Message.

Addr	Fun	Data Start Reg hi	Data Start Reg lo	Data # of Reg hi	Data # of Reg lo	CRC16 Hi	CRC16 Lo
11H	10H	40H	48H	00H	02H	D6H	8EH

7.3 Data Address Table and Application Details

There are several rules to follow in using the PXM 3000 Series meter:

1. Data type:

- "bit" refers to binary.
- "word" refers to a 16-bit unsigned integer using one data address and two bytes of memory; it varies from 0 to 65535.
- "int" refers to a 16-bit integer using one data address and two bytes of memory; it varies from -32768 to 32767.
- "Dword" refers to a 32-bit unsigned integer using two data addresses and four bytes of memory with the high word at the front and low word at the end; it varies from 0 to 4294967295. Rx = high word x 65536 + low word.
- "float" refers to a 32-bit single value using two data addresses and four bytes of memory; it varies from -1.175494E-38 to 3.402823E+38.

2. Relationship Between Communication Value and Numerical Value

It is important to note that the value returned in some Modbus registers have to be scaled to reflect the actual circuit reading. Table 27 shows how they respond to each other.

Note: In the current channel, CTs are optional, which include voltage type CT or current type CT.

A. When a current type CT is selected and the value of CT2 is 1 or 5, and using relationship listed below to count primary value, the value of CT2 should be the original 1 or 5.

B. When a voltage type CT is selected and the value of CT2 is 333, and using relationship listed below to count primary value, the value of CT2 should not be 333, but 1.

C. When you select 100 mv/50 Hz, or 120 mv/60 Hz Rogowski coil, the value of CT2 is 100, and using relationship listed below to count primary value, the value of CT2 should not be 100, but 1.

D. When you select 80/100/200 mA CT, the value of CT2 is 80, 100, or 200, and using relationship listed below to count primary value, the value of CT2 should not be 80, 100, or 200, but 1.

Table 27. Relationship Between Communications and Numerical Values.

Parameters	Relationship	Unit	Format Code
System parameters	Numerical value equal to communication value	No unit	F1
Run time	T = Rx/100	Hour	F2
Clock	Numerical value equal to communication value	Unit of time	F3
Energy (primary)	Ep = Rx/10	kWh	F4
Reactive energy (primary)	Eq = Rx/10	kvarh	F5
Apparent energy (primary)	Es = Rx/10	kVA	F6
Energy (secondary)	Ep = Rx/1000	kWh	F7
Reactive energy (secondary)	Eq = Rx/1000	kvarh	F8
Apparent energy (secondary)	Es = Rx/1000	kVA	F9
Frequency	F = Rx/100	Hz	F10
Voltage (secondary)	U = Rx X (PT1/PT2)/10	V	F11
Current, current demand (secondary)	I = Rx X (CT1/CT2)/1000	A	F12
Power, demand (secondary)	P = Rx X (PT1/PT2) X (CT1/CT2)	W	F13
Reactive power, demand (secondary)	Q = Rx X (PT1/PT2) X (CT1/CT2)	var	F14
Apparent power, demand (secondary)	S = Rx X (PT1/PT2) X (CT1/CT2)	VA	F15
Power factor	PF = Rx/1000	No unit	F16
Unbalance factor	Unbl = (Rx/1000) X 100%	No unit	F17
THD	THD = (Rx/10000) X 100%	No unit	F18
Harmonics	HDn = (Rx/10000) X 100%	No unit	F19
Total odd HD	HDo = (Rx/10000) X 100%	No unit	F20
Total even HD	HDe = (Rx/10000) X 100%	No unit	F21
Crest factor	CF = Rx/1000	No unit	F22
K factor	KF = Rx/10	No unit	F23
THFF	THFF = (Rx/10000) X 100%	No unit	F24
Phase angle	Phase angle = Rx/10	Degree	F25

Note: Regions from the "System parameters settings" to the "Data logging 3 settings" are the regions that can be set and modified. Please use the following guidelines when communicating with the meter.

1. When function code 10H is used, one communication command can only modify the contents in one region, such as "System parameters settings", "System status parameter", "Date and Time table", "Out-of-limit alarms - Global settings", "Out-of-limit alarm - Single settings", "I/O Modules settings", Data logging 1 settings, Data logging 2 settings, and Data logging 3 settings. It can not be accomplished in one communication order to modify contents in both of two or more regions above.
2. When function code 03H is used, the rules and limitations described in 1. above will not be applied.

7.3.1 System Parameter Setting

System parameters determine how the meter works. Please refer to Chapters 3 and 4 for more details.

Function code: 03H for reading, 10H for writing. Data type: word. Format code: F1.

Table 28. System Parameter Settings.

Address	Parameter	Default	Range	Data Type	Property
0ffdH	Frequency	0	0 : 50Hz 1 : 60 Hz 2: 400 Hz	Word	R/W
0ffdH	First Communication Protocol	0	0:MODBUS Protocol; 1:DNP3.0 Protocol	Word	R/W
0fffH	Parity Setting1	3	0: EVEN 1: odd 2:NON2 3:NON1	Word	R/W
1000H	Password	0	0 ~ 9999	Word	R/W
1001H	Communication Address	1	1 ~ 247 (MODBUS); 0 ~ 65534 (DNP3.0)	Word	R/W
1002H	Baud Rate	19200	1200 ~ 115200	Word	R/W
1003H	Voltage Input Wiring Type	0	0:3LN, 1:1LN, 2:2LL, 3:3LL, 4:1LL	Word	R/W
1004H	Current Input Wiring Type	0	0:3CT ,1:1CT, 2:2CT	Word	R/W
1005H	PT1	220.0	50.0 ~ 500000.0	Unit 32	R/W
1007H	PT2	220.0	50.0 ~ 400.0	Word	R/W
1008H	CT1	5	1 ~ 50000	Word	R/W
1009H	CT2	5	1, 5	Word	R/W
100aH	kWh Pulse Constant	1	1 ~ 60000 (Setting Method, See Remarks)	Word	R/W
100bH	kvarh Pulse Constant	1	1 ~ 60000 (Setting Method, See Remarks)	Word	R/W
100cH	LCD Backlight Time	1	0 ~ 120	Word	R/W
100dH	Demand Slid Window Time	15	1 ~ 30	Word	R/W
100eH	Demand Calculating Mode	1	0: Fixed, 1: Sliding Window 2: Thermal, 3: Rolling	Word	R/W
100fH	Clear Demand Memory	0	Only 1 Works	Word	R/W
1010H	Max/Min Clear	55H	Only 0AH Works	Word	R/W
1011H	Run Time Clear	0	Only 1 Works	Word	R/W
1012H	Current I1 Direction	0	0: Positive; 1: Negative	Word	R/W
1013H	Current I2 Direction	0	0: Positive; 1: Negative	Word	R/W
1014H	Current I3 Direction	0	0: Positive; 1: Negative	Word	R/W
1015H	VAR/PF Convention	0	0: IEC, 1: IEEE	Word	R/W
1016H	Energy Clear	0	Only 1 Works	Word	R/W
1017H	Energy Calculating Mode	1	0: Fundamental; 1: Full-wave (also referred to as True RMS)	Word	R/W
1018H	Reactive Power Measuring Mode	0	0: Real, 1: General	Word	R/W
1019H	Energy Display Mode	0	0: Primary, 1: Secondary	Word	R/W
101aH	Ethernet Module Reset	0	0: None, 1: Reset, 2: Load Default and Reset	Word	R/W
101bH	SOE Enable	0	0: None; 1: PXM1K-11; 2: PXM1K-21; 3: PXM1K-31; 4: PXM1K-12; 5: PXM1K-22; 6: PXM1K-32;	Word	R/W
101cH	Pulse Counter Clear	0	0: None; 1: PXM1K-11; 2: PXM1K-21; 3: PXM1K-31; 4: PXM1K-12; 5: PXM-1K-22; 6: PXM1K-32;	Word	R/W
101dH	Basic Parameter Mode	0	0: Primary; 1:Secondary	Word	R/W

Notes: When 0FFEH is 0, first communication protocol is set to MODBUS, while 0FFEH is 1, first communication protocol is DNP3.0. At this time, special DNP3.0 software is needed, and normal software will be invalid.

100AH, 100BH setting method: 1000 * 3600 / (U * I * n * pulse constant) = pulse period (S), pulse period calculated by the pulse constant must be greater than the pulse width (20ms ~ 1000 ms) of IO module DO setting. Wherein n is applied by the User, and if the three-phase signals are added, then n is 3. U and I are generally equal to the User settings PT2, CT2, ie, rated voltage and rated current.

0x1017H: When selecting 400 Hz type,only support true RMS.

7.3.2 System Status Parameters

"System status" indicates what events happened in the meter, what kinds of flags are read by the user, and the index of the storage of events. Flags should be cleared after being read by the controller, otherwise new data will not be stored properly.

Function code: 03H for reading, 10H for writing. Data type: word.

Table 29. System Status Parameters.

Address	Parameter	Format Code	Range	Data Type	Property
101EH	Sealed Nonstandard Parameters		Bit0: 1st Communication Parameters; Bit1: 2nd Communication Parameters; Bit2: Run Time Clear; Bit3: DI Pulse Count; Bit4: TOU; 1: Valid of Corresponding Selection; 0: invalid	Word	R/W
101FH	Seal Status		0x0A: Seal Sealed; Other: Seal Opened.	Word	R/W
1020H	Reserved				
1021H	Alarm Record Clear		0x0A: Clear; Other: Not Clear	Word	R/W
1022H~102DH	Reserved				
102eH	System Status		Bit0: New Alarming or Not; Bit1: New SOE or Not	Word	R
102fH	Baud Rate 2	115200	(fixed)	Word	R/W
1030H	Parity Setting 2		0: EVEN 1: Odd 2:NON2 3:NON1 (fixed)	Word	R/W
1031H	Communication Address2		1 ~ 247	Word	R
1032H	Alarming Group Number	F1	0: No Alarming Record; 1 ~ 16: Last Alarming Record Group Number	Word	R
1033H	SOE Group Number	F1	0: No SOE Record; 1 ~ 20: Last SOE Group Number	Word	R
1034H	Run Time (High)	F2	0 ~ 999999999	Word	R
1035H	Run Time (Low)			Word	R
1036H	Expanded IO Modules Connecting Status		Bit0: PXM1K-11; Bit1: PXM1K-12; Bit2: PXM1K-21; Bit3: PXM1K-22; Bit4: PXM1K-31; Bit5: PXM1K-32; 0: Disconnected 1: Connected	Word	R
1037H	Reserved				
1039H	Ten Year Download Setting Enable		1: Enable is Valid	Word	R/W
103aH	Sharp Demand Clear		0x0A: Clear; Other: Not Clear	Word	R/W
103bH	Peak Demand Clear		0x0A: Clear; Other: Not Clear	Word	R/W
103cH	Valley Demand Clear		0x0A: Clear; Other: Not Clear	Word	R/W
103dH	Normal Demand Clear		0x0A: Clear; Other: Not Clear	Word	R/W
103eH	Total Demand Clear		0x0A: Clear; Other: Not Clear	Word	R/W

Note: Please refer to Chapter 3 and Chapter 4 for more details about parameter settings.

7.3.3 Date and Time Table

Function code: 03H for reading, 10H for presetting.

Table 30. Date and Time Table.

Address	Parameter	Format Code	Range	Data Type	Property
103fH	Week	F3	0~6	Word	R/W
1040H	Year	F3	2000-2099	Word	R/W
1041H	Month	F3	1~12	Word	R/W
1042H	Day	F3	1~31	Word	R/W
1043H	Hour	F3	0~23	Word	R/W
1044H	Minute	F3	0~59	Word	R/W
1045H	Second	F3	0~59	Word	R/W

7.3.4 Over/Under Limit Alarming Setting

This setting consists of global alarm settings and single channel alarm settings. Global alarm settings contain settings of all global variables. There are 16 groups of records with the same format. Function code: 03H for reading, 10H for writing. Please refer to Chapter 4 for more details.

Table 31. Global Alarm Settings.

Address	Parameter	Range	Data Type	Property
1046H	Global Alarming Enable	0:disable; 1:enable	Word	R/W
1047H	Alarming Flash Enable	0:disable; 1:enable	Word	R/W
1048H	Alarming Channel Enable Setting	0~65535; Bit0: channel 1; 1:enable; 0:disable; Bit1: channel 2; ……; Bit15: channel 16	Word	R/W
1049H	Logic "And " Between Alarming Setting	0~255; Bit0: first logic switch; 1:enable; 0:disable; Bit1: second logic switch; ……; Bit7: eighth logic switch	Word	R/W
104aH	Alarming Output to DO1 Setting	0~65535; Bit0: channel 1 output; 1:enable; 0:disable; Bit1: channel 2 output; ……; Bit15: channel 16 output	Word	R/W
104bH	Alarming Output to DO2 Setting	0~65535; The same as previous	Word	R/W
104cH	Alarming Output to DO3 Setting	0~65535; The same as previous	Word	R/W
104dH	Alarming Output to DO4 Setting	0~65535; The same as previous	Word	R/W

Table 32. Single Alarm Settings.

Address	Parameter	Format Code	Range	Data Type	Property
104eH	First Group: Parameter Code	F1	0~79	Word	R/W
104fH	First Group: Comparison Mode	F1	1: greater than; 2: equal to; 3: less than	Word	R/W
1050H	First Group: Setpoint Value	F10~F18	Related with parameters	Word	R/W
1051H	First group: delay	F1	0~3000(*10 ms)	Word	R/W
1052H	First Group: Output to Relay	F1	0: none, 1~8: related relay	Word	R/W
1053H~ 109dH	2nd to 16th Group		Same as the first group	Word	R/W

Table 33. Alarming Parameter Code.

Setting Value	Alarming Object	Setting Value	Alarming Object	Setting Value	Alarming Object
0	Frequency	1	Va	2	Vb
3	Vc	4	Average Phase Voltage	5	Uab
6	Ubc	7	Uca	8	Average Line Voltage
9	Current of Phase A	10	Current of Phase B	11	Current of Phase C
12	Average Current	13	Neutral Current	14	Power of Phase A
15	Power of Phase B	16	Power of Phase C	17	Power of All
18	Reactive Power of Phase A	19	Reactive Power of Phase B	20	Reactive Power of Phase C
21	Reactive Power of All	22	Apparent Power of Phase A	23	Apparent Power of Phase B
24	Apparent Power of Phase C	25	Apparent Power of All	26	PF of A
27	PF of B	28	PF of C	29	PF
30	Voltage Unbalance Factor U_unbl	31	Current Unbalance Factor I_unbl	32	Load Characteristic (R/L/C)
33	THD_V1(V1 or V12)	34	THD_V2(V2 or V31)	35	THD_V3(V3 or V23)
36	Average THD_V	37	THD_I1	38	THD_ I2
39	THD_ I3	40	Average THD_I	41	AI1 Sampling Value
42	AI2 Sampling Value	43	AI3 Sampling Value	44	AI4 Sampling Value
45	Active Power Demand of All	46	Reactive Power Demand of All	47	Apparent Power Demand of All
48	Current Demand of Phase A	49	Current Demand of Phase B	50	Current Demand of Phase C
51	Reversed Phase Sequence	52~79	DI1~DI28		

Note: When reversed phase sequence (51) is selected, whether the value of the comparison mode or setpoint value is set or not, does not affect alarm result and angle of Ub to Ua will be recorded.

Note: When DI (52~79) is selected, whether the value of the comparison mode is set or not, does not affect the alarm result as long as the setpoint value is set to 1, 2 or 3.
1 stands for DI alarm is from OFF to ON, and recovery is from ON to OFF.
2 stands for DI alarm is from ON to OFF, and recovery is from OFF to ON.
3 stands for DI alarm from ON to OFF, and recovery is from OFF to ON and present DI status is recorded.

7.3.5 I/O Modules Settings

I/O module setting changes will be made only if the corresponding I/O modules are installed. No changes will be made otherwise. Please check the I/O module connection status before doing any settings. Function code: 03H for reading, 10H for writing. Please refer to Chapter 5 Extended Modules for more details.

Table 34. I/O Module Settings - PXM1K-11.

Address	Parameter	Default	Range	Data Type	Property
109eH	DI1~6 Type	0	Bit0: DI1, Bit1: DI2, Bit2: DI3, Bit3: DI4, Bit4: DI5, Bit5: DI6, 0: DI,1: pulse counter	Word	R/W
109fH	DI Pulse Constant	0	1~65535	Word	R/W
10a0H	Working Mode of Relays 1 and 2	0	0: control output, 1: alarming output	Word	R/W
10a1H	Output Mode of Relays 1 and 2	0	0: latch, 1: pulse	Word	R/W
10a2H	Pulse Width	50	50~3000 ms	Word	R/W

Table 35. I/O Module Settings - PXM1K-21.

Address	Parameter	Default	Range	Data Type	Property
10a3H	DI7~10 Type	0	Bit0: DI7, Bit1: DI8, Bit2: DI9, Bit3: DI10, 0: DI,1: pulse counter	Word	R/W
10a4H	DI Pulse Constant	0	1~65535	Word	R/W
10a5H	Working Mode of DO	0	0: pulse output; 1: alarming output	Word	R/W
10a6H	DO Pulse Width	20	20~1000 ms	Word	R/W
10a7H	DO1 Output	0	0: none; 1: consumption power (real forward energy); 2: generating power (real reverse energy); 3: absorption reactive power (reactive delivered energy); 4: generating reactive power(reactive received energy)	Word	R/W
10a8H	DO2 Output	0	Same as above	Word	R/W
10a9H	AO1,2 Type	1 or 2	0: 0~20mA, 1: 4~20mA, 2: 0~5V, 3: 1~5V	Word	R/W

Table 36. I/O Module Settings - PXM1K-31.

Address	Parameter	Default	Range	Data Type	Property
10aaH	DI11~14 Type	0	Bit0: DI11,Bit1: DI12, Bit2: DI13, Bit3: DI14, 0: DI, 1: pulse counter	Word	R/W
10abH	DI Pulse Constant	0	1~65535	Word	R/W
10acH	Working Mode of Relays 3 and 4	0	0: control output, 1: alarming output	Word	R/W
10adH	Output Mode of Relays 3 and 4	0	0: latch, 1: pulse	Word	R/W
10aeH	Pulse Width	50	50~3000 ms	Word	R/W
10afH	AI1, 2 Type	1 or 2	0: 0~20 mA, 1: 4~20 mA,, 2: 0~5 V, 3: 1~5 V	Word	R/W

Table 37. I/O Module Settings - PXM1K-12.

Address	Parameter	Default	Range	Data Type	Property
10b0H	DI15~20 t\Type	0	Bit0: DI15, Bit1: DI16, Bit2: DI17, Bit3: DI18, Bit4: DI19, Bit5: DI20 0-DI, 1-pulse counter	Word	R/W
10b1H	DI Pulse Constant (High)	0	1~65535	Word	R/W
10b2H	Working Mode of Relays 5 and 6	0	0: control output, 1: alarming output	Word	R/W
10b3H	Output Mode of Relays 5 and 6	0	0: latch, 1: pulse	Word	R/W
10b4H	Pulse Width	50	50-3000 ms	Word	R/W

Table 38. I/O Module Settings - PXM1K-22.

Address	Parameter	Default	Range	Data Type	Property
10b5H	DI21~24 Type	0	Bit0: DI21, Bit1: DI22, Bit2: DI23, Bit3: DI24 0: DI, 1: pulse counter	Word	R/W
10b6H	DI Pulse Constant	0	1~65535	Word	R/W
10b7H	Working Mode of DO3, 4	0	0: pulse output, 1: alarming output	Word	R/W
10b8H	DO Pulse Width	20	20~1000 ms	Word	R/W
10b9H	DO3 Output	0	0: none; 1: consumption power 2: generating power; 3: consumption reactive power; 4: generating reactive power	Word	R/W
10baH	DO4 Output	0	Same as above	Word	R/W
10bbH	AO3,4 Type	1 or 2	0: 0~20 mA, 1: 4~20 mA, 2: 0~5 V, 3: 1~5 V	Word	R/W

Table 39. I/O Module Settings - PXM1K-32.

Address	Parameter	Default	Range	Data Type	Property
10bcH	DI25~28 Type	0	Bit0: DI25, Bit1: DI26, Bit2: DI27, Bit3: DI28, 0: DI, 1: pulse counter	Word	R/W
10bdH	DI Pulse Constant	0	1~65535	Word	R/W
10beH	Working Mode of Relays 7 and 8	0	0: control output, 1: alarming output	Word	R/W
10bfH	Output Mode of Relays 7 and 8	0	0: latch, 1: pulse	Word	R/W
10c0H	Pulse Width	50	50~3000	Word	R/W
10c1H	AI3, 4 Type	1 or 2	0: 0~20 mA, 1: 4~20 mA, 2: 0~5 V, 3: 1~5 V	Word	R/W

Table 40. Analog Output (AO) Parameter Assignment.

Address	Parameter	Default	Range	Data Type	Property
10c2H	AO1 Transforming Parameter	0	Refer to Table 41	Word	R/W
10c3H	AO2 Transforming Parameter	0	Refer to Table 41	Word	R/W
10c4H	AO3 Transforming Parameter	0	Refer to Table 41	Word	R/W
10c5H	AO4 Transforming Parameter	0	Refer to Table 41	Word	R/W

Table 41. AO Parameter Settings.

Setting Value	Transforming Object
0	Frequency
1	Vc
2	Vb
3	Vc
4	Average phase voltage
5	Uab
6	Ubc
7	Uca
8	Average line voltage
9	Phase A current
10	Phase B current
11	Phase C current
12	Average Current
13	Neutral current
14	Phase A Power
15	Phase B power
16	Phase C power
17	All power
18	Phase A reactive power
19	Phase B reactive power

Setting Value	Transforming Object
20	Phase C reactive power
21	All reactive power
22	Phase A apparent power
23	Phase B apparent power
24	Phase C apparent power
25	All apparent power
26	A PF
27	B PF
28	C PF
29	PF

Table 42. AO Transfer Curve Settings.

Address	Parameter	Default	Range	Data Type	Property
10d0H	AO1 Gradient Number Selection of Input/Output Transfer Curve	1	1: 1 Gradient, 2: 2 Gradient; 3: 3 Gradient	INT	R/W
10d1H	AO1 Following Value Range Setting Start Point		Please see Note	INT	R/W
10d2H	AO1 Following Value Range Setting Point 2			INT	R/W
10d3H	AO1 Following Value Range Setting Point 3			INT	R/W
10d4H	AO1 Following Value Range Setting End Point			INT	R/W
10d5H	AO1 Output Range Setting Start Point		AO type of 0~24 mA or 0~6: 0~4915; AO type of 4~24 mA or 1~6: 819~4915	INT	R/W
10d6H	AO1 Output Range Setting Point 2			INT	R/W
10d7H	AO1 Output Range Setting Point 3			INT	R/W
10d8H	AO1 Output Range Setting End Point			INT	R/W
10d9-10E1H	AO2 Gradient Setting (Same as AO1)		Same as AO1	INT	R/W
10E2H-10EAH	AO3 Gradient Setting (Same as AO1)		Same as AO1	INT	R/W
10EBH-10F3H	AO4 Gradient Setting (Same as AO1)		Same as AO1	INT	R/W

Note: AO Gradient Number Selection of input/output transfer curve. When the number is 1, only AO following value range setting start point, AO following value range setting end point, AO1 output range setting start point, and AO1 output range setting end point should be set.

When the number is 2, only AO following value range setting start point, AO1 following value range setting point 2, AO following value range setting end point, AO1 output range setting start point , AO1 output range setting point 2, and AO1 output range setting end point should be set.

When the number is 3, only AO following value range setting start point, AO1 following value range setting point 2, AO1 following value range setting point 3, and AO following value range setting end point should be set. At the same time, AO1 output range setting start point, AO1 output range setting point 2, AO1 output range setting point 3, and AO1 output range setting end point should be set.

Note: Following value range setting:

AO following value range setting start point, AO1 following value range setting point 2, AO1 following value range setting point 3, and AO following value range setting end point are increasing value, while they should be within range of AO following value. Otherwise, the function of AO will be affected.

Frequency: When selecting 50 or 60 Hz type, frequency range is 45 ~ 65 Hz, real setting value is 4,500 ~ 6,500.

When selecting 400 Hz type, frequency range is 300 ~ 500 Hz, real setting value is 30,000 ~ 50,000.

Table 41. AO Transfer Curve Settings (continued).

Phase voltage V1, V2, V3, and average phase voltage: 0 ~ 480 V, real setting value is 0 ~ 4,800.

Line to line voltage V12, V23, V31, and average line to line voltage: 0 ~ 831 V, real setting value is 0 ~ 8,310.

Current I1, I2, I3, and average current: 0 ~ 10 A, real setting value is 0 ~ 1,0000.

Power Pa, Pb, and Pc: -4,800 ~ 4,800 W, real setting value is -4,800 ~ 4,800.

System power: -14,400~14,400 W, real setting value is -14,400~14,400.

Reactive power Qa, Qb, and Qc: -4,800 ~ 4,800 Var, real setting value is 4,800 ~ 4,800.

System reactive power: -14,400 ~ 14,400 Var, real setting value is -14,400 ~ 14,400.

Apparent power Sa, Sb, and Sc: 0 ~ 4,800 VA, real setting value is 0 ~ 4,800.

System apparent power: 0 ~ 14,400 VA, real setting value is 0 ~ 14,400.

Power factor PFa, PFb, PFc, and system power factor: -1 ~ 1, real setting value is -1,000 ~ 1,000.

Note: AO output range setting:

AO output value range setting start point, AO1 output value range setting point 2, AO1 output value range setting point 3, and AO output value range setting end point are increasing value, while they should be within range of AO output value.

When AO type is 0 ~ 20 mA, the setting value range is 0 ~ 4,915, and the relationship is mA = setting value x 20/4,096.

When AO type is 4 ~ 20 mA, the setting value range is 819 ~ 4,915, and the relationship is mA = setting value x 20/4,096.

When AO type is 0 ~ 5 V, the setting value range is 0~ 4,915, and the relationship is V = setting value x 5/4,096.

When AO type is 1 ~ 5 V, the setting value range is 819~ 4,915, and the relationship is V = setting value x 5/4,096.

7.3.6 100 ms Refresh Metering Parameter

Table 43. 100 ms Refresh Metering Parameters.
Note: Parameters are all secondary values in this table.

Address	Parameter	Code	Relationship	Data Type	Property
3000H~3001H	Frequency	F1	F = Rx	Float	R
3002H~3003H	Phase Voltage V1	F1	U = Rx × (PT1/PT2)	Float	R
3004H~3005H	Phase Voltage V2	F1	U = Rx × (PT1/PT2)	Float	R
3006H~3007H	Phase Voltage V3	F1	U = Rx × (PT1/PT2)	Float	R
3008H~3009H	Average Voltage Vavg	F1	U = Rx × (PT1/PT2)	Float	R
300aH~300bH	Line Voltage V12	F1	U = Rx × (PT1/PT2)	Float	R
300cH~300dH	Line Voltage V23	F1	U = Rx × (PT1/PT2)	Float	R
300eH~300fH	Line Voltage V31	F1	U = Rx × (PT1/PT2)	Float	R
3010H~3011H	Average Line Voltage Vlavg	F1	U = Rx × (PT1/PT2)	Float	R
3012H~3013H	Current I1	F1	I = Rx × (CT1/CT2)	Float	R
3014H~3015H	Current I2	F1	I = Rx × (CT1/CT2)	Float	R
3016H~3017H	Current I3	F1	I = Rx × (CT1/CT2)	Float	R
3018H~3019H	Average Current Iavg	F1	I = Rx × (CT1/CT2)	Float	R
301aH~301bH	Neutral Current In	F1	I = Rx × (CT1/CT2)	Float	R
301cH~301dH	Phase A Power Pa	F1	P = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
301eH~301fH	Phase B Power Pb	F1	P = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
3020H~3021H	Phase C Power Pc	F1	P = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
3022H~3023H	System Power Psum	F1	P = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
3024H~3025H	Phase A Reactive Power Qa	F1	Q = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
3026H~3027H	Phase B Reactive Power Qb	F1	Q = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
3028H~3029H	Phase C Reactive Power Qc	F1	Q = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
302aH~302bH	System Reactive Power Qsum	F1	Q = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
302cH~302dH	Phase A Apparent Power Sa	F1	S = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
3302eH~302fH	Phase B Apparent Power Sb	F1	S = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
3030H~3031H	Phase C Apparent Power Sc	F1	S = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
3032H~3033H	System S Apparent Power Ssum	F1	S = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
3034H~3035H	Phase A Power Factor PFa	F1	PF = Rx	Float	R
3036H~3037H	Phase B Power Factor PFb	F1	PF = Rx	Float	R
3038H~3039H	Phase C Power Factor PFc	F1	PF = Rx	Float	R
303aH~303bH	System Power Factor PFsum	F1	PF = Rx	Float	R

7.3.7 Metering Parameter Address

Basic Analog Measurements

There are two different modes to read basic analog measurements: one is secondary mode, and the other is primary mode. In primary mode, the numerical value in the register of the meter is equal to the real physical value. In secondary mode, the relationship between the numerical value in the register and the real physical value is shown in the Table 44. (Rx is the numerical value in the register of PXM 3000 meter.)

Function code: 03H for reading.

Table 44. Relationship Between the Numerical Value and the Real Physical Value.

Address	Parameter	Code	Relationship	Data Type	Property
4000H~4001H	Frequency	F1	F = Rx	Float	R
4002H~4003H	Phase Voltage V1	F1	U = Rx × (PT1/PT2)	Float	R
4004H~4005H	Phase Voltage V2	F1	U = Rx × (PT1/PT2)	Float	R
4006H~4007H	Phase Voltage V3	F1	U = Rx × (PT1/PT2)	Float	R
4008H~4009H	Average Voltage Vavg	F1	U = Rx × (PT1/PT2)	Float	R
400aH~400bH	Line Voltage V12	F1	U = Rx × (PT1/PT2)	Float	R
400cH~400dH	Line Voltage V23	F1	U = Rx × (PT1/PT2)	Float	R
400eH~400fH	Line Voltage V31	F1	U = Rx × (PT1/PT2)	Float	R
4010H~4011H	Average Line Voltage Vlavg	F1	U = Rx × (PT1/PT2)	Float	R
4012H~4013H	Current I1	F1	I = Rx × (CT1/CT2)	Float	R
4014H~4015H	Current I2	F1	I = Rx × (CT1/CT2)	Float	R
4016H~4017H	Current I3	F1	I = Rx × (CT1/CT2)	Float	R
4018H~4019H	Average Current Iavg	F1	I = Rx × (CT1/CT2)	Float	R
401aH~401bH	Neutral Current In	F1	I = Rx × (CT1/CT2)	Float	R
401cH~401dH	Phase A Power Pa	F1	P = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
401eH~401fH	Phase B Power Pb	F1	P = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
4020H~4021H	Phase C Power Pc	F1	P = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
4022H~4023H	System Power Psum	F1	P = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
4024H~4025H	Phase A Reactive Power Qa	F1	Q = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
4026H~4027H	Phase B Reactive Power Qb	F1	Q = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
4028H~4029H	Phase C Reactive Power Qc	F1	Q = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
402aH~402bH	System Reactive Power Qsum	F1	Q = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
402cH~402dH	Phase A Apparent Power Sa	F1	S = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
402eH~402fH	Phase B Apparent Power Sb	F1	S = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
4030H~4031H	Phase C Apparent Power Sc	F1	S = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
4032H~4033H	System Apparent Power Ssum	F1	S = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
4034H~4035H	Phase A Power Factor PFa	F1	PF = Rx	Float	R
4036H~4037H	Phase B Power Factor PFb	F1	PF = Rx	Float	R
4038H~4039H	Phase C Power Factor PFc	F1	PF = Rx	Float	R
403aH~403bH	System Power Factor PFsum	F1	PF = Rx	Float	R
403cH~403dH	Voltage Unbalance Factor U_unbl	F1	Unbalance = Rx × 100%	Float	R
403eH~403fH	Current Unbalance Factor I_unbl	F1	Unbalance = Rx × 100%	Float	R
4040H~4041H	Load Characteristic (L/C/R)	F1	76.0/67.0/82.0(ASCII)	Float	R
4042H~4043H	Power Demand	F1	P = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
4044H~4045H	Reactive Power Demand	F1	P = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
4046H~4047H	Apparent Power Demand	F1	P = Rx × (PT1/PT2) × (CT1/CT2)	Float	R

Note: When CT2 = 5, calculating use CT2 = 5 in the formula, others use CT2 = 1 in the formula.

Table 45. Standard Registers.

Address	Parameter	Relationship	Data Type	Property
1FFEH-1FFFH	Product ID	200E	Uint32	R
2001H-2001H	Primary/Secondary Status	See Note Following Table	Uint16	R
2002H-2002H	Cause-Of-Status	0-79	Uint16	R
2003H-2004H	Current - IA	I = Rx × (CT1/CT2)	Float	R
2005H-2006H	Current - IB	I = Rx × (CT1/CT2)	Float	R
2007H-2008H	Current - IC	I = Rx × (CT1/CT2)	Float	R
2009H-200AH	Current - IN	I = Rx × (CT1/CT2)	Float	R
200BH-200CH	Current - Average	I = Rx × (CT1/CT2)	Float	R
200DH-200EH	Voltage - VAB	U = Rx × (PT1/PT2)	Float	R
200FH-2010H	Voltage - VBC	U = Rx × (PT1/PT2)	Float	R
2011H-2012H	Voltage - VCA	U = Rx × (PT1/PT2)	Float	R
2013H-2014H	Voltage - VLL Average	U = Rx × (PT1/PT2)	Float	R
2015H-2016H	Voltage - VAN	U = Rx × (PT1/PT2)	Float	R
2017H-2018H	Voltage - VBN	U = Rx × (PT1/PT2)	Float	R
2019H-201AH	Voltage - VCN	U = Rx × (PT1/PT2)	Float	R
201BH-201CH	Voltage - VLN Average	U = Rx × (PT1/PT2)	Float	R
201DH-201EH	Real Power - Watts	P = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
2021H-2022H	Power Factor	PF = Rx	Float	R
2023H-2024H	Frequency	F = Rx	Float	R
2025H-2026H	Real Power - Phase A Watts	P = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
2027H-2028H	Real Power - Phase B Watts	P = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
2029H-202AH	Real Power - Phase C Watts	P = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
202BH-202CH	Apparent Power - Phase A VA	S = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
202DH-202EH	Apparent Power - Phase B VA	S = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
202FH-2030H	Apparent Power - Phase C VA	S = Rx × (PT1/PT2) × (CT1/CT2)	Float	R
2031H-2032H	Power Factor - Phase A	PF = Rx	Float	R
2033H-2034H	Power Factor - Phase B	PF = Rx	Float	R
2035H-2036H	Power Factor - Phase C	PF = Rx	Float	R
2037H-2037H	Forward Energy Scaling = 2 for F4; 0 for F7	0, 2	Uint16	R
2038H-2039H	Forward Energy	F4, F7	Uint32	R
203AH-203AH	Reverse Energy Scaling, = 2 for F4; 0 for F7	0, 2	Uint16	R
203BH-203CH	Reverse Energy	F4, F7	Uint32	R
203DH-203DH	Sum Total Energy Scaling = 2 for F4; 0 for F7	0, 2	Uint16	R
203EH-203FH	Sum Total Energy	F4, F7	Uint32	R

Note: Register 2001H

Primary Status Code	Primary Status Definition	Secondary Status Code	Secondary Status Definition
4	Alarmed	1	Not applicable
9	Operational	7	Powered up

Real Time Energy Measurement

Data stored in this block can be preset or cleared.

Function code: 03H for reading, 10H for writing. Data type: Dword.

It can be configured as primary energy or secondary energy according to the user setting. Please refer to Table 27 for more details about the relationship between numerical value in the register and the real physical value.

Note: Max and min values are always in secondary, so relationship must be applied.

Table 46. Real Time Energy Measurement.

Address	Parameter	Code	Range	Data Type	Property
4048H~4049H	Energy IMP	F4/F7	0~999999999	Dword	R/W
404aH~404bH	Energy EXP	F4/F7	0~999999999	Dword	R/W
404cH~404dH	Reactive Energy IMP	F5/F8	0~999999999	Dword	R/W
404eH~404fH	Reactive Energy EXP	F5/F8	0~999999999	Dword	R/W
4050H~4051H	Energy TOTAL	F4/F7	0~999999999	Dword	R/W
4052H~4053H	Energy NET	F4/F7	0~999999999	Dword	R/W
4054H~4055H	Reactive Energy TOTAL	F5/F8	0~999999999	Dword	R/W
4056H~4057H	Reactive Energy NET	F5/F8	0~999999999	Dword	R/W
4058H~4059H	Apparent Energy	F6/F9	0~999999999	Dword	R/W
4620H~4621H	Phase A Energy IMP	F4/F7	0-999999999	Dword	R/W
4622H~4623H	Phase A Energy EXP	F4/F7	0-999999999	Dword	R/W
4624H~4625H	Phase B Energy IMP	F4/F7	0-999999999	Dword	R/W
4626H~4627H	Phase B Energy EXP	F4/F7	0-999999999	Dword	R/W
4628H~4629H	Phase C Energy IMP	F4/F7	0-999999999	Dword	R/W
462AH~462BH	Phase C Energy EXP	F4/F7	0-999999999	Dword	R/W
462CH~462DH	Phase A Reactive Energy IMP	F5/F8	0-999999999	Dword	R/W
462EH~462FH	Phase A Reactive Energy EXP	F5/F8	0-999999999	Dword	R/W
4630H~4631H	Phase B Reactive Energy IMP	F5/F8	0-999999999	Dword	R/W
4632H~4633H	Phase B Reactive Energy EXP	F5/F8	0-999999999	Dword	R/W
4634H~4635H	Phase C Reactive Energy IMP	F5/F8	0-999999999	Dword	R/W
4636H~4637H	Phase C Reactive Energy EXP	F5/F8	0-999999999	Dword	R/W
4638H~4639H	Phase A Apparent Energy	F6/F9	0-999999999	Dword	R/W
463AH~463BH	Phase B Apparent Energy	F6/F9	0-999999999	Dword	R/W
463CH~463DH	Phase C Apparent Energy	F6/F9	0-999999999	Dword	R/W

Harmonics

THD, harmonics, odd HD, even HD, crest factor, THFF, K factor, etc., are all stored here. The data type is "Word". Voltage parameters refer to line voltage when it is set to "2 LL/3 LL" and phase voltage for others. Function code: 03H for reading.

Table 47. Harmonics.

Address	Parameter	Code	Range	Data Type	Property
The following are the THD of voltage and current.					
405aH	THD_V1 of V1(V12)	F18	> = 0	Word	R
405bH	THD_V1 of V2(V31)	F18	> = 0	Word	R
405cH	THD_V1 of V3(V23)	F18	> = 0	Word	R
405dH	Average THD_V	F18	> = 0	Word	R
405eH	THD_I1	F18	> = 0	Word	R
405fH	THD_I2	F18	> = 0	Word	R
4060H	THD_I3	F18	> = 0	Word	R
4061H	Average THD_I	F18	> = 0	Word	R
Voltage harmonics, even HD, odd HD, and crest factor are shown as below.					
4062H~407fH	Harmonics of V1(V12) (the 2nd to 31st)	F19	> = 0	Word	R
4500H~451fH	Harmonics of V1(V12) (the 32nd to 63rd)	F19	> = 0	Word	R
4080H	Odd HD of V1(V12)	F20	> = 0	Word	R
4081H	Even HD of V1(V12)	F21	> = 0	Word	R
4082H	Crest Factor of V1 (V12)	F22	0~65535	Word	R
4083H	THFF of V1 (V12)	F24	> = 0	Word	R
4520H~453fH	Harmonics of V2 (V31) (the 32nd to 63rd)	Same as V1		Word	R
40A2H	Odd HD of V2(V31)	Same as V1		Word	R
40A3H	Even HD of V2(V31)	Same as V1		Word	R
40A4H	Crest Factor of V2(V31)	Same as V1		Word	R
40A5H	THFF of V2(V31)	Same as V1		Word	R
40A6H~40C3H	Harmonics of V3(V23) (the 2nd to 31st)	Same as V1		Word	R
4540H~455fH	Harmonics of V3 (V23) (the 32nd to 63rd)	Same as V1		Word	R
40C4H	Odd HD of V3(V23)	Same as V1		Word	R
40C5H	Even HD of V3(V23)	Same as V1		Word	R
40C6H	Crest Factor of V3(V23)	Same as V1		Word	R
40C7H	THFF of V3(V23)	Same as V1		Word	R
Current harmonics, even HD, odd HD, and K factor are shown as below.					
40c8H~40e5H	Harmonics of I1 (the 2nd to 31st)	F19	> = 0	Word	R
4560H~457fH	Harmonics of I1 (the 32nd to 63rd)	F19	> = 0	Word	R
40e6H	Odd HD of I1	F20	> = 0	Word	R
40e7H	Even HD of I1	F21	> = 0	Word	R
40e8H	K Factor of I1	F23	0~65535	Word	R
40E9H~4106H	Harmonics of I2 (the 2nd to 31st)	Same as I1		Word	R
4580H~459fH	Harmonics of I2 (the 32nd to 63rd)	Same as I1		Word	R
4107H	Odd HD of I2	Same as I1		Word	R
4108H	Even HD of I2	Same as I1		Word	R
4109H	K Factor of I2	Same as I1		Word	R
410AH~41 H	Harmonics of I3 (the 2nd to 31st)	Same as I1		Word	R
45a0H~45bfH	Harmonics of I3 (the 32nd to 63rd)	Same as I1		Word	R

Table 47. Harmonics. (continued).

Address	Parameter	Code	Range	Data Type	Property
4128H	Odd HD of I3	Same as I1		Word	R
4129H	Even HD of I3	Same as I1		Word	R
412AH	K Factor of I3	Same as I1		Word	R

Note: When selecting 400 Hz type, harmonics support 2nd ~ 15th.

MAX/MIN Records

MAX/MIN value and time stamp. Function code: 03H for reading.

Table 48. MIN/MAX Values and Time Stamp.

Address	Parameter	Code	Range	Data Type	Property
4136H	MAX of V1	F11	-32768~32767	int	R
4137H~413cH	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
413dH	MAX of V2	F11	-32768~32767	int	R
413eH~4143H	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
4144H	MAX of V3	F11	-32768~32767	int	R
4145H~414aH	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
414bH	MAX of V12	F11	-32768~32767	int	R
414cH~4151H	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
4152H	MAX of V23	F11	-32768~32767	int	R
4153H~4158H	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
4159H	MAX of V31	F11	-32768~32767	int	R
415aH~415fH	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
4160H	MAX of I1	F12	-32768~32767	int	R
4161H~4166H	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
4167H	MAX of I2	F12	-32768~32767	int	R
4168H~416dH	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
416eH	MAX of I3	F12	-32768~32767	int	R
416fH~4174H	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
4175H	MAX of system power	F13	-32768~32767	int	R
4176H~417bH	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
417cH	MAX of system reactive power	F14	-32768~32767	int	R
417dH~4182H	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
4183H	MAX of system apparent power	F15	-32768~32767	int	R
4184H~4189H	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
418aH	MAX of power factor	F16	-32768~32767	int	R
418bH~4190H	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
4191H	MAX of frequency	F10	-32768~32767	int	R
4192H~4197H	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
4198H	MAX of power demand	F13	-32768~32767	int	R
4199H~419eH	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
419fH	MAX of reactive power demand	F14	-32768~32767	int	R

Table 48. MIN/MAX Values and Time Stamp (continued).

Address	Parameter	Code	Range	Data Type	Property
41a0H~41a5H	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
41a6H	MAX of apparent power demand	F15	-32768~32767	int	R
41a7H~41acH	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
41adH	MAX of voltage unbalance factor	F17	-32768~32767	int	R
41aeH~41b3H	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
41b4H	MAX of current unbalance factor	F17	-32768~32767	int	R
41b5H~41baH	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
41bbH	MAX of V1(V12) THD	F18	-32768~32767	int	R
41bcH~41c1H	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
41c2H	MAX of V2(V31) THD	F18	-32768~32767	int	R
41c3H~41c8H	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
41c9H	MAX of V3(V23) THD	F18	-32768~32767	int	R
41caH~41cfH	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
41d0H	MAX of I1 THD	F18	-32768~32767	int	R
41d1H~41d6H	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
41d7H	MAX of I2 THD	F18	-32768~32767	int	R
41d8H~41ddH	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
41deH	MAX of I3 THD	F18	-32768~32767	int	R
41dfH~41e4H	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R

The addresses for the MIN value of the above parameters are located in 41e5H to 4293H. They have the same format as the MAX value.

Note: The MAX and MIN Frequency value should use word data type (0~65535).

Sequence Component

U1 (U12) and I1 consist of a real part and a complex part. They have a positive sequence, negative sequence, and zero sequence. Data type is "int". Function Code: 03H for reading.

Table 49. Sequence Component.

Address	Parameter	Code	Range	Data Type	Property
4294H	Positive sequence real part of UA	F11	-32768~32767	int	R
4295H	Positive sequence complex part of UA	F11	-32768~32767	int	R
4296H	Negative sequence real part of UA	F11	-32768~32767	int	R
4297H	Negative sequence complex part of UA	F11	-32768~32767	int	R
4298H	Zero sequence real part of UA	F11	-32768~32767	int	R
4299H	Zero sequence complex part of UA	F11	-32768~32767	int	R
429aH	Positive sequence real part of IA	F12	-32768~32767	int	R
429bH	Positive sequence complex part of IA	F12	-32768~32767	int	R
429cH	Negative sequence real part of IA	F12	-32768~32767	int	R
429dH	Negative sequence complex part of IA	F12	-32768~32767	int	R
429eH	Zero sequence real part of IA	F12	-32768~32767	int	R
429fH	Zero sequence complex part of IA	F12	-32768~32767	int	R

Phase Angle

All voltage and current's phase angles corresponding to V1 (V12) are stored here. The user can find out the phase sequence according to them. Data type is "Word". Function Code: 03H for reading.

Table 50. Phase Angle.

Address	Parameter	Code	Range	Data Type	Property
42a0H	Phase angle of V2 to V1 Phase angle of V23 to V12	F25	0~3600	Word	R
42a1H	phase angle of V3 to V1; phase angle of V31 to V12	F25	0~3600	Word	R
42a2H	phase angle of I1 to V1; phase angle of I1 to V12	F25	0~3600	Word	R
42a3H	Phase angle of I2 to V1; Phase angle of I2 to V12	F25	0~3600	Word	R
42a4H	Phase angle of I3 to V1; Phase angle of I3 to V12	F25	0~3600	Word	R
42a5H	Reserved			Word	R
42a6H	Reserved			Word	R
42a7H	Reserved			Word	R
42a8H	Reserved			Word	R

Alarming Records

There are 16 groups of records with the same format. Function Code: 03H for reading, 10H for writing. Please refer to Chapter 4 for more details.

Table 51. Alarm Records.

Address	Parameter	Code	Range	Data Type	Property
42a9H	First group: alarming status	F1	0~65535	Word	R
42aaH	First group: alarming parameter code	F1	0~79	Word	R
42abH	First group: over/under limit or reset value	F10~F18	Related with parameters	Word	R
42acH~42b2H	First group: time stamp: yyyy:mm:dd:hh:mm:ss:ms	F3		Word	R
42b3H~42bcH	Second group	Same as the first group			
42bdH~42c6H	Third group	Same as the first group			
42c7H~42d0H	Fourth group	Same as the first group			
42d1H~42daH	Fifth group	Same as the first group			
42dbH~42e4H	Sixth group	Same as the first group			
42e5H~42eeH	Seventh group	Same as the first group			
42efH~42f8H	Eighth group	Same as the first group			
42f9H~4302H	Ninth group	Same as the first group			
4303H~430cH	Tenth group	Same as the first group			
430dH~4316H	Eleventh group	Same as the first group			
4317H~4320H	Twelfth group	Same as the first group			
4321H~432aH	Thirteenth group	Same as the first group			
432bH~4334H	Fourteenth group	Same as the first group			
4335H~433eH	Fifteenth group	Same as the first group			
433fH~4348H	Sixteenth group	Same as the first group			

Counting Number of I/O Modules

DI are arranged according to expanded I/O module addresses. The user can check the counting number of the DI along with those modules. The DI counting records are stored in non-volatile memory and will not be erased during power off. They can be reset via communication and the front panel. Data type is "Dword". Function Code: 03H for reading.

Table 52. DI Counting Number.

Address	Parameter	Code	Range	Data Type	Property
PXM1K-11					
4349H~434aH	DI1 pulse counter number	F1	0~4294967295	Dword	R
434bH~434cH	DI2 pulse counter number	F1	0~4294967295	Dword	R
434dH~434eH	DI3 pulse counter number	F1	0~4294967295	Dword	R
434fH~4350H	DI4 pulse counter number	F1	0~4294967295	Dword	R
4351H~4352H	DI5 pulse counter number	F1	0~4294967295	Dword	R
4353H~4354H	DI6 pulse counter number	F1	0~4294967295	Dword	R
PXM1K-21					
4355H~4356H	DI7 pulse counter number	F1	0~4294967295	Dword	R
4357H~4358H	DI8 pulse counter number	F1	0~4294967295	Dword	R
4359H~435aH	DI9 pulse counter number	F1	0~4294967295	Dword	R
435bH~435cH	DI10 pulse counter number	F1	0~4294967295	Dword	R
PXM1K-31					
435dH~435eH	DI11 pulse counter number	F1	0~4294967295	Dword	R
435fH~4360H	DI12 pulse counter number	F1	0~4294967295	Dword	R
4361H~4362H	DI13 pulse counter number	F1	0~4294967295	Dword	R
4363H~4364H	DI14 pulse counter number	F1	0~4294967295	Dword	R
PXM1K-12					
4365H~4366H	DI15 pulse counter number	F1	0~4294967295	Dword	R
4367H~4368H	DI16 pulse counter number	F1	0~4294967295	Dword	R
4369H~436aH	DI17 pulse counter number	F1	0~4294967295	Dword	R
436bH~436cH	DI18 pulse counter number	F1	0~4294967295	Dword	R
436dH~436eH	DI19 pulse counter number	F1	0~4294967295	Dword	R
436fH~4370H	DI20 pulse counter number	F1	0~4294967295	Dword	R
PXM1K-22					
4371H~4372H	DI21 pulse counter number	F1	0~4294967295	Dword	R
4373H~4374H	DI22 pulse counter number	F1	0~4294967295	Dword	R
4375H~4376H	DI23 pulse counter number	F1	0~4294967295	Dword	R
4377H~4378H	DI24 pulse counter number	F1	0~4294967295	Dword	R
PXM1K-32					
4379H~437aH	DI25 pulse counter number	F1	0~4294967295	Dword	R
437bH~437cH	DI26 pulse counter number	F1	0~4294967295	Dword	R
437dH~437eH	DI27 pulse counter number	F1	0~4294967295	Dword	R
437fH~4380H	DI28 pulse counter number	F1	0~4294967295	Dword	R

Analog Input (AI)

The Modbus register reading of AI is mapped to the range of 0 ~ 4,095 according to its sampling value using an algorithm. Data type is "Word". Function Code: 03H for reading. Please refer to Chapter 5 - "Extended Modules" for more details.

Table 53. AI Input Value.

Address	Parameter	Code	Range	Data Type	Property
4385H	AI1 sampling value	F1	0~4095	Word	R
4386H	AI2 sampling value	F1	0~4095	Word	R
4387H	AI3 sampling value	F1	0~4095	Word	R
4388H	AI4 sampling value	F1	0~4095	Word	R

Analog Output (AO)

The Modbus register reading of AO is the actual value of the output. There are two output options for AO: V or mA. Out-of-limit or Data type is "Float". Function Code: 03H for reading. Please refer to Chapter 5 - "Extended Modules" for more details.

Table 54. AO Output.

Address	Parameter	Code	Range	Data Type	Property
4389H - 438aH	Value of A01	F1		Float	R
438bH - 438cH	Value of A02	F1		Float	R
438dH - 438eH	Value of A03	F1		Float	R
438fH - 4390H	Value of A04	F1		Float	R

Sequence of Event (SOE) Records

There are 20 groups of records with the same format. Function Code: 03H for reading. Before gathering SOE records, the selected I/O module must be SOE enabled. If the SOE is enabled but I/O module is not connected, SOE record logs will not be collected. Please refer to Chapter 5 - "Extended Modules" for more details.

Table 55. SOE Records.

Address	Parameter	Code	Range	Data Type	Property
4399H~439fH	First group: time stamp: yyyy:mm:dd:hh:mm:ss:ms	F3		Word	R
43a0H	First group: DI status	F1		Word	R
43a1H~4438H	2nd to 20th group			Word	R
4439H	I/O module of SOE	F1	0: none; 1: PXM1K-11; 2: PXM1K-21; 3: PXM1K-31; 4: PXM1K-12; 5: PXM1K-22; 6: PXM1K-32	Word	R

Current Demand

Includes real-time current demand, the maximum current demand, and time of occurrence. Function Code: 03H for reading.

Table 56. Current Demand.

Address	Parameter	Code	Range	Data Type	Property
4600H - 4601H	Phase A current demand	F1	I = Rx × (CT1/CT2)	Float	R
4602H - 4603H	Phase B current demand	F1	I = Rx × (CT1/CT2)	Float	R
4604H - 4605H	Phase C current demand	F1	I = Rx × (CT1/CT2)	Float	R
4606H	Max of Phase A current demand	F12	0-65535	int	R
4607 - 460cH	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
460dH	Max of Phase B current demand	F12	0-65535	int	R
460e - 4613H	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R
4614H	Max of Phase C current demand	F12	0-65535	int	R
4615 - 461AH	Time stamp: yyyy:mm:dd:hh:mm:ss	F3	Time	int	R

Digital Input (DI) Status

Current DI status, if the related I/O module is not connected, will be set to 0. Function Code: 02H for reading.

Table 57. DI Status.

Address	Parameter	Range	Data Type
PXM1K-11			
0000H	DI1	1 = ON, 0 = OFF	Bit
0001H	DI2	1 = ON, 0 = OFF	Bit
0002H	DI3	1 = ON, 0 = OFF	Bit
0003H	DI4	1 = ON, 0 = OFF	Bit
0004H	DI5	1 = ON, 0 = OFF	Bit
0005H	DI6	1 = ON, 0 = OFF	Bit
PXM1K-21			
0006H	DI7	1 = ON, 0 = OFF	Bit
0007H	DI8	1 = ON, 0 = OFF	Bit
0008H	DI9	1 = ON, 0 = OFF	Bit
0009H	DI10	1 = ON, 0 = OFF	Bit
PXM1K-31			
000aH	DI11	1 = ON, 0 = OFF	Bit
000bH	DI12	1 = ON, 0 = OFF	Bit
000cH	DI13	1 = ON, 0 = OFF	Bit
000dH	DI14	1 = ON, 0 = OFF	Bit
PXM1K-12			
000eH	DI15	1 = ON, 0 = OFF	Bit
000fH	DI16	1 = ON, 0 = OFF	Bit
0010H	DI17	1 = ON, 0 = OFF	Bit
0011H	DI18	1 = ON, 0 = OFF	Bit
0012H	DI19	1 = ON, 0 = OFF	Bit
0013H	DI20	1 = ON, 0 = OFF	Bit
PXM1K-22			
0014H	DI21	1 = ON, 0 = OFF	Bit

Table 57. DI Status. (continued).

Address	Parameter	Range	Data Type
0015H	DI22	1 = ON, 0 = OFF	Bit
0016H	DI23	1 = ON, 0 = OFF	Bit
0017H	DI24	1 = ON, 0 = OFF	Bit
PXM1K-32			
0018H	DI25	1 = ON, 0 = OFF	Bit
0019H	DI26	1 = ON, 0 = OFF	Bit
001aH	DI27	1 = ON, 0 = OFF	Bit
001bH	DI28	1 = ON, 0 = OFF	Bit

Relay Status

Function Code: 01H for reading, 05H for controlling output.

Table 58. Relay Status.

Address	Parameter	Range	Data type
PXM1K-11			
0000H	Relay1	1 = ON, 0 = OFF	Bit
0001H	Relay2	1 = ON, 0 = OFF	Bit
PXM1K-31			
0002H	Relay3	1 = ON, 0 = OFF	Bit
0003H	Relay4	1 = ON, 0 = OFF	Bit
PXM1K-12			
0004H	Relay5	1 = ON, 0 = OFF	Bit
0005H	Relay6	1 = ON, 0 = OFF	Bit
PXM1K-32			
0006H	Relay7	1 = ON, 0 = OFF	Bit
0007H	Relay8	1 = ON, 0 = OFF	Bit

7.3.8 Data Logging

Data Logging Setting

In order to generate historical logs for the selected parameters, users should program the meter so that the selected parameters from the corresponding Modbus registers can be copied to the historical log record. Since certain parameters occupy two registers, to supplement this, the programmable settings for the historical logs contain a list of descriptors. Each descriptor lists the number of Modbus registers for the specified parameter. By combining these two lists, the historical log record can be interpreted.

For example:

Registers 4002H and 4003H are programmed to be recorded by the historical log. Since two registers are used, the corresponding descriptor is set as 2. These registers program the log to record "Volts AN."

The historical log programmable settings are comprised of three blocks, one for each log. Each log works identically to each other. Therefore, only historical log 1 is described here. All register addresses in this section are shown within the address range of historical log 1.

1100H - 11DFH (Historical Log 1)

11C0H - 127FH (Historical Log 2)

1280H - 133FH (Historical Log 3)

Block Size: 192 registers per log (384 bytes)

Data Log Settings' Address Map.

Table 59. Data Log Settings' Address Map.

1100H - 1101H	1102H – 1176H	1177H – 11b1H	11B2H - 11B5H
Header	Register List	Item Descriptor List	Logging Timer setting

The following are the details.

1. Header:

Registers - 1100H-1101H

Size - 2 Registers

Table 60. Header Description

Address	1100H		1101H	
Byte	0 (low byte)	1 (high byte)	2 (low byte)	3 (high byte)
Value	Sectors	Registers	Interval	

Registers: The number of registers to log in the record range from 0 - 117. The size of the record in memory is [12 + (Registers x 2)].

Sectors: The number of memory sectors allocated to this log. Each sector is 64 kbyte in size. 63 sectors are available for allocation among the three historical logs. Valid allocation range is from 0 - 63 (when the sector is set 0, this log is disabled).

Interval: The data capture interval for the historical log records. Valid time interval can be set from 0 - 1,440 minutes. When the interval is set to 0, this log is disabled.

Note: When Sectors or Register or Interval is zero, this log is disabled.

2. Register List:

Registers: 1102H - 1176H

Size: One or two register(s) per parameter; 117 available registers per historical log.

The register list controls which Modbus registers are recorded in each historical log record. Since many parameters, such as voltage, energy, etc., take up more than one register, multiple registers are allocated for those parameters.

For example:

In order to record "Volts AN" into the historical log, Volts AN's Modbus address (4002H and 4003H) are assigned and programmed to the log record list so that information can be stored into the historical log registers.

- Each unused register item should be set to 0000H or FFFFH to indicate no parameters are associated with them.
- The actual size of the record, and the number of items in the register list which are used, is determined by the registers in the header.
- Valid register address ranges that can be recorded in the historical log registers are 4000H-412BH, 4294H-42A8H, 4349H-4398H, and 4500H-461BH.

3. Item Descriptor List:

Registers: 1177H–11B1H

Size: One byte per item, 117 bytes (59 registers)

While the register list describes what to log, the item descriptor list describes how to interpret that information. Each descriptor describes how many Modbus addresses are used to describe a parameter. Either one or two addresses will be used for each parameter.

For example:

If the first descriptor is two, and the second descriptor is one, then the first two register items belong to the 1st descriptor, and the 3rd register item belongs to the 2nd descriptor.

Note: As can be seen from the example above, it is not a one-to-one relation between the register list and the descriptor list. A single descriptor may refer to two register items.

4. Logging Timer Setting

If data logging only records one period of data, or only starts from one specific time, the corresponding time and logging mode should be set so the data logging function is successful.

Modbus address 11B2H is used as the logging mode select. When value is 0, the logging mode is set to mode 1, without time setting. When value is 1, the logging mode is set to mode 2, so 11B3H-11B5H (start year, month, day, hour, minute, and second) and 11B6H-11B8H (end year, month, day, hour, minute, and second) should be set. When the value is 2, the logging mode is set to mode 3, so only 11B4H (hour) and 11B5H (minute) should be set. Refer to Chapter 4 for more information on how data logging works.

Registers 11B3H-11B5H (start time)

11B6H-11B8H (end time)

Size 6 Registers

Table 61. Logging Timer Description

Byte	0	1	2	3	4	5
value	month	year	hour	day	second	minute

Log Status Block

The log status block describes the current status of the log in question.

Table 62. Log Status Block.

Address	Parameter	Range	Data Type	Property
6100H ~ 6101H	Max Records	0– 294903	Dword	R
6102H ~ 6103H	Used Records	1– 294903	Dword	R
6104H	Record Size	14 – 246	Word	R
6105H	Reserved		Word	R
6106H ~ 6108H	First Record Time Stamp		Word	R
6109H ~ 610bH	Last Record Time Stamp		Word	R
6200H ~ 620bH	Data Logging 2 status	Same as the first group		
6300H ~ 630bH	Data Logging 3 status	Same as the first group		

Max Records: The maximum number of records the log can hold given the record size and sector allocation.

Used Records: The number of records stored in the log. This number will equal the Max Records when the log has filled. This value will be set to 1 when the log is reset.

Record Size: The number of bytes in this record, including the time stamp. The record's format in the meter is:

Record number (4 bytes) + time stamp (6 bytes) + [data1 ~ dataN] (2 Nbytes) + CRC(2 bytes).

First Record Time stamp: Time stamp of the oldest record.

Last Record Time stamp: Time stamp of the newest record.

Log Retrieval Block

The log retrieval block consists of two parts: the header and the window. The header is used to verify the data shown within the requested log window. The window is a sliding block of data that can be used to access any record in the specified log.

Registers 6000H-6003H

Size 4 Registers

Table 63. Log Retrieval.

Address	Parameter	Property	Format	Description
6000H	Log type	R/W	Nnnnnnnn	log type
			ssssssss	reserve
6001H	Record number,	R/W	Nnnnnnnn	record number
	status		wwwwwwww	status
6002H ~ 6003H	offset	R/W		
6004H ~ 607eH	window	R		

Log type: The log to be retrieved. Write this value to set which log is being retrieved.

0 - Historical Log 1

1 - Historical Log 2

2 - Historical Log 3

Records number: The number of records that fit within a window. This value is configurable. Any number less than a full window may be used. This number tells the retrieving program how many records to expect to be retrieved in the window:

(Record number x record size) = bytes used in the window.

This value should be ((123 x 2) / Record Size), rounded down. The greater this number is, the faster the retrieving speed is.

For example, with a record size of 50:

Records number = ([123 x 2] / 50) = 4.92 ≈ 4.

Status: The status of the current window. Since the time to prepare a window may exceed an acceptable Modbus delay (one second), this acts as a ready status flag to notify when the window is ready for retrieval. When this value indicates that the window is not ready, the data in the window should be ignored. Window Status is read-only; any writes are ignored.

This value also indicates the memory erasing status when setting the date logging settings.

bH Window is Ready

FFH Window is Not Ready

aaH Memory is Erasing

bbH Memory Erasing is Finished

CXH Register List is Set as Error.

X: bit0 1, Register List is Set as Error in data logging 1;

bit1 1, Register List is Set as Error in data logging 2;

bit2 1, Register List is Set as Error in data logging 3.

For example, 0 x C6H, Register Lists are an Error in data logging 2 and 3.

Offset: The offset of the record number of the first record in the data window and the record number of the "first record time stamp". Setting this value controls which records will be available in the data window. When the log is retrieved, the first (oldest) record is "latched". This means that offset 0 will always point to the oldest record at the time of latching.

Window: The actual data of the records, arranged according to the above settings.

Note: If the logging timer is disabled, the first recording sector will be erased when the log is full. Therefore, the user should not read the whole log when the used record number is near to the max record number. Under this condition, the user should read the "Used Records" field and compare it to the previous "Used Records" field from the last reading before retrieving the information and reading the window.

If the current "Used Records" field is greater than the "Used Records" field from the last reading, and if the "Offset" field is less than the difference between the current and previous "Used Records" field, the first sector has been erased and the difference between the "Used Records" field should be subtracted from the recording number.

If the "Offset" field is greater than the difference between the current and previous "Used Records" field, the "Offset" number should be subtracted from the recording number.

To avoid this situation, the user should read the log before it is almost full.

For example: Data logging 1 has 3 sectors and each has 448 records, and the total records are 1,344. If the user presses the "Read All" button when the "Used Records" number is at 1,340, and if the first sector is erased before the information is transferred to the computer, the data stored in this sector is erased permanently and cannot be retrieved. If the records from the first sector can be retrieved before it gets erased, the new value of "Offset" will be equal to the original "Offset" field minus the value of the difference between the current and previous "Used Records" fields.

Data Logging Operation Examples

The following example illustrates a data logging operation. The example makes the following assumptions:

- The log is Historical Log 1.
- The Log contains VAN, VBN, VCN (12 bytes), the interval is 1min, the sectors are 10, the registers are 6, and the logging timer function is disabled.
- Retrieval is starting at record offset 0 (oldest record).
- No new records are recorded to the log during the log retrieval process.

A. Data Logging Settings

Now set the data log 1 according to the assumptions:

1. Set the data log with VAN, VBN, VCN. Here we should set their modbus address 0 x 4002, 0 x 4003, 0 x 4004, 0 x 4005, 0 x 4006 and 0 x 4007 to 0 x 1102, 0 x 1103, 0 x 1104, 0 x 1105, 0 x 1106, and 0 x 1107. The descriptor is 2, so set the 0 x 0202 and 0 x 0200 to 0 x 1177 and 0 x 1178.
2. The register is 6 and sector is 10, so we set 0 x 060A to 0 x 1100.
3. The interval is 1min , so set the 0 x 0001 to 0 x 1101.
4. The logging timer function is disabled, so set the 0 to 0 x 11b9.

B. Log Retrieval Procedure

The following procedure documents how to retrieve a single log from the oldest record to the newest record.

1. Compute the number of records per window, as follows:

   Records Per Window = (246 \ RecordSize) = 246\24 = 10.

2. Write the Records per window and Record offset. In this example, set the 0 x 0A0B and 0 x 0000 to 0 x 6001d and 0 x 6002.

   This step tells the meter what data to return in the window.

3. Read the record window status from 0 x 6001.
   - If the Window Status is 0 x FF, go to Step 2.
   - If the Window Status is 0 x 0B, read the data window.

4. Read the data window and compute the next Expected Record offset.
   - Compute the next expected record offset by adding the Records Per Window and go to Step 2.
   - If there are no remaining records after the current record window, stop reading.

7.3.9 Time of Use (TOU)

Data Address of TOU Energy

The data address saves the parameter of energy, which includes the data address of the last month TOU energy, data address of current month TOU energy, data address of TOU parameter setting, and data address of TOU default parameter. Except for the data address of the TOU default parameter, the data address is read with 03 code, preset with 16 code.

Figure 79. Division Plan of TOU Energy.

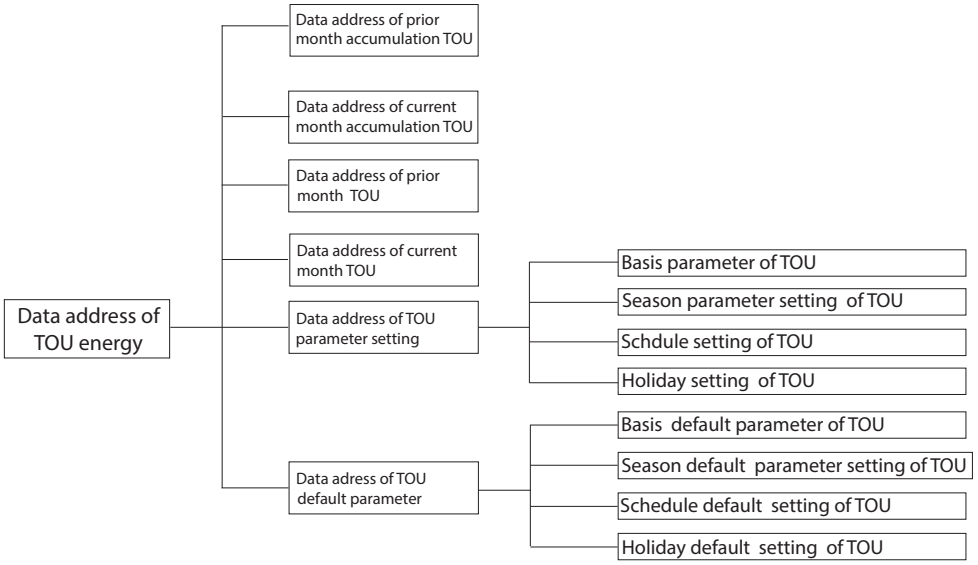


Table 64. Data Address of TOU.

Address	Parameter	Range	Data Type	Type of Access
Current month accumulation TOU energy				
7200H~7201H	Ep_imp(sharp)	0~99999999.9	Dword	R/W
7202H~7203H	Ep_exp(sharp)	0~99999999.9	Dword	R/W
7204H~7205H	Eq_im(sharp)	0~99999999.9	Dword	R/W
7206H~7207H	Eq_exp(sharp)	0~99999999.9	Dword	R/W
7208H~7209H	Es(sharp)	0~99999999.9	Dword	R/W
720AH~720BH	Ep_imp(peak)	0~99999999.9	Dword	R/W
720CH~720DH	Ep_exp(peak)	0~99999999.9	Dword	R/W
720EH~720FH	Eq_imp(peak)	0~99999999.9	Dword	R/W
7210H~7211H	Eq_exp(peak)	0~99999999.9	Dword	R/W
7212H~7213H	Es(peak)	0~99999999.9	Dword	R/W
7214H~7215H	Ep_imp(valley)	0~99999999.9	Dword	R/W
7216H~7217H	Ep_exp(valley)	0~99999999.9	Dword	R/W
7218H~7219H	Eq_imp(valley)	0~99999999.9	Dword	R/W
721AH~721BH	Eq_exp(valley)	0~99999999.9	Dword	R/W
721CH~721DH	Es(valley)	0~99999999.9	Dword	R/W
721EH~721FH	Ep_imp(normal)	0~99999999.9	Dword	R/W
7220H~7221H	Ep_exp(normal)	0~99999999.9	Dword	R/W
7222H~7223H	Eq_imp(normal)	0~99999999.9	Dword	R/W
7224H~7225H	Eq_exp(normal)	0~99999999.9	Dword	R/W
7226H~7227H	Es(normal)	0~99999999.9	Dword	R/W
7228H~7229H	Ep_imp(sum)	0~99999999.9	Dword	R/W
722AH~722BH	Ep_exp(sum)	0~99999999.9	Dword	R/W
722CH~722DH	Eq_imp(sum)	0~99999999.9	Dword	R/W
722EH~722FH	Eq_exp(sum)	0~99999999.9	Dword	R/W
7230H~7231H	Es(sum)	0~99999999.9	Dword	R/W
Prior month accumulation TOU energy				
7232H~7233H	Ep_imp(sharp)	0~99999999.9	Dword	R/W
7234H~7235H	Ep_exp(sharp)	0~99999999.9	Dword	R/W
7236H~7237H	Eq_imp(sharp)	0~99999999.9	Dword	R/W
7238H~7239H	Eq_exp(sharp)	0~99999999.9	Dword	R/W
723AH~723BH	Es(sharp)	0~99999999.9	Dword	R/W
723CH~723DH	Ep_imp(peak)	0~99999999.9	Dword	R/W
723EH~723FH	Ep_exp(peak)	0~99999999.9	Dword	R/W
7240H~7241H	Eq_imp(peak)	0~99999999.9	Dword	R/W
7242H~7243H	Eq_exp(peak)	0~99999999.9	Dword	R/W
7244H~7245H	Es(peak)	0~99999999.9	Dword	R/W
7246H~7247H	Ep_imp(valley)	0~99999999.9	Dword	R/W
7248H~7249H	Ep_exp(valley)	0~99999999.9	Dword	R/W
724AH~724BH	Eq_imp(valley)	0~99999999.9	Dword	R/W
724CH~724DH	Eq_exp(valley)	0~99999999.9	Dword	R/W
724EH~724FH	Es(valley)	0~99999999.9	Dword	R/W

Table 63. Data Address of TOU (continued).

Address	Parameter	Range	Data Type	Type of Access
7250H~7251H	Ep_imp(normal)	0~99999999.9	Dword	R/W
7252H~7253H	Ep_exp(normal)	0~99999999.9	Dword	R/W
7254H~7255H	Eq_imp(normal)	0~99999999.9	Dword	R/W
7256H~7257H	Eq_exp(normal)	0~99999999.9	Dword	R/W
7258H~7259H	Es(normal)	0~99999999.9	Dword	R/W
725AH~725BH	Ep_imp(sum)	0~99999999.9	Dword	R/W
725CH~725DH	Ep_exp(sum)	0~99999999.9	Dword	R/W
725EH~725FH	Eq_imp(sum)	0~99999999.9	Dword	R/W
7260H~7261H	Eq_exp(sum)	0~99999999.9	Dword	R/W
7262H~7263H	Es(sum)	0~99999999.9	Dword	R/W

The address area includes the max of Ep_imp, Ep_exp, Eq_im, Eq_exp, Es, Current demand, and time stamp, when tariff setting parameter is sharp, peak, valley, and normal. Function: 03H Read.

Table 65. Data Address of Peak Demand

Note: The following Format Codes apply:F12 for current, F13 for Active power, F14 for Reactive power and F15 for Apparent power.

Address	Parameter	Range	Data Type	Type of Access
7500H~7503H	Max of Ep_imp (sharp) demand and time stamp; (format: power; year/mon; day/hour; min/sec)	-32768~32767	Int	R
7504H~7507H	Max of Ep_exp(sharp) demand and time stamp	-32768~32767	Int	R
7508H~750bH	Max of Eq_im(sharp) demand and time stamp	-32768~32767	Int	R
750cH~750fH	Max of Eq_exp(sharp) demand and time stamp	-32768~32767	Int	R
7510H~7513H	Max of Es(sharp) demand and time stamp	-32768~32767	Int	R
7514H~7517H	Max of Ia(sharp) demand and time stamp	-32768~32767	Int	R
7518H~751bH	Max of Ib(sharp) demand and time stamp	-32768~32767	Int	R
751cH~751fH	Max of Ic(sharp) demand and time stamp	-32768~32767	Int	R
7520H~7523H	Max of Ep_imp(peak) demand and time stamp	-32768~32767	Int	R
7524H~7527H	Max of Ep_exp(peak) demand and time stamp	-32768~32767	Int	R
7528H~752bH	Max of Eq_im(peak) demand and time stamp	-32768~32767	Int	R
752cH~752fH	Max of Eq_exp(peak) demand and time stamp	-32768~32767	Int	R
7530H~7533H	Max of Es(peak) demand and time stamp	-32768~32767	Int	R
7534H~7537H	Max of Ia (peak)demand and time stamp	-32768~32767	Int	R
7538H~753bH	Max of Ib (peak)demand and time stamp	-32768~32767	Int	R
753cH~753fH	Max of Ic (peak) demand and time stamp	-32768~32767	Int	R
7540H~7543H	Max of Ep_imp(valley) demand and time stamp	-32768~32767	Int	R
7544H~7547H	Max of Ep_exp(valley) demand and time stamp	-32768~32767	Int	R
7548H~754bH	Max of Eq_im(valley) demand and time stamp	-32768~32767	Int	R
754cH~754fH	Max of Eq_exp(valley) demand and time stamp	-32768~32767	Int	R
7550H~7553H	Max of Es(valley) demand and time stamp	-32768~32767	Int	R
7554H~7557H	Max of Ia (valley)demand and time stamp	-32768~32767	Int	R
7558H~755bH	Max of Ib (valley)demand and time stamp	-32768~32767	Int	R
755cH~755fH	Max of Ic (valley) demand and time stamp	-32768~32767	Int	R
7560H~7563H	Max of Ep_imp(normal) demand and time stamp	-32768~32767	Int	R
7564H~7567H	Max of Ep_exp(normal) demand and time stamp	-32768~32767	Int	R
7568H~756bH	Max of Eq_im(normal) demand and time stamp	-32768~32767	Int	R

Table 64. (continued).

Address	Parameter	Range	Data Type	Type of Access
756cH~756fH	Max of Eq_exp(normal) demand and time stamp	-32768~32767	Int	R
7570H~7573H	Max of Es(normal) demand and time stamp	-32768~32767	Int	R
7574H~7577H	Max of Ia (normal)demand and time stamp	-32768~32767	Int	R
7578H~757bH	Max of Ib (normal)demand and time stamp	-32768~32767	Int	R
757cH~757fH	Max of Ic (normal) demand and time stamp	-32768~32767	Int	R
7580H~7583H	Max of Ep_imp(all) demand and time stamp	-32768~32767	Int	R
7584H~7587H	Max of Ep_exp(all) demand and time stamp	-32768~32767	Int	R
7588H~758bH	Max of Eq_im(all) demand and time stamp	-32768~32767	Int	R
758cH~758fH	Max of Eq_exp(all) demand and time stamp	-32768~32767	Int	R
7590H~7593H	Max of Es(all) demand and time stamp	-32768~32767	Int	R
7594H~7597H	Max of Ia(all)demand and time stamp	-32768~32767	Int	R
7598H~759bH	Max of Ib(all)demand and time stamp	-32768~32767	Int	R
759cH~759fH	Max of Ic(all) demand and time stamp	-32768~32767	Int	R

The address area includes the daylight saving time (DST) setting. Function: 03H Read, 10H Preset.

Table 66. DST Settings.

Address	Parameter	Range	Data Type	Type of Access
7700H	DST enable	0: disable; 1: enable	Word	R/W
7701H	DST format	0: format 1; 1: format 2	Word	R/W
Format 1				
7702H	DST Start Mon	1~12	Word	R/W
7703H	DST Start Day	1~31	Word	R/W
7704H	DST Start Hour	0~23	Word	R/W
7705H	DST Start Min	0~59	Word	R/W
7706H	DST Start Adjust time (Unit: Min)	1~120 Default: 60	Word	R/W
7707H	DST Ending Mon	1~12	Word	R/W
7708H	DST Ending Day	1~31	Word	R/W
7709H	DST Ending Hour	0~23	Word	R/W
770AH	DST Ending Min	0~59	Word	R/W
770BH	DST Ending Adjust time (Unit: Min)	1~120 Default: 60	Word	R/W
Format 2				
770CH	DST Start Mon	1~12	Word	R/W
770DH	DST Start week	0~6 0: Sunday; 1~6 Monday to Saturday	Word	R/W
770EH	DST Start First few weeks	1~5	Word	R/W
770FH	DST Start Hour	0~23	Word	R/W
7710H	DST Start Min	0~59	Word	R/W
7711H	DST Start Adjust time (Unit: Min)	1~120 Default: 60	Word	R/W
7712H	DST Ending Mon	1~12	Word	R/W
7713H	DST Ending Week	0~6 0: Sunday; 1~6 Monday to Saturday	Word	R/W
7714H	DST Ending First few weeks	1~5	Word	R/W
7715H	DST Ending Hour	0~23	Word	R/W
7716H	DST Ending Min	0~59	Word	R/W
7717H	DST Ending Adjust time (Unit: Min)	1~120 Default: 60	Word	R/W

Data address of the TOU parameter setting includes the basis parameter of TOU, time zone setting parameter of TOU, time table setting parameter of TOU, and holiday setting parameter of TOU. Function: 03 code, 10 reset.

Table 67. Data Address of TOU.

Address	Parameter	Range	Data Type	Type of Access
Basis parameter of TOU				
7800H	Season Number	0~12	Word	R/W
7801H	Schedule Number	0~14	Word	R/W
7802H	Segment Number	0~14	Word	R/W
7803H	Tariff Number	0~3	Word	R/W
7804H	Weekend Setting (bit0-Sunday; bit1~bit6 : Monday~Saturday bit = 1 means using energy, bit = 0 means not using energy)	0~127	Word	R/W
7805H	Weekend Schedule	0-14	Word	R/W
7806H	Holiday Number	0-30	Word	R/W
7807H	Time of Use factory setting	1:enable	Word	R/W
7808H	Choice of calculation auto reset; (0: End of Month)	1:enable	Word	R/W
7809H	TOU auto reset fixed date:day; (default is 1)		Word	R/W
780AH	TOU auto reset fixed date:hour (default is 0)	1~31	Word	R/W
780BH	TOU auto reset fixed date:minute (default is 0)	0~23	Word	R/W
780CH	TOU auto reset fixed date:second (default is 0)	0~59	Word	R/W
780DH	TOU auto reset fixed date: second (default is 0)	0~59	Word	R/W
780EH	Error code (default)	0: The setting of parameter is correct; 1: Tariff setting error; 2: Schedule setting error; 4: Segment setting error; 8: Season setting error; 16: Parameter of season setting error 32: Holiday setting error; 64: Parameter of holiday setting error; 256: Tariff of schedule setting error; 512: Time of schedule setting error; 1024: Period of schedule setting error; 048: Period of weekend setting error; 4096: Weekend setting error;		
Season Setting				
7820H~7822H	Data and Season Table of the 1st Season		Word	R/W
7823H~7825H	Data and Season Table of the 2nd Season		Word	R/W
7826H~7828H	Data and Season Table of the 3rd Season		Word	R/W
7829H~782BH	Data and Season Table of the 4th Season		Word	R/W
782CH~782EH	Data and Season Table of the 5th Season		Word	R/W
782FH~7831H	Data and Season Table of the 6th Season		Word	R/W
7832H~7834H	Data and Season Table of the 7th Season		Word	R/W
7835H~7837H	Data and Season Table of the 8th Season		Word	R/W
7838H~783AH	Data and Season Table of the 9th Season		Word	R/W
783BH~783DH	Data and Season Table of the 10th Season		Word	R/W
783EH~7840H	Data and Season Table of the 11th Season		Word	R/W
7841H~7843H	Data and Season Table of the 12th Season		Word	R/W
Schedule Setting				
7844H~7846H	1st Segment and Tariff Number of the 1st schedule		Word	R/W
7847H~7849H	2nd Segment and Tariff Number of the 1st schedule		Word	R/W
784AH~784CH	3rd Segment and Tariff Number of the 1st schedule		Word	R/W

Table 66 Data Address of TOU (continued).

Address	Parameter	Range	Data Type	Type of Access
784DH~784FH	4th Segment and Tariff Number of the 1st schedule		Word	R/W
7850H~7852H	5th Segment and Tariff Number of the 1st schedule		Word	R/W
7853H~7855H	6th Segment and Tariff Number of the 1st schedule		Word	R/W
7856H~7858H	7th Segment and Tariff Number of the 1st schedule		Word	R/W
7859H~785BH	8th Segment and Tariff Number of the 1st schedule		Word	R/W
785CH~785EH	9th Segment and Tariff Number of the 1st schedule		Word	R/W
785FH~7861H	10th Segment and Tariff Number of the 1st schedule		Word	R/W
7862H~7864H	11th Segment and Tariff Number of the 1st schedule		Word	R/W
7865H~7867H	12th Segment and Tariff Number of the 1st schedule		Word	R/W
7868H~786AH	13th Segment and Tariff Number of the 1st schedule		Word	R/W
786BH~786DH	14th Segment and Tariff Number of the 1st schedule		Word	R/W
786EH~7897H	From 1st to 14th Segment and Tariff Number of the 2nd schedule	The same as 1st schedule	Word	R/W
7898H~78C1H	From 1st to 14th Segment and Tariff Number of the 3rd schedule	The same as 1st schedule	Word	R/W
78C2H~78EBH	From 1st to 14th Segment and Tariff Number of the 4th schedule	The same as 1st schedule	Word	R/W
78ECH~7915H	From 1st to 14th Segment and Tariff Number of the 5th schedule	The same as 1st schedule	Word	R/W
7916H~793FH	From 1st to 14th Segment and Tariff Number of the 6th schedule	The same as 1st schedule	Word	R/W
7940H~7969H	From 1st to 14th Segment and Tariff Number of the 7th schedule	The same as 1st schedule	Word	R/W
796AH~7993H	From 1st to 14th Segment and Tariff Number of the 8th schedule	The same as 1st schedule	Word	R/W
7994H~79BDH	From 1st to 14th Segment and Tariff Number of the 9th schedule	The same as 1st schedule	Word	R/W
79BEH~79E7H	From 1st to 14th Segment and Tariff Number of the 10th schedule	The same as 1st schedule	Word	R/W
79E8H~7A11H	From 1st to 14th Segment and Tariff Number of the 11th schedule	The same as 1st schedule	Word	R/W
7A12H~7A3BH	From 1st to 14th Segment and Tariff Number of the 12th schedule	The same as 1st schedule	Word	R/W
7A3CH~7A65H	From 1st to 14th Segment and Tariff Number of the 13th schedule	The same as 1st schedule	Word	R/W
7A66H~7A8FH	From 1st to 14th Segment and Tariff Number of the 14th schedule	The same as 1st schedule	Word	R/W
Holiday Setting				
7A90H~7A92H	Data and the Schedule of the 1st holiday		Word	R/W
7A93H~7A95H	Data and the schedule of the 2nd holiday		Word	R/W
7A96H~7A98H	Data and the schedule of the 3rd holiday		Word	R/W
7A99H~7A9BH	Data and the schedule of the 4th holiday		Word	R/W
7A9CH~7A9EH	Data and the schedule of the 5th holiday		Word	R/W
7A9FH~7AA1H	Data and the schedule of the 6th holiday		Word	R/W
7AA2H~7AA4H	Data and the schedule of the 7th holiday		Word	R/W
7AA5H~7AA7H	Data and the schedule of the 8th holiday		Word	R/W
7AA8H~7AAAH	Data and the schedule of the 9th holiday		Word	R/W
7AABH~7AADH	Data and the schedule of the 10th holiday		Word	R/W
7AAEH~7AB0H	Data and the schedule of the 11th holiday		Word	R/W
7AB1H~7AB3H	Data and the schedule of the 12th holiday		Word	R/W
7AB4H~7AB6H	Data and the schedule of the 13th holiday		Word	R/W
7AB7H~7AB9H	Data and the schedule of the 14th holiday		Word	R/W
7ABAH~7ABCH	Data and the schedule of the 15th holiday		Word	R/W
7ABDH~7ABFH	Data and the schedule of the 16th holiday		Word	R/W
7AC0H~7AC2H	Data and the schedule of the 17th holiday		Word	R/W
7AC3H~7AC5H	Data and the schedule of the 18th holiday		Word	R/W
7AC6H~7AC8H	Data and the schedule of the 19th holiday		Word	R/W
7AC9H~7ACBH	Data and the schedule of the 20th holiday		Word	R/W

Table 66. Data Address of TOU (continued).

Address	Parameter	Range	Data Type	Type of Access
7ACCH~7ACEH	Data and the schedule of the 21st holiday		Word	R/W
7ACFH~7AD1H	Data and the schedule of the 22nd holiday		Word	R/W
7AD2H~7AD4H	Data and the schedule of the 23rd holiday		Word	R/W
7AD5H~7AD7H	Data and the schedule of the 24th holiday		Word	R/W
7AD8H~7ADAH	Data and the schedule of the 25th holiday		Word	R/W
7ADBH~7ADDH	Data and the schedule of the 26th holiday		Word	R/W
7ADEH~7AE0H	Data and the schedule of the 27th holiday		Word	R/W
7AE1H~7AE3H	Data and the schedule of the 28th holiday		Word	R/W
7AE4H~7AE6H	Data and the schedule of the 29th holiday		Word	R/W
7AE7H~7AE9H	Data and the schedule of the 30th holiday		Word	R/W
7AEAH	Holiday setting enable		Word	R/W
7AEBH	Start year holiday setting		Word	R/W
7AECH	End year holiday setting		Word	R/W

The address area includes Ten-Year Holiday setting. Function: 03H Read 10H Preset.

Table 68. Data Address of Ten-Year Holiday.

The 1st Year Holiday Address Function: 03H Read 10H Preset

Address	Parameter	Data Type	Type of Access
7B00H~7B02H	The 1st holiday and schedule (format: month / day / schedule)	Word	R/W
7B03H~7B05H	The 2nd holiday and schedule	Word	R/W
7B06H~7B08H	The 3rd holiday and schedule	Word	R/W
7B09H~7B0BH	The 4th holiday and schedule	Word	R/W
7B0CH~7B0EH	The 5th holiday and schedule	Word	R/W
7B0FH~7B11H	The 6th holiday and schedule	Word	R/W
7B12H~7B14H	The 7th holiday and schedule	Word	R/W
7B15H~7B17H	The 8th holiday and schedule	Word	R/W
7B18H~7B1AH	The 9th holiday and schedule	Word	R/W
7B1BH~7B1DH	The 10th holiday and schedule	Word	R/W
7B1EH~7B20H	The 11th holiday and schedule	Word	R/W
7B21H~7B23H	The 12th holiday and schedule	Word	R/W
7B24H~7B26H	The 13th holiday and schedule	Word	R/W
7B27H~7B29H	The 14th holiday and schedule	Word	R/W
7B2AH~7B2CH	The 15th holiday and schedule	Word	R/W
7B2DH~7B2FH	The 16th holiday and schedule	Word	R/W
7B30H~7B32H	The 17th holiday and schedule	Word	R/W
7B33H~7B35H	The 18th holiday and schedule	Word	R/W
7B36H~7B38H	The 19th holiday and schedule	Word	R/W
7B39H~7B3BH	The 20th holiday and schedule	Word	R/W
7B3CH~7B3EH	The 21st holiday and schedule	Word	R/W
7B3FH~7B41H	The 22nd holiday and schedule	Word	R/W
7B42H~7B44H	The 23rd holiday and schedule	Word	R/W
7B45H~7B47H	The 24th holiday and schedule	Word	R/W
7B48H~7B4AH	The 25th holiday and schedule	Word	R/W

Table 67. Data Address of Ten-Year Holiday (continued).

The 1st Year Holiday Address Function: 03H Read 10H Preset

Address	Parameter	Data Type	Type of Access
7B4BH~7B4DH	The 26th holiday and schedule	Word	R/W
7B4EH~7B50H	The 27th holiday and schedule	Word	R/W
7B51H~7B53H	The 28th holiday and schedule	Word	R/W
7B54H~7B56H	The 29th holiday and schedule	Word	R/W
7B57H~7B59H	The 30th holiday and schedule	Word	R/W
7B5AH	The 1st setting year	Word	R/W
7B5BH	Holiday number of the 1st year	Word	R/W

The 2nd Year Holiday Address Function: 03H Read 10H Preset

Address	Parameter	Data Type	Type of Access
7B5CH~7B5EH	The 1st holiday and schedule (format: month / day /schedule)	Word	R/W
7B5FH~7B61H	The 2nd holiday and schedule	Word	R/W
7B62H~7B64H	The 3rd holiday and schedule	Word	R/W
7B65H~7B67H	The 4th holiday and schedule	Word	R/W
7B68H~7B6AH	The 5th holiday and schedule	Word	R/W
7B6BH~7B6DH	The 6th holiday and schedule	Word	R/W
7B6EH~7B70H	The 7th holiday and schedule	Word	R/W
7B71H~7B73H	The 8th holiday and schedule	Word	R/W
7B74H~7B76H	The 9th holiday and schedule	Word	R/W
7B77H~7B79H	The 10th holiday and schedule	Word	R/W
7B7AH~7B7CH	The 11th holiday and schedule	Word	R/W
7B7DH~7B7FH	The 12th holiday and schedule	Word	R/W
7B80H~7B82H	The 13th holiday and schedule	Word	R/W
7B83H~7B85H	The 14th holiday and schedule	Word	R/W
7B86H~7B88H	The 15th holiday and schedule	Word	R/W
7B89H~7B8BH	The 16th holiday and schedule	Word	R/W
7B8CH~7B8EH	The 17th holiday and schedule	Word	R/W
7B8FH~7B91H	The 18th holiday and schedule	Word	R/W
7B92H~7B94H	The 19th holiday and schedule	Word	R/W
7B95H~7B97H	The 20th holiday and schedule	Word	R/W
7B98H~7B9AH	The 21st holiday and schedule	Word	R/W
7B9BH~7B9DH	The 22nd holiday and schedule	Word	R/W
7B9EH~7BA0H	The 23rd holiday and schedule	Word	R/W
7BA1H~7BA3H	The 24th holiday and schedule	Word	R/W
7BA4H~7BA6H	The 25th holiday and schedule	Word	R/W
7BA7H~7BA9H	The 26th holiday and schedule	Word	R/W
7BAAH~7BACH	The 27th holiday and schedule	Word	R/W
7BADH~7BAFH	The 28th holiday and schedule	Word	R/W
7BB0H~7BB2H	The 29th holiday and schedule	Word	R/W
7BB3H~7BB5H	The 30th holiday and schedule	Word	R/W
7BB6H	The 2th setting year	Word	R/W
7BB7H	Holiday number of the 2th year	Word	R/W

Table 67. Data Address of Ten-Year Holiday (continued).

The 3rd Year Holiday Address Function: 03H Read 10H Preset

Address	Parameter	Data Type	Type of Access
7BB8H~7BBAH	The 1th holiday and schedule (format: month / day / schedule)	Word	R/W
7BBBH~7BBDH	The 2nd holiday and schedule	Word	R/W
7BBEH~7BC0H	The 3rd holiday and schedule	Word	R/W
7BC1H~7BC3H	The 4th holiday and schedule	Word	R/W
7BC4H~7BC6H	The 5th holiday and schedule	Word	R/W
7BC7H~7BC9H	The 6th holiday and schedule	Word	R/W
7BCAH~7BCCH	The 7th holiday and schedule	Word	R/W
7BCDH~7BCFH	The 8th holiday and schedule	Word	R/W
7BD0H~7BD2H	The 9th holiday and schedule	Word	R/W
7BD3H~7BD5H	The 10th holiday and schedule	Word	R/W
7BD6H~7BD8H	The 11th holiday and schedule	Word	R/W
7BD9H~7BDBH	The 12th holiday and schedule	Word	R/W
7BDCH~7BDEH	The 13th holiday and schedule	Word	R/W
7BDFH~7BE1H	The 14th holiday and schedule	Word	R/W
7BE2H~7BE4H	The 15th holiday and schedule	Word	R/W
7BE5H~7BE7H	The 16th holiday and schedule	Word	R/W
7BE8H~7BEAH	The 17th holiday and schedule	Word	R/W
7BEBH~7BEDH	The 18th holiday and schedule	Word	R/W
7BEEH~7BF0H	The 19th holiday and schedule	Word	R/W
7BF1H~7BF3H	The 20th holiday and schedule	Word	R/W
7BF4H~7BF6H	The 21st holiday and schedule	Word	R/W
7BF7H~7BF9H	The 22nd holiday and schedule	Word	R/W
7BFAH~7BFCH	The 23rd holiday and schedule	Word	R/W
7BFDH~7BFFH	The 24th holiday and schedule	Word	R/W
7C00H~7C02H	The 25th holiday and schedule	Word	R/W
7C03H~7C05H	The 26th holiday and schedule	Word	R/W
7C06H~7C08H	The 27th holiday and schedule	Word	R/W
7C09H~7C0BH	The 28th holiday and schedule	Word	R/W
7C0CH~7C0EH	The 29th holiday and schedule	Word	R/W
7C0FH~7C11H	The 30th holiday and schedule	Word	R/W
7C12H	The 3rd setting year	Word	R/W
7C13H	Holiday number of the 3rd year	Word	R/W

The 4th Year Holiday Address Function: 03H Read 10H Preset

Address	Parameter	Data Type	Type of Access
7C14H~7C16H	The 1st holiday and schedule (format: month / day / schedule)	Word	R/W
7C17H~7C19H	The 2nd holiday and schedule	Word	R/W
7C1AH~7C1CH	The 3rd holiday and schedule	Word	R/W
7C1DH~7C1FH	The 4th holiday and schedule	Word	R/W
7C20H~7C22H	The 5th holiday and schedule	Word	R/W
7C23H~7C25H	The 6th holiday and schedule	Word	R/W
7C26H~7C28H	The 7th holiday and schedule	Word	R/W
7C29H~7C2BH	The 8th holiday and schedule	Word	R/W
7C2CH~7C2EH	The 9th holiday and schedule	Word	R/W

Table 67. Data Address of Ten-Year Holiday (continued).

The 4th Year Holiday Address Function: 03H Read 10H Preset

Address	Parameter	Data Type	Type of Access
7C2FH~7C31H	The 10th holiday and schedule	Word	R/W
7C32H~7C34H	The 11th holiday and schedule	Word	R/W
7C35H~7C37H	The 12th holiday and schedule	Word	R/W
7C38H~7C3AH	The 13th holiday and schedule	Word	R/W
7C3BH~7C3DH	The 14th holiday and schedule	Word	R/W
7C3EH~7C40H	The 15th holiday and schedule	Word	R/W
7C41H~7C43H	The 16th holiday and schedule	Word	R/W
7C44H~7C46H	The 17th holiday and schedule	Word	R/W
7C47H~7C49H	The 18th holiday and schedule	Word	R/W
7C4AH~7C4CH	The 19th holiday and schedule	Word	R/W
7C4DH~7C4FH	The 20th holiday and schedule	Word	R/W
7C50H~7C52H	The 21st holiday and schedule	Word	R/W
7C53H~7C55H	The 22nd holiday and schedule	Word	R/W
7C56H~7C58H	The 23rd holiday and schedule	Word	R/W
7C59H~7C5BH	The 24th holiday and schedule	Word	R/W
7C5CH~7C5EH	The 25th holiday and schedule	Word	R/W
7C5FH~7C61H	The 26th holiday and schedule	Word	R/W
7C62H~7C64H	The 27th holiday and schedule	Word	R/W
7C65H~7C67H	The 28th holiday and schedule	Word	R/W
7C68H~7C6AH	The 29th holiday and schedule	Word	R/W
7C6BH~7C6DH	The 30th holiday and schedule	Word	R/W
7C6EH	The 4th setting year	Word	R/W
7C6FH	Holiday number of the 4th year	Word	R/W

The 5th Year Holiday Address Function: 03H Read 10H Preset

Address	Parameter	Data Type	Type of Access
7C70H~7C72H	The 1st holiday and schedule (format: month /day /schedule)	Word	R/W
7C73H~7C75H	The 2nd holiday and schedule	Word	R/W
7C76H~7C78H	The 3rd holiday and schedule	Word	R/W
7C79H~7C7BH	The 4th holiday and schedule	Word	R/W
7C7CH~7C7EH	The 5th holiday and schedule	Word	R/W
7C7FH~7C81H	The 6th holiday and schedule	Word	R/W
7C82H~7C84H	The 7th holiday and schedule	Word	R/W
7C85H~7C87H	The 8th holiday and schedule	Word	R/W
7C88H~7C8AH	The 9th holiday and schedule	Word	R/W
7C8BH~7C8DH	The 10th holiday and schedule	Word	R/W
7C8EH~7C90H	The 11th holiday and schedule	Word	R/W
7C91H~7C93H	The 12th holiday and schedule	Word	R/W
7C94H~7C96H	The 13th holiday and schedule	Word	R/W
7C97H~7C99H	The 14th holiday and schedule	Word	R/W
7C9AH~7C9CH	The 15th holiday and schedule	Word	R/W
7C9DH~7C9FH	The 16th holiday and schedule	Word	R/W
7CA0H~7CA2H	The 17th holiday and schedule	Word	R/W

Table 67. Data Address of Ten-Year Holiday (continued).

The 5th Year Holiday Address Function: 03H Read 10H Preset

Address	Parameter	Data Type	Type of Access
7CA3H~7CA5H	The 18th holiday and schedule	Word	R/W
7CA6H~7CA8H	The 19th holiday and schedule	Word	R/W
7CA9H~7CABH	The 20th holiday and schedule	Word	R/W
7CACH~7CAEH	The 21st holiday and schedule	Word	R/W
7CAFH~7CB1H	The 22nd holiday and schedule	Word	R/W
7CB2H~7CB4H	The 23rd holiday and schedule	Word	R/W
7CB5H~7CB7H	The 24th holiday and schedule	Word	R/W
7CB8H~7CBAH	The 25th holiday and schedule	Word	R/W
7CBBH~7CBDH	The 26th holiday and schedule	Word	R/W
7CBEH~7CC0H	The 27th holiday and schedule	Word	R/W
7CC1H~7CC3H	The 28th holiday and schedule	Word	R/W
7CC4H~7CC6H	The 29th holiday and schedule	Word	R/W
7CC7H~7CC9H	The 30th holiday and schedule	Word	R/W
7CCAH	The 5th setting year	Word	R/W
7CCBH	Holiday number of the 5th year	Word	R/W

The 6th Year Holiday Address Function: 03H Read 10H Preset

Address	Parameter	Data Type	Type of Access
7CCCH~7CCEH	The 1st holiday and schedule (format: month / day / schedule)	Word	R/W
7CCFH~7CD1H	The 2nd holiday and schedule	Word	R/W
7CD2H~7CD4H	The 3rd holiday and schedule	Word	R/W
7CD5H~7CD7H	The 4th holiday and schedule	Word	R/W
7CD8H~7CDAH	The 5th holiday and schedule	Word	R/W
7CDBH~7CDDH	The 6th holiday and schedule	Word	R/W
7CDEH~7CE0H	The 7th holiday and schedule	Word	R/W
7CE1H~7CE3H	The 8th holiday and schedule	Word	R/W
7CD4H~7CE6H	The 9th holiday and schedule	Word	R/W
7CE7H~7CE9H	The 10th holiday and schedule	Word	R/W
7CEAH~7CECH	The 11th holiday and schedule	Word	R/W
7CEDH~7CEFH	The 12th holiday and schedule	Word	R/W
7CF0H~7CF2H	The 13th holiday and schedule	Word	R/W
7CF3H~7CF5H	The 14th holiday and schedule	Word	R/W
7CF6H~7CF8H	The 15th holiday and schedule	Word	R/W
7CF9H~7CFBH	The 16th holiday and schedule	Word	R/W
7CFCH~7CFEH	The 17th holiday and schedule	Word	R/W
7CFFH~7D01H	The 18th holiday and schedule	Word	R/W
7D02H~7D04H	The 19th holiday and schedule	Word	R/W
7D05H~7D07H	The 20th holiday and schedule	Word	R/W
7D08H~7D0AH	The 21st holiday and schedule	Word	R/W
7D0BH~7D0DH	The 22nd holiday and schedule	Word	R/W
7D0EH~7D10H	The 23rd holiday and schedule	Word	R/W
7D11H~7D13H	The 24th holiday and schedule	Word	R/W
7D14H~7D16H	The 25th holiday and schedule	Word	R/W
7D17H~7D19H	The 26th holiday and schedule	Word	R/W

Table 67. Data Address of Ten-Year Holiday (continued).

The 6th Year Holiday Address Function: 03H Read 10H Preset

Address	Parameter	Data Type	Type of Access
7D1AH~7D1CH	The 27th holiday and schedule	Word	R/W
7D1DH~7D1FH	The 28th holiday and schedule	Word	R/W
7D20H~7D22H	The 29th holiday and schedule	Word	R/W
7D23H~7D25H	The 30th holiday and schedule	Word	R/W
7D26H	The 6th setting year	Word	R/W
7D27H	Holiday number of the 6th year	Word	R/W

The 7th Year Holiday Address Function: 03H Read 10H Preset

Address	Parameter	Data Type	Type of Access
7D28H~7D2AH	The 1st holiday and schedule (format: month / day / schedule)	Word	R/W
7D2BH~7D2DH	The 2nd holiday and schedule	Word	R/W
7D2EH~7D30H	The 3rd holiday and schedule	Word	R/W
7D31H~7D33H	The 4th holiday and schedule	Word	R/W
7D34H~7D36H	The 5th holiday and schedule	Word	R/W
7D37H~7D39H	The 6th holiday and schedule	Word	R/W
7D3AH~7D3CH	The 7th holiday and schedule	Word	R/W
7D3DH~7D3FH	The 8th holiday and schedule	Word	R/W
7D40H~7D42H	The 9th holiday and schedule	Word	R/W
7D43H~7D45H	The 10th holiday and schedule	Word	R/W
7D46H~7D48H	The 11th holiday and schedule	Word	R/W
7D49H~7D4BH	The 12th holiday and schedule	Word	R/W
7D4CH~7D4EH	The 13th holiday and schedule	Word	R/W
7D4FH~7D51H	The 14th holiday and schedule	Word	R/W
7D52H~7D54H	The 15th holiday and schedule	Word	R/W
7D55H~7D57H	The 16th holiday and schedule	Word	R/W
7D58H~7D5AH	The 17th holiday and schedule	Word	R/W
7D5BH~7D5DH	The 18th holiday and schedule	Word	R/W
7D5EH~7D60H	The 19th holiday and schedule	Word	R/W
7D61H~7D63H	The 20th holiday and schedule	Word	R/W
7D64H~7D66H	The 21st holiday and schedule	Word	R/W
7D67H~7D69H	The 22nd holiday and schedule	Word	R/W
7D6AH~7D6CH	The 23rd holiday and schedule	Word	R/W
7D6DH~7D6FH	The 24th holiday and schedule	Word	R/W
7D70H~7D72H	The 25th holiday and schedule	Word	R/W
7D73H~7D75H	The 26th holiday and schedule	Word	R/W
7D76H~7D78H	The 27th holiday and schedule	Word	R/W
7D79H~7D7BH	The 28th holiday and schedule	Word	R/W
7D7CH~7D7EH	The 29th holiday and schedule	Word	R/W
7D7FH~7D81H	The 30th holiday and schedule	Word	R/W
7D82H	The 7th setting year	Word	R/W
7D83H	Holiday number of the 7th year	Word	R/W

Table 67. Data Address of Ten-Year Holiday (continued).

The 8th Year Holiday Address Function: 03H Read 10H Preset

Address	Parameter	Data Type	Type of Access
7D84H~7D86H	The 1st holiday and schedule (format: month / day / time table)	Word	R/W
7D87H~7D89H	The 2nd holiday and schedule	Word	R/W
7D8AH~7D8CH	The 3rd holiday and schedule	Word	R/W
7D8DH~7D8FH	The 4th holiday and schedule	Word	R/W
7D90H~7D92H	The 5th holiday and schedule	Word	R/W
7D93H~7D95H	The 6th holiday and schedule	Word	R/W
7D96H~7D98H	The 7th holiday and schedule	Word	R/W
7D99H~7D9BH	The 8th holiday and schedule	Word	R/W
7D9CH~7D9EH	The 9th holiday and schedule	Word	R/W
7D9FH~7DA1H	The 10th holiday and schedule	Word	R/W
7DA2H~7DA4H	The 11th holiday and schedule	Word	R/W
7DA5H~7DA7H	The 12th holiday and schedule	Word	R/W
7DA8H~7DAAH	The 13th holiday and schedule	Word	R/W
7DABH~7DADH	The 14th holiday and schedule	Word	R/W
7DAEH~7DB0H	The 15th holiday and schedule	Word	R/W
7DB1H~7DB3H	The 16th holiday and schedule	Word	R/W
7DB4H~7DB6H	The 17th holiday and schedule	Word	R/W
7DB7H~7DB9H	The 18th holiday and schedule	Word	R/W
7DBAH~7DBCH	The 19th holiday and schedule	Word	R/W
7DBDH~7DBFH	The 20th holiday and schedule	Word	R/W
7DC0H~7DC2H	The 21st holiday and schedule	Word	R/W
7DC3H~7DC5H	The 22nd holiday and schedule	Word	R/W
7DC6H~7DC8H	The 23rd holiday and schedule	Word	R/W
7DC9H~7DCBH	The 24th holiday and schedule	Word	R/W
7DCCH~7DCEH	The 25th holiday and schedule	Word	R/W
7DCFH~7DD1H	The 26th holiday and schedule	Word	R/W
7DD2H~7DD4H	The 27th holiday and schedule	Word	R/W
7DD5H~7DD7H	The 28th holiday and schedule	Word	R/W
7DD8H~7DDAH	The 29th holiday and schedule	Word	R/W
7DDBH~7DDDH	The 30th holiday and schedule	Word	R/W
7DDEH	The 8th setting year	Word	R/W
7DDFH	Holiday number of the 8th year	Word	R/W

The 9th Year Holiday Address Function: 03H Read 10H Preset

Address	Parameter	Data Type	Type of Access
7DE0H~7DE2H	The 1st holiday and schedule (format: month /day /schedule)	Word	R/W
7DE3H~7DE5H	The 2nd holiday and schedule	Word	R/W
7DE6H~7DE8H	The 3rd holiday and schedule	Word	R/W
7DE9H~7DEBH	The 4th holiday and schedule	Word	R/W
7DECH~7DEEH	The 5th holiday and schedule	Word	R/W
7DEFH~7DF1H	The 6th holiday and schedule	Word	R/W
7DF2H~7DF4H	The 7th holiday and schedule	Word	R/W
7DF5H~7DF7H	The 8th holiday and schedule	Word	R/W
7DF8H~7DFAH	The 9th holiday and schedule	Word	R/W

Table 67. Data Address of Ten-Year Holiday (continued).

The 9th Year Holiday Address Function: 03H Read 10H Preset

Address	Parameter	Data Type	Type of Access
7DFBH~7DFDH	The 10th holiday and schedule	Word	R/W
7DFEH~7E00H	The 11th holiday and schedule	Word	R/W
7E01H~7E03H	The 12th holiday and schedule	Word	R/W
7E04H~7E06H	The 13th holiday and schedule	Word	R/W
7E07H~7E09H	The 14th holiday and schedule	Word	R/W
7E0AH~7E0CH	The 15th holiday and schedule	Word	R/W
7E0DH~7E0FH	The 16th holiday and schedule	Word	R/W
7E10H~7E12H	The 17th holiday and schedule	Word	R/W
7E13H~7E15H	The 18th holiday and schedule	Word	R/W
7E16H~7E18H	The 19th holiday and schedule	Word	R/W
7E19H~7E1BH	The 20th holiday and schedule	Word	R/W
7E1CH~7E1EH	The 21st holiday and schedule	Word	R/W
7E1FH~7E21H	The 22nd holiday and schedule	Word	R/W
7E22H~7E24H	The 23rd holiday and schedule	Word	R/W
7E25H~7E27H	The 24th holiday and schedule	Word	R/W
7E28H~7E2AH	The 25th holiday and schedule	Word	R/W
7E2BH~7E2DH	The 26th holiday and schedule	Word	R/W
7E2EH~7E30H	The 27th holiday and schedule	Word	R/W
7E31H~7E33H	The 28th holiday and schedule	Word	R/W
7E34H~7E36H	The 29th holiday and schedule	Word	R/W
7E37H~7E39H	The 30th holiday and schedule	Word	R/W
7E3AH	The 9th setting year	Word	R/W
7E3BH	Holiday number of the 9th year	Word	R/W

The 10th Year Holiday Address Function: 03H Read 10H Preset

Address	Parameter	Data Type	Type of Access
7E3CH~7E3EH	The 1st holiday and schedule (format: month /day /schedule)	Word	R/W
7E3FH~7E41H	The 2nd holiday and schedule	Word	R/W
7E42H~7E44H	The 3rd holiday and schedule	Word	R/W
7E45H~7E47H	The 4th holiday and schedule	Word	R/W
7E48H~7E4AH	The 5th holiday and schedule	Word	R/W
7E4BH~7E4DH	The 6th holiday and schedule	Word	R/W
7E4EH~7E50H	The 7th holiday and schedule	Word	R/W
7E51H~7E53H	The 8th holiday and schedule	Word	R/W
7E54H~7E56H	The 9th holiday and schedule	Word	R/W
7E57H~7E59H	The 10th holiday and schedule	Word	R/W
7E5AH~7E5CH	The 11th holiday and schedule	Word	R/W
7E5DH~7E5FH	The 12th holiday and schedule	Word	R/W
7E60H~7E62H	The 13th holiday and schedule	Word	R/W
7E63H~7E65H	The 14th holiday and schedule	Word	R/W
7E66H~7E68H	The 15th holiday and schedule	Word	R/W
7E69H~7E6BH	The 16th holiday and schedule	Word	R/W
7E6CH~7E6EH	The 17th holiday and schedule	Word	R/W
7E6FH~7E71H	The 18th holiday and schedule	Word	R/W

Table 67. Data Address of Ten-Year Holiday (continued).

The 10th Year Holiday Address Function: 03H Read 10H Preset

Address	Parameter	Data Type	Type of Access
7E72H~7E74H	The 19th holiday and schedule	Word	R/W
7E75H~7E77H	The 20th holiday and schedule	Word	R/W
7E78H~7E7AH	The 21st holiday and schedule	Word	R/W
7E7BH~7E7DH	The 22nd holiday and schedule	Word	R/W
7E7EH~7E80H	The 23rd holiday and schedule	Word	R/W
7E81H~7E83H	The 24th holiday and schedule	Word	R/W
7E84H~7E86H	The 25th holiday and schedule	Word	R/W
7E87H~7E89H	The 26th holiday and schedule	Word	R/W
7E8AH~7E8CH	The 27th holiday and schedule	Word	R/W
7E8DH~7E8FH	The 28th holiday and schedule	Word	R/W
7E90H~7E92H	The 29th holiday and schedule	Word	R/W
7E93H~7E95H	The 30th holiday and schedule	Word	R/W
7E96H	The 10th setting year	Word	R/W
7E97H	Holiday number of the 10th year	Word	R/W

Table 69. Event Trigger Settings

Address	Parameter	Data Type	Property	Range	Default	Factory Setting
8000H	Manual Triggering Waveform	Word	R/W	0XAA: Enable; 0: Disable	0	0
8001H	DI Triggering - PXM1K-11	Word	R/W	bit1bit0: DI1; bit3bit2: DI2; bit5bit4: DI3; bit7bit6: DI4; bit9bit8: DI5; bit11bit10: DI6; 00: Disable; 01: From OFF to ON; 10: From ON to OFF; 11: Any DI state change;	0	0
8002H	DI Triggering - PXM1K-21	Word	R/W	Bit1bit0: DI7; bit3bit2 : DI8; bit5bit4: DI9; bit7bit6: DI10; The same as above	0	0
8003H	DI Triggering - PXM1K-31	Word	R/W	Bit1bit0: DI11; bit3bit2 : DI12; bit5bit4: DI13; bit7bit6: DI14; The same as above	0	0
8004H	Voltage Rated Value	Word	R/W	50 – 400 V or 50 – 690 V (only in 3LL)	400	400
8005H	Voltage Sag Triggering Waveform	Word	R/W	1: Enable; 0: Disable	0	0
8006H	Voltage Sag Threshold	Word	R/W	20 - 100(%)	50	50
8007H	Voltage Sag Half Cycle Count	Word	R/W	4 - 200 half cycles	10	10
8008H	Voltage Swell Triggering Waveform	Word	R/W	1: Enable; 0: Disable	0	0
8009H	Voltage Swell Threshold	Word	R/W	50 - 140(%)	100	100
800AH	Reserved	Word	R/W			
800BH	Current Rated Value	Word	R/W	500 - 5,000 (mA)	5,000	5,000
800CH	Over-current Triggering Waveform	Word	R/W	1: Enable; 0: Disable	0	0
800DH	Over-current	Word	R/W	50 - 150(%)	100	100
	Threshold					
800EH	Clear Waveform	Word	R/W	0X55 Enable	0	0
800FH	Clear Power Quality Event	Word	R/W	0X55 Enable	0	0

Note: In 3LL and 2LL, the voltage rated value is line voltage; in 3LN, the voltage rated value is phase voltage.

Waveform Capture Data Retrieve Address

Waveform Capture includes the timestamp, triggering condition, and waveform data. Every group uses the same data format. Only one group of waveforms are saved in the registers. When retrieving the waveform, first write the 1 - 8 group number into 0X801FH, then read the registers after the 8 group to acquire the waveform corresponding to the written group number.

The relationship between voltage waveform value and real value:

1. 5 A, 1 A: Real Value (Unit: A) = Waveform Value/1683.153;
2. 333 mV: Real Value (Unit: A) = Waveform Value/K (firmware above 3.21, K=14427.15; other: K = 15869.87);
3. 100 mV (Rogowski coil): Real Value (Unit: A) = Waveform Value/K (firmware above 3.21, K = 20291.1; firmware 3.20, K = 22068.8, other: K = 15869.87);
4. mA CT: Real Value (Unit:A) = Waveform Value/K (80 mA CT:K = 7414.289; 100 mA: K = 9267.440; 200 mA: K = 18514.68).

The relationship between the current waveform value and the real value:

Real Value (unit: V) = Waveform Value x 0.01463866.

The voltage and current value obtained from the waveform are the PT or CT secondary side value.

Read: FC03, Preset: FC16. For more information, please refer to Chapter 5.7.

Table 70. Event List

Address	Parameter	Data Type	Property	Range	Default	Factory Settings
8CFDH	Newest Event Group Number	Word	R	1 - 50,000; 0: No Data	0	0
8CFEH	Event for Retrieving Starting Group Number	Word	R/W	1 - 50,000 **Note:** Only valid smaller or equal to the Newest Event Group Number.	1	1
8CFFH						
No.1 Event						
8D00H	Timestamp High Byte – Year Low Byte - Month	Word	R	Time		
8D01H	Timestamp High Byte – Day Low Byte - Hour	Word	R	Time		
8D02H	Timestamp High Byte – Minute Low Byte - Second	Word	R	Time		
8D03H	Timestamp: Millisecond	Word	R	Time		
8D04H	Voltage Sag or Voltage Swell Condition	Word	R	0: Disabled; 1: Voltage Sag; 2: Voltage Swell		
8D05H	Rated Value	Word	R	50 – 400 V or 50 – 690 V (Only in 3LL)		
8D06H	Threshold	Word	R	Voltage Sag: 20 - 100 (%) Voltage Swell: 50 - 140 (%)		
8D07H	Half-cycle Count	Word	R	Voltage Sag Event: 4 - 200; Voltage Swell Event: 0		
8D08H-8D0FH	No. 2 Event	Word	R			
8D10H-8D17H	No. 3 Event	Word	R			
8D18H-8D1FH	No. 4 Event	Word	R			
8D20H-8D27H	No. 5 Event	Word	R			
8D28H-8D2FH	No. 6 Event	Word	R			

Power Quality Event Retrieve Address

A power quality event includes timestamp, triggering condition, and related settings. Every group uses the same data format. Only ten groups of data are saved in the registers. When retrieving the event data, its parameters must be correctly set in order to get correct information.

Read: FC03, Preset: FC16. For more information, please refer to Chapter 5.7.

Appendix A – Technical Data and Specification

Input Ratings

Energy Accuracy (PXM 3000)

Active Power	
(According to IEC 62053-22)	Class 0.2S
(According to ANSI C12.20)	Class 0.2
Reactive	
(According to IEC 62053-23)	Class 2

Harmonic Resolution

Metered Value	2nd ~ 63rd (50 or 60 Hz type)
	2nd ~ 15^{th}(400 Hz type)

Voltage Input

Nominal Full Scale	400 Vac L-N, 690 Vac L-L
Withstand	1,500 Vac continuous; 3,250 Vac, 50/60 Hz for 1 minute
Metering Frequency	45 ~ 65 Hz
Pickup Voltage	10 Vac
Burden	0.05 VA (typical) @ 5 Arms
Accuracy	0.1% IEC61557-12 from 20%-120% of Nominal Full Scale

Current inputs (each channel)
Nominal secondary sensor settings:

Current Sensor Input Options	5A	1A
Nominal Configuration Selection	5A	1A
Metering range (%of nominal)	200%	200%
Pickup current (%of nominal)	0.1%	0.1%
Withstand	20 A rms continuous, 100 A rms for 1 second, non-recurring	
Burden	0.05 VA (typical) @ 5 A rms	
Accuracy	0.1% IEC61557-12 from 10%-120% of Nominal Selection	

Accuracy

Parameters		Accuracy	Resolution	Range
Voltage		0.1%	0.1 V	10 V ~ 1000 kV
Current		0.1%	0.001 A	5 mA ~ 50000 A
Power		0.2%	1 W	-9999 MW ~ 9999 MW
Reactive Power		0.2%	1 Var	-9999 MVar ~ 9999 MVar
Apparent Power		0.2%	1 VA	0 ~ 9999 MVA
Power Demand		0.2%	1 W	-9999 MW ~ 9999 MW
Reactive Power Demand		0.2%	1 Var	-9999 MVar ~ 9999 MVar
Apparent Power Demand		0.2%	1 VA	0 ~ 9999 MVA
Power Factor		0.2%	0.001	-1.000 ~ 1.000
Frequency		0.2%	0.01Hz	45.00 ~ 65.00 Hz (50Hz or 60Hz type) 300.00 ~ 500.00 (400 type)
Energy	Primary	0.2%	0.1 kWh	0 ~ 99999999.9 kWh
	Secondary	0.2%	0.001 kWh	0 ~ 999999.999 kWh
Reactive Energy	Primary	0.2%	0.1 kvarh	0 ~ 99999999.9 kvarh
	Secondary	0.2%	0.001 kvarh	0 ~ 999999.999 kvarh
Apparent Energy	Primary	0.2%	0.1 kVAh	0 ~ 99999999.9 kVAh
	Secondary	0.2%	0.001 kVAh	0 ~ 999999.999 kVAh
Harmonics		1.0%	0.1%	> = 0.0%
Phase Angle		2.0%	0.1°	0.0° ~ 359.9°
Imbalance Factor		2.0%	0.1%	0.0% ~ 100%
Running Time		<1 second/day	0.01 h	0 ~ 9999999.99 h
Temperature Drift		<100 ppm/°C		

Control Power

AC/DC Control Power

Operating Range	100 - 300 Vac, 50/60 Hz; 100 - 300 Vdc
Burden	10 W
Withstand	3250 Vac, 50/60 Hz 1 min
	Installation Category III (Distribution)

I/O Options

Digital Input

Input Voltage Range	20 - 160 Vac/dc
Input Current (Max)	2 mA
Start Voltage	15 V
Stop Voltage	5 V
Pulse Frequency (Max)	100 Hz, 50% Duty Ratio
SOE Resolution	2 ms

Digital Output (DO) (Photo-MOS)

Voltage Range	0 - 250 Vac/dc
Load Current	100 mA (Max)
Output Frequency	25 Hz, 50% Duty Ratio
Isolation Voltage	2500 Vac

Relay Output (RO)

Switching Voltage (Max)	250 Vac, 30 Vdc
Load Current	5 A (resistive), 2 A (inductive)
Set Time	10 ms (Max)
Contact Resistance	30 mΩ (Max)
Isolation Voltage	2500 Vac
Mechanical Life	1.5×10^7

Analog Output (AO)

Output Range	0 - 5 V/1 - 5 V, 0 - 20 mA/4-20 mA (Optional)
Accuracy	0.5%
Temperature Drift	50 ppm/°C typical
Isolation Voltage	500 Vdc
Open Circuit Voltage	15 V

Analog Input (AI)

Input Range	0 - 5 V/1-5 V, 0 - 20 mA/4 - 20 mA (Optional)
Accuracy	0.2%
Temperature Drift	50 ppm/°C typical
Isolation Voltage	500 Vdc

Power Supply for DI (24Vdc)

Output Voltage	24 Vdc
Output Current	42 mA
Load (Max)	21 DIs

General Specifications

Product Specifications

Dimensions (mm)	96 x 96 x 51 (Cut-out 92 x 92 or 4-inch Round)
Protection Level	IP54 (Front), IP30 (Cover)
Weight (g) [oz]	350 g [11.25 oz]
Temperature	-25°C ~ 70°C (-13°F ~ 158°F), Metering -40°C~85°C (-40°F ~ 185°F), Storage
Humidity	5% ~ 95% Non-condensing
Power Supply	100 - 300 Vac, 50/60 Hz; 100 - 300 Vdc; Category III, Pollution Degree 2
Power Consumption	10 W (including PXM 3000-GC and optional modules)

General Specificatons (Continued)

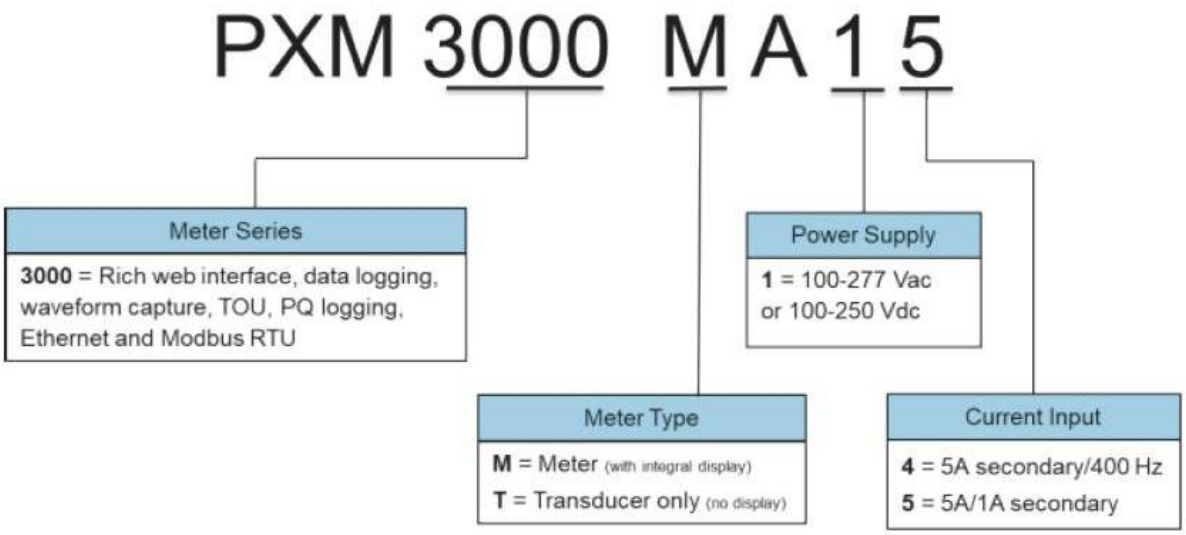
Standard Compliance

Measurement Standard	IEC 62053-22; ANSI C12.20
Environmental Standard	IEC 60068-2
Safety Standard	IEC 61010-1, UL 61010-1
EMC Standard	IEC 61000-4/-2-3-4-5-6-8-11, CISPR 22
Outlines Standard	DIN 43700, ANSI C39.1

Communication

RS-485 (Standard)	2-wire connection
	MODBUS RTU Protocol
	Up to 115200 baud rate

Appendix B – Ordering Information



Notes:

Eaton
1000 Eaton Boulevard
Cleveland, OH 44122
United States
Eaton.com

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